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WordPub

BECE

(Integrated) Science

Past Questions & Solutions

1990 to 2013

Junior High School

Years 1, 2 & 3

Compiled by:

WORD PUBLISHERS

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WHATSAPP US ON 0201054428

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- ✓ Teachers, parents and educational institutions – for your tireless efforts at ensuring quality education for the 21st century child.

- ✓ Junior high school students – this is for you.

DEDICATION

To the Lord Jesus Christ, our soon-coming King

CLICK ON ANY YEAR OF YOUR CHOICE TO STUDY.
FROM THE LIST BELOW

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June 2013

INTEGRATED SCIENCE 1

OBJECTIVE TEST

1 hour

1. The S.I unit of temperature is

A. candela
B. degree Celsius
C. joule
D. kelvin

2. Air is an example of

A. gas in gas mixture
B. liquid in liquid mixture
C. solid in liquid mixture
D. solid in solid mixture

3. Which of the following activities are involved in the rearing of tilapia?

I. Feeding of fish
II. Pond stocking
III. Pest control

A. I and II only
B. I and III only
C. II and III only
D. I, II and III

4. The part of the flower that contains nectar is called

A. anther
B. petal
C. ovary
D. sepal

5. Which of the following methods protects pure iron from rusting by coating with zinc?

A. Alloying
B. Galvanizing
C. Greasing
D. Painting

6. Leaching of nutrients from the soil leads to soil

A. acidity

B. alkalinity
C. erosion
D. porosity

7. The hereditary material that is passed on from parents to offspring is known as
A. gene
B. sperm
C. red blood cell
D. white blood cell

8. The solar system is made up of the
A. earth, moon and stars
B. sun, comets and meteors
C. sun, earth and moon
D. sun, planets and other heavenly bodies

9. The type of management system which allows farm animals to roam about freely is known as
A. extensive system
B. intensive system
C. semi-extensive system
D. semi-intensive system

10. External respiration in living organisms is also known as
A. aerobic respiration
B. anaerobic respiration
C. gaseous respiration
D. tissue respiration

11. A rigid bar which is capable of turning about a fixed point is a / an
A. inclined plane
B. lever
C. screw
D. wheel and axle

12. A positively charged ion is called
A. anion
B. cation
C. neutron
D. proton

13. One way of ensuring that organisms are **not** endangered is to
A. allow them to be hunted
B. allow their predators to outnumber them
C. protect their habitats
D. pollute their habitats

14. A good thermometric liquid must

A. be colourless
B. boil at 100°C and freeze at 0°C
C. cling to the walls of the glass
D. expand evenly and regularly

15. Which of the following metals will produce a gas when placed in lime juice?

A. Copper
B. Lead
C. Magnesium
D. Silver

16. The physical property of a soil which is determined by the feel method is soil

A. structure
B. temperature
C. texture
D. water

17. The reason why gaps are left in the joints of railway lines is to allow for

A. cooling
B. contraction
C. expansion
D. maintenance

18. Which of the following effects are caused by rusting?

I. Loss of strength
II. Loss of structure
III. Loss of electrical conductivity

A. I and II only
B. I and III only
C. II and III only
D. I, II and III only

19. Leguminous crops may often be cultivated to add

A. carbon to the soil
B. nitrogen to the soil
C. protein to the soil
D. potassium to the soil

20. A place where an organism can live and interbreed successfully is called

A. community
B. ecosystem
C. environment

D. habitat

21. Which of the following methods of treating water makes it soft?
A. Addition of alum
B. Addition of sodium carbonate
C. Chlorination
D. Filtration

22. Dehusking and shelling are both activities carried out in the processing of
A. cowpea
B. groundnut
C. maize
D. sorghum

23. Typhoid fever is transmitted through
A. contact with contaminated skin wounds
B. eating contaminated food
C. eating uncooked food
D. drinking treated water

24. One advantage of friction is that it
A. enables cutting tools to be sharpened
B. increases the efficiency of machines
C. produces a lot of heat in machines
D. wears off the soles of shoes

25. Sickle is a farm tool used for
A. harvesting rice
B. trimming hedges
C. transplanting seedlings
D. watering crops

26. One characteristic of the image formed in a pin-hole camera is that the image is
A. diminished
B. magnified
C. upright
D. virtual

27. Transplanting of seedlings is usually done in the evening because
A. darkness promotes rapid growth
B. pest attack is minimal
C. transpiration is minimal
D. seedlings require less nutrients

28. Which of the following statements about molecules is/ are correct? Molecules

I. are chemically combined group of atoms
II. are physically combined group of atoms
III. can exist on their own

A. I only
B. II only
C. I and III only
D. I, II and III

29. Fruits which are dispersed by wind are likely to be
A. dry
B. hairy
C. juicy
D. sticky

30. Which of the following materials allows electric current to pass through easily?
A. Aluminium
B. Carbon
C. Glass
D. Water

31. Water contains two elements, hydrogen and oxygen, in the ratio of
A. 1:2
B. 1:3
C. 2:1
D. 3:1

32. In the digestive system of ruminants, vitamins are synthesized in the
A. abomasum
B. omasum
C. reticulum
D. rumen

33. The **best** way of protecting oneself from high blood pressure is to
A. consume alcoholic drinks
B. eat high carbohydrate diets
C. engage in smoking
D. exercise regularly

34. The property of metals which makes them to be easily drawn into thin wires is known as
A. conductivity
B. ductility
C. malleability
D. resistivity

35. An atom of an element is represented as $^{27}_{13}X$. How many neutrons are in the nucleus of the atom?

A. 13
B. 14
C. 27
D. 40

36. Viable seeds are ones that

A. are able to germinate under suitable conditions
B. are eaten by animals after planting
C. contain a lot of oil
D. have hard seed coat

37. Which of the following processes can occur at all temperatures?

A. Boiling
B. Evaporation
C. Melting
D. Sublimation

38. Which of the following human activities maintains the carbon cycle?

A. Bush burning
B. Felling of trees
C. Release of fumes from factories
D. Replanting of trees felled as timber

39. The farming system which involves the growing of one type of crop on the same piece of land every season is known as

A. mixed cropping
B. mixed farming
C. monocropping
D. monoculture

40. The presence of chlorophyll in green plants is a necessary condition for photosynthesis because it

A. absorbs oxygen
B. absorbs solar energy
C. produces carbon dioxide
D. produces water vapour

June 2013

INTEGRATED SCIENCE 1

SOLUTIONS

OBJECTIVE TEST

1.	D.	Kelvin
2.	A.	gas in gas mixture
3.	D.	I, II and III
4.	B.	petal
5.	B.	Galvanizing
6.	A.	acidity
7.	A.	gene
8.	D.	sun, planets and other heavenly bodies
9.	A.	extensive system
10.	C.	gaseous respiration
11.	B.	lever
12.	B.	cation
13.	C.	protect their habitats
14.	D.	expand evenly and regularly
15.	C.	Magnesium
16.	C.	texture
17.	C.	expansion
18.	D.	I, II and III only
19.	B.	nitrogen to the soil
20.	D.	habitat

21. B. Addition of sodium carbonate

22. B. groundnut

23. B. eating contaminated food

24. A. enables cutting tools to be sharpened

25. A. harvesting rice

26. A. diminished

27. C. transpiration is minimal

28. C. I and III only

29. B. hairy

30. A. Aluminium

31. C. 2:1

32. D. rumen

33. D. exercise regularly

34. B. ductility

35. B. 14

36. A. are able to germinate under suitable conditions

37. B. Evaporation

38. D. Replanting of trees felled as timber

39. C. monocropping

40. B. absorbs solar energy

June 2013

INTEGRATED SCIENCE 2

Essay

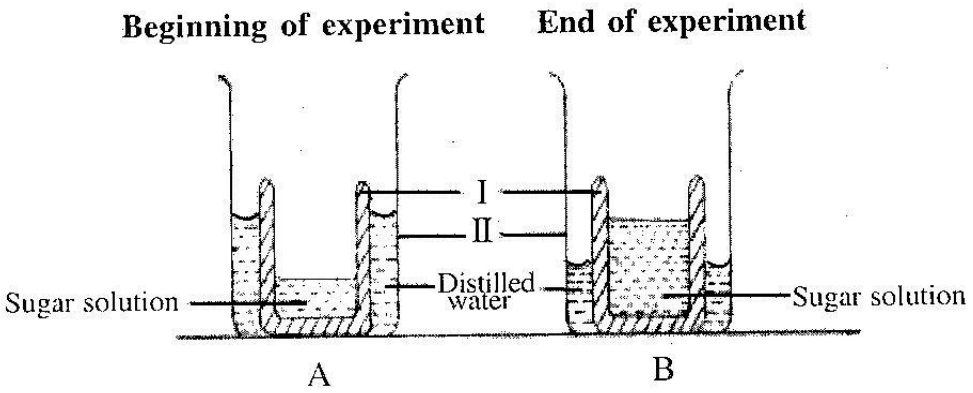
1 ¼ hours

PART I
[40 marks]

Answer Question **1**

1. (a) The diagrams below are illustrations of an experiment to demonstrate a biological principle

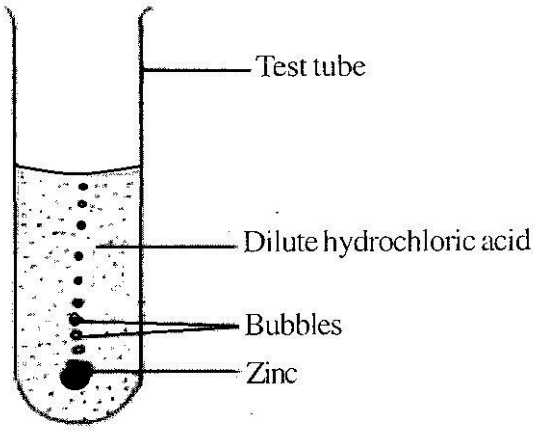
Study the diagrams carefully and answer the questions that follow



(i) Name the parts labelled **I** and **II**
(ii) State **two** difference between the set-ups **A** and **B**
(iii) What is the role played by the part labelled I in the experiment?
(iv) Name the biological principle being demonstrated in this experiment
(v) State **one** way in which plants benefit from the principle named in (iv)
(vi) State **one** way in which animals benefit from the principle named in (iv)

[10 marks]

(b) In an experiment to investigate the reactivity of zinc, a piece of the metal was dropped into a test tube containing dilute hydrochloric acid. The experimental set-up is illustrated below.

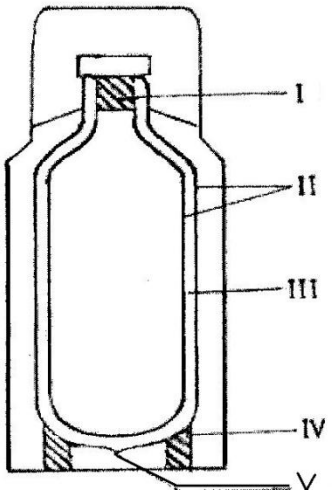


Study the set-up carefully and answer the questions that follow.

(i) Write a balanced chemical equation for the reaction that occurred in the experiment.
(ii) Name the gas evolved.
(iii) List **two** metals which can react in a similar way as the zinc
(iv) List **two** metals which cannot react in a similar way as the zinc.
(v) Name **two** glass apparatus which could have been used instead of the test tube.

[10 marks]

(c) The diagram below is an illustration of a thermos flask

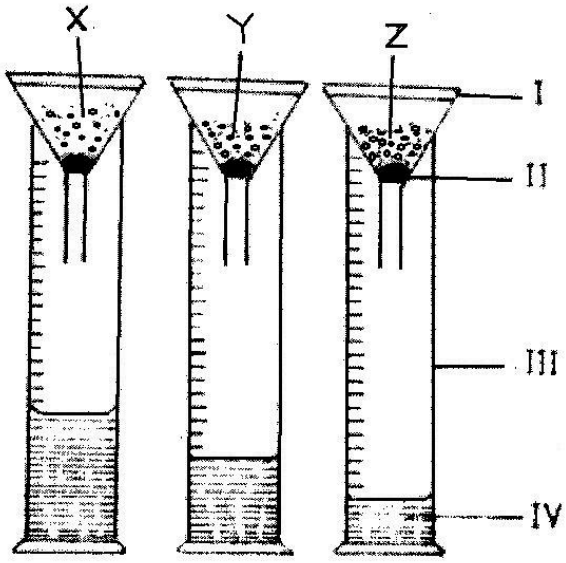


Study the diagram carefully and answer the questions that follow

(i) Name the parts labelled **I, II, III, IV** and **V**
(ii) How does the device minimize heat loss or gain through
  (α) conduction ?
  (β) convection ?
  (γ) radiation ?
(iii) State **one** use of the thermos flask

[10 marks]

(d) The diagrams below illustrate an experimental set-up on a physical property of soil using three soil types, **X, Y** and **Z**



Study the set-up carefully and answer the questions that follow:

(i) Name the parts of the set-up labelled **I, II, III** and **IV**

(ii) Which of the three soil types has the

(α) **highest** water holding capacity?

(β) **least** water holding capacity?

(iii) Name **each** of the **three** soil types **X, Y** and **Z**

(iv) Suggest a suitable title for the experiment

[10 marks]

PART II
[60 marks]

Answer **four** *questions* **only** *from this part*

2. (a) List the **three** particles which make up matter [3 marks]

(b) State **four** hereditary features in humans [4 marks]

(c) State the energy transformation that takes place in **each** of the following activities:

(i) dry cell in use;

(ii) solar panel in use;

(iii) electric stove in use;

(iv) hammering of a piece of metal

[4 marks]

(d) State **two** ways **each** in which **each** of the following cultural practices is important in vegetable production:

(i) staking

(ii) pruning

[4 marks]

3. (a) (i) What is *indiscriminate sex?*

(ii) Give **two** reasons why teenagers indulge in indiscriminate sex.

[4 marks]

(b) Name **two** sources **each** of

(i) natural light

(ii) artificial light

[4 marks]

(c) State **three** ways in which soil texture is important in crop production.

[3 marks]

(d) Write down the systematic name of **each** of the following chemical compounds:

(i) FeS

(ii) CO

(iii) Cu_2O

(iv) NaOH [4 marks]

4. (a) (i) What do the symbols **L, N** and **E** represent in an electric plug?

(ii) What is the function of the fuse box in household electrical wiring?

[4 marks]

(b) Mention **four** classes of insect pest based on their feeding habits [4 marks]

(c) Classify the **first** four elements of the periodic table as *metals* and *non-metals*

[4 marks]

(d) Name the **three** types of blood vessels in humans

[3 marks]

5. (a) (i) State the difference between *organic fertilizer* and *inorganic fertilizer*

(ii) State two effects of *inorganic fertilizer* on the environment

[4 marks]

(b) Classify the following substances as *acids* or *bases*.
(i) unripe lemon juice;
(ii) wood ash;
(iii) liquid in a car battery;
(iv) bicarbonate of soda [4 marks]

(c) (i) What is a *fruit*?
(ii) State **two** differences between a *fruit* and a *seed*. [4 marks]

(d) State the effect of heat on **each** of the following substances:
(i) plastics
(ii) alcohol
(iii) metal rod
[3 marks]

6. (a) Classify the following chemical substances based on their uses under the headings as shown in the table below:
Milk of magnesia, alcohol, paracetamol, sodium hydroxide, N.P.K.

Agriculture	**Industry**	**Medicine**

[5 marks]

(b) State **one** use **each** of the following instruments used in the study of the weather:
(i) rain gauge;
(ii) hygrometer;
(iii) anemometer. [3 marks]

(c) (i) Name **two** types of transistors.
(ii) Draw and label the circuit symbols of the transistors named in (i)
[4 marks]

(d) State **three** reasons why vegetable farming is important.
[3 marks]

June 2013

INTEGRATED SCIENCE 2

SOLUTIONS

Essay

1. (a) (i) **PART** **NAME**

I - A hollow yam or potato or pawpaw cup or semi-permeable container

II - A beaker or trough

(ii) Differences between set-ups A and B

Set-up A	Set-up B
The sugar solution is concentrated	The sugar solution is dilute
The level of the distilled water is high	The level of the distilled water is low
The amount of sugar solution is smaller	The amount of sugar solution is greater

(iii) Role played by part I

It serves as a semi-permeable membrane – to allow the movement of water molecules from the water into the sugar solution.

(iv) Biological principle demonstrated

Osmosis

(v) How plants benefit from the named principle (osmosis)

- Absorption of water by the roots of plants
- Transportation of water from the roots to the other parts of the plant
- Movement of water from one plant cell to the other

(vi) How animals benefit from osmosis

- Movement of water into the cytoplasm of some organisms such as amoeba
- Re-absorption of water in the kidney tubules of mammals

(b) (i) Balanced equation for the reaction

$Zn + 2HCl \longrightarrow ZnCl_2 + H_2$

(ii) The gas evolved

Hydrogen gas or H_2 (gas)

(iii) Metals that can react in a similar way as the zinc

Lithium, Magnesium, Potassium, Sodium, Calcium, Aluminium, Iron

(iv) Metals that cannot react in a similar way as the zinc

Copper, silver, gold, mercury, platinum

(v) Glass apparatus that could have been used.

Beaker, conical flask, measuring cylinder, flat bottomed flask, gas jar

(c) (i) Parts

I - Cork or plastic/rubber stopper
II - Silvered or shiny surfaced double wall
III - Vacuum or empty space
IV - Cork support or plastic/rubber support
V - Vacuum seal

(ii) How the device minimizes heat loss or gain through

(α) conduction

Minimized by the cork or plastic/rubber stopper and Cork support or plastic/rubber support

(β) convection

Minimized by the vacuum

(γ) radiation

Minimized by the silvered or shiny surfaced double wall

(iii) Use of the thermos flask

- Maintains the temperature of its contents for a relatively long time, ie, it keeps its hot contents hot and its cold contents cold
- It prevents heat loss or heat gain of its contents for a long period.

(d) (i) PARTS

I - funnel
II - cotton wool
III - measuring cylinder
IV - water

(ii) **(α)** Highest water holding capacity - **Z**

(β) Least water holding capacity - **X**

(iii) **SOIL TYPES**

X - Sandy

Y - Loamy

Z - Clayey

(iv) **Suitable title for experiment**

- Experiment to compare the water-holding capacities of sandy, loamy and clayey soils
- Experiment to demonstrate the drainage abilities of sandy, loamy and clayey soils.

2. **(a)** **Particles which make up matter**

- Protons, Neutrons and Electrons

(b) **Hereditary features in humans**

- Shape of nose
- Size of ear
- Height
- Colour of skin
- Mass
- Colour or eyes
- Temperament
- Intelligence

(c) **Energy transformations in the following activities:**

(i) **dry cell in use;**

Chemical energy ⟶ electrical energy

(ii) **solar panel in use;**

Solar / Light energy ⟶ electrical energy

(iii) **electric stove in use;**

Electrical energy ⟶ heat energy + light energy

(iv) **hammering of a piece of metal**

Kinetic energy ⟶sound energy + light energy

(d) **Importance of the following cultural practices in vegetable production:**

(i) **staking**

- Gives the plants the ability to grow without bending or breaking
- Prevents fruits from getting rotten (as they lie on the ground)

- Enables the plant to get sufficient sunlight needed for healthy growth
- Creates space in-between plants for easier movement of the farmer.

(ii) pruning

- Increases the quality and quantity of crop yield
- Helps to check the spread of pest and diseases
- Enables the plant to get sufficient sunlight needed for healthy growth
- Enhances healthier growth of the plant
- Improves ventilation for the plant

3. (a) (i) *Indiscriminate sex*

Having sexual intercourse with multiple (two or more) partners and usually without protection (use of condom)

(ii) Reasons why teenagers indulge in indiscriminate sex.

- Curiosity
- Peer pressure
- Poverty
- Illiteracy or ignorance of consequences
- Lack of sufficient recreational avenues
- Broken homes
- Irresponsible parenting
- Low self esteem
- Fallen moral standards of society.
- Lack of self-control

(b) Sources of

(i) natural light

- Sun
- Stars
- Lightning
- Volcanic eruption
- Firefly
- Glow worm
- Anglerfish
- lanternfish

(ii) artificial light

- Moon
- Electric bulb
- Torch
- Matches

- Vehicle headlamps
- Light-emitting diodes (LEDs)
- Light-emitting electrochemical cells (LECs)

(c) Ways in which soil texture is important in crop production.

- It affects soil fertility
- It affects its water-holding capacity
- It affects its nutrient retention
- It affects its plant-holding capacity
- It affects the ability of the soil to resist erosion.
- It affects soil temperature

(d) Systematic name the following chemical compounds:

(i) FeS - Iron (II) sulphide

(ii) CO - Carbon (II) oxide **or** Carbon monoxide (preferred IUPAC name)

(iii) Cu_2O - Copper (I) oxide

(iv) NaOH - Sodium hydroxide

4. (a) (i) What the symbols L, N and E represent in an electric plug

L - live pin

N - neutral pin

E - earth pin

(ii) The function of the fuse box in household electrical wiring

- The fuse box houses and controls the incoming electrical service and distribution to circuits within the house.
- It provides protection against power fluctuation through the use of fuses

(b) Four classes of insect pest based on their feeding habits

- Piercing & sucking insects (eg, aphids, mosquitoes)
- Chewing insects (eg, grasshoppers, beetles, weevils)
- Siphoning insects (eg, moths, butterflies)
- Sponging insects (eg, housefly)

(c) Classification of the first four elements of the periodic table

METALS	NON-METALS
Lithium	Hydrogen
Beryllium	Helium

(d) The three types of blood vessels in humans

- Capillaries
- Veins
- Arteries

5. (a) (i) Difference between *organic fertilizer* and *inorganic fertilizer*

ORGANIC FERTILIZER	INORGANIC FERTILIZER
Made from animal or plant matter (natural sources)	Made from chemicals (artificial sources)

(ii) Effects of *inorganic fertilizer* on the environment

- Damage and destruction of plant and animal life, when used in excess
- Contamination of water bodies when rain washes chemicals into them.
- Contributes to the formation of acid rain, when fossil fuels are burnt during their production
- Contributes to the greenhouse effect, , when fossil fuels are burnt during their production
- Nitrogen-containing fertilizers can cause soil acidification

(b) Classification of the following substances as *acids* or *bases*.

ACIDS	BASES
Unripe lemon juice	Wood ash
Liquid in a car battery	bicarbonate of soda

(c) (i) A *fruit*?

A mature ovary containing seed

(ii) Differences between a *fruit* and a *seed*.

FRUIT	SEED
A mature ovary	A mature ovule
Contains a seed	Contains an embryo
Pericarp is formed from the wall of the ripened ovary	The seed coat (testa) is formed from integuments of ovule.
Cannot germinate under any condition	Can germinate under the right conditions.

(d) The effect of heat on each of the following substances:

(i) plastics - Causes melting
(ii) alcohol - Causes boiling or faster evaporation
(iii) metal rod - Causes expansion

6. (a) Classification of:
Milk of magnesia, alcohol, paracetamol, sodium hydroxide, N.P.K.

Agriculture	Industry	Medicine
N.P.K	Alcohol Sodium hydroxide	Milk of magnesia Paracetamol

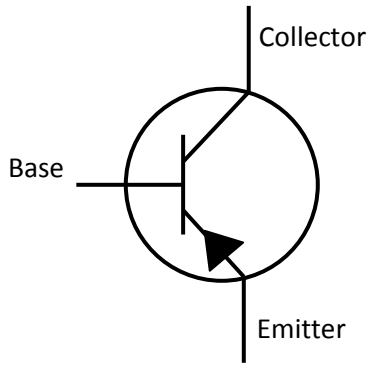
(b) One use each of the following instruments

(i) rain gauge - To measure the amount of rainfall
(ii) hygrometer - To measure the atmospheric humidity
(iii) anemometer. - To measure the speed of wind.

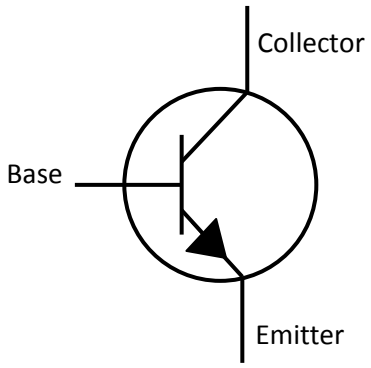
(c) (i) Types of transistors.

- PNP transistor
- NPN transistor

(ii) Circuit symbol of **PNP** transistor



(ii) Circuit symbol of **NPN** transistor



(d) Reasons why vegetable farming is important.

- Source of employment - Income generation or wealth creation
- Foreign exchange earner for the nation, when exported.
- Provision of nutrition for man - Vegetables are very rich in essential vitamins, mineral salts and proteins.
- Certain vegetables help to conserve soil fertility
- Provides raw material for the food processing industry
- Certain vegetables are used in the pharmaceutical industry
- Vegetable take a shorter time to mature, hence it provides greater returns per given time and space.

April 2012

INTEGRATED SCIENCE 1

OBJECTIVE TEST
1 hour

1. The S.I. unit for measuring the work done by a force is
A. J.
B. K.
C. N.
D. W.

2. The chemical formula of a compound describes the
A. number of molecules in the compound
B. type of bonding in the compound
C. ratio in which the elements are combined
D. state of the compound

3. Which of the following life processes is represented by the equation
Glucose + oxygen → carbon dioxide + water + energy

A. Digestion
B. Excretion
C. Photosynthesis
D. Respiration

4. Each layer of soil profile is known as
A. horizon
B. litter
C. regolith
D. solum

5. When the *p*-*n* junction of a transistor is reverse biased
A. current flows from the *p*-type to the *n*-type
B. no current flows from the *p*-type to the *n*-type
C. conduction of current occurs
D. current flows from the *n*-type to the *p*-type

6. When a solid-liquid mixture is filtered, the liquid that separates out into the container is called
A. filtrate
B. residue

C. sediment
D. solution

7. Which of the following processes involves the solid state of matter?
A. Boiling
B. Condensation
C. Evaporation
D. Melting

8. Which of the following farming systems is **most** effective in maintaining soil fertility?
A. Crop rotation
B. Land rotation
C. Mixed cropping
D. Monoculture

9. The disease in humans which is associated with insufficient intake of calcium is
A. goiter
B. kwashiorkor
C. rickets
D. scurvy

10. The arrow in the circuit symbol of **either** *n-p-n* **or** *p-n-p* transistor is always on the
A. base lead
B. collector lead
C. emitter lead
D. receiver lead

11. Which of the following insect pests of crops has piercing and sucking mouthparts?
A. Aphids
B. Grasshoppers
C. Stemborers
D. Termites

12. A reflex action involves the
A. brain and muscles
B. brain and nerves
C. spinal cord and muscles
D. spinal cord and nerves

13. The type of image formed in a plane mirror is always
A. diminished
B. enlarged
C. real
D. virtual

14. Which of the following statements about acids are **correct**?

I. They turn red litmus paper blue
II. They can be classified as either organic or mineral acids
III. They can be neutralized by bases

A. I and II only
B. I and III only
C. II and III only
D. I, II and III

15. A transistor is said to operate in an active region when

A. one *p*-*n* junction is forward biased and the other is reversed biased
B. base-emitter junctions are reversed biased
C. both *p*-*n* junctions are reversed biased
D. base-collector junction is forward biased

16. *Tuberculosis* is spread

A. through eating of contaminated food
B. when an infected person coughs openly into the air
C. through shaking of hands of infected persons
D. through sharing of contaminated syringes

17. One function of engine oil in the engine of a tractor is to

A. warm the engine
B. enhance air intake
C. ensure proper mixing of fuel
D. lubricate the engine parts

18. *Black pod* is a disease of

A. cocoa
B. coffee
C. guava
D. mango

19. The efficiency of a machine is always less than 100% because part of the energy input is used to

A. stop the machine after working
B. perform useful work on the load
C. overcome friction
D. lift the machine up

20. Which of the following electronic components are used to produce oscillator circuits?

I. Transistor
II. Inductor
III. Capacitor

A. I and II only
B. I and III only
C. II and III only
D. I, II and III

21. Which of the following sources of light is natural?
A. Filament bulb
B. Fluorescent tube
C. Glow worm
D. Fire cracker

22. Which of the following substances are carried by the blood?
I. Hormones
II. Urine
III. Oxygen
IV. Carbon dioxide

23. The practice of starting new organization in response to identified opportunities is termed
A. agribusiness
B. business enterprise
C. entrepreneurship
D. management

24. The sub-atomic particle with zero charge in the nucleus of an atom is called
A. electron
B. ion
C. neutron
D. proton

25. Which of the following management practices greatly helps in record keeping?
A. Culling
B. Debeaking
C. Dehorning
D. Identification

26. In the pin-hole camera, when the size of the pin-hole is increased, the image formed is
A. blurred
B. erect
C. magnified
D. virtual

27. Non-reactive metals are preferred in making ornaments and jewellery because they
A. are attractive
B. do not react with atmospheric oxygen
C. do not retain their lustre

D. are corrosive

28. Producers in an ecosystem are plants that
A. attract insects
B. feed on other plants
C. manufacture their own food
D. trap insects

29. The use of resistant varieties of crop in controlling diseases is described as
A. biological control method
B. chemical control method
C. cultural control method
D. physical control method

30. The chemical formula for aluminium oxide is represented as Al_xO_y. The values of ***x*** and ***y*** are respectively
A. 3 and 2
B. 3 and 1
C. 1 and 3
D. 2 and 3

31. In electrical circuits, the component that protects appliances against very high currents is the
A. capacitor
B. fuse
C. resistor
D. switch

32. One benefit of technology to industrialization is
A. environmental pollution
B. reduction in skilful labour
C. provision of machinery
D. increase in cost of production of goods

33. Which of the following factors contribute to early parenthood?
I. Poverty
II. Illiteracy
III. Lack of parental care
IV. Peer pressure

A. I and II only
B. II and III only
C. III and IV only
D. I, II, III and IV

34. Which of the following examples of fertilizers improves soil texture?

A. Compost
B. NPK
C. Sulphate of ammonia
D. Urea

35. The reaction between hydrochloric acid and sodium hydroxide produces
A. sodium oxide only
B. sodium chloride only
C. sodium oxide and water
D. sodium chloride and water

36. In modern electrical wirings the colour code for the live wire is
A. blue
B. brown
C. green
D. green and yellow

37. An example of soil minor nutrient is
A. calcium
B. iron
C. nitrogen
D. phosphorus

38. The complete development of a human foetus in the womb normally takes
A. 7 months
B. 8 months
C. 9 months
D. 10 months

39. A material that allows a small amount of light energy to pass through it but **cannot** be seen through is referred to as
A. opaque
B. reflective
C. translucent
D. transparent

40. The leads of the transistor responsible for activation is the
A. amplifier
B. base
C. collector
D. emitter

April 2012

INTEGRATED SCIENCE 1

SOLUTIONS

OBJECTIVE TEST

1. C. N.

2. C. ratio in which the elements are combined

3. D. Respiration

4. A. horizon

5. B. no current flows from the *p*-type to the *n*-type

6. A. filtrate

7. D. Melting

8. A. Crop rotation

9. C. rickets

10. D. receiver lead

11. A. Aphids

12. D. spinal cord and nerves

13. D. virtual

14. C. II and III only

15. A. one *p*-*n* junction is forward biased and the other is reversed biased

16. B. when an infected person coughs openly into the air

17. D. lubricate the engine parts

18. A. cocoa

19. C. overcome friction

20. D. I, II and III

21. C. Glow worm

22. D. I, III and IV only

23. C. entrepreneurship

24. C. neutron

25. D. Identification

26. A. blurred

27. B. do not react with atmospheric oxygen

28. C. manufacture their own food

29. A. biological control method

30. D. 2 and 3

31. B. fuse

32. C. provision of machinery

33. D. I, II, III and IV

34. A. Compost

35. D. sodium chloride and water

36. B. brown

37. B. iron

38. C. 9 months

39. C. translucent

40. B. base

April 2012

INTEGRATED SCIENCE 2

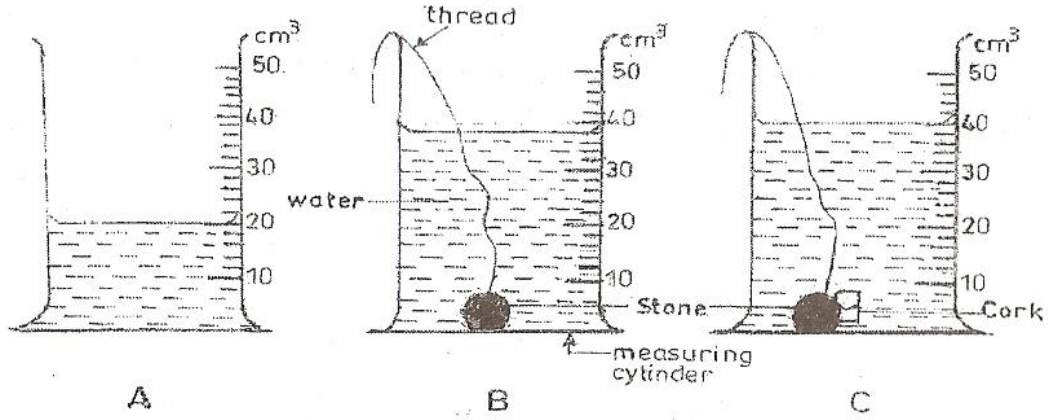
Essay

1 ¼ hours

PART I
[40 marks]

Answer Question **1**

1. (a) The diagrams below are illustrations of an experiment in the laboratory using a piece of stone, a cork of mass 4.0 g and other necessary materials.



The initial volume of water in **A** was read and noted. A string was attached to a piece of stone and the stone lowered gently into the water as shown in **B**. The volume was again read and noted. Finally, the cork of mass 4.0 g was attached to the stone and both materials lowered gently into the water as shown in **C**. The volume was read and noted.

(i) Why did the level of the water rise when the stone was lowered gently into it as shown in diagram **B**?

(ii) Why was it necessary to attach the stone to the cork before lowering it gently into the water as shown in diagram **C**?

(iii) What would have happened if the cork alone were lowered gently into the water?

(iv) What is the volume of the

(α) stone?

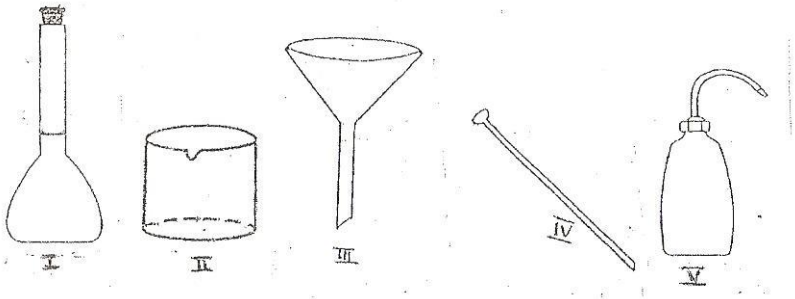
(β) cork?

(v) Calculate the density of the cork.

(vi) Why were the materials lowered gently into the water?

(b) A salt solution was prepared in the laboratory using the set of apparatus illustrated below.

Study the illustrations carefully and use them to answer the questions that follow.

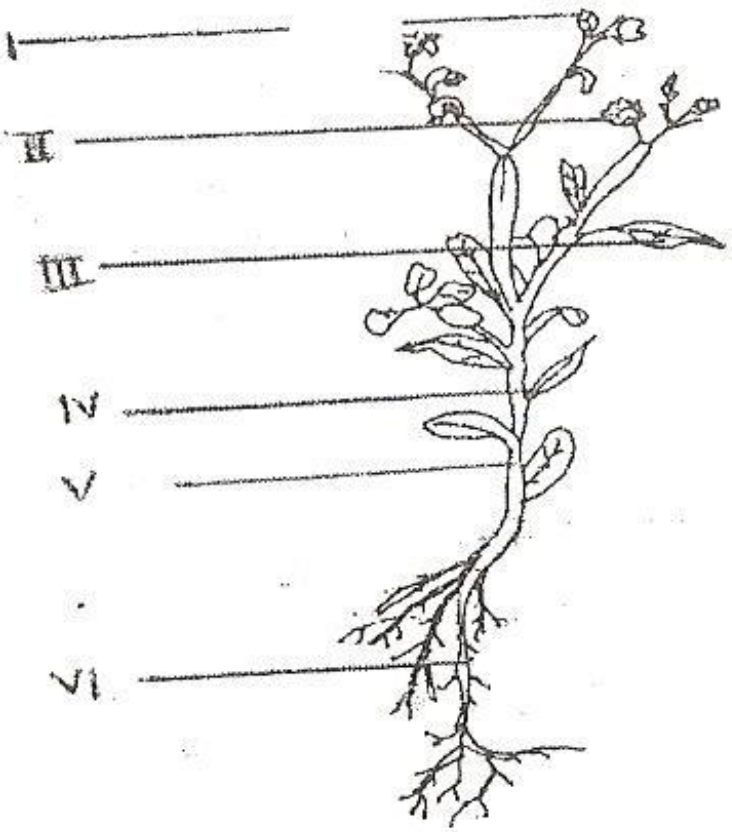


(i) Name **each** of the apparatus labeled I, II, III IV and V.

(ii) State **one** function of **each** of the apparatus labeled **I, II, III, IV** and **V**.

(c) The diagram below is an illustration of the external features of a flowering plant.

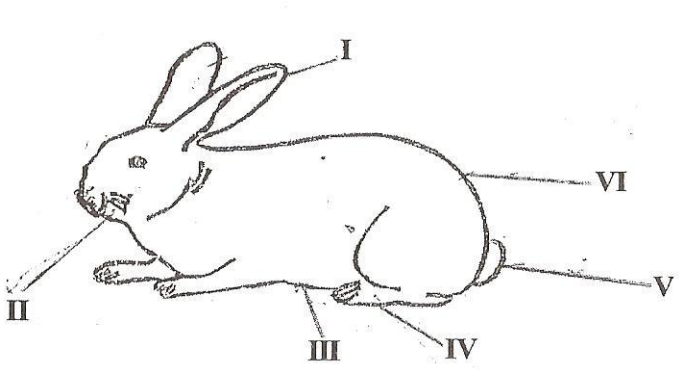
Study it carefully and use it to answer the questions that follow



(i) Name the parts labeled **I, II, III, IV, V** and **VI**
(ii) State **one** function of **each** of the parts labeled **I, II, III, V** and **VI**
(iii) State the **two** main parts of a flowering plant

(d) The diagram below is an illustration of a small farm animal.

Study it carefully and use it to answer the questions that follow



(i) Identify the animal
(ii) Name **each** of the parts of the animal labeled **I, II, III, IV, V** and **VI**
(iii) Name the structure in which the animal is kept
(iv) Mention **three** breeds of the animal.
(v) State **two** management practices to be adopted in order to control diseases and pests in the rearing of the animal.

PART II
[60 marks]

Answer **four** *questions* **only** *from this part*

2. (a) (i) What is *technology*?
(ii) State **one** use of technology in communication

(b) Write and balance **each** of the following chemical equations:
(i) $Fe + O_2 \rightarrow Fe_2O_3$;
(ii) $Na + Cl_2 \rightarrow NaCl$;
(iii) $H_2 + O_2 \rightarrow H_2O$

(c) State **one** function of **each** of the following components of a typical cell:
(i) nucleus;
(ii) chloroplast;
(iii) mitochondrion

(d) Mention **four** cultural practices in vegetable crop production

(e) Name **two** agencies in food safety and quality assurance in Ghana.

3. (a) Explain why a tomato plant is likely to wilt if too much fertilizer is applied to it.

(b) (i) Give **two** differences between *electrical insulators* and *electrical conductors*.
(ii) State **two** effects of illegal electrical connections in the home.

(c) Explain **each** of the following terms as used to describe change of state of matter:
(i) condensation;
(ii) freezing

(d) (i) State **two** diseases of the circulatory system in humans
(ii) Mention **two** ways in which **each** of the diseases you have stated in (*d*) (i) can be prevented.

4. (a) (i) What is a *transistor*?
(ii) Give **two** uses of a transistor

(b) Mention the suitable solvent for **each** of the following solutes:
(i) grease;
(ii) ink stain
(iii) starch

(iv) cube sugar
(v) oil paint
(vi) iodine

(c) (i) What is a *respiratory organ*?
(ii) Name **two** structures of the respiratory system of humans.

(d) (i) What is *agricultural chain*?
(ii) Name **two** types of agricultural chain

5. (a) (i) What are *stars*?
(ii) Arrange in order, starting from the sun, the **first four** planets in the solar system.

(b) State
(i) **two** differences between plants and animals;
(ii) **two** similarities between plants and animals

(c) Explain **each** of the following farming systems:
(i) pastoral farming
(ii) ecological farming

(d) State the properties of water in terms of
(i) odour,
(ii) taste,
(iii) effect on litmus

6. (a) What are
(i) annual plants?
(ii) perennial plants?

(b) Mention **one** danger involved in **each** of the following activities in the laboratory:
(i) eating or drinking water in the laboratory;
(ii) washing hands with unknown liquid in a beaker;
(iii) walking barefooted.

(c) (i) What is a *digestive enzyme*?
(ii) Give **two** examples of digestive enzymes in humans.

(d) Give **two** differences between *conduction* and *radiation* of heat.

April 2012

INTEGRATED SCIENCE 2

SOLUTIONS

OBJECTIVE TEST

1. (a) (i) Why the level of water rose

The additional volume of the stone caused the water to be displaced upwards

Or: due to the space occupied by the stone

(ii) Why it was necessary to attach the stone to the cork

In order for the cork to be totally submerged in the water

Or: in order for the cork to stay under the water.

(iii) What would have happened if the cork alone was lowered

It would have risen back up and floated on the water

Or: it would not have been able to stay under the water.

(iv) (α) Volume of the stone = 2nd reading – 1st reading

= $38\ cm^3 - 20\ cm^3$

= $18\ cm^3$

(β) Volume of the cork = 3rd reading – 2nd reading

= $40cm^3 - 38cm^3$

= $2cm^3$

(v) Density of the cork = $\frac{Mass\ of\ the\ cork}{Volume\ of\ the\ cork}$

= $\frac{4.0\ g}{2\ cm^3}$

= $2g/cm^3$ or $2gcm^{-3}$

(vi) Why the materials were lowered gently

To prevent the water from splashing which could change the volume of the water

Or: In order not to change the first reading (of the initial volume of water)

(b) (i)

I - Measuring flask / volumetric flask / graduated flask

II - Beaker

III - funnel

IV - Stirrer

V - wash bottle

(ii) Functions

Apparatus	Functions
I	✓ Holding a specific volume of liquid ✓ Preparation of standard solutions
II	✓ Measuring volumes of liquids ✓ Holding a liquid substance
III	✓ Transferring liquids from one container into another container (with a small neck or opening) ✓ Pouring liquids without spilling
IV	✓ For stirring a mixture to make it uniform / homogeneous ✓ For stirring to dissolve a solute in a solvent
V	✓ Adding water (or other liquid) to mixtures /solutions / substances ✓ Rinsing laboratory equipment / apparatus

[any one for each]

(c) (i)

I - bud

II - flower

III - leaf

IV - leaf stalk / petiole

V - stem / internode

VI - roots

(ii)

Part	Function(s)
I	✓ Gives rise to a new flower or leaf or shoot
II	✓ Pollination (transfer of pollen grains from an anther to the stigma of the flower) ✓ Sexual reproduction (fusion of the male and female sex cells) ✓ Fruit and seed(s) production after fertilization has occurred
III	✓ Photosynthesis (manufacturing food using water, carbon, sunlight and chlorophyll) ✓ Transpiration (loss of water vapour through stomata) ✓ Gaseous Exchange ✓ Food storage in some plants
V	✓ Supports / holds the branches and leaves of the plant ✓ Transports food from the leaves to other parts ✓ Transports water and mineral salts from the soil to other parts ✓ Stores extra food
VI	✓ Holds the plant firmly in the ground ✓ Absorbs water and mineral salt from the soil for the plant's use. ✓ Stores extra food

[any one for each]

(iii) Sexual Reproductive part and vegetative part

(d) **(i)** rabbit

(ii) **I** - ear

II - whiskers

III - belly

IV - hind limb / leg

V - tail

VI - rump

(iii) hutch

(iv) Alaska, American White, Chinchilla, Polish, Rose, English, Flemish giant, Dutch, Beveren, etc,

[any three]

(v) Practices to control diseases and pests

- Provision of clean drinking water and healthy feed
- Keeping the hutch clean, well ventilated and at a suitable temperature at all times
- Regular veterinary examination / services of the rabbits
- Separating unhealthy or infected animals from the healthy ones
- Vaccination against infections / diseases

[any two]

2. (a) (i) Technology

The use of scientific knowledge to solve problems in everyday life

Or

The application of scientific knowledge and methods to make life / work easier, faster and more comfortable

Or

The study, development, and application of devices, machines, and techniques for manufacturing and productive processes

(ii) Uses of technology in communication

- Making telephone calls
- Sending a text messages
- Sending a fax
- Sending an e-mail
- Online chatting (on social networks)
- Searching for information on the internet
- Typing and printing documents
- Photocopying documents
- Scanning documents

[any one]

(b) (i) $4Fe + 3O_2 \rightarrow 2Fe_2O_3$

(ii) $2Na + Cl_2 \rightarrow 2NaCl$

(iii) $2H_2 + O_2 \rightarrow 2H_2O$

(c)

(i)

Component	Function
Nucleus	▪ Controls the life activities of the cell ▪ Contains the genes (for hereditary traits) ▪ Controls cell reproduction

(ii)	Chloroplast	▪ Contains chlorophyll, which absorbs energy from sunlight /light
(iii)	Mitochondrion	▪ Generate chemical energy for cell functions ▪ Control cell cycle / cell growth and death

(d) Cultural practices in vegetable crop production

- Weeding
- Stirring (the soil)
- Watering / irrigation
- Earthing up
- Mulching
- Thinning out
- Pruning
- Staking
- Supplying-in
- Application of fertilizer

[any four]

(e) Agencies in food safety and quality assurance

- Food and Drugs Board
- Ghana Standards Board
- Ministry of Food and Agriculture
- Ministry of Health
- Plant Protection and Regulatory Services Department

[any two]

3. (a) Why a tomato plant is likely to wilt if too much fertilizer is applied to it.

Nitrogen-rich fertilizers encourage top growth of tomato plants at the expense of root formation. Once hot weather begins, the root system is insufficient to transport the water needed to keep the top green growth alive, and the plant wilts.

(b) (i) Differences

Electrical Insulators	Electrical Conductors.
Does not allow electricity to flow through	Allows electricity to flow through
Has no free electrons	Has free electrons
Has a wide forbidden energy gap	Has no forbidden energy gap
Electrons are firmly bound to the nucleus	Electrons are loosely bound to the nucleus

(ii) Effects of illegal electrical connections in the home.

- Improper working of electrical appliances
- Fire outbreaks
- Destruction of electrical appliances
- Power cuts
- Financial waste

(c) Explanation as used to describe change of state of matter:

(i) Condensation;

The change of the physical state of matter from gaseous state into liquid state

Or

The process by which a vapour loses heat and changes into a liquid

This happens because the vapour molecules slow down due to the heat loss and the distances between molecules reduce.

(ii) Freezing

The change of the physical state of matter from the liquid state to the solid state

Or:

The process by which a liquid loses heat and changes into a solid

This happens because the liquid molecules slow down and stop moving about randomly due to the heat loss and the distances between molecules reduce.

(d) (i) State two diseases of the circulatory system in humans

Haemorrhoid

High blood pressure

Low blood pressure

Leukemia

Arteriosclerosis

[any two]

(ii) Ways in which the diseases stated in (*d*) (i) can be prevented.

DISEASES	PREVENTION
Haemorrhoid	▪ Drinking lots of water ▪ Eating a lot of fruits and vegetables ▪ Freeing the bowels as frequently as possible ▪ Exercising the body regularly ▪ Using hygienic and soft toilet paper
High blood pressure	▪ Reducing salt intake ▪ Avoiding drinking alcoholic beverages ▪ Avoiding smoking ▪ Exercising regularly ▪ Having enough sleep

	▪ Reducing emotional stress ▪ Reducing fat intake ▪ Increasing fruits and vegetable intake
Leukemia	▪ Avoiding exposure to certain radiations ▪ Avoiding exposure to certain chemicals ▪ Avoiding smoking
Arteriosclerosis	▪ Exercising the body regularly ▪ Reducing the intake of animal fat ▪ Taking more fruits and vegetables ▪ Avoiding smoking

4. (a) (i) A *transistor*

A small low-powered solid-state electronic device consisting of a semiconductor and at least three electrodes, the emitter, the base and the collector

(ii) Uses of a transistor

- As current amplifiers
- As current rectifier
- As an electronic switch in integrated circuits
- As a regulator – to regulate current, voltage or power.
- Used in oscillator circuits

(b)

	SOLUTE	**SOLVENT**
(i)	Grease;	Petrol/ kerosene
(ii)	Ink stain	Isopropyl / rubbing alcohol
(iii)	Starch	Cold water
(iv)	Cube sugar	Warm water
(v)	Oil paint	Turpentine
(vi)	Iodine	ethanol / diethyl ether / acetic acid / benzene

(c) (i) A *respiratory organ*

Any organ of a living organism that is involved in the process of respiration

(ii) Structures of the respiratory system of humans.

Nostrils, pharynx, larynx, trachea, bronchi, bronchioles, alveoli, ribs, diaphragm, intercostal muscle, ribs, (lungs)

[any two]

(d) (i) A*gricultural chain*

The linkage of stages that are interconnected in the production and supply of agricultural goods and services

(ii) Types of agricultural chain

- Production chain
- Processing chain
- Value chain
- Supply chain

5. (a) (i) *Stars*

Stars are heavenly bodies that are made up mostly of burning gases

(ii) The **first four** planets in order

Mercury, Venus, Earth, Mars

(b) State

(i) Differences

PLANTS	ANIMALS
Manufacture their own food (photosynthesis)	Do not manufacture their own food
Cannot move freely from place to place	Able to move freely from place to place
Most plants do not respond quickly to stimuli	Respond quickly to stimuli
Have a cell wall	Do not have a cell wall
Have chloroplasts, which contain chlorophyll	Do not have chloroplast
Excess carbohydrates are stored as starch	Excess carbohydrates are stored as glycogen
Absorb carbon dioxide for photosynthesis and release oxygen as waste product	Inhale oxygen for respiration and release carbon dioxide as waste product
Cells have large and permanent vacuoles	Cells have small temporary vacuoles
Growth takes place at specific parts	Growth takes place at all parts

[any two]

(ii) Similarities between plants and animals

- Both plants and animals are living things / organisms
- Both plants and animals respire
- Both plants and animals grow
- Both plants and animals reproduce
- Both plants and animals feed
- Both plants and animals excrete

[any two]

(c) Explanation

(i) Pastoral farming

A farming system in which farm animals such as cattle and sheep are moved from place to place in search of food and water. The animals feed on natural vegetation and drink water from water bodies such as streams, rivers, ponds, etc

(ii) Ecological farming

A method of farming in which the natural vegetation and animals and their environment are preserved

Chemicals and heavy machinery are not used. The soil is fertilized using organic manure, such as animal droppings. Pests and diseases are controlled biologically, ie, using resistant crops and animals

(d) Properties of water

(i) Odour - Water is odourless (it has no odour / smell)

(ii) Taste - Water is tasteless (it has no taste)

(iii) Effect on litmus -Water turns red or blue litmus paper to purple.

6. (a) (i) Annual plants

Plants that complete their life cycle in one season or one year

(ii) Perennial plants

Plants that continue to live for several years (usually three or more)

(b) Dangers involved in

(i) Eating or drinking water in the laboratory;

- The food or drink can get contaminated by toxic substances, which could lead to poisoning
- The food or drink can spill or drop into other substances in the laboratory, which could lead to accidents and injuries.

(ii) Washing hands with unknown liquid in a beaker;

- The unknown liquid could be toxic, which will cause fatal injuries or even death.

- The unknown liquid could be harmful or irritant, which could cause severe harm or injuries or irritation to the hands or body

(iii) Walking barefooted.

- One could step on a slippery substance and slip or fall, causing injuries to the body
- One could step on a toxic or irritant substance, which could cause severe harm or injuries or irritation to the feet or body

(c) (i) *Digestive enzyme*

A biological catalyst which speeds up the digestion of food substances in the body of an animal

Or

A complex chemical, produced in the body of an animal, which speeds up the process of digestion

(ii) Examples of digestive enzymes in humans.

Ptyalin, pancreatic amylase, trypsin, pepsin, rennin, lipase, maltase

[any two]

(d) Differences

Conduction	Radiation
Requires a material medium	Does not require a material medium
Process is relatively slow	Process is relatively very fast
Heat is transferred by contact of molecules or atoms.	Heat is transferred by waves or rays

April 2011

INTEGRATED SCIENCE 1

OBJECTIVE TEST
45 minutes

1. An example of a noble gas is
A. chlorine
B. neon
C. nitrogen
D. oxygen

2. The structure that stores sperms temporarily in the male reproductive system of humans is
A. epididymis
B. scrotal sac
C. sperm duct
D. testes

3. Which of the following properties of alcohol as a thermometric liquid is correct?
A. It is opaque
B. It does not wet glass
C. It has a very low freezing point
D. It has a high freezing point

4. In which of the following vegetation zones of Ghana does millet and sorghum grow well?
A. Coastal savannah
B. Forest zone
C. Guinea savannah
D. Transition zone

5. Which of the following methods is/are used for preserving fish?
I. Canning
II. Frying
III. Smoking

A. I only
B. I and II only
C. II and III only
D. I, II and III

6. Which of the following practices in the home can prevent disease infection?

A. Covering one's food
B. Drinking unclean water
C. Keeping one's surroundings untidy
D. Sharing towels and sponges

7. An example of a semimetal is
A. calcium
B. lithium
C. silicon
D. sodium

8. Which kind of energy transformation takes place in an electric motor?
A. Chemical energy to electrical energy
B. Chemical energy to mechanical energy
C. Electrical energy to light energy
D. Electrical energy to mechanical energy

9. *Landrace* is a breed of
A. cattle
B. goats
C. pigs
D. sheep

10. In flowering plants, the stamen is made up of
A. anther and stigma
B. anther and filament
C. stigma and filament
D. stigma and style

11. Which of the following statements about a transistor is correct?
A. It can be used to amplify current
B. It is similar to three diodes
C. It consists of two leads only
D. It has three junctions

12. An atom has 6 protons and 7 neutrons in its nucleus. What is its mass number?
A. 1
B. 6
C. 7
D. 13

13. A record of daily activities on a farm is termed
A. labour record
B. inventory record
C. farm diary

D. production record

14. Which of the following devices work(s) on the principle of transmission of pressure in fluids?

I. Water pumps
II. Syringes
III. Bicycle brakes

A. I only
B. I and II only
C. II and III only
D. I, II and III

15. Which of the following crops is **not** correctly matched with its group?

A. Cowpea - cereal crop
B. Cocoa - beverage crop
C. Coconut - oil crop
D. Cocoyam - tuber crop

16. Which of the following substances is a salt?

A. H_2SO_4
B. NaOH
C. HCl
D. $CaCl_2$

17. In which part of the digestive system of a fowl does grinding of feed take place?

A. Crop
B. Gizzard
C. Oesophagus
D. Proventriculus

18. The parts of a bony fish that are used to control the level at which the fish swims are known as

A. dorsal and caudal fins
B. caudal and pectoral fins
C. pectoral and pelvic fins
D. dorsal and pectoral fins

19. Million's reagent is used to test for

A. carbohydrates
B. fats
C. proteins
D. vitamins

20. In an *n-p-n* transistor, the *n*-type collector is connected to the positive terminal of the battery thus making the

A. base-collector junction reverse biased

B. base-collector junction forward biased
C. base-emitter junction reverse biased
D. base-emitter junction forward biased

21. Which type of vegetation favours both wet and hot environmental conditions?
A. Strand and mangrove
B. Coastal savannah
C. Guinea savannah
D. Tropical forest

22. During drought, some plants dry out because of high
I. atmospheric temperature
II. humidity
III. rate of evaporation

Which of the statements above is/are correct?
A. III only
B. I and II only
C. I and III only
D. I, II and III

23. Soil erosion on sloppy farmlands is best controlled by
A. cover cropping
B. mulching
C. strip cropping
D. terracing

24. A viable seed is one that
A. germinates under suitable conditions
B. contains oil
C. develops from fertilized ovary
D. has a pericarp

25. Which of the following pairs of structures form part of the female reproductive system of humans?
A. Urethra and uterus
B. Cervix and uterus
C. Ureter and uterus
D. Cervix and ureter

26. Which of the following chemical symbols is that of a metal?
A. Ca
B. Ne
C. P
D. S

27. All the living and non-living things that surround an organism constitute its

A. community
B. ecosystem
C. environment
D. habitat

28. Which of the following crops should be planted after cassava in crop rotation?

A. Cocoyam
B. Cowpea
C. Onion
D. Yam

29. Endoparasites in farm animals can be controlled by

A. drenching
B. dipping
C. dusting
D. spraying

30. Which of the following devices requires the use of transistors in its operation?

A. Computer
B. Electric heater
C. Microphone
D. Wall clock

31. Feel Method is used to determine soil

A. air
b. colour
C. structure
D. texture

32. Which of the following modes of heat transfer is the thermos flask designed to minimize?

I. Conduction
II. Convection
III. Radiation

A. I and II only
B. I and III only
C. II and III only
D. I, II and III

33. An atom of an element is represented as $^{25}_{12}X$. What is the respective number of neutrons and protons in the atom?

A. 12 and 13
B. 12 and 25
C. 13 and 12

D. 25 and 12

34. The anemometer is an instrument used in determining
A. amount of rainfall
B. speed of wind
C. relative humidity
D. intensity of light

35. Which of the following subjects is/are considered as applied science?
I. Biology
II. Medicine
III. Psychology

A. I only
B. I and II only
C. I and III only
D. II and III only

36. *Chinchilla* is a breed of
A. goats
B. pigs
C. rabbits
D. sheep

37. Which of the following characters is **not** acquired through heredity?
A. Language spoken
B. Shape of nose
C. Colour of eyes
D. Temperament

38. The food nutrient which ensures good health in farm animals is
A. carbohydrates
B. minerals
C. proteins
D. vitamins

39. The use of resistant breeds of farm animals to control pests is a
A. biological method
B. chemical method
C. cultural method
D. physical method

40. Which of the following arrangements show the correct order of increasing complexity of structures in living organisms?
A. cells → organs → tissues → systems

B. cells → tissues → organs → systems
C. cells → systems → tissues → organs
D. cells → tissues → systems → organs

April 2011

INTEGRATED SCIENCE 1

SOLUTIONS

OBJECTIVE TEST

1.	B.	neon
2.	A.	epididymis
3.	C.	It has a very low freezing point
4.	C.	Guinea savannah
5.	D.	I, II and III
6.	A.	Covering one's food
7.	C.	silicon
8.	D.	Electrical energy to mechanical energy
9.	C.	pigs
10.	B.	anther and filament
11.	A.	It can be used to amplify current
12.	D.	13
13.	C.	farm diary
14.	D.	I, II and III
15.	A.	Cowpea - cereal crop
16.	D.	$CaCl_2$
17.	B.	Gizzard
18.	C.	pectoral and pelvic fins

19. C. proteins

20. A. base-collector junction reverse biased

21. D. Tropical forest

22. C. I and III only

23. D. terracing

24. A. germinates under suitable conditions

25. B. Cervix and uterus

26. A. Ca

27. C. environment

28. B. Cowpea

29. A. drenching

30. A. Computer

31. D. texture

32. D. I, II and III

33. C. 13 and 12

34. B. speed of wind

35. D. II and III only

36. C. rabbits

37. A. Language spoken

38. D. vitamins

39. A. biological method

40. B. cells → tissues → organs → systems

April 2011

INTEGRATED SCIENCE 2

ESSAY

1 ¼ hours

[100 marks]

This paper is in two parts, **I** *and* **II**

Answer Question **1** *in Part* **I** *and any other* **four** *questions in Part* **II**.

Credit will be given for clarity of expression and orderly presentation of material

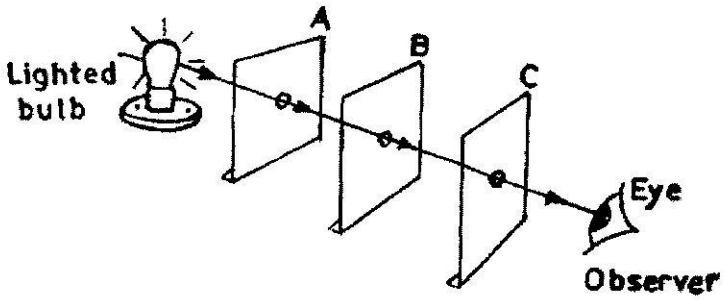
PART I

[40 marks]

Answer **all** *of Question* **1**

1. (a) In an experiment to demonstrate a property of light, three cardboards, **A, B** and **C** with holes in their centres are arranged in a straight line between a lighted bulb and an observer as shown in the illustration below.

Study the illustration carefully and use it to answer the questions that follow



(i) What would the observer see from the position shown?

(ii) What would the observer see when cardboard **B** is slightly displaced from the line?

(iii) Explain the observation made in (a)(ii) above.

(iv) What would be observed when the cardboard **B** is brought back to its original position?

(v) What property of light is being demonstrated in this experiment?

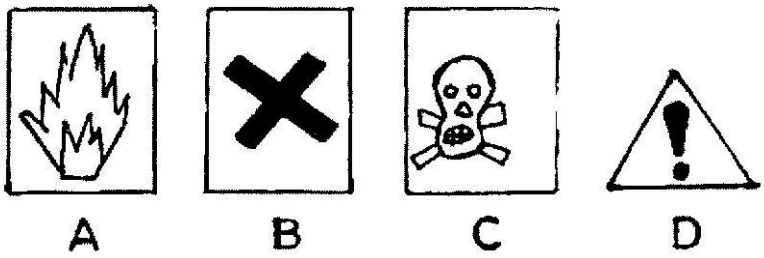
(vi) Mention

(α) **two** natural occurrences that could be explained by the property of light demonstrated.

(β) **one** device that works on the property of light demonstrated.

(b) The diagrams below are illustrations of hazard symbols found in everyday life.

Study them carefully and use them to answer the questions that follow:



(i) What does **each** symbol **A, B, C** and **D** represent?

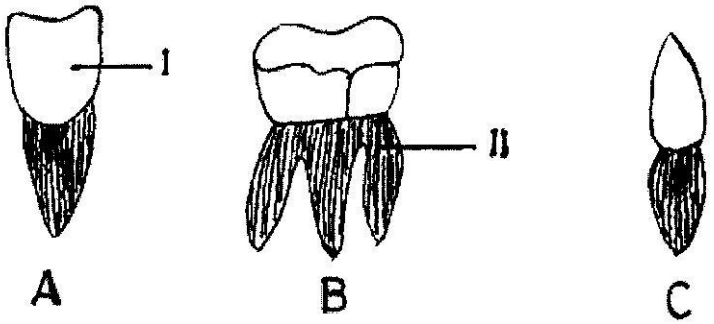
(ii) Name **one** substance **each** that is associated with **each** of the symbols **A, B** and **C**

(iii) Name **one** place where the symbol **D** can be found.

(iv) State **two** advantages of hazard symbols.

(c) The diagrams below are illustrations of the different types of teeth in humans.

Study them carefully and use them to answer the questions that follow:



(i) Identify **each** type of teeth labeled **A, B** and **C**

(ii) Describe the shape of **each** of the teeth labeled **A, B** and **C**

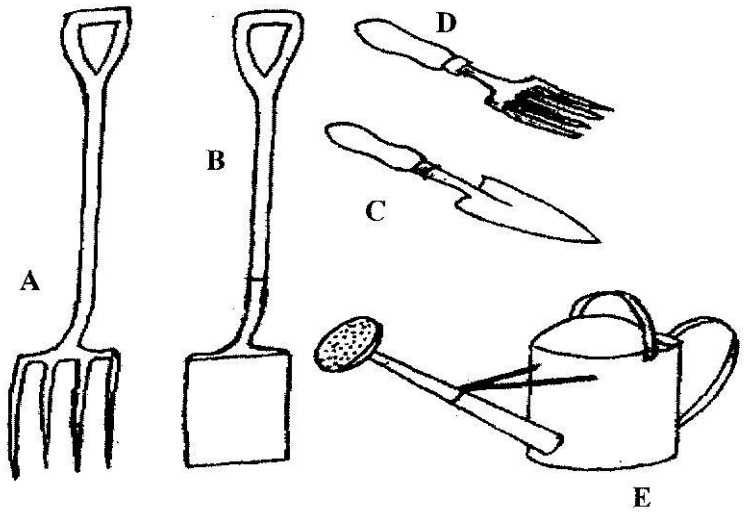
(iii) State **one** function of **each** of the teeth labeled **A, B** and **C**

(iv) Name the parts of the teeth labeled **I** and **II**

[10 marks]

(d) The diagrams below are illustrations of some farm tools.

Study them carefully and use them to answer the questions that follow:



(i) Identify **each** of the tools labeled **A, B, C, D** and **E.**

(ii) Mention **one** use of **each** of the tools labeled **A, B, C, D** and **E.**

PART II

[60 marks]

Answer **four** *questions* **only** *from this part*

2. (a) (i) What are ruminants?

(ii) Give **two** examples of ruminants

(b) (i) What is force?

(ii) State **two** effects of forces on a body

(c) (i) Mention **two** ways in which the carbon cycle can be maintained.

(ii) State **one** environmental effect when the carbon cycle is disrupted

(d) (i) Mention the **three** sub-atomic particles

(ii) State the relative charge on **each** of the **three** sub-atomic particles mentioned in (d) (i) above.

(iii) Name the particle formed when an atom loses an electron

3. (a) (i) What is a mixture?

(ii) Explain why some mixtures are thoroughly stirred before they are used.

(b) (i) What is reflection of light?

(ii) State **two** characteristics of the images formed by plane mirrors

(c) (i) What is a fertilizer?

(ii) Give **one** example of an inorganic fertilizer

(d) (i) What is indiscriminate sex?

(ii) State **two** dangers of indiscriminate sex on humans

4. (a) (i) What is a simple machine?

(ii) Give **two** examples of a simple machine

(b) (i) What is rusting?

(ii) State **two** effects of rusting

(c) (i) What are food nutrients?

(ii) Classify the following food items as carbohydrate, fats and oil or protein:

Beans, palm fruits, meat, margarine, bread and maize.

(d) (i) State **two** effects of malnutrition in farm animals.

(ii) Mention **one** disease of farm animals caused by virus.

5. (a) (i) What is soil erosion?

(ii) Name **two** methods of controlling soil erosion

(b) (i) Explain **each** of the following terms as used in ecology:

(α) Adaptation

(β) Endangered species

(c) (i) Give **one** example of hard water

(ii) Explain why it is advisable to drink water which is hard.

(d) (i) What is magnetic field?

(ii) State **two** methods of making magnets

6. (a) (i) Define **each** of the following terms:

(α) Solvent

(β) Solute

(ii) Name **one** common solvent used in the home.

(b) Explain the following terms as used in animal production:

(i) Ration;

(ii) Dehorning

(c) (i) What is an element

(ii) Write down the symbol of **each** of the following chemical substances:

(α) Potassium

(β) Sulphur

(d) Explain **each** of the following terms:

(i) Mixed farming

(ii) Mixed cropping

April 2011

INTEGRATED SCIENCE 2

SOLUTIONS

ESSAY

1. **(a)** **(i)** Light rays from the lighted bulb

(ii) Sees no light or a portion of cardboard B

(iii) Since light moves in a straight line, the observer sees the light only when the holes are in a straight line with the light source. Since light rays cannot bend around the cardboard B when slightly displaced, the observer does not see the light

(iv) The observer would see the light rays from the lighted bulb again.

(v) The property that light travels in a straight line or rectilinear propagation of light

(vi) **(α)**

Eclipse of the sun (solar eclipse)
Eclipse of the moon (lunar eclipse)
Day and night
Shadow [any two]

(β) camera, microscope, telescope, binoculars, periscope, torch, [any one]

(b) **(i)**

A - highly inflammable or flammable
B - Irritant or can cause harm
C - Poisonous or toxic or deadly
D - Danger ahead

(ii) A - liquefied petroleum gas (LPG), petrol, kerosene, spirit, ethanol [any one]

B - Concentrated Hydrochloric acid [Conc. HCl], concentrated sulphuric acid [conc. H_2SO_4], concentrated magnesium hydroxide [$Mg(OH)_2$], concentrated sodium hydroxide [NaOH] [any one]

C - Dichlorodiphenyltrichloroethane (DDT)[$C_{14}H_9Cl_5$], Potassium cyanide [KCN], Sodium cyanide [Na CN], mercury [any one]

(iii) On roads, at construction sites, very high voltage plants/devices, [any one]

(iv) - They help prevent accidents by making people take precautions.
- They serve as warnings to prevent injuries or death.

(c) **(i)**
A - incisor
B - premolar / molar
C - canine

(ii)
A - chisel shaped
B - almost flat surface with small projections / cusps / ridges
C - pointed

(iii)
A - Biting or cutting
B - chewing or grinding or mashing
C - tearing

(iv)
I - enamel / crown
II - root

(d) **(i)**
A - Garden fork
B - Spade
C - hand trowel
D - hand fork
E - watering can

(ii)
A - Breaking up soil **or** making soil loose **or** ploughing soil **or** turning over soil

B - Collecting soil **or** mixing substance such as animal feed **or** digging ground **or** making surface of ground level

C - earthing up crops **or** transplanting seedlings **or** spreading fertilizer **or** manuring **or** mounding certain crops

D - Breaking up soil **or** making soil loose **or** ploughing soil **or** turning over soil

E - Watering soil / crops

2. **(a)** **(i)** **Ruminants are** cud-chewing hoofed mammals with multiple-chambered stomach.

(ii) **Examples**: camel, goat, sheep, giraffe, cow

(b) **(i)** **A force** is a push or pull exerted on a body.
Or:
A physical influence that tends to change the position or shape of an object with mass

(ii) **Effects of forces on a body:**

- Can cause a moving body to come to rest (stop moving)
- Can cause a body at rest to move
- Can cause a moving body to accelerate
- Can cause a moving body to decelerate
- Can change the direction of a moving body
- Can change the shape of a body [any two]

(c) **(i)** **Ways of maintaining the carbon cycle:**

- Afforestation (planting new trees to replace the ones that have been destroyed or cut down
- Photosynthesis
- Respiration
- Reducing the amount of carbon dioxide emissions from industries
- Avoiding excessive bush burning
- Reducing the amount of fumes from vehicles [any two]

(ii) **Environmental effects**

- Global warming as a result of the depletion of the ozone layer
- Increase in volume of sea water due to melting of icebergs
- Greenhouse effect [any one]

(d) **(i)** Protons, neutrons and electrons

(ii)

PARTICLE	RELATIVE CHARGE
Proton	positive (+1)
Neutron	no charge or neutral (0)
Electron	negative (– 1)

(iii) cation

3. (a) (i) A mixture is

A physical combination of two or more substances

Or:

A substance consisting of two or more substances that have been combined physically

(ii) Why some mixtures are **thoroughly stirred** before they are used:

To attain a uniform mixture **or** to make it homogenous, since the original mixture may not be uniform **or** there may be some suspended particles of the solute

(b) (i) Reflection of light is

The bouncing back or redirection of light when it strikes a surface

(ii) Characteristics of images formed by plane mirrors:

- Same size as object
- Same distance from mirror as object
- Virtual
- Erect / upright
- Laterally inverted [any two]

(c) (i) A fertilizer is:

Any substance usually added to or spread onto soil to increase its ability to support plant growth

Or: A substance added to soil to increase its nutrient content or fertility

(ii) Examples of inorganic fertilizer:

NPK **or** ammonium sulphate **or** urea **or** potassium chloride (or muriate of potash)

(d) (i) Indiscriminate sex

Having sexual intercourse with multiple (two or more) partners and usually without protection (use of condom)

(ii) Dangers of indiscriminate sex

- Contracting sexually transmitted diseases (STDs), such as AIDS, syphilis, etc
- Teenage pregnancy (where teenagers are involved)
- Abortion, which is usually quite dangerous to the mother
- Loss of social respect / reputation
- Loss of self-esteem [any two]

4. (a) (i) A simple machine

A mechanical device that makes work easier and/ or faster

(ii) Examples of simple machines

Bottle opener, wheel barrow, nut cracker, inclined plane, spanner, crowbar, etc

(b) (i) Rusting is

the corrosion (wearing away) of the surface of iron or steel due to the formation of iron oxide

or: the formation of a reddish brown coating of iron oxide on the surface of iron or steel that forms when the metal is exposed to air and moisture

(ii) Effects of rusting:

- Shortage of life span of item
- Loss of beauty of item
- Reduction in effectiveness of implement
- Collapse of buildings [any two]

(c) (i) Food nutrients

Chemical substances, found in food, that the body needs in order to function properly

(ii)

CARBOHYDRATE	FATS AND OILS	PROTEIN
Bread	palm fruits	beans
Maize	margarine	meat

(d) (i) Effects of malnutrition in farm animals

- Deficiency diseases
- Stunted growth
- Low energy and activity (lack of vitality)
- Slow recovery from illnesses
- Death
- Lower reproduction [any two]

(ii) Disease caused by virus

- Rabies
- Bird flu

- Newcastle
- Rinder pest [any one]

5. (a) (i) Soil erosion:

The washing away of the top soil by agents of erosion

Or: The removal of soil material by natural processes, principally running water, glaciers, waves, and wind

(ii) Methods of controlling soil erosion:

- Planting of cover crops
- Planting of grasses
- Planting of wind brakes (trees)
- Ploughing across slopes
- Terracing
- Strip cropping [any two]

(b) (i) (α) Adaptation:

The development of physical and behavioral characteristics that allow organisms to survive and reproduce in their habitats

(β) Endangered species

Species threatened by extinction:

Or: Species whose numbers are so few, or are declining so quickly, that the animal, plant, or other organism may soon become extinct

(c) (i) Example of hard water

Deep well water

Borehole water

(ii) It is advisable to drink hard water because

it contains dissolved mineral salts, such as calcium and magnesium salts, which are necessary to maintain good health and proper functioning of the body.

NB: (Minerals are essential for the healthy growth of teeth and bones. They also help in cellular activities, such as enzyme action, muscle contraction, nerve reaction, and blood clotting)

(d) (i) Magnetic field:

A region of space surrounding a magnet or current-carrying circuit in which the resulting magnetic force can be detected

(ii) Methods of making magnets

- Electrical method
- Single stroking
- Double stroking
- Induction
- Hammering [any two]

6. (a) (i)

(α) Solvent:

A substance that dissolves things:

Or: A substance in which other substances are dissolved, usually a liquid

(β) Solute

A substance that is dissolved in another substance

(ii) Common solvent used in home

Water, Turpentine, liquid soap, alcohol, kerosene [any one]

(b) (i) Ration:

A fixed and limited amount of feed, given to an animal or group of animals at specific times / intervals.

This is done to ensure that the animal has the right amounts of essential food nutrients for healthy growth and development.

(ii) Dehorning:

Removing or preventing the growth of the horns of an animal by surgery or cauterization.

This is done primarily to prevent animals from using them to injure other animals or destroy property

(c) (i) Element:

A substance that is made up of the same kind of atoms

Or:

Any substance that cannot be broken down into a simpler one by a chemical reaction

(ii) (α) Potassium - K

(β) Sulphur - S

(d) (i) Mixed farming:

The cultivation of crops and the rearing of livestock on the same farm at the same time

Or: farming that combines growing crops and rearing livestock on the same farm at the same time

(ii) Mixed cropping

The cultivation of different kinds of crops on the same piece of land at the same time

April 2010

INTEGRATED SCIENCE 1

OBJECTIVE TEST
45 minutes

1. Water that forms lather readily with soap is said to be

A. clean B. hard C. soapy D. soft

2. Which of the following life activities are common to both plants and animals?

I. Feeding
II. Locomotion
III. Respiration

A. I and II only B. I and III only C. II and III only D. I, II and III

3. A suitable instrument that could be used to measure the internal diameter of a bamboo flute is

A. beam balance B. metre rule C. surveyor's tape D. vernier calipers

4. The ability of soils to supply the right amounts of essential nutrients to plants is known as

A. soil consistency B. soil fertility C. soil structure D. soil texture

5. The simplest way of making well water suitable for laundry purposes is by

A. adding alum B. boiling C. chlorination D. filtering

6. Washing down of soil nutrients beyond the reach of roots of plants is referred to as

A. aeration B. infiltration C. leaching D. percolation

7. Which of the following statements explain(s) why an object floats in water?

A floating object

I. is less dense than water
II. is denser than water
III. has the same density as water

A. I only B. II only C. III only D. I and III only

8. An example of leafy vegetable crops is

A. cucumber B. lettuce C. okro D. onion

9. The structure in the respiratory system of humans where gaseous exchange occurs is

A. alveolus B. bronchus C. nostril D. trachea

10. The gas produced when glucose is oxidized during internal respiration is

A. carbon (II) oxide B. carbon (IV) oxide C. hydrogen D. nitrogen

11. The farm animal which has crop as part of its digestive system is the

A. fowl B. goat C. pig D. sheep

12. A beneficial effect of wind on the environment is

A. promotion of photosynthesis in green plants
B. pollination in flowering plants
C. leaching in soils
D. erosion in soils

13. The chemical solution that can be used to test for the presence of protein in food substances is

A. Benedict's solution B. Fehling's solution C. Iodine solution D. Million's reagent

14. Which of the following statements about bushfires is/are true?

I. They cause air pollution
II. They deplete vegetation cover
III. They contribute to global warming

A. I only B. II only C. II and III only D. I, II and III

15. The most effective method of controlling soil erosion on steep slopes is

A. cover cropping B. mulching C. strip cropping D. terracing

16. Which of the following agencies is responsible for providing information on the weather and climatic conditions of an area?

A. Animal Husbandry Department
B. Extension Services Department
C. Information Services Department

D. Meteorological Services Department

17. The farming system which involves the growing of one type of crop on the same piece of land every season is known as

A. mixed cropping B. mixed farming C. mono cropping D. monoculture

18. The fifth planet from the sun in the solar system is

A. Jupiter B. Mars C. Neptune D. Venus

19. The process that takes place when the sperm and the egg of humans fuse together is referred to as

A. fertilization B. menstruation C. ovulation D. reproduction

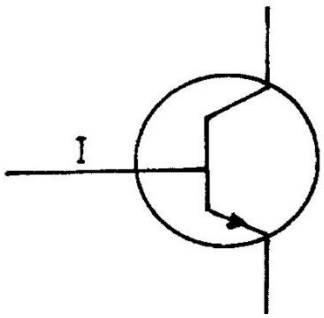
20. The process by which water vapour moves through the stomata of leaves into the atmosphere is known as

A. absorption B. diffusion C. osmosis D. transpiration

21. An example of intensive system of poultry keeping is the

A. deep litter system
B. fold unit system
C. free range system
D. Free-running system

The diagram below represents a circuit symbol in an electronic device. *Study it carefully and use it to answer questions 22 and 23*



22. The symbol represents

A. a light emitting diode
B. *n-p-n* transistor
C. light dependent resistor
D. *p-n-p* transistor

23. The part of the diagram labelled I has

A. more electrons than holes
B. no holes
C. more holes than electrons
D. the same number of holes as electrons

24. A spot made on a white paper with a given food substance turned the spot on the paper translucent. The food substance is likely to contain

A. glucose B. oil C. protein D. starch

25. Which of the following sources of energy is non-renewable?

A. Sun B. Waves C. Wind D. wood

26. Which of the following farm animals is / are housed in a hutch?

I. Goats
II. Pigs
III. Rabbits

A. I only B. III only C. I and II only D. II and III only

27. Which of the following blood vessels carries deoxygenated blood?

A. Aorta B. Pulmonary artery C. Pulmonary vein D. Renal artery

28. The energy transformation that takes place in a car battery is

A. electrical energy → chemical energy → light energy
B. chemical energy → light energy → electrical energy
C. chemical energy → electrical energy → light energy
D. electrical energy → light energy → chemical energy

29. The type of liquid suitable to construct a thermometer to read temperatures of about 150°C is

A. alcohol B. mercury C. turpentine D. water

30. Amino acids are the end-products of the digestion of

A. carbohydrates B. fats C. oils D. proteins

31. The type of teeth used by mammals to cut food materials is

A. canine B. incisor C. molar D. premolar

32. Fishponds are often stocked using

A. anchovies B. fingerlings C. fish eggs D. frys

33. Which of the following processes result(s) in the formation of a new substance?

I. Burning of wood
II. Rusting of iron nail
III. Heating water into vapour

A. I only B. I and II only C. II and III only D. I, II and III

34. The use of resistant breeds of farm animals in controlling diseases is a

A. biological control method
B. chemical control method
C. cultural control method
D. physical control method

35. A load of 10 N is moved through a distance of 2 m. Calculate the work done.

A. 5 J B. 10 J C. 20 J D. 50 J

36. Which of the following components of human blood is dissolved in the plasma?

A. Mineral salts B. Phagocytes C. Platelets D. Red blood cells

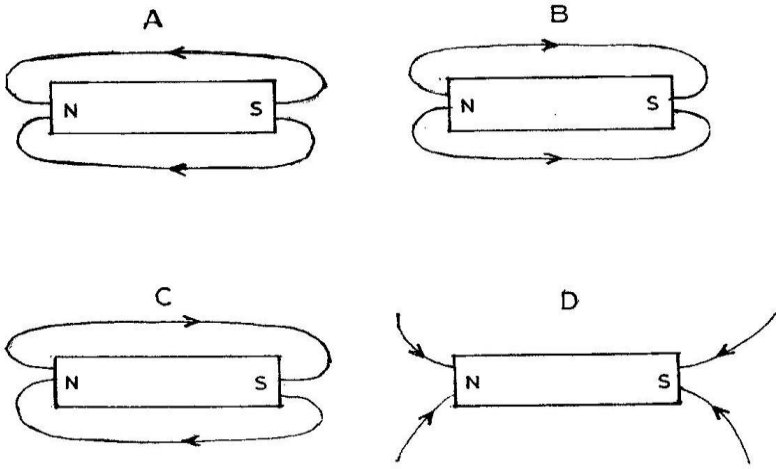
37. An atom of carbon is represented as ${}^{12}_{6}C$. How many neutrons are in the nucleus of the carbon atom?

A. 2 B. 4 C. 6 D. 12

38. In agribusiness, middlemen operate within the

A. production chain B. processing chain C. supply chain D. value chain

39. Which of the following illustrations shows the correct direction of the lines of force around a bar magnet?



40. The human sex cells are produced in the

A. scrotum and uterus
B. testis and ovary
C. penis and vagina
D. scrotum and ovary

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INTEGRATED SCIENCE 1

SOLUTIONS

OBJECTIVE TEST

1.	D.	soft
2.	B.	I and III only
3.	D.	vernier calipers
4.	B.	soil fertility
5.	B.	boiling
6.	C.	leaching
7.	A.	I only
8.	B.	lettuce
9.	A.	alveolus
10.	B.	carbon (IV) oxide
11.	A.	fowl
12.	B.	pollination in flowering plants
13.	D.	Million's reagent
14.	D.	I, II and III
15.	D.	terracing
16.	D.	Meteorological Services Department
17.	D.	monoculture
18.	A.	Jupiter
19.	A.	fertilization
20.	D.	transpiration

21. A. deep litter system

22. B. *n-p-n* transistor

23. C. more holes than electrons

24. B. oil

25. D. wood

26. B. III only

27. B. Pulmonary artery

28. C. chemical energy → electrical energy → light energy

29. B. mercury

30. D. proteins

31. B. incisor

32. B. fingerlings

33. B. I and II only

34. A. biological control method

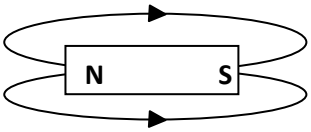
35. C. 20 J

36. A. Mineral salts

37. C. 6

38. C. supply chain

39. B



40. B. testis and ovary

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INTEGRATED SCIENCE 2

1 ¼ hours
ESSAY

[100 marks]

This paper is in two parts, **I** *and* **II**

Answer Question **1** *in Part* **I** *and any other* **four** *questions in Part* **II**.

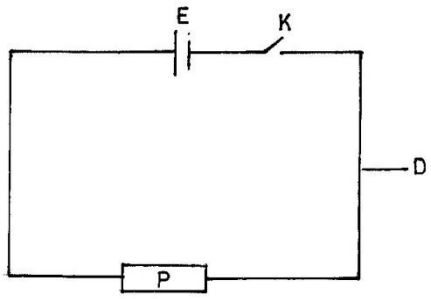
Credit will be given for clarity of expression and orderly presentation of material

PART I
[40 marks]

Answer **all** *of Question* **1**

1. (*a*) The diagram below is used to demonstrate an activity in the laboratory.

Study it carefully and use it to answer the questions that follow:



(i) What does the diagram represent?

(ii) Identify the components labelled **D, E, K** and **P** in the diagram

(iii) State **one** function **each** of the parts labelled **D, E, K** and **P.**

(iv) Mention the energy transformation that occurs in **E** in the diagram when **K** is closed.

[10 marks]

(b) In an experiment, red and blue litmus papers were dipped separately into **three** test tubes each containing one of the test substances listed in the table below.

Test substances	*Observations*		*Conclusion*
	Red litmus paper	***Blue litmus paper***	
Lemon juice			
Calcium hydroxide solution			
Dilute hydrochloric acid			

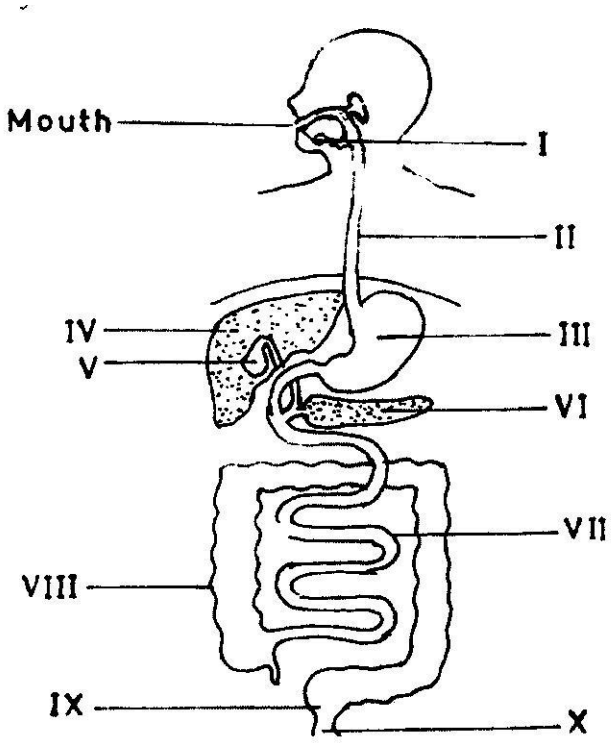
(i) Copy and complete the table by making the necessary **observation** and **conclusion** for **each** substance.

(ii) Name **two** of the test substances that would react with each other to produce salt and water.

(iii) Write down a balanced chemical equation for the reaction in (ii) above.

[10 marks]

(c) The diagram below is an illustration of the human digestive system. *Study it carefully and use it to answer the questions that follow:*



(i) Name the parts labelled **I, II, III** and **IV**

(ii) State **one** function of **each** of the parts labelled **V** and **VI**

(iii) Name the part where the digestion of protein starts.

(iv) Identify the part where

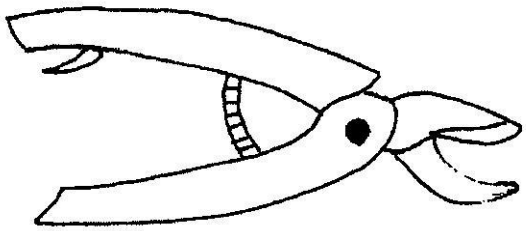
(α) absorption of end-products of digestion takes place

(β) re-absorption of water takes place

(γ) egestion takes place

[10 marks]

(d) The diagram below is an illustration of a simple farm tool.
Study it carefully and use it to answer the questions that follow:



(i) Identify the tool

(ii) State **three** uses of the tool

(iii) Mention **three** ways of maintaining the tool

[10 marks]

PART II

[60 marks]

Answer **four** questions **only** from this part.

2. (a) (i) What is *neutralization reaction*?

(ii) Write a balanced chemical equation for the reaction between **each** of the following pairs of substances:

(α) Sodium metal and dilute hydrochloric acid.

(β) Sodium hydroxide and dilute hydrochloric acid

[6 marks]

(b) Explain *weaning* as used in animal production

[2 marks]

(c) (i) What is *milky way*?

(ii) State **one** use of artificial satellites [3 marks]

(d) (i) What is a *habitat*?

(ii) Give **two** examples of a habitat. [4 marks]

3. (a) (i) Define *pressure*.

(ii) Explain why it is important to sharpen a knife before use

[4 marks]

(b) (i) State **two** differences between *metals* and *non-metals*

(ii) What is an *alloy*?

(iii) Mention the components of **each** of the following alloys:

(α) steel

(β) brass

[6 marks]

(c) Mention **three** conditions suitable for rearing tilapia in a fishpond

[3 marks]

(d) Explain how the streamlined body of a bony fish enables it to live successfully in water

[2 marks]

4. (a) (i) What is a *disease vector*?

(ii) Mention **two** methods of controlling **each** of the following types of pests of farm animals:

(α) ectoparasites

(β) endoparasites

[6 marks]

(b) (i) State **two** symptoms of nitrogen deficiency in a tomato plant.

(ii) Describe *side dressing* as a method of fertilizer application.

[5 marks]

(c) (i) Define *power*

(ii) State the S.I. unit of power.

[2 marks]

(d) Draw the electronic structure of sulphur

{Atomic number of sulphur = 16}

[2 marks]

5. (a) (i) What is *respiration*?

(ii) Name the types of respiration that occur in humans

[4 marks]

(b) List **three** ways of maintaining soil fertility

[3 marks]

(c) (i) Write the systematic name of **each** of the following chemical compounds:

(α) FeS;

(β) SO_2

(γ) CO_2

(ii) Give **one** reason why copper, silver and gold are mostly used in making ornaments and jewellery.

[4 marks]

(d) (i) What is a *fuse*?

(ii) Explain why a fuse is used in an electrical circuit.

[4 marks]

6. (a) (i) What is the difference between *unicellular organism* and *multicellular organism*

(ii) State two reasons why vegetable crops are important to humans

[4 marks]

(b) (i) State **two** elements of climate

(ii) What is the difference between *climate* and *weather*?

[4 marks]

(c) Mention **three** advantages of staking in crop production

[3 marks]

(d) Explain **each** of the following processes:

(i) corrosion;

(ii) sublimation

[4 marks]

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INTEGRATED SCIENCE 2

SOLUTIONS

ESSAY

1. **(a)** **(i)** An electrical circuit

(ii) **The components:**

D - Conductor wire
E - Cell or emf source
K - Switch or key
P - Resistor

(iii) **Function of**

D - Electric current flows through it
Or it conducts electricity

E - Provides the electromotive force (emf)
Or: provides the potential difference or voltage
Or: produces the electric current or electricity
Or: converts chemical energy to electrical energy

K - Opens or closes the circuit
Or: stops or starts the flow of current through the circuit

P - Opposes the flow of current through the circuit
Or: converts electrical energy to other forms

(iv) **energy transformation that occurs in E in the diagram when K**

Chemical energy → Electrical energy

(b) **(i)**

Test substances	***Observations***		***Conclusion***
	Red litmus paper	***Blue litmus paper***	
Lemon juice	Remains red	Turns red	Acidic substance

	or: no colour change		**or**: Contains acid
Calcium hydroxide solution	Turns blue	Remains blue **or**: no colour change	Basic substance **or**: Contains base
Dilute hydrochloric acid	Remains red **or**: no colour change	Turns red	Acidic substance Contains acid

(ii) Reactants to form salt and water:

Calcium hydroxide solution and dilute hydrochloric acid

(iii) Balanced chemical equation:

$Ca(OH)_2 + 2\ HCl \rightarrow CaCl_2 + 2\ H_2O$

(c) (i) The parts:

I - salivary gland
II - oesophagus **or**: gullet
III - stomach
IV - liver

(ii) **Function of:**

Part V - Stores bile **Or**: Concentrates bile

Part VI - Secretes juices containing enzymes into the small intestines
Or: secretes hormones such as insulin into the bloodstream

(iii) Digestion of protein starts at: Part III or The stomach

(iv) (α) absorption of end-products - Part VII - Small intestines

(β) re-absorption of water - Part VIII - Large intestines

(γ) egestion - Part X - Anus

(d) (i) The Tool - Secateurs or a pair of secateus

(ii) Uses of the tool

- Trimming
- Cutting
- Pruning

(iii) Ways of maintaining the tool

- Oiling or greasing metal parts
- Washing and drying after use
- Tightening bolts and nuts when loose
- Sharpening cutting edges when blunt (any three)

2. (a) (i) Neutralization reaction

A chemical reaction between an acid and a base to produce salt and water

(ii) (α) Sodium metal and dilute hydrochloric acid.

$$2\,Na + 2\,HCl \rightarrow 2\,NaCl + H_2$$

(β) Sodium hydroxide and dilute hydrochloric acid

$$NaOH + HCl \rightarrow NaCl + H_2O$$

(b) Weaning

Starting to feed a young animal with food other than its mother's milk

(c) (i) Milky way

the spiral galaxy to which Earth and its solar system belong

(ii) Use of artificial satellites

- Weather studies and forecasting
- Telecommunication
- Sending and receiving TV signals
- Taking photographs from space for various purposes
- Internet networking
- Global Positioning System (GPS) - a worldwide navigation system

[any one of the above]

(d) (i) Habitat

The natural living place of a plant or animal

Or: The natural conditions and environment of an organism

(ii) Examples of habitat

Forest, desert, river or wetlands, tree, etc (any two)

3. (a) (i) Pressure

The force per unit area

Or: Pressure = force / area

(ii) When a knife is sharpened, the area at the cutting edge is reduced; hence greater pressure is attained at a given applied force. Therefore it cuts easily with less applied force.

(b) (i)

METALS	NON-METALS
High melting point	Low melting point
Shiny surface	Dull surface
Malleable	Not malleable
Ductile	Not ductile
Good conductors of heat	Bad conductors of heat
Good conductors of electricity	Bad conductors of electricity

(ii) Alloy

A substance containing a metal and another metal or non-metal uniformly physically combined

Or: A homogeneous mixture of a metal and another metal or non-metal

(iii) (α) Steel - Iron and Carbon

(β) brass - Zinc and copper

(c) Conditions suitable for rearing tilapia in a fishpond

- Suitable salinity (salt concentration) of the pond
- Non-polluted water
- Favourable temperature of water
- Sufficient dissolved oxygen in the water
- Suitable pH of the water

(d) Importance of the streamlined body of a bony fish

Enables the fish to move smoothly or easily in the water

Or: it reduces the friction between the water molecules and the body of the fish.

4. (a) (i) Disease Vector

An organism that transmits disease-causing microorganisms from an infected person or animal to another

(ii) Methods of control

(α) ectoparasites

- Use of pesticides
- Dipping
- Dusting
- Handpicking
- Use of footbath
- Rotational grazing

(β) endoparasites

- Drenching
- Deworming
- Vaccination
- Regular change of feed and litter
-

(b) (i) Symptoms of nitrogen deficiency in a tomato plant

- Stunted growth
- Weak stem or branches
- Leaves turn yellowish
- Fruits are smaller and fewer than normal

(ii) Side dressing (fertilizer application)

Applying the fertilizer on / in the ground at the sides of the crop

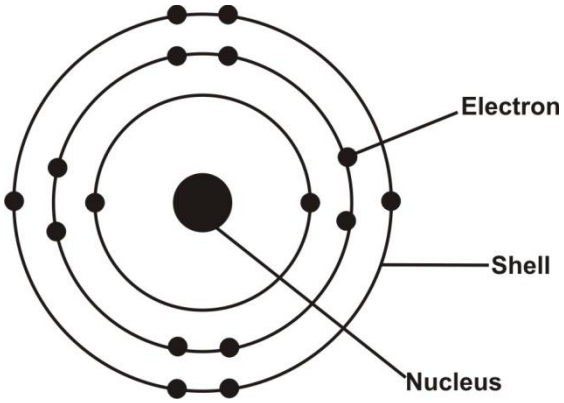
(c) (i) Power

The time rate of doing work **Or**: The work done per unit time

Or: (work done) / (time taken)

(ii) S.I. unit of power = watt or W

(d) The electronic structure of sulphur (atomic no. 16)



5. (a) (i) Respiration

The breakdown of food substances in an organism to release energy, carbon dioxide, water

(ii) Types of respiration

External respiration and internal respiration
Or: Aerobic and anaerobic respiration

(b) Ways of maintaining soil fertility

- Crop rotation
- composting
- mulching
- Fertilizer application
- Bush fallowing
- Green manuring
- Planting of cover crops
- Mixed farming

(c) (i) Systematic names of:

(α) FeS - Iron (II) sulphide

(β) SO_2 - Sulphur (IV) oxide

(γ) CO_2 - Carbon (IV) oxide

(ii) Reasons

They are very malleable, ductile, have high lustre and less reactive.

(d) (i) Fuse

An electrical safety device containing a piece of a metal that melts if the current running through it exceeds a particular level, thereby breaking the circuit.

(ii) Why fuse is used in an electrical circuit

To protect the electrical circuit from damage caused by abnormal power surges or increases

6. (a) (i)

Unicellular organism	Multicellular organism
organism is made up of only one cell	organism is made up of more than one cell

(ii) Why vegetable crops are important to humans

- Good source of vitamins
- Good source of mineral salts
- Make food tasty
- They have high fibre content

(b) (i) Elements of climate

- Rain
- Temperature
- Humidity
- Sunshine
- Atmospheric pressure
- Cloud
- Wind

(ii) Differences

Climate	Weather
atmospheric condition of a place over a long period of time	atmospheric condition of a place over a short period of time

(c) Advantages of *staking* in crop production

- Keeps fruits clean
- Enables easy weeding
- Enables easy harvesting
- Prevents early spoilage of fruits
- Keeps plant from falling during storms
- Keeps plant from falling under the weight of mature fruits

(d) (i) Corrosion explanation

A process by which a substance, especially a metal, is destroyed progressively by chemical action Example: the rusting of iron

(ii) Sublimation explanation

A process in which a substance is converted directly from a solid to a gas or from a gas to a solid without passing through the liquid state

Example: the conversion of naphthalene ball (camphor) from solid to gas directly

April 2009

SCIENCE 1

OBJECTIVE TEST
45 minutes

1. Plants are different from animals because they
A. have protoplasm
B. have chlorophyll
C. are able to respire
D. are able to reproduce sexually

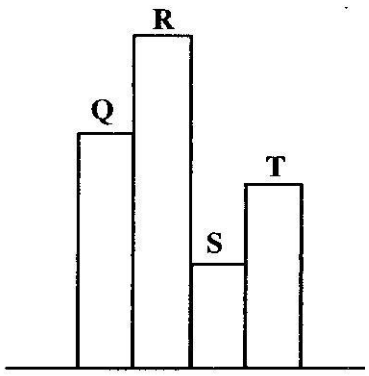
2. Which of the following conditions is **not** required for a current to flow through an electric circuit?
A. The circuit should be closed
B. A battery must be present
C. There must be connecting wires
D. A bulb must be connected in the circuit

3. Which of the following examples is a source of energy?
A. Electricity B. Heat C. Light D. Wood

4. Heat is transferred from the bottom of water in a container to the top by
A. conduction B. convection C. radiation D. conduction and radiation

5. A mercury thermometer works on the principle that
A. solids expand on heating and contract on cooling
B. liquids expand on heating and contract on cooling
C. gases expand on heating and contract on cooling
D. liquids evaporate when heated to a certain temperature.

6. In which of the positions Q, R, S, T in the diagram below will a body have the greatest potential energy?

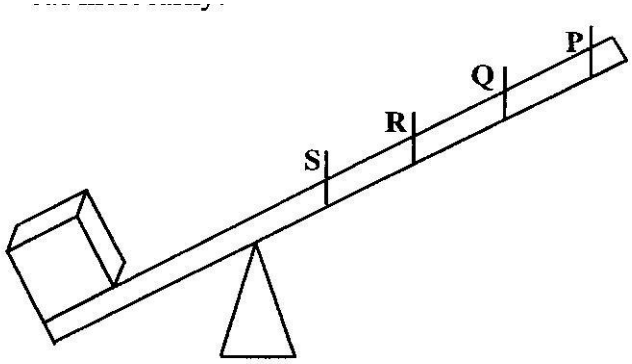


A. **Q** B. **R** C. **S** D. **T**

7. A solar cell produces electrical energy from
A. electromagnets B. heat C. light D. sound

8. In a third class lever the
A. pivot is between the load and effort
B. pivot is absent
C. load is between the effort and pivot
D. effort is between the load and pivot

9. At which positions S, R, Q and P on the lever in the diagram below must a force be applied to lift the load most easily?



A. **P** B. **Q** C. **R** D. **S**

10. Which of the following machines works on the same principle as an inclined plane?
A. Bottle opener B. Wheelbarrow C. See-saw D. Screw

11. Power is defined as the
A. ability to do work
B. force to move an object
C. rate of doing work
D. ability to replace energy used

12. Which of the following food substances is **not** involved in respiration?
A. Amino acids B. Fatty acids C. Glucose D. Vitamins

13. Which of the following organisms is at the beginning of a food chain?
A. Fish B. Grass C. Locust D. Mosquito

14. The human male sex cell that takes part in reproduction is the

A. penis B. scrotum C. sperm D. testis

15. The removal of waste products from the cells of organisms is called

A. assimilation B. circulation C. exhalation D. excretion

16. The central nervous system consists of the

A. spinal cord and spine
B. brain and the nerves
C. brain and the spinal cord
D. skeletal system and skull

17. Which of the following life processes is **not** performed by all living things?

A. Excretion B. Photosynthesis C. Reproduction D. Respiration

18. All veins in the body carry blood that does not contain oxygen with the exception of the veins from the

A. heart B. kidneys C. lungs D. liver

19. The method of purification of water containing very little impurities is

A. boiling B. chlorination C. distillation D. filtration

20. Boiling and chlorination are used in water purification to

A. kill germs
B. help suspended particles to settle
C. remove suspended particles
D. remove hardness of the water

21. The chemical symbol **C** is the symbol for

A. calcium B. chlorine C. carbon D. sodium

22. A substance which is made up of the same kind of atoms is called

A. a compound B. a mixture C. an element D. an ion

23. Which of the following processes results in the formation of new substances?

A. Heating camphor
B. Cooling water to form ice

C. Adding saliva to cooked yam
D. Mixing iron dust and sand together

24. When elements combine to form a compound the
A. constituents can be separated by a physical process
B. volume of the compound is the sum of the volumes of the constituents
C. reaction does not involve a heat change
D. reaction can be represented by a chemical formula

25. Which of the following organisms is likely to increase the oxygen content of the water in a pond?
A. Fish B. Tadpole C. Mosquito larva D. Water lily

26. Which of the following parasites is a plant?
A. Bilharzia B. Dodder C. Lice D. Tapeworm

27. The part of the soil that is most important for the growth of plants is
A. clay B. humus C. sand D. silt

28. Which of the following diseases may be associated with water?
A. chicken pox B. cholera C. small pox D. tetanus

29. The mosquito and housefly are harmful in that they
A. suck blood B. bite and cause pain C. settle on food D. spread diseases

30. Which of the following diseases affects the central nervous system?
A. Cerebro Spinal Meningitis (CSM) B. Poliomyelitis C. Measles D. Yellow Fever

31. Day and night occur because the earth
A. moves round the sun
B. moves from east to west
C. moves round the moon
D. rotates on its axis

32. Bronze is an alloy of
A. iron and carbon B. zinc and copper C. iron and zinc D. copper and tin

33. Which of the following substances causes acid rain when released into the atmosphere?
A. Carbon dioxide B. Water vapour C. Sulphur dioxide D. cement dust

34. Which of the following gases helps in rusting?

A. Carbon dioxide B. Hydrogen C. Oxygen D. Nitrogen

35. The force of attraction between molecules of different substances is called

A. adhesion B. cohesion C. tension D. viscosity

36. A metal block has a mass of 0.1kg. Calculate its volume if the density is 1.0 kgm^{-3}.

A. 0.01 m^3 B. 0.10 m^3 C. 1.00 m^3 D. 10.00 m^3

37. The food manufactured by a plant is distributed to all parts of the plant through the

A. chlorophyll B. phloem C. stomata D. xylem

38. Some insects are able to walk on water because

A. they are lighter than water
B. their skins cannot be penetrated by water
C. the surface of the water acts like an elastic skin
D. the surface of water is curved

39. Oxygen is made available to all cells of the human body by the

A. plasma B. platelets C. white blood cells D. red blood cells

40. The force which pulls all objects towards the centre of the earth is the

A. frictional force
B. electrostatic force
C. gravitational force
D. rotational force

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SCIENCE 1

SOLUTIONS

OBJECTIVE TEST

1.	B.	have chlorophyll
2.	D.	A bulb must be connected in the circuit
3.	D.	Wood
4.	B.	convection
5.	B.	liquids expand on heating and contract on cooling
6.	B.	**R**
7.	C.	light
8.	D.	effort is between the load and pivot
9.	A.	**P**
10.	B.	Wheelbarrow
11.	C.	rate of doing work
12.	D.	Vitamins
13.	B.	Grass
14.	C.	sperm
15.	D.	excretion
16.	C.	brain and the spinal cord
17.	B.	Photosynthesis
18.	C.	lungs
19.	C.	distillation

20. A. kill germs

21. C. carbon

22. C. an element

23. C. Adding saliva to cooked yam

24. D. reaction can be represented by a chemical formula

25. D. Water lily

26. B. Dodder

27. B. humus

28. B. cholera

29. D. spread diseases

30. A. Cerebro Spinal Meningitis (CSM)

31. D. rotates on its axis

32. D. copper and tin

33. C. Sulphur dioxide

34. C. Oxygen

35. A. adhesion

36. B. $0.10\ m^3$

37. B. phloem

38. C. the surface of the water acts like an elastic skin

39. D. red blood cells

40. C. gravitational force

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SCIENCE 2

1 ¼ hours

ESSAY

[75 marks]

This paper is in two sections, **A** *and* **B**

Answer Question **1** *in Section* **A** *and any other* **three** *questions in Section* **B**.

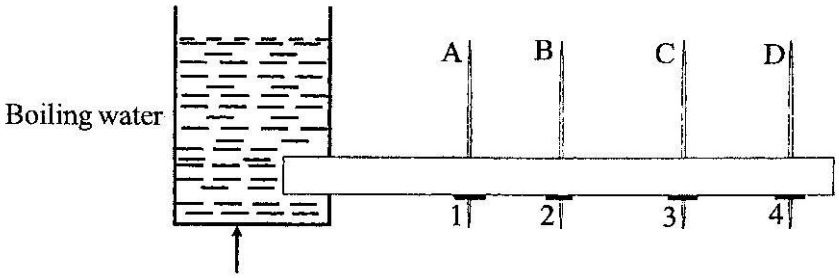
Credit will be given for clarity of expression and orderly presentation of material

SECTION A

[30 marks]

Answer **all** *of Question* **1**

1. (a) In an experiment, four nails 1, 2, 3 and 4 are fixed with candle wax onto a metal bar and one end of the bar is heated by means of boiling water as shown in the diagram below



A, B, C, D are thermometers inserted in holes along the bar to measure the temperatures at the various points.

(i) What is the temperature of the boiling water?

(ii) State the observations that will be made about nails 1, 2, 3 and 4

(iii) State the observations that will be made about the temperatures recorded by thermometers A, B, C and D.

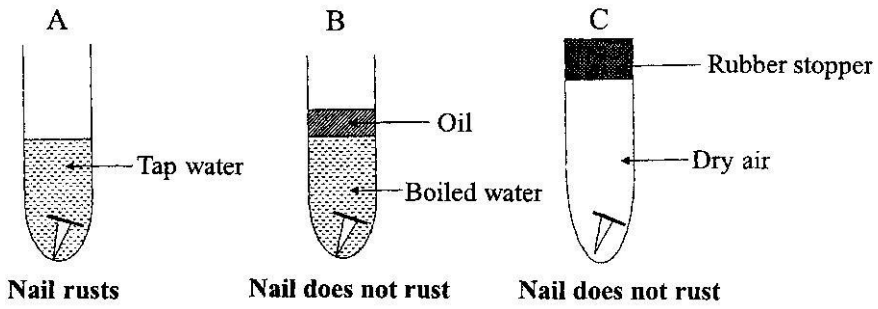
(iv) What mode of heat transfer is demonstrated in the experiment

(v) State one effect of heat that is associated with the experiment

(vi) State the aim of the experiment

[10 marks]

(b) In an experiment, a student took three iron nails and cleaned their surfaces dry and placed them in three separate test tubes in set-ups A, B and C as shown in the diagram.



After three days the nail in set-up A was found to have rusted while the nails in set-ups B and C did not.

Answer the following questions:

(i) Why was the water in set-up B boiled?

(ii) Explain the function of the oil on top of the water in set-up B.

(iii) State the purpose of the rubber stopper in set-up C

(iv) Why did the nail in set-up A rust?

(v) Suggest an aim for the experiment.

(vi) From the experiment, explain why oil or grease is applied on the surface of a metal to prevent rusting.

[10 marks]

(c) In an experiment the following activities were carried out on two green leaves A and B.
Leaf A was from a plant placed in the sunlight for sometime while leaf B was from a plant placed in a dark cupboard for 24 hours.

	Activity
I.	Leaves dipped in boiling water for 1 minute
II.	Leaves dipped in warm alcohol
III.	Leaves washed in cold water
IV.	Leaves dipped into iodine solution

After dipping in the iodine solution, it was observed that leaf A changed colour while leaf B did not change colour.

Answer the following questions:

(i) Explain why each of the activities I, II, III and IV was carried out.

(ii) State the colour change of leaf A

(iii) Explain why leaf A changed colour but leaf B did not.

(iv) Suggest an aim for the experiment.

[10 marks]

SECTION B

[45 marks]

Answer any **three** *questions from this section*

2. (a) Explain each of the following observations in nature:

(i) In the depths of the ocean where it is always dark, there are no green plants.

(ii) On an island where there are no insects and birds, the pawpaw plant produces only flowers but no fruits [6 marks]

(b) Describe how soil is formed. [4 marks]

(c) (i) State the laws of reflection

(ii) Draw a labelled diagram to show the reflection of light on a plane mirror [5 marks]

3. (a) A child is found not to be able to see at night

(i) What deficiency disease may the child be suffering from?

(ii) What food nutrient is the child lacking?

(iii) List two sources of food substances that can provide the nutrient the child lacks.

[4 marks]

(b) (i) What is an atom?

(ii) Name the components of an atom which determines each of the following quantities.

(α) mass of an atom

(β) charge of an atom

(iii) Why is an atom electrically neutral?

[5 marks]

(c) (i) What is friction?
(ii) Give two bad effects of friction.
(iii) Give two situations where friction is an advantage

[6 marks]

4. (a) (i) Explain heredity
(ii) Give two examples of heredity characters

[5 marks]

(b) Explain whether each of the following processes is a chemical change or a physical change:
(i) Rusting;
(ii) Burning;
(iii) Filtration;
(iv) Expansion of copper [4 marks]

(c) (i) What is a simple machine?
(ii) Name two types of simple machines
(iii) Explain why the efficiency of a machine cannot be equal to 100% [6 marks]

5. (a) Study the organisms listed below:

Earthworm;
Sheep;
Cassava plant;
Grasshopper;
Man;
Hen.

Draw a food chain using four of the organisms [4 marks]

(b) (i) Name four types of forces apart from friction.
(ii) Give three effects of a force

[7 marks]

(c) A certain pupil in a classroom can see clearly on the blackboard only when he sits at the front.
(i) State the eye defect the child is suffering from.
(ii) Explain why the child is unable to see when he is far from the blackboard.
(iii) How can the defect be corrected?

[4 marks]

April 2009

SCIENCE 2

SOLUTIONS

ESSAY

1. (a) (i) Temperature = 100°C or 212°F or 373K

(ii) Observations (Nails)

Nails 1, 2, 3 and 4 will drop off one after the other

– Nail 1 drops first, nail 2 second, nail 3 third and nail 4fourth

(iii) Observations (Thermometers)

The readings of Thermometers A, B, C and D will increase progressively till they reach 100°C

– Thermometer A reaches 100°C first, B second, C third and D fourth

(iv) Mode of heat transfer - Conduction

(v) Effects of heat

- Causes increase in temperature (as shown by the thermometer readings)
- Causes change in state (candle wax melts – changes from solid to liquid)

(vi) Aim of experiment

- To show that metals are good conductors of heat
- To show that heat travels through metals

(b) (i) Why water in set-up B was boiled

To remove dissolved air / oxygen from the water

(ii) The function of the oil on top of the water in set-up B

To prevent air (hence, oxygen) from entering / getting dissolved in the water

(iii) The purpose of the rubber stopper in set-up C

To prevent water particles or vapour or moisture from entering and contacting the nail

(v) Aim for the experiment

- To show that moisture/ water and oxygen/ air are necessary for rusting to occur.

(vi) Why grease is applied on metals to prevent rusting

To prevent moisture and air, which contains oxygen, from getting into contact with the metal, since metals containing iron will rust in the presence of air and moisture.

(c) (i) Why each activity was carried out

I - to kill the plant cells or stop the process of photosynthesis
II - remove the green pigment (chlorophyll)
III - to wash away the alcohol and soften the leaf
IV - to test for starch in the leaf

(ii) Colour change of leaf A

From cream / pale yellow / light brown to blue-black

(iii) The blue-black colouration indicates the presence of starch, which means that photosynthesis occurred in **leaf A** to produce starch.

Leaf B did not change in colour because photosynthesis did not occur, and hence, no starch was produced.

2. (a) (i) Reasons for no green plants in the dark ocean depths

No light reaches there, hence no photosynthesis takes place.

(ii) Reasons for no fruits on pawpaw plant, where there are no insects and birds

Since there are no agents of pollination, no pollination takes place; hence, no fertilization takes place. As a result, no fruits are produced.

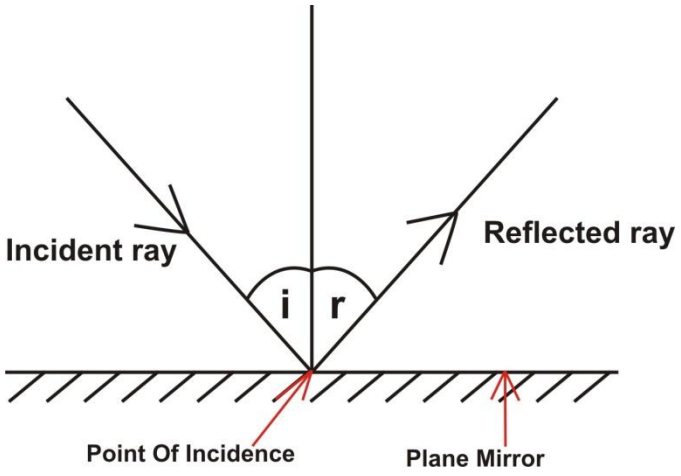
(b) How soil is formed:

Soil is formed from the disintegration/ breakdown/ weathering of rocks. The broken down rock particles combine with water, decayed organic matter, air and micro organisms to form soil

(c) (i) Laws of reflection

- The angle of incidence is equal to the angle of reflection
- The incident ray, the reflected ray and the normal, at the point of incidence, all lie in the same plane

(ii) Labelled diagram



i = angle of incidence, r = angle of reflection

3. (a) (i) Deficiency disease : Night blindness

(ii) Food nutrient : Vitamin A

(iii) Food sources:

Milk, butter, cheese, egg yolk, liver, palm oil, tomatoes, carrots, lettuce, cod-liver oil,

(b) (i) Atom

- The smallest particle of a substance that can take part in a chemical reaction
- The smallest portion into which an element can be divided and still retain its properties

(ii) Components of atom that determine

(α) mass of atom: protons and neutrons

(β) charge of atom: protons and electrons

(iii) Why an atom is electrically neutral:

The numbers of protons and electrons are equal. The positively charged protons neutralize the negatively charged electrons, hence, making the atom electrically neutral.

(c) (i) Friction is

The force that opposes the relative sliding motion of two bodies in contact
Or:
The resistance encountered by an object moving relative to another object with which it is in contact

(ii) Bad effects of friction

- Reduces the efficiency of machines
- Causes wear and tear
- Causes loss of energy through heat generation
- Slows down sliding motion of a body

(iii) Situations where friction is an advantage

- Enables walking, running, etc without slipping
- Enables vehicles to brake effectively
- Enables birds to perch on trees without falling
- Enable wheels to roll on a surface
- Enables our hands to hold items

4. (a) (i) Heredity

The passing on of genetic factors from parents to children through genes:
Or: the transfer of genetically controlled characteristics from one generation to the next in living organisms

(ii) Examples

Intelligence, shape of nose, height, colour of eyes, etc

(b) Explanation of whether chemical or physical change:

(i) Rusting - Chemical change; because a new substance is formed and the change is irreversible
(ii) Burning - Chemical change; because a new substance is formed and the change is irreversible
(iii) Filtration - Physical change; because no new substance is formed and the change is reversible

(iv) Expansion of copper - Physical change; because no new substance is formed and the change is reversible

(c) (i) A simple machine

A mechanical device that makes work easier and/ or faster

(ii) Types of simple machines

Inclined plane, wheel and axle, lever and pulley

(iii) Why the efficiency cannot be 100%

Part of the energy input is used to overcome friction and part also expended as heat.

5. (a) Food chain

Cassava plant → grasshopper → hen → man
Or: Cassava plant → earthworm → hen → man

(b) (i) Types of forces apart from friction:
Gravitational, tensional, centripetal, centrifugal, magnetic, electrostatic, upthrust, cohesive, adhesive

(ii) Effects of a force

- Can cause a moving body to come to rest (stop moving)
- Can cause a body at rest to move
- Can cause a moving body to accelerate
- Can cause a moving body to decelerate
- Can change the direction of a moving body
- Can change the shape of a body

(c) (i) The eye defect: shortsightedness or myopia

(ii) Explanation

Light rays from a distant object are focused in front of the retina, due to the eye lens being thicker than normal or eyeball being too long. Hence the brain does not receive the correct information for interpretation and so the child is unable to see.

(iii) How the defect can be corrected: By the use of a concave or diverging lens

April 2008

SCIENCE 1

OBJECTIVE TEST

45 minutes

1. A metre rule can be used for measuring the
- A. volume of a liquid
- B. area of a ball
- C. length of a table
- D. diameter of a wire

2. Heat travels through vacuum by
- A. conduction
- B. convection
- C. radiation
- D. convection and radiation

3. A mirror is used to direct sunlight onto the wall of a classroom by
- A. dispersion
- B. refraction
- C. reflection
- D. radiation

4. Which of the following energy changes takes place in ringing bell?
- A. Potential to kinetic
- B. Potential to sound
- C. Mechanical to sound
- D. Sound to light

5. A fuse is connected in an electric circuit to
- A. prevent electric shock
- B. increase the current
- C. prevent current from increasing
- D. increase the heat generated

6. Which of these objects is a magnetic substance?
- A. Aluminium
- B. Brass
- C. Copper

D. Iron

7. Which of the following bodies are natural sources of light?

I. Moon
II. Sun
III. Firefly

A. I and II only
B. I and III only
C. II and III only
D. I, II and III only

8. Which of the following statements describes the nature of light?

A. It passes through all objects
B. It is absorbed at a polished surface
C. It travels in a straight line
D. It is always reflected in a medium

9. Metals are able to conduct electricity because they possess

A. free electrons
B. neutrons
C. atoms
D. free protons

10. *Beriberi* is caused by deficiency of vitamin

A. A
B. B
C. E
D. K

11. Which of the following fruits is adapted for dispersal by air?

A. Succulent fruit
B. Brightly coloured fruit
C. Sticky fruit
D. Hairy fruit

12. The food substance needed for growth and repair of tissues is

A. carbohydrate
B. oil
C. protein
D. vitamin

13. The type of teeth used for biting and cutting are the

A. canines
B. incisors

C. molars
D. premolars

14. Water is absorbed by roots in plants and transported to the leaves through the
A. chlorophyll
B. phloem
C. stomata
D. xylem

15. The outer ear consists of the
A. ossicles and eardrum
B. cochlea and pinna
C. ossicles and cochlea
D. pinna and eardrum

16. Changes in pressure in the middle ear is regulated by the
A. ampullae
B. cochlea
C. Eustachian tube
D. ear drum

17. The amount of light that enters the eye is reduced by the size of the
A. lens
B. iris
C. pupil
D. retina

18. In humans, features of parents are passed on to offspring through
A. breastfeeding
B. reproduction
C. good training
D. sharing the same things

19. The taste of water changes when it is boiled because
A. heat is absorbed during boiling
B. evaporation takes place during boiling
C. air is lost during boiling
D. steam is produced during boiling

20. Arrange the following types of water in order of increasing hardness
I. Tap water
II. Sea water
III. Distilled water
IV. Rain water

A. IV, III, II, I
B. III, IV, I, II
C. II, I, IV, III
D. I, II, III, IV

21. Which of these methods is used to separate insoluble solids from liquids?
A. Distillation
B. Evaporation
C. Filtration
D. Winnowing

22. A uniform mixture of two or more metals is called
A. an alloy
B. a compound
C. a solvent
D. a suspension

23. Which of the following liquids **cannot** dissolve an oil paint?
A. Kerosene
B. Petrol
C. Turpentine
D. Water

24. An atom which contains more electrons than protons becomes a
A. positive ion
B. negative ion
C. neutral atom
D. binary compound

25. The following substances are mixtures except
A. air
B. carbon dioxide
C. salt solution
D. smoke

26. The fish is protected from injury by the
A. fins
B. gills
C. operculum
D. scales

27. Which of the following substances is an air pollutant?
A. Water vapour
B. Oxygen
C. Nitrogen

D. Hydrogen sulphide

28. Bathing in lakes and slow moving streams can lead to one getting
A. bilharziasis
B. cholera
C. dysentery
D. river blindness

29. Some plants shed their leaves during the dry season to prevent
A. water loss
B. heat loss
C. catching fire
D. losing nutrients

30. Caterpillar is an example of
A. an animal parasite that feeds on plant
B. a plant parasite that lives on an animal host
C. an animal parasite that feeds on animal host
D. a plant parasite that grows on plant host

31. How long does it take the moon to go completely round the earth?
A. 1 day
B. 28 days
C. 30 days
D. 365 days

32. An unripe orange is said to be acidic because it
A. has low sugar content
B. has bitter taste
C. turns red litmus paper blue
D. turns blue litmus paper red

33. The chemical symbol of potassium is
A. K
B. N
C. P
D. S

34. The property of a metal that makes it possible for it to be drawn into a wire is called
A. conductivity
B. ductility
C. malleability
D. resistivity

35. Which of the following methods of preserving food make use of heat energy?

I. Drying
II. Frying
III. Salting
IV. Canning

A. I and II
B. I and IV
C. III and IV
D. I, II and III

36. Weight is an example of
A. energy
B. force
C. work
D. power

37. Blood is prevented from flowing back into the veins by the
A. arteries
B. capillaries
C. heart
D. valves

38. The force that binds molecules of the same substance together is called
A. adhesion
B. cohesion
C. capillary action
D. surface tension

39. Soluble digested food substances enter the blood of the human body by a process called
A. absorption
B. egestion
C. excretion
D. ingestion

40. Capillary action is applied in the following activities except
A. blotting an ink with a blotting paper
B. absorption of water with a towel
C. rising of water in a very narrow glass tube
D. sticking of water droplets on the surface of a glass tube

April 2008

SCIENCE 1

SOLUTIONS

OBJECTIVE TEST

1.	C.	length of a table
2.	C.	radiation
3.	C.	reflection
4.	C.	Mechanical to sound
5.	C.	prevent current from increasing
6.	D.	Iron
7.	C.	II and III only
8.	C.	It travels in a straight line
9.	A.	free electrons
10.	B.	B
11.	D.	Hairy fruit
12.	C.	protein
13.	B.	incisors
14.	D.	xylem
15.	D.	pinna and eardrum
16.	C.	Eustachian tube
17.	C.	pupil
18.	B.	reproduction
19.	C.	air is lost during boiling

20. B. III, IV, I, II

21. C. Filtration

22. A. an alloy

23. D. Water

24. B. negative ion

25. B. carbon dioxide

26. D. scales

27. D. Hydrogen sulphide

28. A. bilharziasis

29. A. water loss

30. A. an animal parasite that feeds on plant

31. B. 28 days

32. D. turns blue litmus paper red

33. A. K

34. B. ductility

35. A. I and II

36. B. force

37. D. valves

38. B. cohesion

39. A. absorption

40. D. sticking of water droplets on the surface of a glass tube

April 2008

SCIENCE 2

1 ¼ hours

ESSAY

[75 marks]

This paper is in two sections, **A** *and* **B**

Answer Question **1** *in Section* **A** *and any other* **three** *questions in Section* **B**.

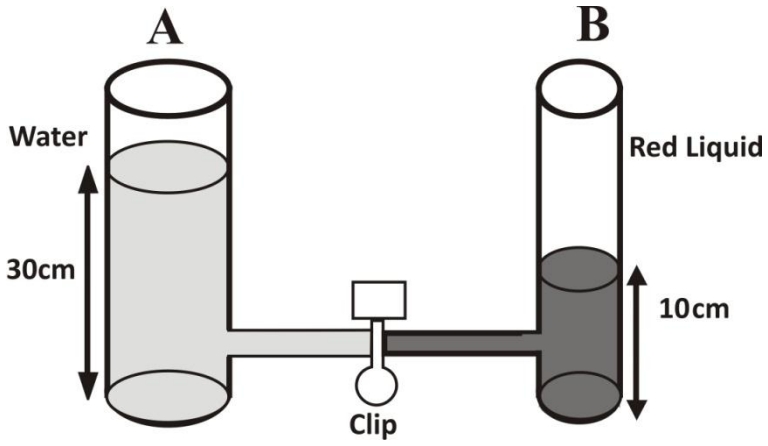
Credit will be given for clarity of expression and orderly presentation of material

SECTION A

[30 marks]

(*Compulsory – Answer* **all** *of Question* **1**)

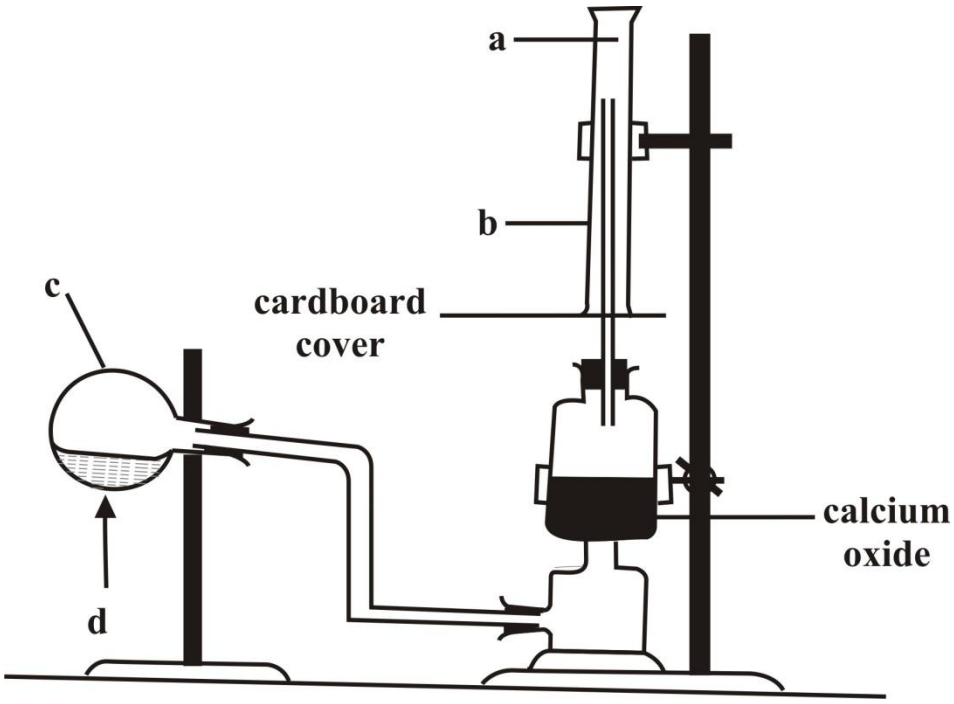
1. (a)



In an experiment, as in the set-up above, two glass containers **A** and **B** of different sizes are joined together with a tube and clipped.
Water is poured into container **A** to a height of 30 cm and a red liquid is poured into **B** to a height of 10 cm. The clip is then removed so that the liquids join together.

(i) State **two** observations that will be made immediately the clip is removed.
(ii) Explain the observations in (i)
(iii) What **two** observations will be made after a long time? Explain.

(b) The set-up below is used in the preparation of ammonia gas in the laboratory.
Study it and answer the questions that follow



(i) Name the parts labeled **a, b** and **c**.
(ii) What is the meaning of the arrow sign **d**?
(iii) What is the function of the calcium oxide?
(iv) Why has c been tilted downwards?
(v) State the method by which the gas is collected.
(vi) How will you test for the gas?
(vii) Give the names and the chemical formulae of the compounds that form the content of c.

(c) In an experiment, a leaf that is partly **green** and partly **yellow** is plucked from a tree and the leaf is
I. boiled for a minute
II. dipped in warm alcohol
III. washed in cold water
IV. dipped in iodine solution

One part of the leaf turns blue-black after the dipping in iodine solution while the other part remains unchanged.
(i) Explain why **each** of the processes **I, II** and **III** is carried out.
(ii) Which part of the leaf turns blue-black? Explain.
(iii) Why does the other part of the leaf **not** change colour?
(iv) What conclusion can you draw from the experiment?

SECTION B

[45 marks]

Answer any **three** *questions from this section*

2. (a) (i) State the difference between an opaque object and a translucent object.
(ii) Give **one** example each of an opaque and a translucent material.

(b) (i) What is the importance of seed dispersal?
(ii) Name **two** types of fruits and state their mode of dispersal.

(c) (i) What is recycling?
(ii) Give **two** advantages of recycling of materials
(iii) List **three** recycled products in Ghana.

3. (a) (i) Explain the term bedwetting?
(ii) State **two** diseases that may result from bedwetting

(b) (i) State the difference between hard water and impure water.
(ii) Give the stages involved in the treatment of water for a community.

(c) A piece of stone has a mass of 36.0 g. When it is put into water in a glass container, the level of water rises from 60.0 cm^3 to 90.0 cm^3.

(i) Calculate the density of the stone.
(ii) Explain what will be observed when the stone is put into another liquid of density 1.4 gcm^3?

4. (a) (i) What is an echo?
(ii) State **two** uses of echoes

(b) (i) State **three** diseases that affect the respiratory system of human.
(ii) Name **one** method **each** by which the diseases you have named can be prevented.

(c) (i) What is biotechnology?
(ii) List **three** products that are obtained from biotechnology.

5. (a) (i) Define pressure
(ii) State **three** applications of pressure in everyday life.

(b) State **four** functions of the liver in digestion

(c) (i) What is a compound?

(ii) Give **two** properties of a compound

(iii) Write down the name and the chemical formula of the compound formed between the following elements:

(α) hydrogen and chlorine

(β) magnesium and oxygen

April 2008

SCIENCE 2

SOLUTIONS

ESSAY

1. **(a)** **(i)** The height of the red liquid will increase as the height of the water reduces

The red liquid mixes with the water as its particles move into the water

(ii) Since the water is higher, it has greater pressure. Therefore it moves in the direction of the lower pressure

Diffusion occurs as the molecules of the red liquid move from a region of higher concentration to the water region, which is of lower concentration

(iii) A liquid uniform mixture, which has the same colour and concentration, will be formed

Since the mixture is essentially uniform, but the containers are of different sizes, the level of liquid in the smaller diameter container, **B**, will be slightly higher than the level in container **A**, due the principle of capillarity.

(b) **(i)**

a - ammonia gas
b - gas jar
c - round bottomed flask

(ii) d - Heat

(iii) The calcium oxide removes the moisture or water content from the gas produced

(iv) c is tilted downwards so that condensed water does not get back into the hot flask

(v) Upward delivery (or downward displacement of air)

(vi) Gas has pungent smell

Or: Moist red litmus paper turns blue in the presence of the gas

Or: The gas produces fumes in the presence of concentrated hydrochloric acid (HCl)

(vii) Any alkali plus ammonium chloride,

eg, $NaOH$ and NH_4Cl **or** KOH and NH_4Cl **or**
$Ca(OH)_2$ and NH_4Cl **or** $Mg(OH)_2$ and NH_4Cl

(c) **(i)** I The leaf is boiled to stop the process of photosynthesis by killing the cells

II. The leaf is dipped in warm alcohol to remove the chlorophyll (the green matter)

III The leaf is washed in cold water to wash away the alcohol and soften it.

(ii) The part that was green

The blue-black colouration indicates the presence of starch, which means that photosynthesis occurred at the green part to produce starch.

(iii) It does not contain starch. This implies that photosynthesis did not occur there due to the absence of chlorophyll, which is necessary to absorb light for photosynthesis.

(iv) Chlorophyll is necessary for photosynthesis to occur.

2. **(a)** **(i)** An **opaque** object does not allow light to pass through, whereas

A **translucent** object allows some amount of light to pass through diffusely.

(ii) Opaque - wooden or metallic materials, mirror, the earth, mammals, etc

Translucent - fabric, lightly-coloured water, oily spot on paper, frosted glass, etc

(b) **(i)** **Importance of seed dispersal**

- Enables plants to grow in other areas;
- Prevents the over-crowding of plants in one area
- Helps to reduce the rapid spread of plant diseases
- Prevents competition for soil nutrients among plants

(ii)

FRUIT		MODE OF DISPERSAL
Tridax, silk cotton,	-	Wind
Cowpea, Crotalaria, Balsam	-	Explosive Mechanism
Coconut,	-	Water
Orange, Guava, Tomatoes, Maize	-	Animals (man and others)

(c) **(i)** **Recycling** - The process of converting waste materials into new useful products

(ii) **Advantages of recycling**

- Employment / income generation for people
- Pollution of the environment by waste materials is reduced
- Reduction of resources for production
- Saving of money that would have been used for controlling waste
- Generation of energy for increased production **[any two]**

(iii) Recycled products in Ghana

Paper, biogas, polythene materials, rubber, particle boards, iron rods, etc **[any three]**

3. (a) (i) Bedwetting - The involuntary passage of urine usually during sleep.

(ii) Bedwetting diseases - ringworm, skin rash, eczema, etc [any two]

(b) (i) Hard water is water that does not easily lather with soap

Impure water is water that contains certain substances that make it unwholesome for drinking.

(ii) Stages for water treatment

- Letting water run through filters to remove suspended materials
- Adding alum to cause sediments to settle
- Adding chlorine to kill germs
- Exposing water to air (aeration) to remove any bad smell

(c) (i) Density = mass ÷ volume

Mass of stone = 36.0g

Volume of stone = $90.0cm^3 - 60.0cm^3$ = $30.0cm^3$

Therefore Density of stone = $36.0g \div 30cm^3$

= $1.2\ g/cm^3$

(ii) Since the density of the stone ($1.2g/\ cm^3$) is less than the density of the other liquid ($1.4g/cm^3$), the stone will float on the liquid

4. (a) (i) An **echo** is the result of the reflection of sound when it strikes a hard surface.

Or: the repetition of a sound caused by the reflection of sound waves from a surface

(ii) Uses of echo

- Determination of distances by bats
- Determination of velocity of sound
- Clinical diagnosis in ultra-sound equipment
- Mineral exploration
- Determination of seabed depth in special equipment
- Detection of possible obstacles by ships **[any two]**

(b) (i) Common cold, Tuberculosis, Whooping cough, Pneumonia, Asthma, Lung cancer

(ii) Common cold : avoid over-exposure to cold air or dusty environment

Tuberculosis: Avoid sharing cutlery, cups and glasses with infected persons

Whooping cough: get vaccinated

Pneumonia: wear warm clothes or avoid exposure to cold conditions

Asthma: keep away from dust, pollen grains, smoke or other substances that one is allergic to

Lung Cancer: avoid smoking

(c) (i) Biotechnology: The practical use of biological processes for man's benefit

(ii) Products obtained from Biotechnology

Cheese, yoghurt, butter, bread, wine, beer, vaccines, insulin, antibiotic drugs, food supplements, biogas, etc **[any three]**

5. (a) (i) Pressure is the force per unit area

Or Pressure = force / area

(ii) Applications of pressure in everyday life

- Pumping a car tyre;
- opening a sealed can with a pointed knife;
- pumping liquid from a low level to a higher level;
- Drawing a drink into the mouth with a straw;
- drawing liquid using a syringe **[any three]**

(b) Functions of the liver in digestion

- Converting glycogen to glucose for use by the body
- Producing bile, which is used to emulsify fats
- Storing excess glucose as glycogen or starch
- Removing excess glucose from blood

(c) **(i)** **Compound**: a substance formed by the chemical combination of elements in fixed proportions

Or: a substance containing two or more elements that are chemically combined

(ii) **Properties of compounds**

- Their properties are different from the individual constituents that form them.
- Heat changes occur when they are being formed
- Their constituents are in a fixed ratio
- Compounds cannot be broken down to obtain their reactants by any known physical means

(iii) **(α)** **Hydrogen + chlorine**

→ hydrochloric acid (HCl)

(β) **Magnesium + oxygen**

→ magnesium oxide (MgO)

April 2007

SCIENCE 1

OBJECTIVE TEST
45 minutes

1. Which of the following properties is true about all matter? Matter has
   A. volume and fixed shape
   B. volume and mass
   C. mass and indefinite volume
   D. density and fixed shape

2. The type of energy **obtained** when an electric bulb is connected to a battery is
   A. chemical energy B. electrical energy C. light energy D. magnetic energy

3. A magnet can attract all these substances except
   A. drawing pins B. glass C. steel D. iron nails

4. The process by which heat travels along a metal is called
   A. convection B. conduction C. expansion D. radiation

5. The instrument used to measure temperature is the
   A. ammeter B. barometer C. manometer D. thermometer

6. A stick placed in water appears to be bent because light travelling from the water to the air is
   A. dispersed B. displaced C. refracted D. reflected

7. A plane mirror could be described as opaque because it
   A. reflects all the light incident on it
   B. absorbs all the light incident on it
   C. allows all the light incident on it to pass through it
   D. allows some of the light incident on it to pass through it and absorbs the rest

8. A suitable machine for loading drums of palm oil onto a truck is
   A. a screw B. a crowbar C. a wheelbarrow D. an inclined plane

9. The human forearm is an example of
   A. a first class lever
   B. a second class lever
   C. a third class lever
   D. both first and third class levers

10. The chemical substances which help in the digestion of food are known as
   A. enzymes B. hormones C. plasma D. vitamins

11. Digestion of proteins starts from the
   A. duodenum B. ileum C. mouth D. stomach

12. A reflex action involves the
   A. brain and nerves
   B. bones and nerves
   C. brain and muscles
   D. spinal cord and nerves

13. Which of the following structures is not part of the central nervous system?
   A. Brain B. Spinal cord C. Nerves D. Veins

14. When the testa of a soaked bean is removed, the seed is seen to be made up of
   A. cotyledons B. embryos C. micropyles D. radicles

15. One function of the root system of plants is to
   A. excrete carbon dioxide
   B. store waste materials
   C. hold the plant firmly in the ground
   D. prepare food for the plant

16. Which of the following diseases is caused by a virus?
   A. Cholera
   B. Cerebro-Spinal Meningitis (CSM)
   C. Malaria
   D. Tuberculosis (TB)

17. The different types of teeth that an animal has gives an indication of the nature of its
   A. digestion B. environment C. feeding D. reproduction

18. Exchange of gases takes place in the respiratory system through the
A. bronchi B. lungs C. nostrils D. trachea

19. Water that requires more soap in order to form lather is described as
A. clean B. hard C. pure D. soft

20. One best way of making water safe for drinking is by
A. adding washing soda
B. filtering
C. adding common salt
D. boiling

21. The process whereby a hot saturated solution is cooled to obtain the solute is called
A. condensation
B. crystallization
C. distillation
D. evaporation

22. If a mixture of water and powdered charcoal is allowed to stand for a long time, the charcoal
A. rises to the top
B. settles at the bottom
C. dissolves completely in the water
D. continues to remain suspended in the water

23. The chemical symbol of potassium is
A. Na B. K C. Pb D. S

24. The atom of an element is represented as $^{35}_{17}\mathrm{Cl}$. How many electrons are in the atom?
A. 52 B. 35 C. 18 D. 17

25. Which of the following agricultural practices prevents loss of water from the soil?
A. Addition of manure
B. Mulching
C. Addition of humus
D. Composting

26. Soil conservation is important because it prevents loss of
A. carbon dioxide B. oxygen C. minerals D. humus

27. Foot rot disease in animals is caused by

A. bacteria B. fungi C. protozoa D. viruses

28. A patient with symptoms of diarrhoea and vomiting may be suffering from

A. cholera B. dysentery C. influenza D. tuberculosis

29. Which of the following human activities pollutes the environment?

I. Release of cement dust into the atmosphere
II. Dumping of refuse into water bodies
III. Burning of rubbish

A. I and II only B. II and III only C. I and III only D. I, II and III

30. Which of the following features of a parent would not be inherited by the children?

A. Colour of eye B. Shape of nose C. Height D. Scar

31. Which of the following bodies is at the centre of the solar system?

A. Earth B. Mars C. Moon D. Sun

32. The chemicals used in the preparation of ammonia in the laboratory are

A. ammonium chloride and calcium hydroxide
B. ammonium nitrate and sodium chloride
C. ammonium sulphate and calcium chloride
D. ammonium sulphate and sodium chloride

33. Steel is an alloy of

A. iron and carbon B. zinc and copper C. iron and zinc D. brass and zinc

34. The property of a metal that makes it possible for it to be beaten into different shapes is called

A. conductivity B. ductility C. malleability D. resistivity

35. Movement of the particles of a gas from a region of higher concentration to a region of lower concentration is known as

A. capillarity B. diffusion C. osmosis D. viscosity

36. Oxygen in the blood finally reaches the parts of the body through the

A. arteries B. alveoli C. capillaries D. veins

37. An aluminium cube with side measuring 2 m has a mass of 24 kg. Calculate the density of the aluminium.
A. 48 kgm^{-3} B. 24 kgm^{-3} C. 12 kgm^{-3} D. 3 kgm^{-3}

38. The force that opposes the motion of an object on another object is called
A. electrical force B. frictional force C. gravitational force D. magnetic force

39. Water is prevented from passing through the tiny holes in an umbrella by
A. diffusion B. frictional force C. capillarity D. surface tension

40. Which of the following methods of preserving food makes use of heat energy from the sun?
A. Drying B. Frying C. Salting D. Refrigeration

April 2007

SCIENCE 1

SOLUTIONS

OBJECTIVE TEST

1.	B.	volume and mass
2.	C.	light energy
3.	B.	glass
4.	B.	conduction
5.	D.	thermometer
6.	C.	refracted
7.	A.	reflects all the light incident on it
8.	D.	an inclined plane
9.	C.	a third class lever
10.	A.	enzymes
11.	D.	stomach
12.	D.	spinal cord and nerves
13.	D.	Veins
14.	A.	cotyledons
15.	C.	hold the plant firmly in the ground
16.	B.	Cerebro-Spinal Meningitis (CSM)
17.	C.	feeding
18.	B.	lungs
19.	B.	hard

20. D. boiling

21. B. crystallization

22. B. settles at the bottom

23. B. K

24. D. 17

25. B. Mulching

26. C. minerals

27. B. fungi

28. A. cholera

29. D. I, II and III

30. D. Scar

31. D. Sun

32. A. ammonium chloride and calcium hydroxide

33. A. iron and carbon

34. C. malleability

35. B. diffusion

36. C. capillaries

37. D. 3 kgm^{-3}

38. B. frictional force

39. D. surface tension

40. A. Drying

April 2007

SCIENCE 2

1 ¼ hours

ESSAY

[75 marks]

This paper is in two sections, **A** *and* **B**

Answer Question **1** *in Section* **A** *and any other* **three** *questions in Section* **B**.

Credit will be given for clarity of expression and orderly presentation of material

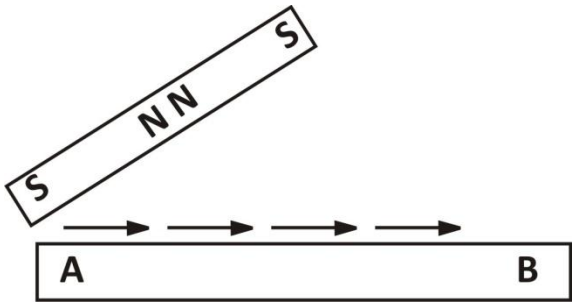
SECTION A

[30 marks]

(*Compulsory – Answer* **all** *of Question* **1**)

1.

a) In an experiment, an iron bar is magnetized by dragging a magnet over the surface of the bar from end A to end B several times as shown in the diagram below.



(i) Mention the method of magnetization

(ii) Give the polarity of the ends A and B of the bar after the magnetization.

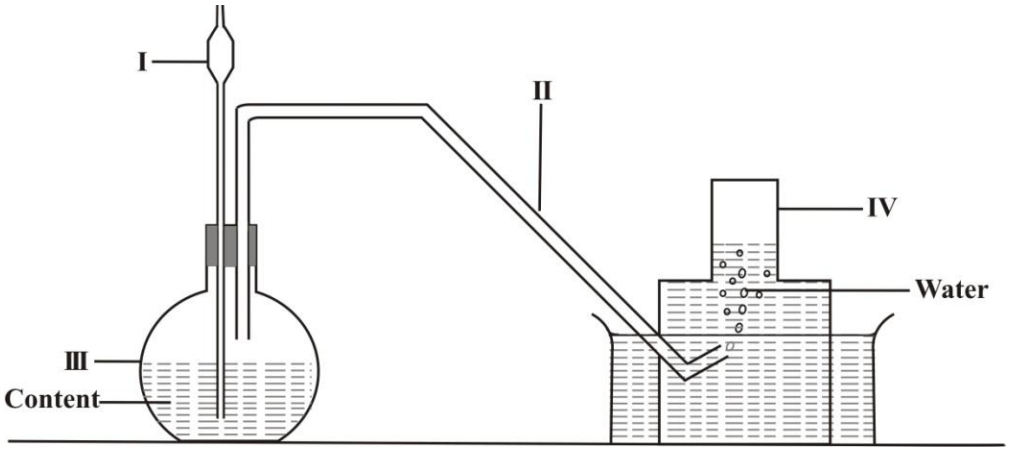
(iii) How will you test that the bar AB is a magnet?

(iv) Indicate how you will test for the poles of A and B of the bar

(v) Give one precaution that should be taken during the magnetization process.

(vi) Name one other method of magnetization

b) The set-up below is used to prepare gases in the laboratory



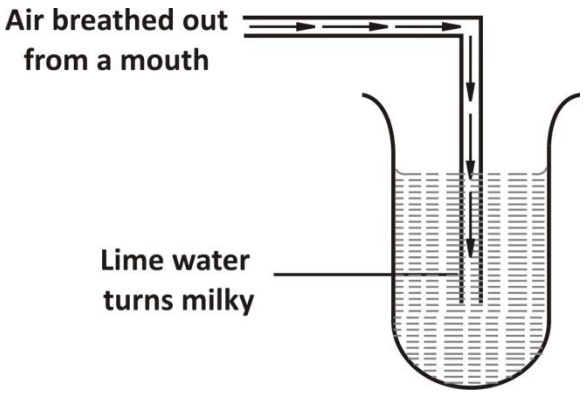
(i) Give the names of the parts labelled I, II, III and IV

(ii) Write down the two names of the method of gas collection

(iii) Explain how the gas collects over the water

(iv) Name two gases that can be prepared using the set-up

c) The set-up below shows air being breathed out through the mouth into test-tube containing lime water



(i) Why does the lime water turn milky?

(ii) Identify the milky substance produced.

(iii) Write a balanced chemical equation for the reaction.

(iv) Name two other substances present in breathed-out air.

(v) What is the aim of the experiment?

SECTION B

[45 marks]

Answer any **three** *questions from this section.*

2.

a) What is a living cell?

b) Give one function of each of the following components of a living cell:
   (i) cell membrane
   (ii) chloroplast
   (iii) nucleus

c) Make sketches to show the arrangement of the particles in each of the three states of matter.

d) What instrument would you use to measure each of the following quantities:
   (i) volume of a liquid
   (ii) mass of a stone
   (iii) time
   (iv) speed of the wind?

3.

a) Name three deficiency diseases associated with diet and state their causes.

b) (i) What is hard water?
   (ii) Mention three ways by which hard water can be made soft.

c) (i) Explain how sound is produced.
   (ii) State one example each, of the following instruments
      α) wind instrument
      β) string instrument

4.

a) State two examples of a
   (i) carnivore
   (ii) herbivore

b) What is the difference between egestion and excretion?

c) Explain briefly why
   (i) gold is used to make necklaces and earrings.
   (ii) steel is used instead of iron to make car bodies

d) (i) What is surface tension?
(ii) Explain why the surface of water in a container is not flat but curved.

5.

a) Copy and complete the table below:

Disease	**Causative Agent**	**Prevention/Control**
Cholera	Vibrio spp.	Eating hot food
i) ……..	Plasmodium	ii) …….
iii) ……	Louse	Personal hygiene
Ringworm	iv) ……..	v)……

b) Name two different particles of matter which carry
(i) a negative charge
(ii) a positive charge
(iii) no charge

c) A simple machine moves a load of 20N through a distance of 2m. If it uses an effort of 25N which moves through a distance of 4 m, calculate the
(i) work input
(ii) work output
(iii) efficiency of the machine

April 2007

SCIENCE 2

SOLUTIONS

ESSAY

1. **(a)** **(i)** Single Stroking or single stroke

(ii) A: South pole

B: North pole

(iii) Testing

- Bring the bar close to iron material such as nails, pins, office clips, iron filings, etc. The materials get attached to the bar.
- Hold the bar close to a plotting compass. The compass is deflected
- Tie a string / thread at the middle of the bar and suspend it horizontally. It comes to rest in the North-South direction.
- Tie a string / thread at the middle of the bar and suspend it horizontally. Bring a magnet close to one pole(end) of the bar. It either gets attracted or repelled.

[any one]

(iv) The south pole of a known magnet is brought near either end of the new bar magnet. There will be repulsion at end A, and an attraction at end B. Where there is repulsion, it indicates that that end is also a south pole; and where there is attraction, it indicates a north pole.

If the north pole of a known magnet is brought near the ends, there will be attraction at end A and repulsion at end B, indicating that end A is a south pole, while B is north.

(v) The magnet should be dragged over the surface of bar AB in the same direction always.

The magnet must be completely removed far away from the bar AB at the end of each stroke.

(vi) Other method of magnetization

- Electrical method
- Hammering
- Double stroke
- Induction **[any one]**

(b) **(i)**

I : thistle funnel
II : delivery tube
III : flat-bottomed flask
IV : gas jar

(ii) Upward delivery

Downward displacement of water

(iii) The gas molecules, which are lighter, rise above the water molecules, thereby exerting pressure on the water molecules and forcing it down

(iv) Hydrogen, Carbon dioxide / oxygen

(c) **(i)** The presence of carbon dioxide (contained in the exhaled air)

(ii) Calcium carbonate **or** Calcium trioxocarbonate (IV) **or** $CaCO_3$

(iii) $Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$

(iv) Nitrogen, water vapour, inert gases and unused oxygen

(v) To show that Carbon dioxide is present in exhaled air

Or: To show that Carbon dioxide is a by-product of respiration

2. **(a)** A living cell is the smallest independently functioning unit in the structure of a living organism

Or: the most basic unit of a living organism that can function independently.

(b) **(i)** **Cell membrane**: protects it from changes in the chemical and physical environment;

Regulates the movement of molecules into and out of the cell

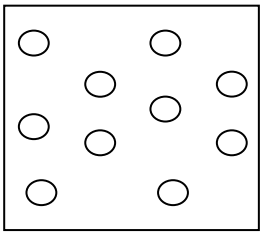
(ii) **Chloroplast**: contains chlorophyll and other pigments;

It is the place where photosynthesis occurs within the cells of plants.

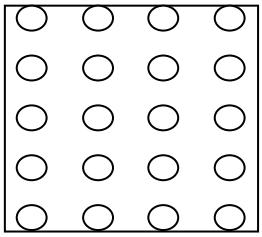
(iii) Nucleus: The nucleus carries the cell's genetic information;

It directs most cell activities

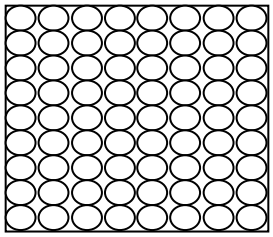
(c) Arrangement of particles in gas, liquid and solid



Gas



Liquid



Solid

(d) (i) Volume of a liquid: measuring cylinder or burette or volumetric flask

(ii) mass of a stone: top pan balance or beam balance

(iii) time: stop clock or clock or watch

(iv) speed of the wind: anemometer

3. (a)

Beriberi:	Lack of vitamin B
Kwashiorkor:	Lack of protein
Rickets:	Lack of vitamin D
Scurvy:	Lack of vitamin C
Night blindness:	Lack of vitamin A
Goitre:	Lack of iodine

[any three]

(b) (i) Hard water is water that does not lather easily with soap

(ii) Softening Boiling, Addition of washing soda and Distillation

(c) (i) Sound is produced when a vibrating body causes corresponding vibration in a mass of air around it

(ii) (α) Wind instrument: flute, trumpet, horn, clarinet, etc, [any one]

(β) String instrument: guitar, harp, banjo, violin, cello, piano, etc, [any one]

4. (a) (i) Carnivore: lion, tiger, crocodile, wolf, hyena, etc, [any two]

(ii) **Herbivore:** goat, cow, grasshopper, grasscutter, sheep, etc [any two]

(b) **Egestion** is the elimination (removal) of undigested food substances from the organism, while,

Excretion is the elimination (removal) of metabolic waste substances from the organism.

(c) **(i)** **Gold**

- Is highly non-reactive, hence it does not easily get stained.
- Is highly malleable, which makes it easy to mold
- Has high Lustre, which makes it very attractive

(ii) **Steel**, an alloy of iron and carbon, stronger than iron and does not rust easily, unlike iron which rusts in the presence of air and moisture.

(d) **(i)** **Surface Tension:**

A condition existing at the surface of a liquid, resembling a stretched elastic skin

Or: the property of a liquid that makes its surface behave like a stretched elastic skin.

(ii) There exist an adhesive force of attraction between the water molecules and the walls of the container, which is greater than the cohesive forces among the water molecules. This makes the water cling to the walls. This results in the curved surface.

5. **(a)**

Disease	**Causative Agent**	**Prevention / Control**
Cholera	Vibrio spp.	Eating hot food
i) Malaria	Plasmodium	ii) ▪ Using mosquito net ▪ Using mosquito repellent ▪ Using anti-malaria drugs ▪ Clearing mosquito breeding grounds such as choked gutters, stagnant pools of water, etc
iii) Relapsing fever or typhus or anaemia	Louse	Personal hygiene
Ringworm	iv) fungus	v) ▪ Keep skin clean and dry ▪ Avoid contact with infected persons or animals

		▪ Avoid sharing personal items like clothes, combs or towels ▪ Use antifungal medication

b) **(i)** **a negative charge:** an electron, negative beta particle (negatron), anion

(ii) **a positive charge:** a proton, positive beta particle (positron), alpha particle, cation

(iii) **no charge:** a neutron, atom, molecule

c) **(i)** Work input = effort × effort distance

= 25 N × 4 m

= 100 Nm or 100 J

(ii) Work output = load × load distance

= 20 N × 2 m

= 40 Nm or 40 J

(iii) Efficiency of machine = (Work output / Work input) × 100%

= (40 J / 100J) × 100%

= 40%

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SCIENCE

SECTION A
OBJECTIVE TEST
45 minutes

1. The property of a rock that makes it different from water is that it has
- A. fixed volume
- B. fixed shape
- C. density
- D. mass

2. A plane mirror can be described as opaque because it
- A. absorbs all the light incident on it.
- B. reflects all the light incident on it
- C. allows all the light incident on it to pass through it
- D. partly absorbs and partly allows the light incident on it to pass through it

3. Wood is used to make the handles of cooking utensils because it is a
- A. bad conductor of heat
- B. good absorber of heat
- C. good reflector of heat
- D. bad radiator of heat

4. Ventilation holes are made at the upper part of a door to allow air to rise by
- A. convection
- B. conduction
- C. radiation
- D. reflection

5. When the north poles of two magnets are brought together they
- A. attract each other
- B. repel each other
- C. first attract and then repel each other
- D. have no effect on each other

6. A fuse is used in an electrical circuit to
- A. measure the energy used up
- B. protect appliances
- C. provide more current

D. make sure current flows continuously

7. The angle between an incident ray and a reflecting surface is 20°. Determine the angle of reflection.
A. 60°
B. 70°
C. 90°
D. 110°

8. A screw is a machine which works in the same way as
A. a lever
B. an inclined plane
C. a wheel and axle
D. a pulley

9. Power is defined as the rate at which
A. energy is used
B. temperature rises
C. a body travels
D. a body cools down

10. An example of a lever in which the effort is applied at the middle is a
A. wheelbarrow
B. pair of forceps
C. pair of scissors
D. beam balance

11. A simpler way of maintaining the efficiency of a machine is by
A. oiling its parts
B. fitting new parts on the machine
C. making sure it is always working
D. making sure it works at low speed.

12. The part of a seed which grows to become the shoot of a plant is the
A. cotyledon
B. plumule
C. radicle
D. testa

13. One disease that affects the nervous system of humans is
A. filariasis
B. measles
C. poliomyelitis
D. typhoid

14. Which of the following life processes leads to the release of energy?

A. Absorption of food
B. Digestion of food
C. Photosynthesis
D. Respiration

15. Mineral salts in dead organisms are released into the soil by a process called
A. decomposition
B. diffusion
C. leaching
D. osmosis

16. Spectacles fitted with concave lenses are worn by people suffering from
A. loss of accommodation
B. total blindness
C. long sightedness
D. short sightedness

17. Which of these organs in humans releases carbon dioxide as a waste product?
A. Kidney
B. Liver
C. Lung
D. Skin

18. Farmers do not plant the stems of maize because the stems
A. do not have buds
B. are difficult to obtain
C. dry up too quickly
D. do not contain food

19. Which of the following ways of treating water makes the water soft?
A. Adding alum
B. Adding washing soda
C. Chlorinating
D. Filtering

20. Water that needs a small amount of soap to form lather is described as
A. clean
B. hard
C. pure
D. soft

21. Common salt (sodium chloride) is obtained from sea water by
A. condensation
B. evaporation
C. precipitation

D. sublimation

22. The chemical formula of a compound describes the
A. ratio in which the elements are combined
B. number of molecules in a compound
C. nature of the bonding in the compound
D. state of matter of the compound

23. Which of the following solvents can be used to remove grease from the hand?
A. Alcohol
B. Palm kernel oil
C. Turpentine
D. Water

24. How many different elements are present in the compound H_2SO_4?
A. 2
B. 3
C. 6
D. 7

25. *Producers* in an ecosystem are plants that
A. feed on other plants
B. feed on dead materials
C. manufacture their own food
D. attract

26. The disease caused by the *Plasmodium* parasite is
A. cholera
B. malaria
C. onchocerciasis
D. schistosomiasis

27. Vaccination is carried out to
A. enable sores to heal faster
B. increase the red blood cells of a person
C. enable a person's white blood cells to fight against diseases
D. kill germs in a person's body

28. Which of the following farming practices is likely to make the soil poor in nutrients?
A. Adding chemical fertilizer during each planting season
B. Planting cover crops on the land
C. Tilling the land before planting
D. Burning the vegetation after clearing

29. The type of cloud found **closest** to the earth's surface is

A. cirrus
B. cumulus
C. stratus
D. stratocumulus

30. The amount of water vapour in the atmosphere is called
A. fog
B. humidity
C. rain
D. temperature

31. The planet which is **farthest** away from the sun is
A. Jupiter
B. Mercury
C. Mars
D. Pluto

32. Which of the following gases is involved in the process of rusting?
A. H_2
B. CO_2
C. O_2
D. N_2

33. A non-reactive metal can be identified from other metals because its surface is
A. hard
B. shiny
C. smooth
D. rough

34. The property to a metal that makes it possible for it to be drawn into a wire is called
A. conductivity
B. ductility
C. malleability
D. resistivity

35. The **main** reason why food is processed is to make it
A. taste better
B. preservable for a longer period
C. possible to eat without cooking
D. richer in nutrients

36. The process by which kerosene rises up in the wick of a lantern is known as
A. capillarity
B. diffusion
C. surface tension

D. osmosis

37. Which of the following statements about friction is **not** true?

A. It wears off materials
B. It produces heat
C. It opposes motion
D. It increases speed

38. Arteries are blood vessels which carry

A. blood to the heart
B. carbon dioxide to the air
C. blood away from the heart
D. oxygen to the lungs

39. Which of the following structures is **not** involved in blood circulation?

A. Capillary
B. Heart
C. Muscle
D. Vein

40. A substance of mass 10 kg has a density of 2.0 kg m^{-3}. Calculate its volume.

A. 0.2 m^3
B. 5.0 m^3
C. 10.0 m^3
D. 20.0 m^3

April 2006

SCIENCE Section A

SOLUTIONS

OBJECTIVE TEST

1.	B.	fixed shape
2.	B.	reflects all the light incident on it
3.	A.	bad conductor of heat
4.	A.	convection
5.	B.	repel each other
6.	B.	protect appliances
7.	B.	70°
8.	B.	an inclined plane
9.	A.	energy is used
10.	B.	pair of forceps
11.	A.	oiling its parts
12.	B.	plumule
13.	C.	poliomyelitis
14.	D.	Respiration
15.	A.	decomposition
16.	D.	short sightedness
17.	C.	Lung
18.	A.	do not have buds
19.	B.	Adding washing soda

20. D. soft

21. B. evaporation

22. A. ratio in which the elements are combined

23. C. Turpentine

24. B. 3

25. C. manufacture their own food

26. B. malaria

27. C. enable a person's white blood cells to fight against diseases

28. D. Burning the vegetation after clearing

29. C. stratus

30. B. humidity

31. D. Pluto

32. C. O_2

33. B. shiny

34. B. ductility

35. B. preservable for a longer period

36. A. capillarity

37. D. It increases speed

38. C. blood away from the heart

39. C. Muscle

40. B. 5.0 m^3

April 2006

SCIENCE

SECTION B

1 hour 15 minutes

ESSAY

[75 marks]

This section is in two parts, **I** *and* **II**

Answer Question **1** *of Part* **I** *and any other* **three** *questions in Part* **II**.

Credit will be given for clarity of expression and orderly presentation of material

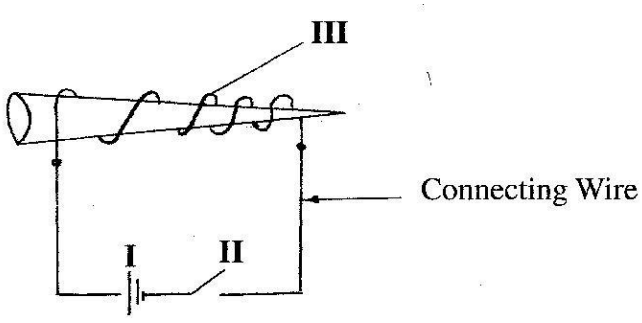
PART I

[30 marks]

(*Compulsory – Answer* **all** *of Question* 1)

1. (a) The diagram below shows the electrical method of magnetizing a nail.

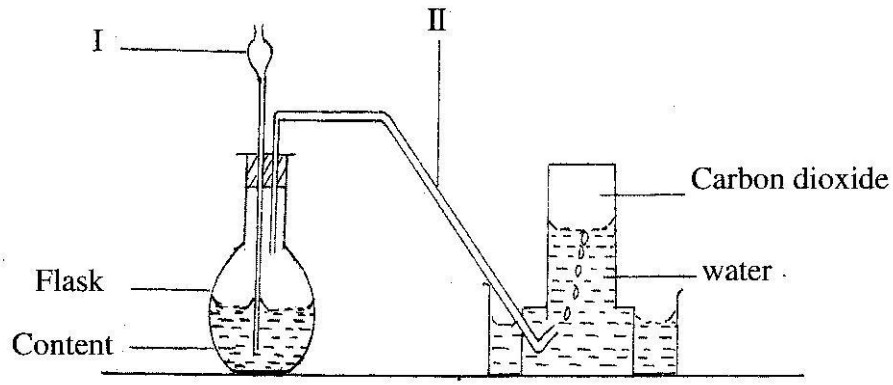
Study the diagram and use it to answer the questions that follow.



(i) Give the names of the parts of the circuit labelled I, II and III
(ii) List two substances that can be made into a magnet.
(iii) State two other methods of making magnets.
(iv) Name one material that is used in making an electrical wire
(v) Give two properties of the material you named in (iv) which makes it useful as a wire.

[10 marks]

(b) The diagram below shows the set-up for the preparation of carbon dioxide in the laboratory.



(i) Name the parts of the set-up labelled **I** and **II**

(ii) Give the name of the method of gas collection shown in the diagram.

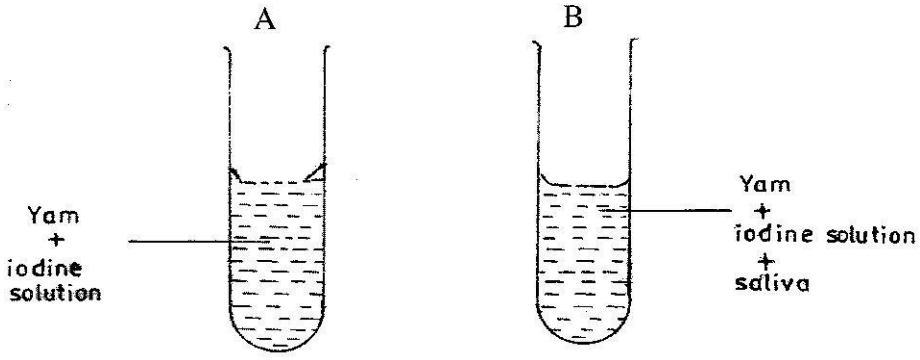
(iii) Give one property of gases collected over water

(iv) What will happen if component **I** does **not** dip into the contents of the flask?

(v) List **two** compounds that form the content of the flask.

(vi) Write down the systematic name of carbon dioxide. [10 marks]

(c)



In an experiment, yam pap is put into two test tubes A and B containing iodine solution. The test tubes are warmed slightly to a temperature of 37°C and saliva is put into test tube B

(i) State the colour of the content of test tube A

(ii) State the colour changes of the contents in test tube B after about 3 minutes.

(iii) Fehling's solution is added to the contents of test tube B after the 3 minutes and it turns brick-red. What food substance is present?

(iv) Give two functions of saliva in eating.

(v) Why was it necessary to warm the contents of the test tubes to about 37°C?

(vi) Give two aims of the experiment. [10 marks]

PART II

[45 marks]

(*Answer* **three** *questions* **only** *from this part*)

2. (a) (i) State two conditions that are required for photosynthesis to take place.

(ii) How would you show that starch is formed during photosynthesis?

(b) Write a balanced chemical equation to show the reaction between hydrogen and oxygen.

(c) (i) What is a satellite?

(ii) Give one example of a natural satellite.

(iii) List two uses of artificial satellites.

3. (a) Define

(i) self-pollination

(ii) cross-pollination

(b) State two ways in which cross-pollinated plants are better than self-pollinated plants.

(c) Sodium chloride is prepared by the reaction between dilute hydrochloric acid and sodium hydroxide.

(i) Write a balanced equation for the reaction

(ii) What is the name given to this reaction?

(d) Describe an experiment to show that pressure acts in all directions in a liquid

4. (a) (i) What is a vector of a disease

(ii) State **two** methods **each** by which the vectors of the following diseases can be destroyed:

α) river blindness;

β) malaria

(b) Write down the systematic names of the following compounds:

(i) $CaCO_3$
(ii) FeS
(iii) NaCl
(iv) NaOH

(c) (i) Give the two properties that are common to all states of matter.

(ii) In an experiment to determine density, a stone of mass 60 g is put into a bowl containing water

If the level of the water rises from the 60 cm^3 mark to 75 cm^3, determine the density of the stone.

5. (a) Mention five differences between plants and animals

(b) Define each of the following terms, giving one example in each case
(i) compound
(ii) element

(c) (i) Give two differences between an electrical insulator and a conductor.

(ii) Draw and label a simple electrical circuit made up of a cell and a switch connected to two electrical bulbs connected in parallel

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SCIENCE Section B

SOLUTIONS

ESSAY

1. **(a)** **(i)** **I** - Cell

II - Key or switch

III - Solenoid or coil

(ii) **Substances:** Iron (Fe), Steel, Nickel (Ni), Cobalt (Co) [any two]

(iii) **Methods of making magnets**

- Single stroking
- Double stroking
- Induction
- Hammering [any two]

(iv) **Material for electrical wire:**

Copper (Cu), aluminum (Al), steel, iron (Fe), gold (Au), silver (Ag), and platinum (Pt)

[any one]

(v) **Property of material**

- ductile,
- malleable,
- low resistivity
- high tensile strength [any two]

(b) **(i)** **I** - thistle funnel

II - delivery tube

(ii) Upward delivery or downward displacement of water

(iii) **They are**

- insoluble in water;
- less dense than water [any one]

(iv) The gas produced will escape through the thistle funnel

(v) **Content of flask**

- Calcium carbonate [$CaCO_3$] - (reactant)
- Hydrochloric acid [HCl] - (reactant)
- Calcium chloride [$CaCl_2$] - (product)
- Water [H_2O] - (product)

(vi) Carbon (IV) oxide

(c) **(i)** Blue black

(ii) pale blue black

(iii) maltose or reducing sugar

(iv) **Functions of saliva**

- makes the food softer for easier chewing and swallowing
- Contains the enzyme ptyalin, which breaks down carbohydrates to reducing sugar

(v) To attain the normal body temperature, at which the enzyme in saliva can work efficiently

(vi) **Aims of experiment**

- To show that saliva contains enzymes that digest carbohydrates or starch
- To show that digestion of carbohydrates starts in the mouth, where saliva is produced
- To show that yam contains starch

2. **(a)** **(i)** The presence of water, carbon dioxide, chlorophyll and light

(ii) **Experiment**

- Pluck a leaf from a living plant that has been in sunlight for 3 or more hours.
- Boil the leaf to kill the cells and stop the process of photosynthesis
- Place the leaf in warm alcohol for some minutes to remove the chlorophyll
- Place the leaf in warm water to wash away the alcohol and soften the leaf
- Place a few drops of iodine solution on the leaf

Observation and Conclusion

It is observed that the leaf turns blue-black. This colouration indicates the presence of starch. This show that starch is formed during photosynthesis

(b) $H_2 + O_2 \rightarrow 2H_2O$

(c) **(i)** A satellite is any heavenly body that orbits (revolves around) a larger one.

(ii) The moon is a natural satellite of the earth;

The planets are natural satellites of the sun

(iii) **Uses of artificial satellites**

- Weather studies and forecasting
- Telecommunication
- Sending and receiving TV signals
- Taking photographs from space for various purposes
- Internet networking
- Global Positioning System (GPS) - a worldwide navigation system

3. **(a)** **(i)** **Self-pollination** is
The transfer of pollen grains from an anther of a flower to the stigma of the same flower

(ii) **Cross-pollination** is
The transfer of pollen grains from the anther of a flower to the stigma of another flower on another plant of the same kind

(b) Produce offspring that are more resistant to diseases

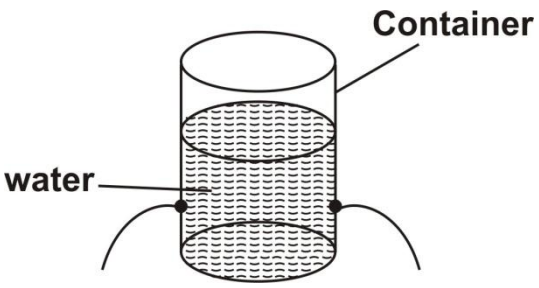
Produce offspring that are stronger and healthier than parents

Causes genetic variations in offspring that make them better then parents [any two]

(c) **(i)** $HCl + NaOH \rightarrow NaCl + H_2O$

(ii) Neutralization reaction

(d) Experiment



- Get a suitable container and make holes of equal sizes at the same height around it (as shown in the diagram above
- Pour water (or other liquid) into the container to fill it
- The water (or liquid) comes out through the holes in all directions and with the same pressure
- This shows that pressure acts in all directions in a liquid

4. (a) (i) **A vector** is
an agent that carries and spreads organisms that cause disease from infected organisms to non-infected ones.

(α) river blindness

- Killing the vector, the black fly, by spraying chemicals onto the river banks or wherever they are located
- Destroying the breeding grounds of the black fly by clearing the bushes along the banks of rivers where they are found.

(β) malaria

- Killing the mosquito by spraying appropriate insecticide at places where they are located
- Killing the mosquitoes by spraying room with insecticide.
- Destroying the breeding grounds of mosquitoes by clearing choked gutters
- Destroying the hiding places of mosquitoes by clearing bushes
- Killing the larvae of mosquitoes by pouring oil on the surface of stagnant water
- Killing mosquitoes by using insecticide treated mosquito nets
- Killing mosquito larvae in ponds by stocking with fish that feed on the larvae.

(b) (i) Calcium trioxocarbonate (IV)

(ii) Iron (II) sulphide

(iii) Sodium Chloride

(iv) Sodium hydroxide

(c) (i) **Properties**

- Matter has mass
- Matter has volume (occupy space)

(d) Density = mass ÷ volume

Mass of stone = 60kg

Volume of stone = $75cm^3 - 60cm^3$

= $15cm^3$

Density = $60\ g \div 15\ cm^3$

= $4\ g/cm^3$ or $4\ g\ cm^{-3}$

5.

(a) Differences

PLANTS	ANIMALS
Manufacture their own food (photosynthesis)	Do not manufacture their own food
Cannot move freely from place to place	Able to move freely from place to place
Most plants do not respond quickly to stimuli	Respond quickly to stimuli
Have a cell wall	Do not have a cell wall
Have chloroplasts, which contain chlorophyll	Do not have chloroplast
Excess carbohydrates are stored as starch	Excess carbohydrates are stored as glycogen
Absorb carbon dioxide for photosynthesis and release oxygen as waste product	Inhale oxygen for respiration and release carbon dioxide as waste product
Cells have large and permanent vacuoles	Cells have small temporary vacuoles
Growth takes place at specific parts	Growth takes place at all parts

[any five]

(b) (i) A compound is a substance formed by the chemical combination of two or more elements in definite proportions

Example, HCl, or Nacl, or $Ca(OH)_2$, or $CaCl_2$, or NH_3, or $MgCO_3$, etc

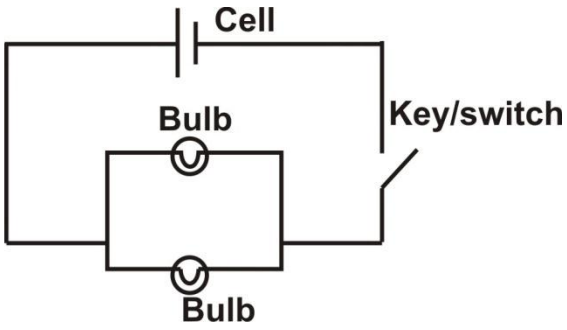
(ii) An element is a substance that is made up of only one kind of atoms.

Example are: Calcium, or Sodium, or Magnesium, or Argon, or Sulphur, etc

(c) (i)

Electrical Insulator	**Electrical Conductor**
Does not conduct electricity	Conducts electricity
There are no free electrons to conduct electricity	There are free electrons which make electrical conduction possible
Narrow conduction energy band	Wide conduction energy band
Wide forbidden gap between valence and conduction band	No forbidden gap between valence and conduction band

(ii) simple electrical circuit



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SCIENCE

SECTION A
OBJECTIVE TEST
45 minutes

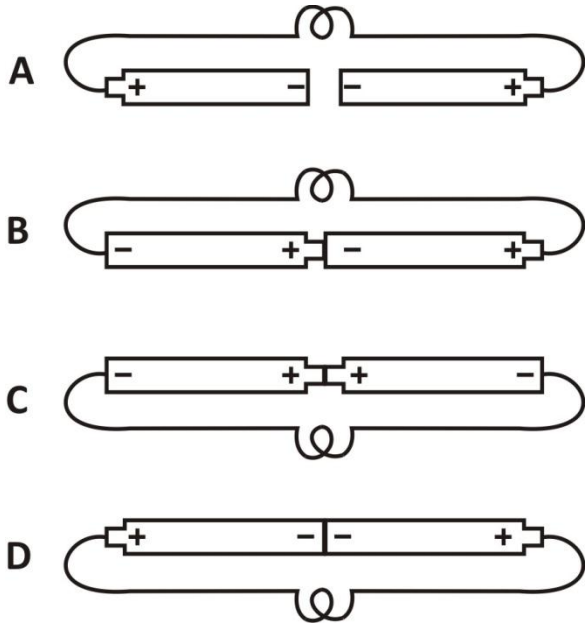
1. Hydrogen is represented by the chemical symbol H_2. This symbol represents
A. two molecules of hydrogen
B. two atoms of hydrogen
C. two elements of hydrogen
D. two ions of hydrogen

2. A metal expands when there is
A. an increase in heat energy
B. an increase in number of electrons
C. a decrease in potential energy
D. a decrease in distance between atoms

3. A safety device which opens an electrical circuit to prevent too much current from passing through the circuit is the
A. earth wire
B. lightning conductor
C. fuse
D. switch

4. An object is placed 20 cm in front of a plane mirror. The distance between the object and the image is
A. 20 cm
B. 40 cm
C. 60 cm
D. 80 cm

5. In which of the following circuit diagrams below will the bulb light?



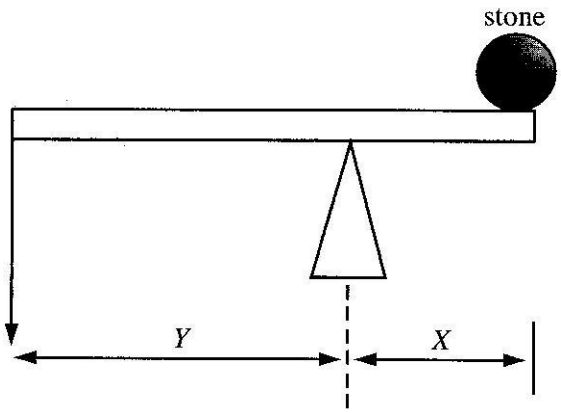
6. The source of all forms of energy can be traced to
- A. clouds
- B. earthquakes
- C. tides
- D. sunlight

7. When a mango is falling from a tree, its potential energy is changed to
- A. sound energy
- B. heat energy
- C. kinetic energy
- D. chemical energy

8. Electricity is used by appliances to do work. This means that electricity is a form of
- A. energy
- B. force
- C. generator
- D. machine

9. A force of 2N moves a body through a distance of 10 m. Calculate the work done.
- A. 5 J
- B. 8 J
- C. 12 J
- D. 20 J

The diagram below shows a lever system used to move a stone. Use it to answer Questions 10 and 11



10. The distance X is the
- A. effort distance
- B. load distance
- C. fulcrum
- D. lever arm

11. Less effort is required to move the stone when the
- A. distance Y is equal to X
- B. distance Y is greater than X
- C. distance Y is less than X
- D. stone is at the pivot

12. An omnivore is an animal that feeds on
- A. raw food
- B. fresh foods
- C. cooked foods
- D. all kinds of food

13. A meal containing all the essential nutrients in the right amounts is said to be
- A. delicious
- B. balanced
- C. well-cooked
- D. rich in fibre

14. The disease associated with insufficient intake of proteins in children is
- A. goiter
- B. kwashiorkor
- C. rickets
- D. scurvy

15. When testing for proteins in a food substance using Fehling's solution, the expected colour change is
- A. blue
- B. purple
- C. blue-black

D. brick-red

16. The substance that enables green plants to trap sunlight for the manufacture of food is

A. chlorophyll
B. chloroplast
C. phloem
D. xylem

17. In man, the conversion of poisonous substances into harmless forms takes place in the

A. bladder
B. duodenum
C. kidney
D. liver

18. Which of the following life processes is represented by the equation below?

Glucose + Oxygen → Water + Carbon dioxide + Energy

A. Digestion
B. Excretion
C. Photosynthesis
D. Respiration

19. Chlorine gas is passed through water during purification to

A. kill germs
B. cause suspended particles to settle
C. soften the water
D. give taste to the water

20. Arrange the following sources of water in the order of increasing contamination

I. Rain
II. Stream
III. Well
IV. Borehole

A. I, II, IV, III
B. II, III, I, IV
C. I, IV, III, II
D. III, II, IV, I

21. A solution in which no more solute will dissolve at a given temperature is said to be

A. concentrated
B. dilute
C. homogeneous
D. saturated

22. A clear solution of sugar was cooled from 100°C to 25°C. Some solid sugar was seen to have formed out of the solution after cooling. This shows that sugar

A. does not dissolve in cold water
B. does not dissolve in hot water
C. dissolves more in hot water than in cold water
D. dissolves more in cold water than in hot water

23. Water is sometimes referred to as a universal solvent because it

A. is the purest liquid on earth
B. is the commonest liquid on earth
C. is found in all living cells
D. dissolves most substances

24. The charge of sulphur in the compound SO_2 is

A. -2
B. +2
C. -4
D. +4

25. The part of the soil that supports plant growth is that part which

A. dissolves in water
B. retains less water
C. contains the humus
D. is closer to the roots

26. Plants do not grow well in gravel because the

A. air spaces are too big to hold water
B. air spaces are too small for the roots
C. particles cannot absorb air
D. particles cannot contain nutrients.

27. Which of the following parasites lives in the blood of humans?

A. Louse
B. Plasmodium
C. Tapeworm
D. Tick

28. The carrier of the malaria parasite is the

A. black fly
B. tsetse fly
C. female anopheles mosquito
D. male anopheles mosquito

29. Which of the following practices is a method of controlling guinea worm disease?

A. Clearing bushes around homes
B. Ensuring that rivers flow rapidly
C. Protecting one's feet before stepping into rivers
D. Protecting one's self against mosquito bites.

30. The humidity of the atmosphere is measured with
A. an anemometer
B. a barometer
C. a hygrometer
D. a hydrometer

31. How long does it take the moon to move completely around the earth?
A. 1 day
B. 28 days
C. 30 days
D. 365 days

32. Steel is an example of a
A. solid in solid mixture
B. solid in liquid mixture
C. gas in solid mixture
D. liquid in liquid mixture

33. Which of the following elements is a metal?
A. Carbon
B. Nitrogen
C. Sulphur
D. Sodium

34. Metals that are usually used to make ornaments have low
A. conductivity
B. ductility
C. malleability
D. reactivity

35. The food processing method in which germs are prevented from multiplying by applying a low temperature is
A. canning
B. drying
C. pickling
D. refrigeration

36. The function of blood platelets is to
A. carry carbon dioxide.
B. fight against diseases

C. help in clotting
D. help maintain constant body temperature

37. A fish is able to swim with little resistance in water because it has
A. fins
B. gills
C. a streamlined body
D. the ability to make use of a limited oxygen supply.

38. When a person jumps up, he/she is able to come down because of the
A. pull of the earth's gravitational force
B. pull of the earth's magnetic force
C. resistance of the earth's atmosphere to the upward motion
D. pressure of the earth's atmosphere on the person

39. A metal displaces 5.0 cm^3 of water when completely immersed in water. If the mass of the metal is 35.0 g, calculate its density.
A. 7.0 g cm^{-3}
B. 30. 0 g cm^{-3}
C. 40.0 g cm^{-3}
D. 175.0 g cm^{-3}

40. Oxygen from the air is able to get into the blood by
A. capillarity
B. diffusion
C. osmosis
D. suction

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SCIENCE Section A

SOLUTIONS

OBJECTIVE TEST

1.	B.	two atoms of hydrogen
2.	A.	an increase in heat energy
3.	C.	fuse
4.	B.	40 cm
5.	B.	
6.	D.	sunlight
7.	C.	kinetic energy
8.	A.	energy
9.	D.	20 J
10.	B.	load distance
11.	B.	distance Y is greater than X
12.	D.	all kinds of food
13.	B.	balanced
14.	B.	kwashiorkor
15.	D.	brick-red
16.	A.	chlorophyll
17.	D.	liver
18.	D.	Respiration
19.	A.	kill germs

20. C. I, IV, III, II

21. D. saturated

22. C. dissolves more in hot water than in cold water

23. D. dissolves most substances

24. D. + 4

25. C. contains the humus

26. D. particles cannot contain nutrients

27. B. Plasmodium

28. C. female anopheles mosquito

29. C. Protecting one's feet before stepping into rivers

30. C. a hygrometer

31. B. 28 days

32. A. solid in solid mixture

33. D. Sodium

34. D. reactivity

35. D. refrigeration

36. C. help in clotting

37. C. a streamlined body

38. A. pull of the earth's gravitational force

39. A. 7.0 g cm^{-3}

40. B. diffusion

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SCIENCE

SECTION B
1 hour 15 minutes
ESSAY

[75 marks]

This section is in two parts, **I** *and* **II**

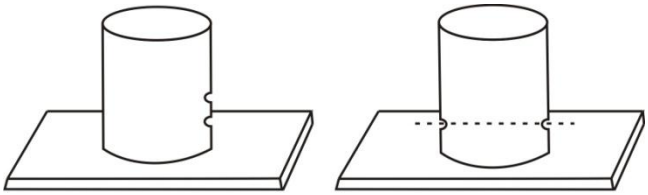
Answer Question **1** *of Part* **I** *and any other* **three** *questions in Part* **II**.

Credit will be given for clarity of expression and orderly presentation of material

PART I
[30 marks]

(*Answer* **all** *of Question* 1)

1. (a)



In an experiment, a pupil took two empty Milo tins and made holes in their sides as shown in the diagram above. The pupil then filled the Milo tins with water.

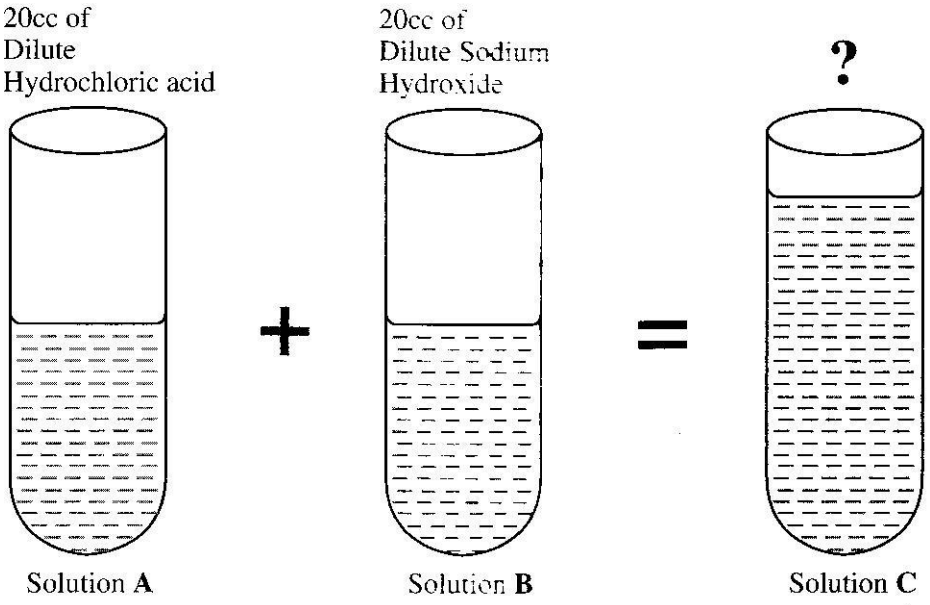
(i) Draw and label the diagrams to show what the pupil will observe in set-up **A** and set-up **B**

(ii) Explain the observations in set-up **A** and set-up **B**

(iii) What is the aim of set-up **A**?

(iv) What is the aim of set-up **B**?

(b) In an experiment, equal volumes of dilute hydrochloric acid (solution **A**) and dilute sodium hydroxide (solution **B**) are mixed together to form solution **C**.



(i) What is the volume of solution **C**?

(ii) Red litmus paper and blue litmus paper are dipped in turns into solutions **A**, **B** and **C**. State the observations you will make in all **six** cases

(iii) Give the name of the reaction that took place between solution **A** and solution **B**

(iv) Identify solution **C**

(v) State what will happen when solution **C** is heated.

(c) The table below gives the steps that were followed in an experiment to test for starch in a green leaf freshly taken from a tree.

	Stages in the test for starch
I.	Leaf is put into boiling water for 1 minute
II.	Leaf is dipped in alcohol warmed in a hot water bath
III.	Leaf is washed in cold water
IV.	Leaf is dipped into iodine solution

(i) State the reason for carrying out **each** of the activities in stages **I**, **II** and **III**.

(ii) What happens when the leaf is dipped in iodine solution?

(iii) Give the colour changes of the leaf from the beginning of the experiment to the end of the experiment.

(iv) Why is the alcohol warmed indirectly in a water bath?

(v) Explain what will be observed if the test is carried out on a leaf taken from a plant kept in a dark room for 1 day

[10 marks]

PART II

[45 marks]

Answer **three** *questions from this part*

2. (a) (i) Define *respiration*.

(ii) Give **two** differences between *photosynthesis* and *respiration*.

(b) (i) List the chemicals used in the preparation of carbon dioxide in the laboratory.

(ii) How would you test for carbon dioxide in the laboratory?

(c) (i) State **four** effects of a force on a body.

(ii) A horizontal force of 250 N is applied to pull a piece of wood lying on a smooth surface through a distance of 20 m. Calculate the work done.

3. (a) (i) Give **two** differences between a vein and an artery.

(ii) State **two** functions of red blood cells.

(iii) Mention **two** diseases associated with the circulatory system.

(b) (i) Classify the following processes into a *physical change* and a *chemical change*

Cooking of food;
Melting of candle;
Melting of ice;
Rusting of iron;
Dissolution of sugar in water;
Burning of wood into ash.

(ii) Explain the basis of the classification in (b) (i)

(c) (i) State the laws of reflection

(ii) Draw a ray diagram of light incident at an angle of 40° on the surface of a plane mirror.

4. (a) (i) Explain *vegetative reproduction*

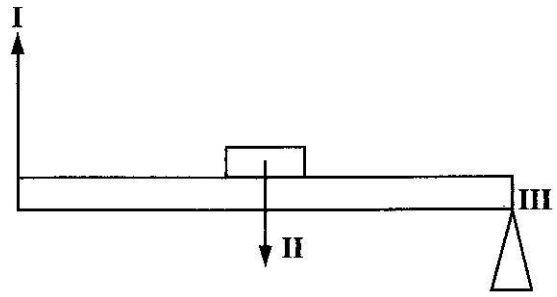
(ii) State **two** differences between *sexual reproduction* and *vegetative reproduction*

(iii) Give **one** advantage of vegetative reproduction

(b) (i) Name suitable solvents for the following substances:
Oil paint, fat, common salt, plant pigments.

(ii) What is the name given to a substance dissolved by a solvent?

(c)



The diagram above represents a simple machine

(i) What class of simple machine does it represent?

(ii) Name the parts labeled **I**, **II**, and **III**.

(iii) Give **two** examples of machines used in everyday life which work on the same principle as the system illustrated above.

5. (a) (i) Give **two** differences between a *parasite* and a *vector*.

(ii) Write down the names of **three** endo-parasites and their hosts.

(b) An atom has eight protons and nine neutrons.

(i) Draw the atomic structure of the atom.

(ii) If the atom gains two extra electrons what will be the charge of the ion formed?
(Show working)

(c) (i) Three insoluble liquids **X**, **Y** and **Z** are shaken together in a bottle and allowed to settle. The densities of the liquids are as follows:
X = 1.5 g/cm^3; **Y** = 3.0 g/cm^3 and **Z** = 1.0 g/cm^3.

Draw a sketch to show the various layers into which the liquids settle.

(ii) The density of petrol is 0.9 g/cm^3. Calculate the volume of 300g of petrol.

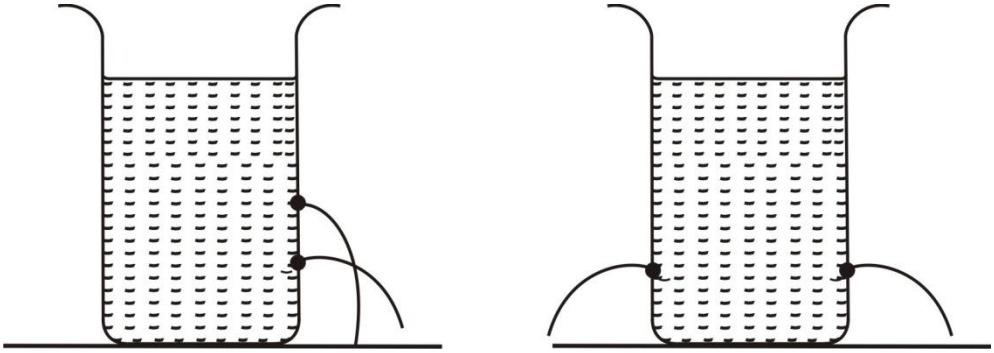
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SCIENCE Section B

SOLUTIONS

ESSAY

1. **(a)** **(i)**



(ii) **Set-up A** : The water comes out with greater force at the lower hole than at the higher because the water pressure at the lower depth is higher than at the higher depth

Set-up B: The water comes out with the same force at both holes because the water pressures at both holes are equal, since they are at equal level/height

(iii) **Set-up A:** To demonstrate that pressure in a liquid increases with increasing depth

(iv) **Set-up B:** To demonstrate that pressure in a liquid at the same depth is equal

(b) **(i)** **Volume of solution C** = 20 cc + 20 cc

= 40 cc

(ii)

ACTIVITY		**OBSERVATION**
Red litmus paper in Solution **A**	-	Red litmus paper remains red

Blue litmus paper in Solution **A**	-	Blue litmus paper turns red
Red litmus paper in Solution **B**	-	Red litmus paper turns blue
Blue litmus paper in Solution **B**	-	Blue litmus paper remains blue
Red litmus paper in Solution **C**	-	Red litmus paper turns purple
Blue litmus paper in Solution **C**	-	Blue litmus paper turns purple

(iii) Neutralization reaction

(iv) Common salt solution (NaCl dissolved in water)

(v) When solution C is heated, water will evaporate and

- Solution becomes more concentrated **or:**
- solid crystals of common salt will be left in the container (if heating continues).

(c) **(i)**

I	-	to kill the plant cells and stop the process of photosynthesis
II	-	to remove the green pigment (chlorophyll)
III	-	to wash away the alcohol and soften the leaf

(ii) The colour changes to blue-black

(iii) green → pale yellow / light brown / cream → blue black

(iv) to avoid direct contact with the flame, which can cause an explosion, since alcohol is highly inflammable.

(v) The leaf would **not** turn blue black when dipped in the iodine solution. This indicates that starch is absent, which is because photosynthesis has not taken place, since the leaf could not absorb light / sunlight, which is necessary for photosynthesis to take place.

2. **(a)** **(i)** **Respiration**

the process in which oxygen is delivered to tissues or cells in an organism and carbon dioxide, water and energy are given off.

(ii) **Differences**

Photosynthesis	Respiration
Occurs in green plants only	Occurs in both plants and animals
Carbon dioxide and water are absorbed, while oxygen is released	Oxygen is absorbed, while carbon dioxide and water are released
Energy is used	Energy is released
Occurs only in the presence of light / sunlight	Occurs both in the presence and absence of light / sunlight
Usually occurs during the day	Occurs at all times

(b) **(i)** Dilute hydrochloric acid (HCl) and

calcium carbonate ($CaCO_3$) [systematic name: calcium trioxocarbonate (IV)]

(ii) Pass the carbon dioxide through lime water [$Ca(OH)_2$]; The lime water turns milky.

(c) **(i)** **Effects of a force on a body**

- Can cause a moving body to come to rest (stop moving)
- Can cause a body at rest to move
- Can cause a moving body to accelerate
- Can cause a moving body to decelerate
- Can change the direction of a moving body
- Can change the shape of a body

(ii) Work done = force × distance

= 250 N × 20 m

= 5000 Nm or 5000 J

3. **(a)** **(i)** **Differences**

Vein	Artery
Carries deoxygenated blood (except the pulmonary vein)	Carries oxygenated blood (except the pulmonary artery)

Carries blood to the heart	Carries blood away from the heart
Blood flows under low pressure	Blood flows under high pressure
Is usually located close to the skin	Is usually located deep inside the body tissues
Has thin walls	Has thick wall
Has valves, which stop the backflow of blood	Has no valves

[any two]

(ii) **Functions** of red blood cells

- Carry oxygen from the lungs to the cells of the body
- Helps to maintain a uniform body temperature

(iii) **Diseases** associated with the circulatory system

- AIDS
- Sickle cell anaemia
- High blood pressure / hypertension
- Anaemia
- Haemorrhage
- Malaria
- Leukemia

(b) **(i)**

Physical change	**Chemical change**
melting of candle	cooking of food
melting of ice	rusting of iron
dissolution of sugar in water	burning of wood into ash

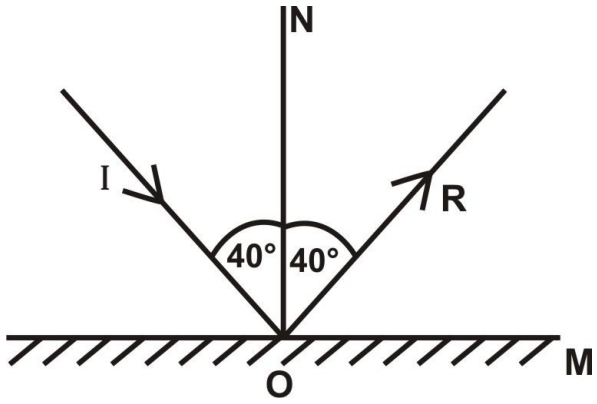
(ii) **Explanation of classification:**

Cooking of food - New substance formed; change is irreversible
Melting of candle - No new substance is formed; change is reversible
Melting of ice - No new substance is formed; change is reversible
Rusting of iron - New substance formed; change is irreversible
Dissolution of sugar in water - No new substance is formed; change is reversible
Burning of wood into ash - New substance formed; change is irreversible

(c) (i) Laws of reflection

- The angle of incidence is equal to the angle of reflection
- The incident ray, the reflected ray and the normal, at the point of incidence, all lie in the same plane

(ii) Ray diagram



4. (a) (i) Vegetative reproduction

Method by which plants reproduce without the union of cells or nuclei of cells

Or:

The process whereby plants produce offspring from growing parts / buds of the parent plant

(ii) Differences

Sexual	Vegetative
Involves the fusion of male and female sex cells	Does not involve the fusion of sex cells
Produces offspring that can be genetically different from either parent	Produces offspring that are genetically identical to parents

(iii) Advantage of vegetative reproduction

- The organism grows at a faster rate
- Results in the preservation of good qualities of parent
- Just one parent is necessary and sufficient for reproduction to occur

[any one]

(b) (i) <u>SUBSTANCE</u> <u>SOLVENT</u>

Oil paint	turpentine or kerosene
Fat	ether
Common salt	water
Plant pigments	alcohol

(ii) Substance dissolved by a solvent = Solute

(c) **(i)** **Second class**

(ii)
I - Effort
II - Load
III - Pivot / fulcrum

(iii) **Examples of second class lever:**
The wheel barrow, nut cracker, garden cart, bottle opener. **[any two]**

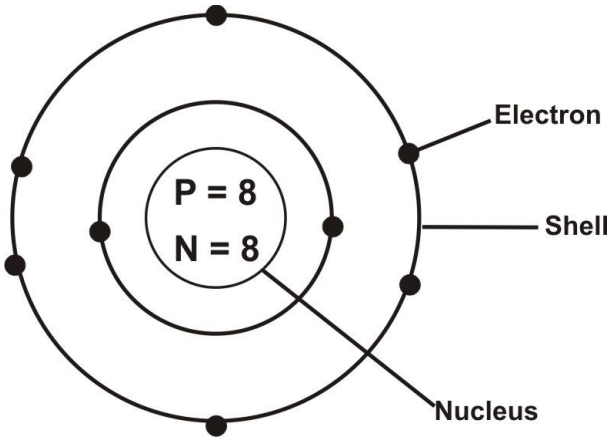
5. **(a)** **(i)** **Differences**

Parasite	**Vector**
Can be either a plant or animal	Are animals
A causative organism of disease.	Carries a causative organism of diseases
Has special features to adapt to its host	Has no special features
Resides(lives) in/ on its host	Resides(lives) on its own

(ii)

ENDO-PARASITE	**HOST**
Tapeworm	humans / mammals
Plasmodium	humans/ mammals
Roundworm	humans/ mammals
Liver fluke	humans/ mammals
Guinea worm	humans/ mammals
Schistosome	humans/ mammals

(b) (i) Atomic Structure



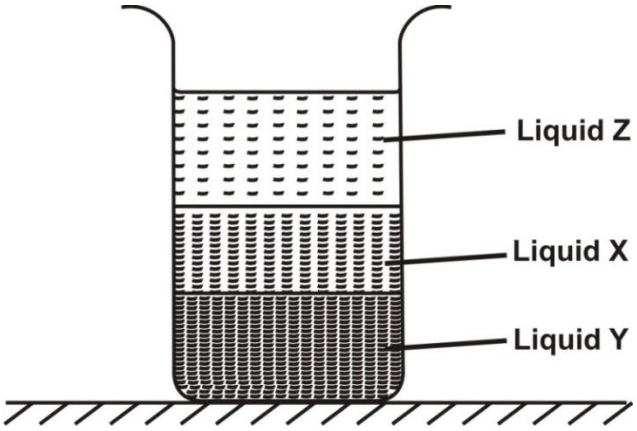
(ii) 8 protons + 8 electrons + 2electrons

= + 8 + (-8) + (- 2)

= 0 + (- 2) = - 2

The charge of the ion formed = **- 2**

(c) (i)



(ii) Volume = mass ÷ density

= 300 g ÷ 0.9 g /cm^3

= 300 / 0.9 cm^3

= 3000 / 9 cm^3

= $333\ ^1/_3\ cm^3$ or $333.33\ cm^3$

April 2004

SCIENCE

SECTION A
OBJECTIVE TEST
45 minutes

1. Which of the following processes of conversion of the states of matter is correct?

A.	Solid	(heat)	→	gas	(heat)	→	liquid
B.	Solid	(cool)	→	liquid	(cool)	→	gas
C.	Gas	(heat)	→	liquid	(heat)	→	solid
D.	Gas	(cool)	→	liquid	(cool)	→	solid

2. The types of energy produced when the hands are vigorously rubbed against each other are

A. chemical and electrical energy
B. heat and sound energy
C. heat and light energy
D. electrical and light energy

3. When a thermometer is put in hot water, the mercury level rises because the mercury increases in

A. density B. mass C. volume D. weight

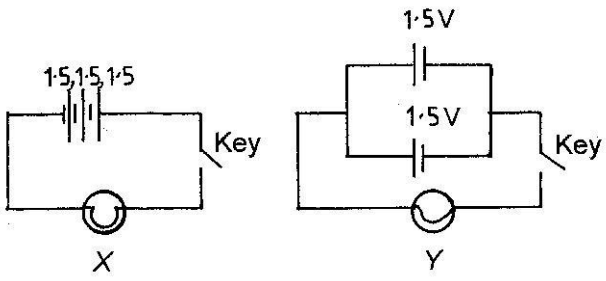
4. The modes of heat transfer involved in the process of heating water in a bucket from the bottom until it boils are

A. conduction and convection
B. conduction and radiation
C. convection and radiation
D. conduction, convection and radiation

5. Which of the following items converts chemical energy to electrical energy?

A. Dry cell B. Electric bulb C. Loudspeaker D. Microphone

Use the diagrams below to answer Questions **6** *and* **7**



6. Which of the following statements is correct about the circuit diagrams above? The cells in diagram

A. X are in series and the key is open
B. X are in parallel and the key is open
C. Y are in series and the key is closed
D. Y are in parallel and the key is closed

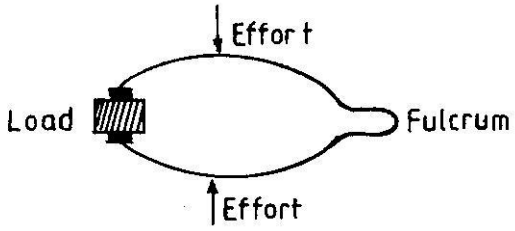
7. The total *emf* of the cells in diagram Y is

A. 4.50 V B. 3.00 V C. 1.50 V D. 0.75 V

8. It is easier to move a heavy load with a crowbar when the

A. effort distance is shorter than the load distance
B. effort distance is longer than the load distance
C. effort distance is equal to the load distance
D. effort is equal to the load

9.



The type of lever shown above is a

A. first class lever
B. second class lever
C. third class lever
D. combination of first and second class levers

10. A balanced diet is one which is made up of

A. right proportions of protein, carbohydrates and oil
B. equal amounts of protein, carbohydrates and oil
C. enough water and iodated salt
D. vegetables, fruits and water

11. Benedict's solution was added to a mixture in a test tube and it turned brick red when heated. The mixture is likely to contain

A. glucose B. oil C. protein D. vitamin

12. The main food substance present in the albumen of an egg is

A. carbohydrates B. fat C. protein D. vitamin

13. One characteristic which is not common to all living things is the ability to

A. grow B. move about C. respire D. respond to stimuli

14. The ovules in a flower develop to form the

A. fruit B. leaves C. seeds D. stem

15. Onions are planted by means of

A. bulbs B. corms C. suckers D. rhizomes

16. The testis in mammals produces

A. blood B. lymph C. sperms D. urine

17. Which of the following structures takes part in human digestion?

A. Caecum B. Kidney C. Larynx D. Pancreas

18. Which of the following diseases is contracted through sex?

A. HIV/AIDS B. Asthma C. Tuberculosis D. Goitre

19. Chlorine gas is an example of

A. an atom B. an element C. a compound D. a molecule

20. The bad smell that comes out of a urinal is due to the presence of

A. ammonia B. hydrogen C. carbon dioxide D. nitrogen

21. The systematic name of the compound FeS is

A. iron (I) sulphide
B. iron (II) sulphide
C. iron (III) sulphide
D. iron (IV) sulphide

22. Which of the following elements is a liquid at room temperature?

A. Carbon B. Mercury C. Silver D. Sulphur

23. The type of cloud that gives rain is

A. cumulus B. cirrus C. nimbus D. stratus

24. Water drains faster through sand than clay because

A. sand particles are rougher
B. sand contains more air spaces

C. clay particles are smoother
D. clay particles are bigger

25. Which of the following organisms is **not** an animal parasite?
A. tick B. tapeworm C. plasmodium D. weevil

26. The housefly is an agent for the spread of
A. cholera B. malaria C. measles D. tuberculosis

27. The **best** method for checking erosion on a slope is
A. cover cropping
B. mixed cropping
C. contour ploughing
D. mulching

28. It is not advisable to sleep in a closed dark room with green plants because, the plants
A. produce heat
B. compete with human beings for oxygen
C. absorb water vapour
D. produce carbon dioxide

29. The process whereby soil is formed by the breakdown of rocks is called
A. composting B. erosion C. leaching D. weathering

30. Kerosene is able to reach the other end of a wick by
A. diffusion
B. suction pressure
C. capillary action
D. osmosis

31. The force, which opposes the motion of one body on another body is called
A. adhesion B. cohesion C. friction D. tension

32. The work done when a force moves a body through a distance of 12 m is 720 J. The force applied is
A. 8640 N B. 732 N C. 708 N D. 60 N

33. Food substances are transported from the leaves to various parts of a plant through the
A. cambium B. epidermis C. phloem D. xylem

34. Which of the following substances are carried by the blood?
I. Nutrients
II. Urine

III. Oxygen
IV. Carbon dioxide

A. II and III only
B. I, II and IV only
C. I and III only
D. I, III and IV only

35. A uniform mixture of two or more metals is called
A. an alloy
B. a colloid
C. an aerosol
D. a compound

36. Gold is usually used to make jewellery because it is
A. precious
B. expensive
C. less reactive with air
D. a good conductor of heat

37. Which of the following substances is a non-metal?
A. Diamond B. Mercury C. Sodium D. Steel

38. Alum is added to water during treatment to
A. kill germs
B. help suspended particles to settle
C. make the water colourless
D. give taste to the water

39. Which of the following types of water would be **most** contaminated by waste substances?
A. Rain water
B. Stream water
C. Bore hole water
D. Pipe borne water

40. The method of preserving food by drying is similar to salting because in **both** cases germs are
A. deprived of the air needed for growth
B. deprived of the water needed for growth
C. destroyed at high temperatures
D. destroyed at low temperatures

April 2004

SCIENCE Section A

SOLUTIONS

OBJECTIVE TEST

1.	D.	Gas (cool) → liquid (cool) → solid
2.	B.	heat and sound energy
3.	C.	volume
4.	A.	conduction and convection
5.	A.	Dry cell
6.	A.	X are in series and the key is open
7.	C.	1.50 V
8.	B.	effort distance is longer than the load distance
9.	C.	third class lever
10.	A.	right proportions of protein, carbohydrates and oil
11.	A.	glucose
12.	C.	protein
13.	B.	move about
14.	C.	seeds
15.	A.	bulbs
16.	C.	sperms
17.	D.	Pancreas
18.	A.	HIV/AIDS
19.	B.	an element
20.	A.	ammonia

21. B. iron (II) sulphide

22. B. Mercury

23. C. nimbus

24. B. sand contains more air spaces

25. D. weevil

26. A. cholera

27. C. contour ploughing

28. B. compete with human beings for oxygen

29. D. weathering

30. C. capillary action

31. C. friction

32. D. 60 N

33. C. phloem

34. D. I, III and IV only

35. A. an alloy

36. C. less reactive with air

37. A. Diamond

38. B. help suspended particles to settle

39. B. Stream water

40. B. deprived of the water needed for growth

April 2004

SCIENCE

SECTION B

1 hour 15 minutes

ESSAY

[75 marks]

This section is in two parts, **I** *and* **II**

Answer Question **1** *of Part* **I** *and any other* **three** *questions in Part* **II**.

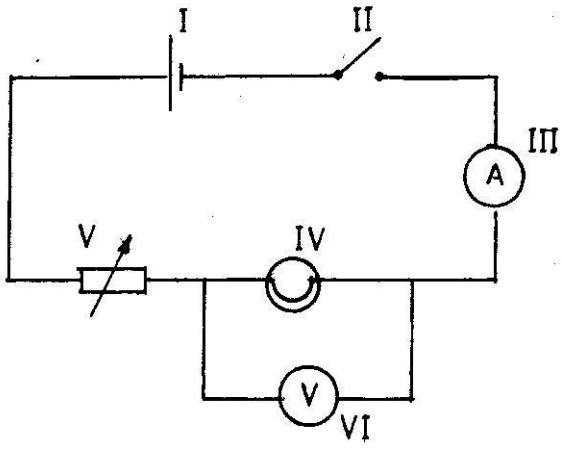
Credit will be given for clarity of expression and orderly presentation of material

PART I

[30 marks]

(*Compulsory – Answer* **all** *of Question* 1)

1. (a) Study the simple circuit diagram shown below and answer the questions that follow



(i) Identify the components labelled **I, II, III, IV, V, VI**

(ii) Which component is used to close the circuit?

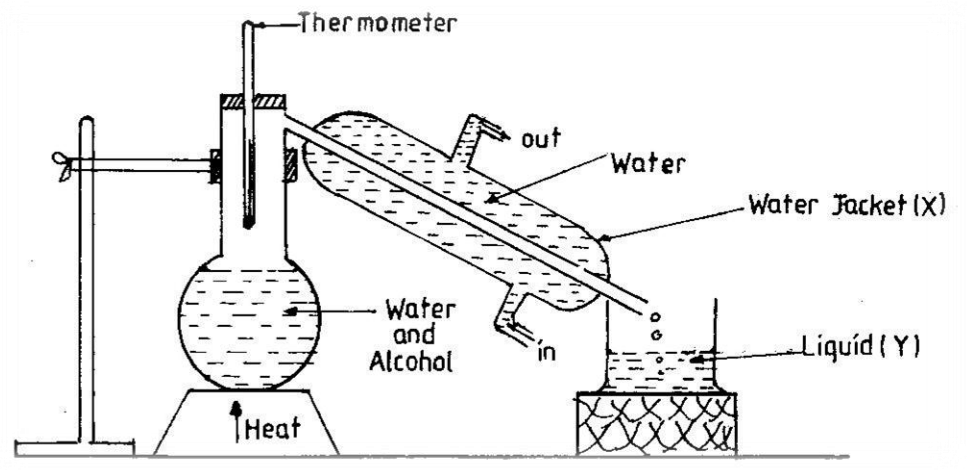
(iii) State the observation that will be made when the circuit is closed.

(iv) State the energy transformations that take place when the circuit is closed

(v) Give the name of the circuit connection between components **IV, VI**

[10 marks]

(b) The diagram below is an illustration of an experimental set-up used for separating a mixture of water and alcohol. Study the diagram carefully and answer the questions that follow.



(i) State the method of separation shown in the set-up

(ii) Identify liquid **Y**, with reasons.

(iii) Give the functions of the thermometer and the water jacket **X**

(iv) What physical processes are involved in the method of separating the mixture?

(v) State the physical property which makes it possible to separate the water and alcohol

[10 marks]

(c) A student performed tests on food substances **A, B** and **C** and made the following observations

Food Substance	Test	Observation
A	Few drops of iodine solution was added to **A**	The iodine solution turns blue-black
B	A drop of **B** was applied to a white sheet of paper	A translucent patch was seen on the paper
C	Benedict's solution was added to **C** and the mixture boiled	Benedicts's solution turns from blue to brick-red

(i) Identify food substances **A, B** and **C**.

(ii) Give the products of digestion of **A, B** and **C**

(iii) In which parts of the alimentary canal does the digestion of each of food substances **A, B** and **C** start?

(iv) In which part of the alimentary canal is food substance **C** absorbed after digestion?

[10 marks]

PART II

[45 marks]

(*Answer* **three** *questions from this part*)

2. (a) (i) List **four** parts of a plant that are used for asexual reproduction

(ii) Give **two** reasons why sexual reproduction is more advantageous than asexual reproduction.

(b) (i) Explain why atoms are electrically neutral

(ii) Write the systematic names of the following substances:

(α) CuO,

(β) SO_2

[5 marks]

(c) (i) What is *dispersion of light*?

(ii) Draw and label a diagram of dispersion of light through a triangular glass prism.

[4 marks]

3. (a) (i) What is *soil erosion*?

(ii) Give **two** types of soil erosion

[4 marks]

(b) Write the name and chemical formula of the compounds formed when the following elements combine:

(i) zinc and oxygen

(ii) calcium and chlorine

[4 marks]

(c) (i) Distinguish between heat and temperature

(ii) Draw and label the mercury-in-glass thermometer.

(iii) What mode of heat transfer is involved in the use of a thermometer?

[7 marks]

4. (a) (i) What is a *hormone*?

(ii) Copy and complete the following table

Name of hormone	*Organ secreting hormone*	*Function of hormone*
Adrenalin Thyroxin Insulin		

[6 marks]

(b) Ammonia gas is prepared in the laboratory by heating ammonium chloride and calcium hydroxide.

(i) Write a balanced equation for the reaction.

(ii) How is ammonia gas tested in the laboratory?

(iii) State **one** use of ammonia.

(c) (i) Name **four** types of forces.

(ii) A boy used a catapult to kill a bird which was on a tree and it fell to the ground. Give **two** forces involved in this process

[3 marks]

5. (a) (i) Distinguish between a rhizome and a stem tuber

(ii) Give **one** example **each** of a rhizome and a stem tuber

[5 marks]

(b) State the method which could be used to separate **each** of the following mixtures and explain your answer in **each** case:

(i) salt and water,

(ii) sawdust and iron filings,

(iii) powdered charcoal and water

[6 marks]

(c) (i) Define *density of a substance*

(ii) Explain how a body which sinks in water could be made to float.

[4 marks]

April 2004

SCIENCE Section B

SOLUTIONS

ESSAY

1. **(a)** **(i)**

I - Cell
II - key / switch
III - Ammeter
IV - Electric bulb
V - Variable resistor
VI - Voltmeter

(ii) The key / switch

(iii) The bulb will light, the ammeter will deflect, the voltmeter will deflect

(iv) Chemical → Electrical → light (and heat)

(v) Parallel connection

(b) **(i)** Distillation

(ii) Liquid Y - Alcohol

This is because it is more volatile and has a lower boiling point than water. Therefore, it boils first, leaving the water behind.

(iii) The thermometer measures the temperature within the flask

The water jacket X cools the gaseous alcohol to condense it back to liquid form

(iv) Heating and cooling / vapourisation and condensation

(v) Boiling points of water and alcohol which are clearly different.

(c) **(i)**

A - Starch or carbohydrate
B - fats and oils
C - reducing sugar or simple sugar

(ii) **Products of digestion**

A - glucose
B - fatty acids and glycerol
C - glucose

(iii) **A** - Mouth
B - The duodenum (first part of the small intestines)
C - Mouth

(iv) C is absorbed after digestion in - The ileum (third & last part of the small intestines

2. **(a)** **(i)** Four **parts** of a plant used for asexual reproduction
Rhizome, bulb, sucker, stem cutting, tuber, corm, bud [any four]

(ii) Two **Advantages** of sexual reproduction over asexual reproduction
- Offspring are usually more resistant to diseases
- Offspring are usually more stronger and healthier
- Offspring have different and usually better characteristics from parents

[any two]

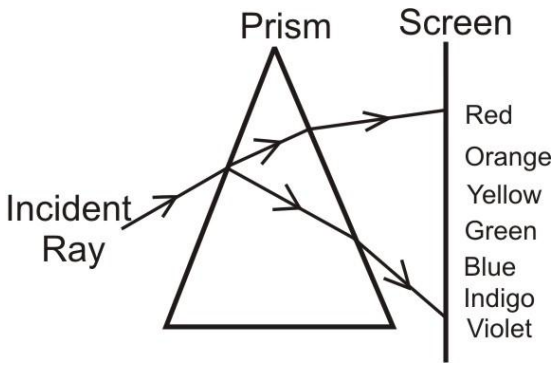
(b) **(i)** **Why atoms are electrically neutral**
The numbers of protons and electrons are equal. The positively charged protons neutralize the negatively charged electrons, hence, making the atom electrically neutral

(ii) **Systematic names of:**

(α) CuO - Copper (II) Oxide
(β) SO_2 - Sulphur (IV) oxide

(c) **(i)** **Dispersion of light**
The separation of light into different wavelengths or its component colours when it passes through a given medium

(ii) Labeled diagram showing dispersion of light through a triangular glass prism



3. **(a)** **(i)** **Soil erosion**

The washing away of the top soil by agents of erosion

Or:

The removal of soil material by natural processes, principally running water, glaciers, waves, and wind

(ii) **Two types** of soil erosion

Sheet, Gully, Rill, Splash [any two]

(b) **(i)** **Zinc and oxygen**

Name: Zinc oxide

Chemical formula: (ZnO)

(ii) **Calcium and Chlorine**

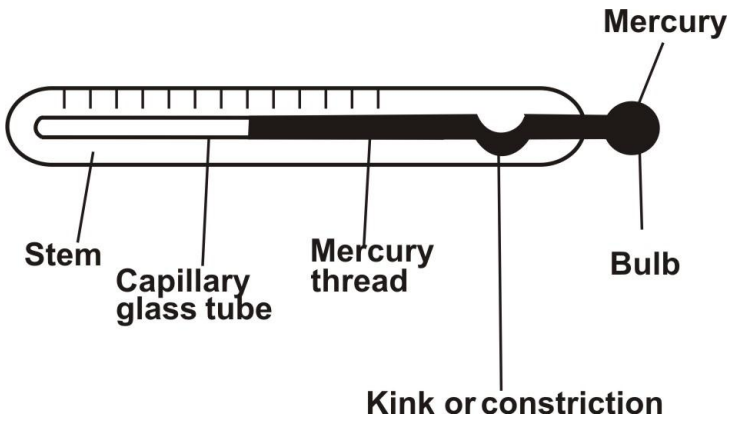
Name: Calcium chloride

Chemical formula: $CaCl_2$

(c) **(i)**

Heat	Temperature
The energy possessed by a body that results in its hotness or coldness	A measure of the hotness or coldness of the body
Unit = joule (J)	Unit = kelvin (K) or degee Celsius (°C) or degree Fahrenheit (°F)

(ii) **Labeled diagram** of mercury-in-glass thermometer



(iii) **Mode of heat transfer** = Conduction

4. (a) (i) Hormone

A chemical secreted by an endocrine gland or some nerve cells that regulates the function of a specific tissue or organ

(ii)

Name of hormone	*Organ secreting hormone*	*Function of hormone*
Adrenalin	Adrenal gland	increases the speed and force of the heart beat or increases the energy of the body.
Thyroxin	thyroid gland	stimulates metabolism for normal growth and development of the body
Insulin	Pancreas	regulates the glucose (sugar) level in blood

(b) (i) Balanced equation for the laboratory preparation of ammonia

$NH_4Cl + Ca(OH)_2 \rightarrow NH_3 + CaCl_2 + H_2O$

(ii) Test for ammonia gas

Hydrochloric acid is introduced into the gas. White fumes are produced in the presence of the HCl.

Or: Gas is smelled. Ammonia has a very pungent / choky smell.

(iii) Uses of ammonia

- As a refrigerant (used in refrigerators)
- Preparation of fertilizers
- Flying of balloons
- Preparation of medicines/ drugs
- Preparation of explosives [any one]

(c) (i) Types of forces

Frictional, tensional, gravitational, centripetal, centrifugal, magnetic, upthrust [any four]

(ii) Forces involved

- Tensional force (in the stretched catapult);
- frictional force (bird perched on the tree)
- gravitational force (bird falling to the ground); [any two]

5. (a) (i) Differences

Rhizome	Stem tuber
Grows horizontally below the ground	Grows vertically/ diagonally below the ground
Buds are covered by scaly leaves	Buds are not covered by scaly leaves
Both nodes and internodes present	Only nodes present

(ii) **Examples** of
Rhizome: Ginger, canna lily
Stem Tuber: Potato, yam

(b) **Method of separation and explanation**

(i) **Salt and water** - Evaporation
The temperatures at which water and salt evaporate are wide apart. Therefore, as the mixture is heated, the water evaporates, leaving the salt crystals behind

(ii) **Sawdust and iron filings** - Magnetization
The iron filings is a magnetic substance, while saw dust is not. Hence, the magnet would attract the iron filings, leaving the saw dust behind

(iii) **powdered charcoal and water** - Decantation or filtration
Powdered charcoal particles are insoluble in water; therefore they can be separated from water by filtration or decantation, if the particles have settled at the bottom

(c) **(i)** **Density** of a substance
The mass per unit volume of the substance
Or: Density = mass / volume

(ii) **Explanation**
When the volume of a body increases at a constant mass, its density reduces. Therefore, the volume of the body is increased by increasing its hollowness / expanse such that its resultant density is less than that of water. Then the body will float on water.

April 2003

SCIENCE

SECTION A
OBJECTIVE TEST
45 minutes

1. The method used to separate an insoluble solid from a liquid is

A. crystallization
B. distillation
C. filtration
D. sublimation

2. The scent of a perfume sprayed at one corner of a room fills the entire room through

A. conduction
B. diffusion
C. osmosis
D. radiation

3. Which of the following processes is a physical change?

A. Rusting of iron
B. Burning of wood
C. Formation of water from hydrogen and oxygen
D. Solidification of water into ice

4. Atoms of the same element have the same

I. number of electrons
II. number of protons
III. chemical properties

Which of the above statements are true?

A. I and II only
B. II and III only
C. I and III only
D. I, II and III

5. The unit of measurement of energy is

A. joule
B. kelvin

C. newton
D. watt

6. Which of the following energy transformations takes place at a hydroelectric power station?
A. Potential energy → kinetic energy → electrical energy
B. Potential energy → sound energy → electrical energy
C. Kinetic energy → electrical energy → light energy
D. Electrical energy → light energy → heat energy

7. A drawing of magnetic lines of force provides information on the
A. source of the field
B. type of magnet
C. strength of the field
D. length of the magnet

8. A bottle of water removed from a refrigerator soon becomes covered with droplets of water because the
A. bottle is porous
B. bottle is full of water
C. water in the surrounding atmosphere is colder than the water in the bottle
D. water in the surrounding atmosphere is warmer than the water in the bottle

9. The volume of a metal ball was determined by dropping it into a measuring cylinder containing 20 cm^3 of water and the water level rose to 35 cm^3. If the mass of the ball is 35 g, calculate its density.
A. 1.0 g cm^{-3}
B. 1.6 g cm^{-3}
C. 1.8 g cm^{-3}
D. 2.3 g cm^{-3}

10. A boy stepped on a banana peel on the ground and fell because the
A. boy was not wearing shoes
B. ground was too hard
C. banana peel reduced the friction on the ground
D. banana peel increased the friction on the ground

11. Which of the following conditions describes a first class lever?
A. The load is between the effort and pivot
B. The pivot is between the effort and the load
C. The effort is between the pivot and the load
D. The load arm is equal to the effort arm

12. A man did 75 J of work by lifting a 50 N load from the floor on to a shelf. Calculate the height of the shelf
A. 0.67 m
B. 1.50 m
C. 25.00 m

D. 125.00 m

13. The lion is a carnivore because it eats
A. flesh
B. grass
C. fruits
D. uncooked food

14. The structure which helps a fish to breathe under water is the
A. gill
B. air sac
C. operculum
D. mouth

15. The thorns found on some plants are used for
A. breathing
B. protection
C. absorbing water from the air
D. resisting bush fires

16. The following characteristics are features of all living things **except** ability to
A. bring forth young ones
B. excrete
C. grow
D. move from place to place

17. Which of the following parasites lives on field animals
A. Aphid
B. Plasmodium
C. Tick
D. Tapeworm

18. The greatest danger faced by mammals living on land is
A. becoming too cold
B. becoming too hot
C. losing water through evaporation
D. gaining too much water through absorption

19. The insect responsible for the spread of malaria is the
A. female anopheles mosquito
B. housefly
C. male anopheles mosquito
D. tsetse fly

20. Which of the following substances is added to the soil to reduce its acidity?

A. Charcoal
B. Humus
C. Lime
D. Manure

Fig. **1** *shows the arrangement of parts of the soil when some soil sample was shaken with water in a glass cylinder and allowed to settle.*
Use it to answer Questions **21** *and* **22**

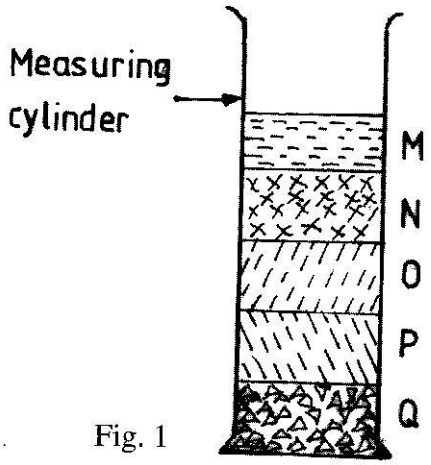


Fig. 1

21. Which of the labeled layers contains organic matter?
A. **M** B. **N** C. **O** D. **P**

22. The layer which does not have the ability to retain much water is
A. **N** B. **O** C. **P** D. **Q**

23. Which of the following characteristics applies to only plants?
A. Excretion of waste materials
B. Response to stimuli
C. Ability to manufacture food from simple substances
D. Ability to respire

24. The waste products of respiration are
A. glucose and oxygen
B. carbon dioxide and water
C. carbon dioxide and heat
D. oxygen and water

25. The function of the lung is to
A. maintain a constant amount of water in the body

B. absorb oxygen from the blood
C. absorb carbon dioxide from the blood
D. maintain the temperature of the body constant

26. Which of the following plants reproduce(s) vegetatively?
I. Banana
II. Mango
III. Ginger

A. II only
B. I and II only
C. I and III only
D. III only

27. The process which takes place in all living cells is
A. digestion
B. fertilization
C. photosynthesis
D. respiration

28. The food substance found in yam is
A. carbohydrate
B. oil
C. protein
D. vitamin

29. The end-products of the digestion of fats and oils are
A. amino acids and glucose
B. amino acids and glycerol
C. fatty acids and glycerol
D. fatty acids and amino acids

30. Which of the following factors promote plant germination?
I. Adequate temperature
II. Moisture
III. Good soil

A. I and II only
B. I and III only
C. II and III only
D. I, II and III

31. The female part of a flower is the
A. calyx B. pistil C. petal D. stamen

32. Which of the following substances breaks down fats into tiny droplets during digestion

A. Bile
B. Gastric juice
C. Pancreatic juice
D. Saliva

33. Blood in the urine is a symptom of

A. bilharzia
B. cholera
C. gonorrhoea
D. typhoid

34. A free movement of the bowels is aided by

A. sleeping at least 6 hours a day
B. washing one's body at least once a day
C. eating fatty foods
D. eating foods containing fibre

35. Which of the following diseases is spread through the air?

A. AIDS B. Measles C. Typhoid D. Yaws

36. How many days in a month does the moon appear?

A. 7 days B. 14 days C. 21 days D. 28 days

37. The earth moves completely round the sun once in every

A. 1 day B. 28 days C. 30 days D. 365 days

38. A smaller heavenly body, which orbits a bigger one could be described as a

A. comet B. meteor C. moon D. sun

39. Which of the following parasites is a plant?

A. Aphid B. Dodder C. Flea D. Jigger

40. Which of the following substances is **not** a constituent of a balanced diet?

A. Carbohydrate B. Protein C. Table salt D. Vitamin

April 2003

SCIENCE Section A

SOLUTIONS

OBJECTIVE TEST

1.	C.	filtration
2.	B.	diffusion
3.	D.	Solidification of water into ice
4.	D.	I, II and III
5.	A.	joule
6.	A.	Potential energy → kinetic energy → electrical energy
7.	C.	strength of the field
8.	D.	water in the surrounding atmosphere is warmer than the water in the bottle
9.	D.	2.3 g cm^{-3}
10.	C.	banana peel reduced the friction on the ground
11.	B.	The pivot is between the effort and the load
12.	B.	1.50 m
13.	A.	flesh
14.	A.	gill
15.	B.	protection
16.	D.	move from place to place
17.	C.	Tick
18.	C.	losing water through evaporation
19.	A.	female anopheles mosquito

20. C. Lime

21. A. **M**

22. D. **Q**

23. C. Ability to manufacture food from simple substances

24. B. carbon dioxide and water

25. C. absorb carbon dioxide from the blood

26. C. I and III only

27. D. respiration

28. A. carbohydrate

29. C. fatty acids and glycerol

30. A. I and II only

31. B. pistil

32. A. Bile

33. A. bilharzia

34. D. eating foods containing fibre

35. B. Measles

36. D. 28 days

37. D. 365 days

38. C. moon

39. B. Dodder

40. C. Table salt

April 2003

SCIENCE

SECTION B
ESSAY
1 hour

[60 marks]

Answer **three** *questions* **only** *from this section.*
Illustrate your answers wherever possible, with large, clear and fully labelled diagrams.
Credit will be given for clarity of expression and orderly presentation of material.
All questions carry equal marks

1. (a) (i) Define *photosynthesis*
(ii) Give **two** ways in which photosynthesis is important to animals

[5 marks]

(b) Name **three** sources of chemical energy

[3 marks]

(c) Describe how you would separate a mixture of sodium chloride and sand

[4 marks]

(d) Complete and balance the following chemical reaction:
$CaCO_3 + HCl \rightarrow$ [4 marks]

(e) Two dry cells, a switch and a bulb are connected in series. Draw and label a circuit diagram of the arrangement

[4 marks]

2. (a) State **two** precautions that should be taken to prevent the spread of **each** of the following diseases:
(i) cholera
(ii) smallpox

[4 marks]

(b) Describe how you would test for starch in a leaf.

[6 marks]

(c) Draw and label the structure of an atom containing three electrons. Indicate the charges of the particles.

[6 marks]

(d) (i) Explain why an inclined plane is classified as a machine

(ii) Give **two** reasons why the output energy of a machine is always **less** than the input energy.

[4 marks]

3. (a) Name the **two** processes that lead to the formation of seeds in a flowering plant

[2 marks]

(b) (i) Give **five** components of blood

(ii) State **three** functions of blood

[8 marks]

(c) (i) What is an *electrical insulator*?

(ii) Mention **three** examples of insulators

(iii) Give **two** characteristics of an insulator that explain its behaviour

[7 marks]

(d) Classify the following substances under the three states of matter:
petrol, palm oil, soap, carbon dioxide, corn flour, smoke

[3 marks]

4. (a) (i) What is a *deficiency disease*?

(ii) Give **two** deficiency diseases and their causes.

[6 marks]

(b) (i) State the importance of excretion in living organisms.

(ii) Name **two** excretory organs of the human body and the substances they excrete.

[6 marks]

(c) (i) Explain why gases are more compressible than solids.

(ii) Give **two** areas where the compressible nature of a gas is applied

[4 marks]

(d) A piece of stone is dropped from a height to the ground.
State the energies possessed by the stone

(i) just before it was released from the maximum point;

(ii) midway between the fall;

(iii) just before it hit the ground.

[4 marks]

April 2003

SCIENCE Section B

SOLUTIONS

ESSAY

1. (a) (i) Photosynthesis:

The process whereby green plants prepare food using carbon dioxide and water in the presence of chlorophyll and light

Or:

a process by which green plants turn carbon dioxide and water into carbohydrates and oxygen, using light energy trapped by chlorophyll

(ii) Importance of photosynthesis to man

- Produces oxygen for respiration
- Produces food (carbohydrates) for nutrition
- Decreases the carbon dioxide content in the atmosphere to reduce pollution

[any two]

(b) Sources of chemical energy:

- Car battery
- Electric cell
- Food (carbohydrates and fats and oils)
- Fossil fuels (eg, petrol, kerosene, coal, etc)

[any three]

(c) Separating a mixture of sodium chloride and sand

I. Add water to the mixture and stir to dissolve the salt

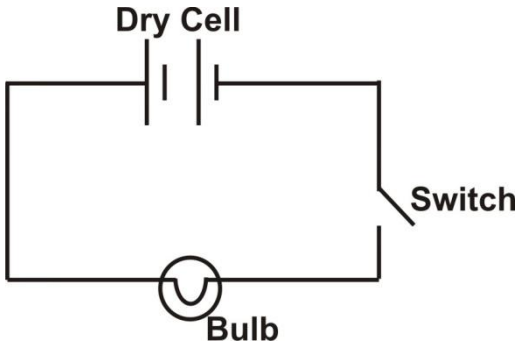
II. Pour the mixture through a filter paper in a funnel to separate the sand (residue) from the salt solution (filtrate)

III. Heat the salt solution till all the water evaporates, leaving the salt crystals behind

IV. Dry the sand and the salt obtained

(d) $CaCO_3 + 2\ HCl \rightarrow CaCl_2 + CO_2 + H_2O$

(e) Labelled Circuit diagram



2. **(a)** **Precautions** to prevent

(i) Cholera

- Eating hot meals,
- drinking clean potable water,
- washing hands with soap after visiting the toilet
- covering meals to prevent flies from settling on them
- keeping our environment clean and free from filth
- ensuring that rubbish bins are properly covered

[any two]

(ii) Smallpox

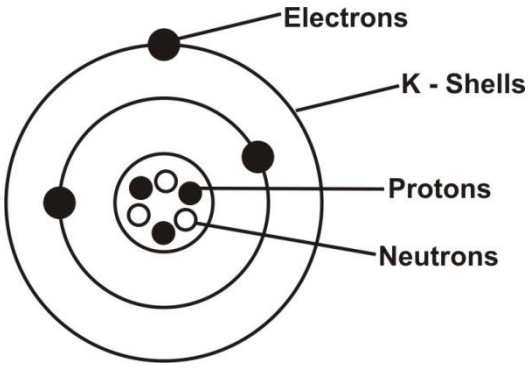
- vaccination
- quarantining (isolating / separating) smallpox patients
- Staying away from dusty areas
- Keeping clothes neat and disinfected
- Ensuring sufficient ventilation in rooms

[any two]

(b) **Test for starch** in a leaf

Stages in the test for starch	
I.	Leaf is put into boiling water for 1 minute
II.	Leaf is dipped in alcohol warmed in a hot water bath
III.	Leaf is washed in cold water
IV.	Leaf is dipped into iodine solution
	The blue-black colouration of the leaf indicates the presence of starch

(c) **Structure of atom** containing three electrons.



(d) **(i)** **Explanation** of why an inclined plane is classified as a machine

An inclined plane is a machine because it helps make work easier by reducing the effort needed and increasing the effort distance instead.

(ii) **Reasons** why the output energy is always less than the input energy because

- Part of the input energy is used to overcome friction
- Part of the input energy is expended as heat
- Part of the input energy is used to overcome gravity

[any two]

3. **(a)** **Processes** that lead to formation of seeds

Pollination and fertilization

(b) **(i)** **Components of blood**

- Red blood cells (erythrocytes)
- White blood cells (phagocytes)
- Platelets
- Dissolved food nutrients, eg, glucose, amino acids
- Metabolic waste, eg, urea, carbon dioxide
- Water
- Hormones
- Mineral salts [any five]

(ii) **Functions** of blood

- Transports oxygen to the cells of the body
- Transports metabolic waste materials away from the cells of the body
- Transports digested food nutrients to parts of the body
- Helps to maintain a uniform body temperature
- Helps to protect the body against infections and diseases
- Transports hormones
- Forms clots to stop bleeding / to prevent excessive blood loss

[any three]

(c) **(i)** **Electrical insulator**

A material that does not allow electrical currents to pass through

(ii) **Examples** of insulators

Plastic, polythene, wood, rubber, paper, fabric, glass, paper

(iii) **Characteristics** of electrical insulator

- The absence of free electrons
- Low charge density
- Large forbidden energy gap
- High electrical resistivity
- Low electrical conductivity

[any two]

(d)

SOLID:	corn flour, soap
LIQUID	petrol, palm oil
GAS:	smoke, carbon dioxide

4. **(a)** **(i)** **Deficiency disease**

A disease that is caused by the lack of a given food nutrient

(ii) Deficiency diseases and their causes

Deficiency disease	Cause
Kwashiorkor	Protein deficiency
Night blindness	Vitamin A deficiency
Anaemia	Iron deficiency
Rickets	Vitamin D or Calcium or Phosphorus deficiency
Beriberi	Vitamin B1 deficiency
Goitre	Iodine deficiency
Scurvy	Vitamin C or Ascorbic Acid deficiency

[any two]

(b) **(i)** **Importance of excretion**

- To get rid of toxic materials, which can cause harm to the body

(ii)

EXCRETORY ORGAN	SUBSTANCE EXCRETED
Skin	sweat
Lungs	carbon dioxide
Kidney	urea

[any two]

(c) **(i)** Molecules of a gas are widely spaced out, while molecules of a solid are closely packed together. Hence, the molecules of a gas have much more space to fit into than those of a solid, which are already close to themselves.

(ii) **Areas where compressible nature of a gas is applied**

- Filling of gas cylinders
- Use of a tyre pump
- Blowing/ pumping air into a balloon

[any two]

(d) **Energies possessed by a stone dropped from a height to the ground**

(i) potential energy

(ii) potential energy and kinetic energy

(iii) kinetic energy

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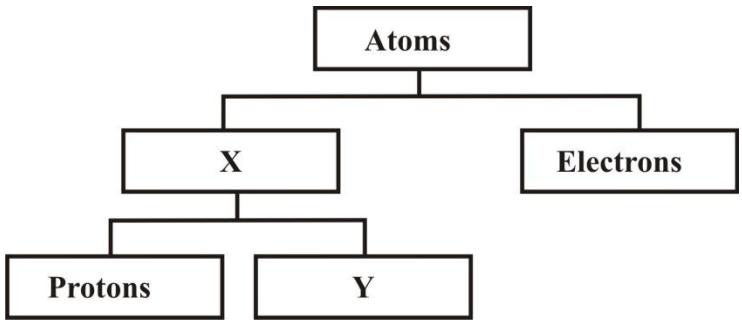
SCIENCE

SECTION A
OBJECTIVE TEST
45 minutes

1. Which of the following substances is made up of only one kind of atom?
- A. Iron
- B. Salt
- C. Steel
- D. Water

2. Ice is able to float on water because the ice is
- A. colder than water
- B. denser than water
- C. has a greater mass for the same volume of water
- D. has a greater volume for the same mass of water

3. Which of the following statements about the state of matter is correct?
- A. Molecules in a solid are disorderly arranged
- B. Solids are highly compressible
- C. Liquids have definite form and volume
- D. Molecules in a gas move faster than those in a liquid.



4. The above diagram illustrates the structure of an atom. The components labeled X and Y are respectively
- A. neutrons and shells
- B. ions and neutrons
- C. neutrons and nucleus

D. nucleus and neutrons

5. The force which opposes the motion of one object on another object is known as
A. friction
B. gravity
C. tension
D. weight

6. When the brakes of a bicycle in motion are applied for a long time the wheels become hot. This is because of a change of energy from
A. kinetic energy to potential energy
B. potential energy to kinetic energy
C. potential energy to heat energy
D. kinetic energy to heat energy

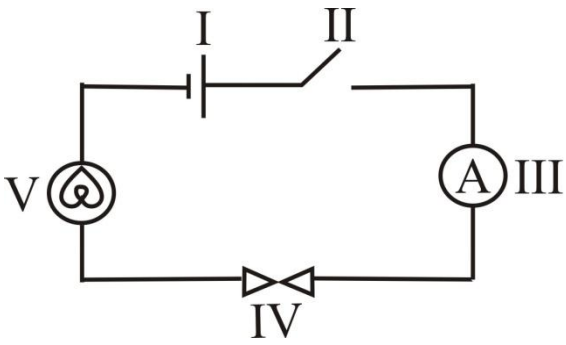
7. A man applied a force of 50 N to push a wheelbarrow through a distance of 6 m. Calculate the work done.
A. 300 J
B. 56 J
C. 44 J
D. 30 J

8. In which of the following situations is work said to be done?
A. Walking with a load on the head in an upright position.
B. Getting up from a chair
C. Pushing against a wall which does not move.
D. Sitting still behind a desk.

9. The energy stored in food is
A. an electrostatic energy
B. a chemical energy
C. a solar energy
D. a thermal energy

10. A machine which has an energy loss of 10% will have an efficiency of
A. 100%
B. 90%
C. 55%
D. 10%

The figure below shows a simple electric circuit. Use it to answer questions 11 to 13



11. The switch is represented by
- A. IV
- B. III
- C. II
- D. I

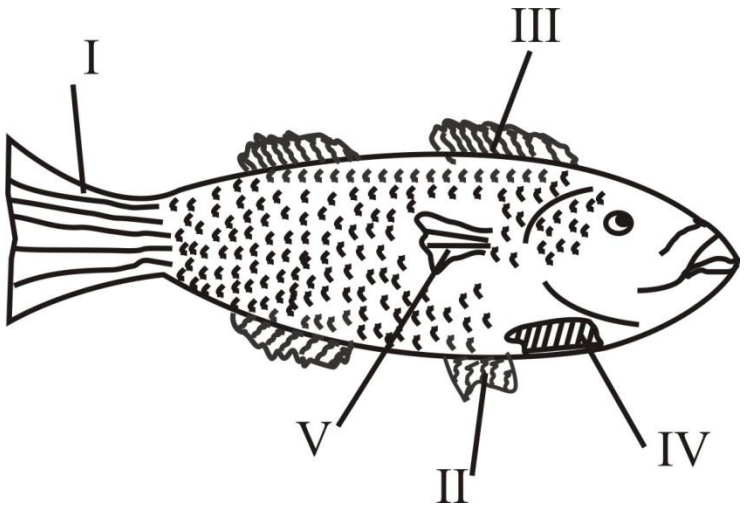
12. The component labeled V is a/an
- A. ammeter
- B. battery
- C. lamp
- D. voltmeter

13. The purpose of the component labeled IV is to
- A. supply energy to the circuit
- B. measure the voltage of the battery
- C. prevent too much current from flowing
- D. close the circuit

14. Which of the following diseases is spread by a mosquito?
- A. Blindness
- B. Cholera
- C. Dysentery
- D. Elephantiasis

15. The intermediate host of the worm that causes Bilharzia is
- A. man
- B. mosquito
- C. tsetse fly
- D. water snail

The figure below is a diagram of a fish. Use it answer questions 16 and 17



16. The function of the part labeled **IV** is to enable the fish to

A. sink in water
B. absorb oxygen from the water
C. protect itself from attack
D. detect smell in water

17. The part labeled **V** is used for

A. balancing
B. steering
C. surfacing
D. paddling

18. Which of the following arrangements is a probable food chain?

A. Grass → Grasshopper → Toad → Snake → Crow
B. Grasshopper → Grass → Goat → Lion → Man
C. Crow → Snake → Toad → Grasshopper → Grass
D. Grass → Goat → Lion → Man → Snake

19. One disease which can be transmitted through eating half cooked meat is

A. guinea worm
B. hook worm
C. thread worm
D. tape worm

20. Which of the following characteristics are common to all living things?

I. Response to stimuli
II. Respiration
III. Movement from place to place
IV. Growth

A. I and II only

B. I and III only
C. I, II and III only
D. I, II and IV only

21. The part of sugarcane that is usually used for planting is the
A. corm
B. leaf
C. stem
D. sucker

22. Excess carbohydrate in the human body is stored as
A. amino acid
B. fatty acid
C. glycogen
D. glucose

23. Carbon dioxide in the blood is removed from the body through the
A. kidneys
B. lungs
C. mouth
D. skin

24. The end-product of protein digestion is
A. amino acid
B. glycerol
C. glycogen
D. peptide

25. The liquid component of the blood is known as
A. cytoplasm
B. haemoglobin
C. plasma
D. platelets

26. Digested food substances are absorbed into the blood stream in the
A. kidneys
B. liver
C. stomach
D. iluem

27. The part of the human ear which collect and direct sound waves is the
A. ear drum
B. ear lobe
C. ear canal
D. semi-circular canal

28. The process by which waste substances are removed from the body is called
A. egestion
B. excretion
C. perspiration
D. respiration

29. The embryo of a mammal develops from
A. eggs fertilized inside the body
B. eggs fertilized outside the body
C. self fertilized eggs
D. only female sex cells

30. Which of the following gases is given out during photosynthesis?
A. Carbon dioxide
B. Carbon monoxide
C. Nitrogen
D. Oxygen

31. The component of the soil which is made up of the finest particles is
A. clay
B. humus
C. loam
D. sand

32. The soil type which is able to retain the highest amount of water is
A. clay
B. loam
C. gravel
D. sand

33. The formation of humus is aided by
A. earthworms
B. insects
C. rats
D. wind

34. The major cause of pollution at a stone quarry is
A. dust
B. human waste
C. acid disposal
D. smoke

35. Ringworm is a skin disease, which is caused by
A. bacteria

B. fungi
C. insects
D. worms

36. Which of the following habits will help in the control of diseases contracted through air?
I. Not throwing human waste into rivers
II. Washing of hands after visiting the toilet
III. Not spitting about in public places
IV. Keeping foods always covered

A. I and II only
B. II and III only
C. III only
D. I, III and IV only

37. Soup which is not well-heated gets spoilt when kept for sometime because the
A. salt in the soup is too small
B. water in the soup do not all evaporate
C. heat in the soup is not enough to kill all bacteria
D. water condenses back into the soup.

38. The measure of the amount of water vapour in the air is known as
A. cloud
B. dew
C. humidity
D. temperature

39. Which of the following heavenly bodies is a star?
A. Earth
B. Moon
C. Sun
D. Milky Way

40. The planet in our solar system which has the largest orbit is
A. Jupiter
B. Mars
C. Pluto
D. Venus

July 2002

SCIENCE Section A

SOLUTIONS

OBJECTIVE TEST

1.	A.	Iron
2.	D.	has a greater volume for the same mass of water
3.	D.	Molecules in a gas move faster than those in a liquid
4.	D.	nucleus and neutrons
5.	A.	friction
6.	D.	kinetic energy to heat energy
7.	A.	300 J
8.	A / B	
9.	B.	a chemical energy
10.	B.	90%
11.	C.	II
12.	C.	lamp
13.	C.	prevent too much current from flowing
14.	D.	Elephantiasis
15.	D.	water snail
16.	B.	absorb oxygen from the water
17.	B.	steering
18.	A.	Grass → Grasshopper → Toad → Snake → Crow
19.	D.	tape worm
20.	D.	I, II and IV only

21. C. stem

22. C. glycogen

23. B. lungs

24. A. amino acid

25. C. plasma

26. D. iluem

27. C. ear canal

28. B. excretion

29. A. eggs fertilized inside the body

30. D. Oxygen

31. A. clay

32. A. clay

33. A. earthworms

34. A. dust

35. B. fungi

36. D. I, III and IV only

37. C. heat in the soup is not enough to kill all bacteria

38. C. humidity

39. C. Sun

40. C. Pluto

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SCIENCE

SECTION B
ESSAY
1 hour

[60 marks]

Answer **three** *questions* **only** *from this section.*
Illustrate your answers wherever possible, with large, clear and fully labelled diagrams.
Credit will be given for clarity of expression and orderly presentation of material.
All questions carry equal marks

1. (a) Classify each of the following substances as colloids, suspension or solution:
Smoke, breast milk, powdered milk in water, sea water, mist, steel.

(b) (i) Define valency
(ii) State the valencies of the following elements and radicals:
(α) Iron
(β) Zinc
(γ) Hydroxyl
(λ) Carbonate

(c) Give one difference between plants and animals in terms of
(i) feeding
(ii) excretion
(iii) sensitivity

(d) (i) Draw and label a diagram to show how an image is formed in a pinhole camera.
(ii) State the characteristics of the image formed.

2. (a) State three differences between molecules in a solid state and liquid state.

(b) (i) What is a solution?
(ii) Difference between solutions and mixtures in terms of particle sizes.

(c) Name the elements which make up the following compounds:
(i) water
(ii) common salt
(iii) ammonia

(d) Copy and complete the table below

Disease	**Vector**	**Causative agent**	**Symptoms**
Malaria			
Sleeping sickness			

(e) (i) What is a man-made satellite?
(ii) Give two examples of natural satellites.
(iii) Mention two applications of man-made satellites

3. (a) Name the four main parts of a flower.
(b) Give two external features of plants pollinated by wind.
(c) Construct a food chain using any four of the organisms listed below.
Goat, lizard, bird, snake, earthworm, grasscutter, grass, ant, cassava

(d) (i) The energy input of a machine is 7.5 J and the output energy is 5.0 J. Calculate the efficiency of the machine.
(ii) Explain why the efficiency of a machine is always less than 1.

4. (a) (i) What is an embryo?
(ii) Where does an embryo develop in the female mammal?
(iii) Describe briefly how a developing embryo breathes in oxygen.

(b) Name three plant nutrients

(c) (i) What is an electrical insulator?
(ii) Explain how a comb rubbed vigorously on dry hair attracts small pieces of paper.

(d) (i) List the chemical substances that are used in the preparation of ammonia gas in the laboratory.
(ii) Write down balanced chemical equation for the reaction.

July 2002

SCIENCE Section B

SOLUTIONS

ESSAY

1. **(a)** **Classification**:

COLLOIDS	SUSPENSION	SOLUTION
smoke breast milk mist	powdered milk in water	sea water steel

(b) **(i)** **Valency:** The combining power of an atom/ ion/ radical (determined by the absolute value of the charge)

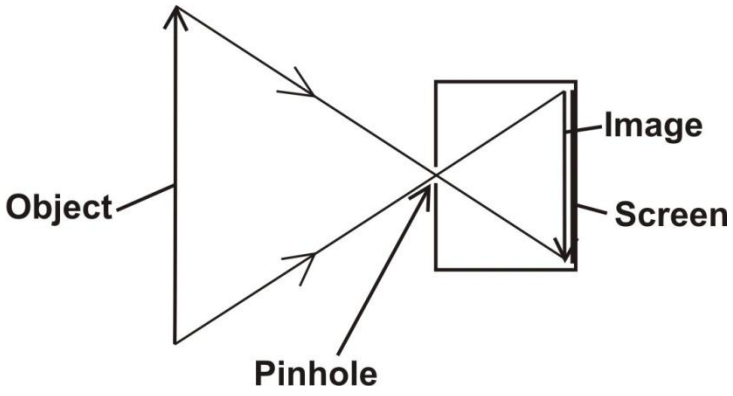
(ii)

	SUBSTANCE	**VALENCIES** :
(α)	Iron:	2 or 3
(β)	Zinc:	2
(γ)	Hydroxyl:	1
(λ)	Carbonate:	2

(c)

	PLANTS	ANIMALS
Feeding	Prepare their own food	Depend on plants and other animals for food
Excretion	Excretory materials are stored in the plants	Excretory materials are removed from the body
Sensitivity	Most plants do not show sensitivity	Animals exhibit high levels of sensitivity

(d) **(i)** **Diagram** showing formation of image in pinhole camera



(ii) **Characteristics** of image formed in pinhole camera

- Real (formed on screen)
- Inverted (turn upside-down)
- Diminished (smaller than object)

2. **(a)**

SOLID	LIQUID
Molecules are closely packed together	Molecules are widely spaced
Molecules vibrate about a fixed position	Molecules are free to move about
Molecules have low energy	Molecules have high energy
Strong forces of attraction among molecules	Weak forces of attraction among molecules

(b) **(i)** **Solution**

A liquid which has one or more other substances uniformly dissolved in it.

Or:

A fluid consisting of two or more substances mixed together and uniformly dispersed

(ii) The particles in solutions are much smaller and evenly distributed

The particles in mixtures may be much larger and possibly unevenly distributed

(c) **Elements** that make up:

(i) water: hydrogen and oxygen

(ii) common salt: sodium and chlorine

(iii) ammonia: nitrogen and hydrogen

(d)

Disease	Vector	Causative agent	Symptoms
Malaria	female anopheles mosquito	plasmodium (protozoan)	▪ fever ▪ body pains ▪ feeling cold or hot ▪ headache ▪ nausea ▪ lack of appetite
Sleeping sickness	tsetse fly	trypanosome (protozoan)	▪ accelerated heartbeat ▪ rash ▪ fever ▪ excessive sleeping ▪ lack of appetite

(e) **(i)** **Man-made satellite**

An artificial heavenly body put into orbit around a larger heavenly body by man/ humans

(ii) **Examples of natural satellites:**

The moon (as a satellite of the earth)

The earth (as a satellite of the sun)

(iii) **Applications of man-made satellites**

- Weather studies and forecasting
- Telecommunication
- Sending and receiving TV signals
- Taking photographs from space for various purposes
- Internet networking
- Global Positioning System (GPS) - a worldwide navigation system

3. **(a)** **Four main parts of a flower**

- Petal or corolla (group of petals)
- Sepal or calyx (group of sepals)
- Pistil (style, stigma and ovary)
- Stamen (anther and filament)

(b) **External features** of plants pollinated by wind

- Dull coloured petals
- Low or no nectar
- Low or no scent
- Plentiful and light pollen grains
- Highly sticky and hairy stigma

[any two]

(c) **Food chain** using: Goat, lizard, bird, snake, earthworm, grasscutter, grass, ant, cassava

Grass → earthworm → bird → snake

Cassava → ant → lizard → snake
Grass → grasscutter → snake → bird
Cassava → ant → bird → snake

(d) (i) **Efficiency** = (Energy output / Energy input) × 100%
= (5.0 J / 7.5 J) × 100%
= (50 / 75) × 100%
= (2 / 3) × 100%
= 0.6667 × 100%
= 66.67%
Or 0.67

(ii) Part of the energy input is used to overcome friction and also expended as heat.

4. (a) (i) **Embryo**
An organism at the developmental stage after the zygote stage (and for a mammal, before the foetal stage)
Or:
The organism formed after fertilization has occurred.

(ii) Where an embryo develops
Uterus or womb

(iii) How a developing embryo breathes in oxygen
Through the placenta, oxygen diffuses from the mother's blood into the embryo.

(b) **Plant nutrients**
Phosphorus, Calcium, Nitrogen, Potassium, Sulphur, Magnesium, Zinc [any three]

(c) (i) **An electrical insulator**
A substance or material that does not allow electric current to pass through

(ii) **How a comb rubbed** on hair attracts small pieces of paper
The comb becomes charged when vigorously rubbed on dry hair. When the comb is brought near small pieces of paper, they also become charged oppositely by induction. The pieces of paper are then attracted to the charged comb as a result of the opposite charge.

(d) (i) **Chemicals for ammonia gas preparation**
Ammonium chloride and Calcium hydroxide

(ii) **Balanced chemical equation**

$NH_4Cl + Ca(OH)_2 \rightarrow NH_3 + CaCl_2 + H_2O$

April 2002

SCIENCE

SECTION A
OBJECTIVE TEST
45 minutes

1. The gas usually used in fire extinguishers is
A. carbon dioxide
B. carbon monoxide
C. hydrogen
D. oxygen

2. The physical change(s) that take(s) place when a piece of ice is heated continuously for a long time can be represented by
A. solid → vapour
B. vapour → solid → liquid
C. solid → liquid → vapour
D. liquid → vapour

3. The chemical formula H_2 represents
A. two atoms of hydrogen
B. two neutrons of hydrogen
C. two ions of hydrogen
D. two molecules of hydrogen

4. The dissolution of sugar in water is a physical change because the sugar
A. cannot be seen in water
B. cannot be separated from the water
C. can be separated from the water
D. can be tasted in the water

5. Which of the following statements is true at the boiling point of water?
A. The mass of the water remains the same
B. The temperature increases
C. The temperature remains constant
D. the volume of the water increases

6. When matter changes state from gas to liquid there is

A. a decrease in mass
B. a decrease in volume
C. an increase in pressure
D. an increase in temperature

7. The energy possessed by a compressed spring is a
A. chemical energy
B. kinetic energy
C. magnetic energy
D. potential energy

8. A ball is dropped from a height to the ground. The energy possessed by the ball just before it strikes the ground is called
A. heat energy
B. kinetic energy
C. potential energy
D. sound energy

9. Which of the following statements about a force are true?
I. It causes moving objects to stop
II. It causes stationary objects to move
III. It can change the direction of motion
IV. It can change the shape of an object

A. I and II only
B. III and IV only
C. I, II and III only
D. I, II, III and IV

10. A body of mass 50 kg falls through a height of 5m. If the acceleration due to gravity is 10 ms^{-2}, calculate the loss in potential energy.
A. 50 J
B. 250 J
C. 500 J
D. 2500 J

11. Which of the following substances can be used as an electrical insulator?
A. Copper
B. Graphite
C. Plastic
D. Zinc

12. Gaps are left between railway lines to
A. prevent contraction
B. prevent rusting

C. allow for expansion
D. enable a train to stop

13. A beak and claws are protective structures of a
A. crab
B. hawk
C. lion
D. snake

14. An example of a parasite which lives in the intestines of an animal is
A. guinea worn
B. jigger
C. plasmodium
D. tapeworm

15. The best way to prevent guinea worm disease is by
A. boiling drinking water
B. putting camphor in drinking water
C. putting drinking water in a refrigerator
D. sieving drinking water

16. Malaria parasites enter the human body through
A. breathing contaminated air
B. drinking contaminated water
C. mosquito bites
D. tsetse fly bites

17. Human beings are omnivorous because they eat
A. both cooked and raw food
B. only flesh
C. only fruits
D. both plants and animals

18. The characteristics that distinguish living things from non-living things is that, living things
A. are able to move, non-living things are not
B. are of different sizes, non-living things have same size
C. can manufacture their own food, non-living things cannot
D. have a definite form, non-living things have irregular forms

19. Animals depend on plants for food because
A. animals do not need to drink when they eat plant food
B. only plants can make their food from simple substances
C. only plant food can satisfy animals
D. plants are available everywhere.

20. Which of the following animals has a body temperature which changes with the temperature of the surroundings?

A. Butterfly
B. Crow
C. Rabbit
D. Toad

21. The farming practice which would improve soil fertility is

A. clean weeding
B. crop rotation
C. continuous cropping
D. mono cropping

22. The best soil for plant growth consists of correct proportions of

A. clay, gravel and sand
B. clay and humus
C. clay, humus and sand
D. clay and sand

23. The by-product of photosynthesis is

A. carbon dioxide
B. starch
C. oxygen
D. water

24. Blood circulates to all parts of the human body with the help of the

A. heart
B. kidney
C. lungs
D. liver

25. The end-product of the digestion of carbohydrates is

A. amino acids
B. fatty acids
C. glucose
D. glycogen

26. The central nervous system is made up of the

A. brain and nerves
B. brain and spinal cord
C. veins and nerves
D. skull and spine

27. Which of the following organs are used for excretion?

I. Lung

II. Skin
III. Kidney
IV. Heart

A. I and IV only
B. II and III only
C. III and IV only
D. I, II and III only

28. The plant structures used for gaseous exchange are the
A. leaves
B. tendrils
C. thorns
D. roots

29. Digestion of starch in humans starts from the
A. gullet
B. ileum
C. mouth
D. stomach

30. The male sex cell is the
A. penis
B. scrotum
C. sperm
D. testis

31. Which of the following seeds is dispersed by explosive mechanism?
A. Coconut
B. Cotton
C. Okro
D. Orange

32. The ear is the organ for detecting
A. colour
B. heat
C. scent
D. sound

33. Which of the following substance is a digestive juice?
A. Blood
B. Bile
C. Sweat
D. Tear

34. An example of a disease transmitted through air is
A. bilharzias
B. cholera
C. measles
D. typhoid

35. Which of the following diseases can be caused by a housefly?
A. Dysentery
B. Elephantiasis
C. Scabies
D. Sleeping sickness

36. The part of the body of an animal that is usually affected by tuberculosis is the
A. heart
B. lung
C. liver
D. intestine

37. The irregular emptying of the bowels is the cause of
A. constipation
B. diarrhoea
C. headache
D. stomach ache

38. The farther away a planet is from the sun, the
A. colder the planet
B. hotter the planet
C. shorter the time it takes to go round the sun
D. shorter the length between day and night

39. The instrument used to measure relative humidity is the
A. anemometer
B. barometer
C. hygrometer
D. manometer

40. How many days does the moon take to go round the earth?
A. 31
B. 30
C. 29
D. 27

April 2002

SCIENCE Section A

SOLUTIONS

OBJECTIVE TEST

1.	A.	carbon dioxide
2.	C.	solid → liquid → vapour
3.	A.	two atoms of hydrogen
4.	C.	can be separated from the water
5.	C.	The temperature remains constant
6.	B.	a decrease in volume
7.	D.	potential energy
8.	B.	kinetic energy
9.	D.	I, II, III and IV
10.	D.	2500 J
11.	C.	Plastic
12.	C.	allow for expansion
13.	B.	hawk
14.	D.	tapeworm
15.	A.	boiling drinking water
16.	C.	mosquito bites
17.	D.	both plants and animals
18.	C.	can manufacture their own food, non-living things cannot
19.	B.	only plants can make their food from simple substances

20. D. Toad

21. B. crop rotation

22. C. clay, humus and sand

23. C. oxygen

24. A. heart

25. C. glucose

26. B. brain and spinal cord

27. D. I, II and III only

28. A. leaves

29. C. mouth

30. C. sperm

31. C. Okro

32. D. sound

33. B. Bile

34. C. measles

35. A. Dysentery

36. B. lung

37. A. constipation

38. A. colder the planet

39. C. hygrometer

40. D. 27

April 2002

SCIENCE

SECTION B
ESSAY
1 hour

[60 marks]

Answer **three** *questions* **only** *from this section.*
Illustrate your answers wherever possible, with large, clear and fully labelled diagrams.
Credit will be given for clarity of expression and orderly presentation of material.
All questions carry equal marks

1. (a) Name four characteristics of non-living things.

(b) Mention two functions of each of the following parts of a plant:
(i) leaves
(ii) roots

(c) Classify the following actions as voluntary or involuntary:
(i) eating
(ii) walking
(iii) sneezing
(iv) laughing

(d) (i) Explain the difference between a physical change and a chemical change.
(ii) Write down the chemical equation for the preparation of carbon dioxide in the laboratory.

(e) (i) What is a machine?
(ii) Write down an expression for the efficiency of a machine.

2. (a) (i) Mention four female sexual characteristics of a human being.
(ii) Name two sense organs in mammals
(iii) Give one function of each of the sense organs named

(b) Define each of the following terms:

(i) compound
(ii) element

(c) List the particles that make up an atom

(d) State the energy changes that take place in each of the following processes.
(i) An orange falls from a tree to the ground
(ii) A carpenter hits a nail with a raised hammer.

3. (a) (i) State three human activities that can cause an increase in soil erosion
(ii) Give three ways of preventing soil erosion in agriculture.

(b) Describe briefly how a mixture of sand and common salt could be separated

(c) (i) Define work
(ii) State the unit of work

(d) (i) Distinguish between heat and temperature
(ii) State the units of heat and temperature

4. (a) Name the organism that causes each of the following diseases:
(i) cholera
(ii) malaria
(iii) tuberculosis
(iv) ringworm

(b) State one method of preventing each of the diseases in (a) above.

(c) A ray of light hits the surface of a plane mirror at an angle of 55°.
(i) Draw a ray diagram showing the normal, incident ray and the reflected ray.
(ii) What is the angle of reflection for the ray.

(d) State two properties of the image formed in a plane mirror.

(e) Write the chemical names of each of the following compounds:
(i) $MgCl_2$
(ii) FeS
(iii) NH_4OH
(iv) AgCl

April 2002

SCIENCE Section B

SOLUTIONS

ESSAY

1. **(a)** **Characteristics** of non-living things

- They do not respire
- They do not excrete
- They do not feed
- They do not grow
- They do not reproduce on their own
- They do not respond to stimuli on their own
- They do not move on their own

[any four]

(b) **(i)** **Functions of leaves**

- Photosynthesis (preparation of food using carbon dioxide and water)
- Transpiration (loss of water vapour through the stomata)
- Respiration (intake of oxygen to produce energy)
- Guttation (oozing out of excess water from the surface of leaf)
- Food storage in certain leaves

(ii) **Functions of roots**

- Absorbs water and mineral salts from the soil
- Some roots store food prepared by the plant
- Holds the plant firmly in place

(c)

(i)	eating	-	voluntary
(ii)	walking	-	voluntary
(iii)	sneezing	-	involuntary
(iv)	laughing	-	voluntary

(d) **(i)**

Physical change	-	No new substance is formed and it is reversible
Chemical change	-	A new substance is formed and it is irreversible

(ii) **Equation** for the laboratory preparation of Carbon dioxide.

$CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$

(e) **(i)** What is a **machine**?

A mechanical device that makes work easier and/ or faster

(ii) **Efficiency of a machine**

= (Work output / Work input) × 100%

Or: (Mechanical Advantage / Velocity Ratio) × 100%

2. **(a)** **(i)** **Female sexual** characteristics

The presence of:

- Ovaries
- Womb
- Fallopian tube
- Cervix
- Vagina
- vulva
- Menstruation
- Ovulation
- Broader hips
- larger breasts

[any four]

(ii) **Sense organs in mammals**

Tongue, Skin, Eyes, Nostrils, Ears

(iii) **Functions**

- Tongue - Tasting
- Skin - feeling (of touch)
- Eyes - seeing
- Nostrils - smelling
- Ears - hearing

(b) **(i)** **Compound**

A substance formed by the chemical combination of two or more elements in a definite proportion

Or:

A substance consisting of two or more elements that are combined chemically

(ii) **Element**

A substance that is made up of the same kind of atoms

Or:

Any substance that cannot be broken down into a simpler one by a chemical reaction

(c) **Particles** that make up an atom

Protons, neutrons and electrons

(d) **Energy** changes as:

(i) An orange falls from a tree to the ground

Potential → Kinetic → Sound + Heat

(ii) A carpenter hits a nail with a raised hammer.

Chemical → potential → kinetic → sound + heat + light

3. **(a)** **(i)** **Increase in soil erosion**

- Removal of cover crops
- Bush burning
- Ploughing along slopes
- Felling of trees
- Sand-winning [any three]

(ii) **Preventing soil erosion**

- Planting of cover crops
- Planting more trees
- Practicing strip cropping
- Ploughing across slopes
- Terracing [any three]

(b) **How a mixture of sand and common salt** can be separated

- Add water to the mixture and stir to dissolve the salt
- Pour the mixture through a filter paper in a funnel to separate the sand (residue) from the salt solution (filtrate)
- Heat the salt solution till all the water evaporates, leaving the salt crystals behind
- Dry the sand and the salt obtained

(c) **(i)** **Work**

Work is done when an applied force moves an object in the direction of the force.

Or:

The transfer of energy, measured as the product of the force applied to a body and the distance moved by that body in the direction of the force

Or:

Work $= f \times d$, where f = force applied

and d = distance moved in the direction of the applied force

(ii) **Unit** of work = joules or J

(d) **(i)** **Difference:**

Heat	Temperature

The energy possessed by a body that results in its hotness or coldness	A measure of the hotness or coldness of the body

(ii) **Units** of:

Heat = joules (J)

Temperature = Kelvin (K) or degrees Celsius (°C) or degrees Fahrenheit (°F)

4. **(a)**

DISEASE-CAUSING ORGANISM

(i) **Cholera** - bacteria (*Vibrio cholerae*)

(ii) **malaria** - protozoa (*plasmodium*)

(iii) **tuberculosis** - bacteria (*Mycobacterium tuberculosis)*

(**iv)** **ringworm** - fungi (*Trichophyton*, *Epidermophyton*, and *Microsporum*)

(b)

(i) **Cholera** - MODE OF PREVENTION

- Eating hot meals,
- drinking clean potable water,
- washing hands with soap after visiting the toilet

[any one]

(ii) **malaria** -

- Sleeping under mosquito nets,
- applying mosquito repellents on the body,
- burying empty cans,
- clearing bushy areas,
- spraying insecticides in room,
- clearing choked drains, [any one]

(iii) **tuberculosis** -

- Vaccination
- Ensuring adequate ventilation
- Avoid sharing cutlery, cups and glasses with infected persons

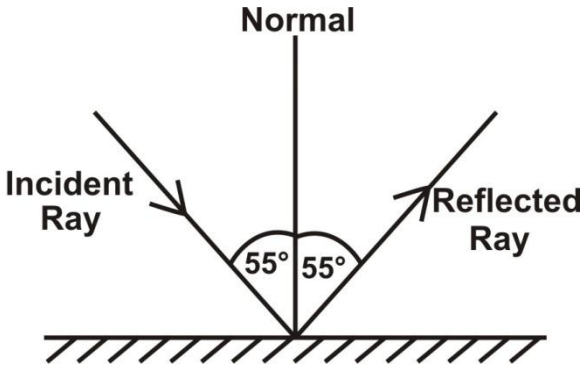
[any one]

(iv) **ringworm** -

- Keep skin clean and dry
- Avoid contact with infected persons or animals
- Avoid sharing personal items like clothes, combs or towels

[any one]

(c) **(i)** **Ray diagram**



(ii) Angle of reflection = angle of incidence
= 90° - 55°
= 35°

(d) **Properties** of image in a plane mirror

- Same size as object
- Same distance from mirror as object
- Virtual
- Erect / upright
- Laterally inverted [any two]

(e)

			CHEMICAL NAMES
(i)	**$MgCl_2$**	-	Magnesium chloride
(ii)	**FeS**	-	Iron (II) sulphide
(iii)	**NH_4OH**	-	Ammonium hydroxide
(iv)	**AgCl**	-	Silver chloride

April 2001

SCIENCE

SECTION A
OBJECTIVE TEST
45 minutes

1. The main pollutant from domestic fires is
A. carbon monoxide
B. sulphur dioxide
C. water vapour
D. oxygen
E. ammonia

2. Plantain is propagated vegetatively by
A. corms
B. stem cuttings
C. suckers
D. rhizomes
E. tubers

3. Which of the following gases supports combustion?
A. carbon dioxide
B. carbon monoxide
C. hydrogen
D. nitrogen
E. Oxygen

4. The main function of petals in a flower is to
A. protect the ovary
B. produce seeds
C. attract insects
D. secrete nectar
E. produce sweet scent

5. The organ in the human body which is responsible for the removal of urea is the
A. heart
B. kidney
C. lung

D. liver
E. bladder

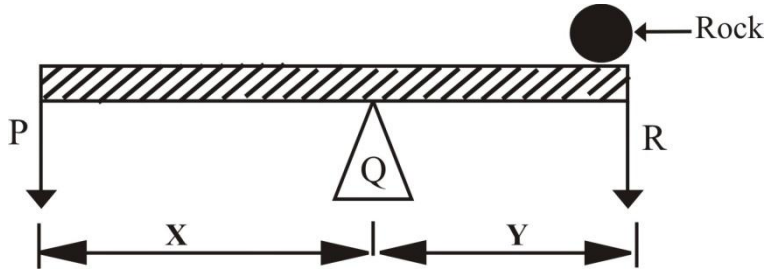
6. The sun and the planets form the
A. atmosphere
B. constellation
C. milky way
D. solar system
E. universe

7. The bilharzia worm can enter the human body through the
A. anus
B. ear
C. mouth
D. nostrils
E. skin

8. Brass is an alloy of copper and
A. aluminium
B. carbon
C. iron
D. tin
E. zinc

9. Which of the following diseases is contracted through infected wounds?
A. Chicken pox
B. Diphtheria
C. Mumps
D. Tetanus
E. Whooping cough

Use the figure below to answer Questions 10 and 11



A rigid bar is balanced horizontally at a point by placing a rock on the bar as shown

10. Which of the parts labeled represents the pivot?

A. **P**
B. **Q**
C. **R**
D. **X**
E. **Y**

11. The effort is represented by

A. **P**
B. **Q**
C. **R**
D. **X**
E. **Y**

12. Which of the following parts of the human body are sense organs?

I. Tongue
II. Hair
III. Nose
IV. Skin

A. I and IV only
B. II and III only
C. I, II and III only
D. I, III and IV only
E. I, II, III and IV

13. The function of the cotyledon is to

A. produce the radicle
B. store food
C. protect the seed
D. produce the plumule
E. produce the fruit

14. Which of the following substances are components of a fertile soil?

A. Nitrogen and phosphorous
B. Phosphorous and sulphur
C. Phosphorous, calcium and sulphur
D. Calcium, sulphur, nitrogen and phosphorous
E. Nitrogen, calcium and phosphorous

15. Which of the following appliances transform(s) electrical energy to heat energy?

I. Swish oven
II. Hair dryer
III. Refrigerator motor

A. I only
B. I and II only
C. II and III only
D. I and III only
E. I, II and III

16. A ray of light makes an angle of 20° with the surface of a plane mirror. Determine the angle of reflection.
A. 20°
B. 50°
C. 70°
D. 90°
E. 120°

17. Seasonal changes are the result of the
A. rotation and tilting of the earth
B. revolution and tilting of the earth
C. revolution and rotation of the earth
D. revolution and tilting of the moon
E. revolution and rotation of the moon

18. Crystals of sugar were obtained when a hot solution of sugar was cooled to room temperature. What conclusion can be drawn from this observation?
A. Sugar does not dissolve in cold water
B. Sugar does not dissolve in hot water
C. Sugar dissolves more in hot water than in cold water
D. Sugar dissolves less in hot water than in cold water
E. Sugar dissolves more in cold water than in cold water

19. Which of the following processes about all living things is / are true?
I. Reproduction
II. Respiration
III. Locomotion
IV. Excretion

A. I only
B. I and II only
C. III and IV only
D. I, II and III only
E. I, II and IV only

20. Tendrils in plants are adaptations that enable the plant to
A. climb
B. expire
C. store food
D. reproduce

E. propagate vegetatively

21. The density of a substance is $2.5 gcm^{-3}$. Calculate the volume of the substance if its mass is 25.0 g.

A. 0.1 cm^3
B. 10.0 cm^3
C. 22.5 cm^3
D. 27.5 cm^3
E. 62.5 cm^3

22. The food substance that promotes good health is

A. carbohydrate
B. fat
C. protein
D. vitamin
E. water

23. The energy that causes the turbine of hydro-electric plant to rotate is

A. chemical energy
B. potential energy
C. electric energy
D. kinetic energy
E. heat energy

24. Which of the following pairs of organs are parts of the central nervous system?

A. Spinal cord and the eyes
B. Brain and spinal cord
C. Brain and the ears
D. Eyes and the ears
E. Eyes and the brain

25. Plants manufacture their food using

I. carbon dioxide
II. oxygen
III. light
IV. chlorophyll

A. I only
B. II only
C. I and II only
D. I, III and IV only
E. I, II, III and IV

26. Which of the following crops improves soil fertility when cultivated continuously on the same piece of land?

A. Carrot

B. Cocoa
C. Cassava
D. Maize
E. Peas

27. It is necessary to keep household refuse in covered containers in order to prevent
A. the spread of tuberculosis
B. the decomposition of the refuse
C. the spread of measles
D. entry of flies
E. entry of bacteria

28. The fraction of the earth's surface that faces the sun at any particular time is
A. 1/4
B. 1/3
C. 1/2
D. 2/3
E. 3/4

29. Which of the following is a micro-organism?
A. Bread mould
B. Bacteria
C. Leech
D. Termite
E. Tick

30. Which of the following statements about machines is/are correct?
I. Machines help us to do work more easily
II. A force applied at one point of the machine overcomes a load at another point of the machine
III. All machines have engines

A. I only
B. II only
C. III only
D. I and II only
E. II and III only

31. The female part of a flower is called the
A. filament
B. petal
C. pollen grain
D. sepal
E. style

32. Which of the following substances is a compound?

A. Hydrogen
B. Nitrogen
C. Oxygen
D. Sulphur
E. Water

33. A solution in which no more solute will dissolve at a particular temperature is said to be
A. concentrated
B. dilute
C. saturated
D. insoluble
E. unsaturated

34. The process of respiration could be represented by the following equation:
Glucose + oxygen → water + carbon dioxide + **X**
What does **X** represent?

A. Alcohol
B. Hydrogen
C. Heat
D. Mineral salts
E. Protein

35. When oil and water are shaken together they form a mixture called
A. a solution
B. an emulsion
C. a solute
D. a solvent
E. a suspension

36. The organism which lives in the human blood as a parasite is
A. louse
B. plasmodium
C. roundworm
D. tapeworm
E. tick

37. The fusion of an egg cell with sperm forms
A. an ovum
B. an embryo
C. a zygote
D. an ovary
E. a placenta

38. The instrument used to measure current in an electric circuit is the

A. ammeter
B. anemometer
C. barometer
D. thermometer
E. voltmeter

39. The atom of an element has 4 protons and 5 neutrons in its nucleus. How many shells are occupied in the atom?

A. 1
B. 2
C. 3
D. 4
E. 5

40. The gaps left between railway lines is to allow for

A. expansion
B. cooling
C. maintenance
D. easy stopping
E. gear changing

April 2001

SCIENCE Section A

SOLUTIONS

OBJECTIVE TEST

1.	A.	carbon monoxide
2.	C.	suckers
3.	E.	Oxygen
4.	C.	attract insects
5.	B.	kidney
6.	D.	solar system
7.	E.	skin
8.	E.	zinc
9.	D.	Tetanus
10.	B.	**Q**
11.	A.	**P**
12.	D.	I, III and IV only
13.	B.	store food
14.	D.	Calcium, sulphur, nitrogen and phosphorous
15.	B.	I and II only
16.	C.	70°
17.	B.	revolution and tilting of the earth
18.	C.	Sugar dissolves more in hot water than in cold water
19.	E.	I, II and IV only

20. A. climb

21. B. $10.0\ cm^3$

22. D. vitamin

23. D. kinetic energy

24. B. Brain and spinal cord

25. D. I, III and IV only

26. E. Peas

27. D. entry of flies

28. C. 1/2

29. B. Bacteria

30. D. I and II only

31. E. style

32. E. Water

33. C. saturated

34. C. Heat

35. B. an emulsion

36. B. plasmodium

37. C. a zygote

38. A. ammeter

39. B. 2

40. A. expansion

April 2001

SCIENCE

SECTION B
ESSAY
1 hour

[60 marks]

Answer **three** *questions* **only** *from this section.*
Illustrate your answers wherever possible, with large, clear and fully labelled diagrams.
Credit will be given for clarity of expression and orderly presentation of material.
All questions carry equal marks

1.

a) (i) What is surface tension?

(ii) Name two household devices that make use of surface tension

(iii) Explain how the principle of surface tension operates in one of the devices you have named

b) (i) What is constipation?

(ii) State four ways by which constipation can be prevented

c) A block of mass 10kg is suspended from a spring. Calculate the force acting on the block. [g = $10ms^{-2}$]

d) An amount of sodium hydroxide solution was put into a beaker and a piece of litmus paper was added. Dilute hydrochloric acid was added a little at a time with stirring until the litmus paper just turned purple. The resulting solution was evaporated to dryness.

(i) What was the colour of the litmus paper in the sodium hydroxide solution?

(ii) What does the colour change of the litmus paper indicate?

(iii) What substance was left after the evaporation?

(iv) Write a balanced equation for the reaction that took place.

2.

a) State and explain three ways by which plants protect themselves from danger

b) (i) What is an anion?

(ii) Give the names of the products formed when the following pairs of compounds react together:

α)NH_4OH and dilute HCL

β) $CaCO_3$ and dilute HCL

c) (i) What is electromagnetism?

(ii) With the aid of a circuit diagram describe how a piece of metal can be magnetized

3.

a) State (i) three characteristics of roots

(ii) two functions of roots

b) State two conditions necessary for the germination of seeds

c) Describe briefly how the volume of an irregular shaped metal ball could be determined

d) Write down the chemical formula for each of the following compounds:

(i) sodium chloride

(ii) copper sulphate

(iii) magnesium sulphate

(iv) potassium carbonate

e) State the method that could be used to separate each of the following compounds:

(i) iodine and sand

(ii) ethanol and water

4.

a) (i) What is digestion?

(ii) Describe briefly how food is digested in the stomach

b) (i) What is meant by eclipse of the sun?

(ii) With the aid of a labelled diagram, distinguish between total eclipse and partial eclipse.

c) (i) State the difference between a physical change and a chemical change

(ii) Give one example of each of a physical change and a chemical change

April 2001

SCIENCE Section B

SOLUTIONS

ESSAY

1. (a) (i) Surface tension

Condition existing at the surface of a liquid, resembling a stretched elastic skin

Or:

The property of a liquid that makes its surface behave like a stretched elastic skin

(ii) Household devices that use surface tension:

Umbrella, raincoat, tents, etc [any two]

(iii) Explanation

The cohesive forces that exist among the water molecules results in the water behaving like a stretched elastic skin over the pores of the household devices, therefore preventing water from passing through.

(b) (i) Constipation

A condition in which a person or animal has difficulty in eliminating solid waste from the body and the feces are hard and dry

(ii) Preventing Constipation

- Drinking a lot of water / liquids
- Eating fruits and vegetables regularly
- Eating foods that contain high amounts of fiber, such as wheat, barley, etc
- Getting enough sleep
- Exercising regularly / daily

[any four]

(c) Force acting on block

Force = mass × acceleration due gravity

= 10 kg × 10 ms^{-2}

= 100 $kgms^{-2}$ or 100 N

(d) (i) Colour of litmus paper in sodium hydroxide: blue

(ii) Colour change (to purple) **indicates** that

the contents of the beaker has changed from a basic nature to a neutral nature
or: that a neutralization reaction has occurred
or: the neutralization process is complete.

(iii) Substance left
Common salt or sodium chloride

(iv) Balanced equation
$NaOH + HCl \rightarrow NaCl + H_2O$

2. (a) Ways by which **plants protect themselves** from danger

- Unpleasant smell from certain plants makes them repulsive to animals.
- Unpleasant taste of certain plants makes them undesirable to taste / eat.
- Sharp thorns / spines of certain plants make them difficult to touch.
- Poisonous parts of certain plants can cause irritation / pain / infection / death.
- Sticky secretions or parts of certain plants makes them uncomfortable to touch

[any three]

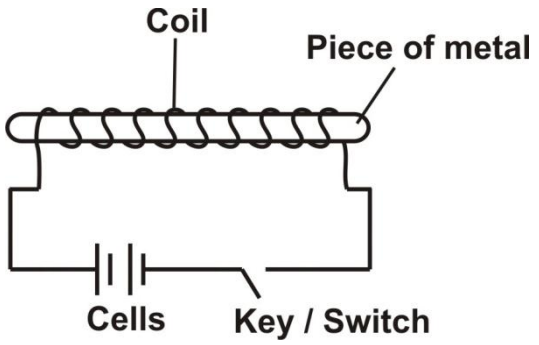
(b) (i) Anion: A negatively charged atom
Or: An atom that has gained extra electrons

(ii)

	NAMES OF PRODUCTS FORMED
α) NH_4OH and dilute HCL	$NH_4Cl + H_2O$ or: ammonium chloride and water
β) $CaCO_3$ and dilute HCL	$CaCl_2 + CO_2 + H_2O$ or: calcium chloride, carbon dioxide and water

(c) (i) Electromagnetism: Magnetism produced by an electric current
or: The process of making a magnet using an electric current

(ii) Circuit diagram



3. (a) (i) Characteristics of **roots**

- Usually found beneath the ground
- Usually grow downwards
- Grow in the opposite direction to light
- Develop from the embryo's radicle
- Usually coloured cream or brown
- Root hairs present

[any three]

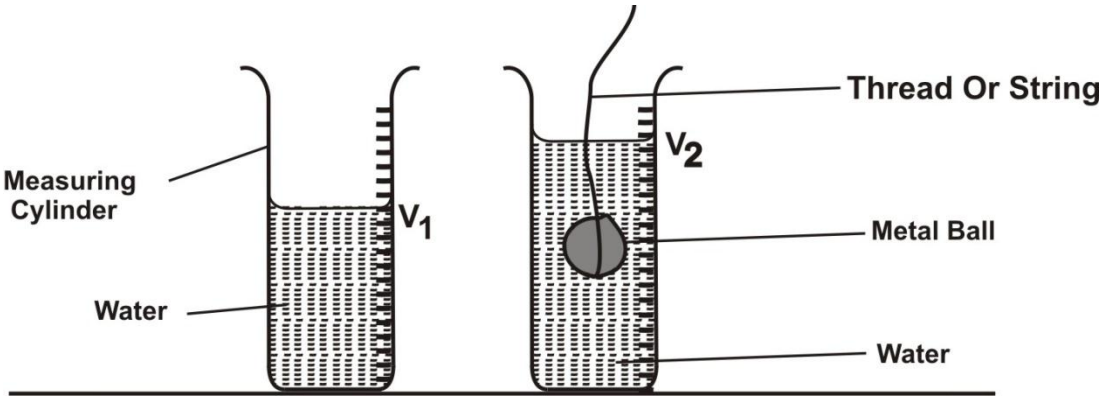
(ii) **Functions** of **roots**

- Absorbs water and mineral salts from the soil
- Holds the plant firmly in place
- Some roots store food prepared by the plant

(b) **Conditions** necessary for germination of seeds

- viability of seed
- suitable temperature
- water / moisture
- air

(c) **Determination of volume** of irregular ball



As shown in the diagram above,

I. Fill a measuring cylinder with water to a suitable volume.

II. Record the volume (V_1)

III. Tie a thread / string around the irregular shaped metal ball and gently lower into the water in the cylinder.

IV. Record the new volume (V_2)

V. Find difference in volumes ($V_2 - V_1$)

Volume of metal ball = $(V_2 - V_1)$

(d)

		CHEMICAL FORMULA
(i)	**sodium chloride**	NaCl
(ii)	**copper sulphate**	$CuSO_4$
(iii)	**magnesium sulphate**	$MgSO_4$

(iv) **potassium carbonate** K_2CO_3

(e) SEPARATING METHOD

(i) **iodine and sand** Sublimation

(ii) **ethanol and water** Distillation

4. **(a)** **(i)** **Digestion**

The breakdown of food in the body into simpler forms that can be absorbed into the bloodstream and used

(ii) **How food is digested in the stomach**

- The digestion of protein begins in the stomach.
- The walls of the stomach secrete a liquid, gastric juice, which contains HCl and protein digesting enzymes.
- The HCl creates the suitable medium for the enzymes to work properly.
- The protein-digesting enzymes break down proteins into simpler forms.

(b) **(i)** **Eclipse of the sun**

The situation where the Moon blocks all or part of the Sun's light from reaching the Earth's surface, because it passes directly between the Earth and the Sun

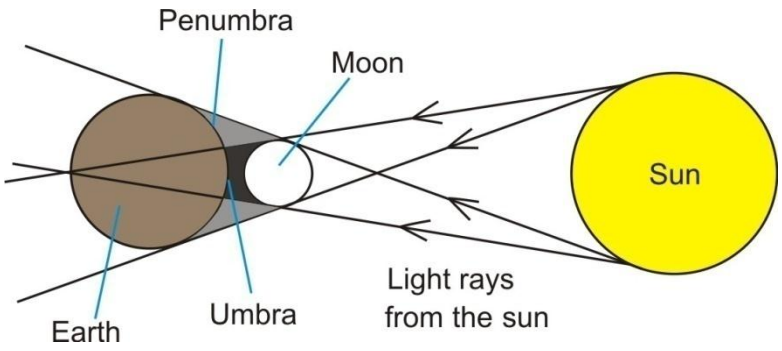
Or:

The case where the moon's shadow is cast on the earth as it comes between the sun and the earth.

Or:

When the earth comes between the sun and the moon in a straight line such that the moon's shadow is cast on the earth

(ii) **Diagram** showing partial and total eclipse



(c) **(i)** **Difference** between physical and chemical change

Physical change: No new substance is formed; and the change is reversible

Chemical change: New substance formed; change is irreversible

(ii) **Examples** of physical and chemical change

Physical change	Chemical change
melting of candle	cooking of yam
melting of ice	rusting of iron
dissolution of salt in water	burning of wood into ash
boiling of water	action of saliva on cooked starch

[any one]

August 2000

SCIENCE

SECTION A
OBJECTIVE TEST
45 minutes

1. Which of the following organs is used by the fish for movement?
- A. eye
- B. fin
- C. gill
- D. tail
- E. mouth

2. A group of stars found in the universe is called
- A. galaxy
- B. meteor
- C. meteorite
- D. planet
- E. solar system

3. The basic unit of matter is the
- A. atom
- B. compound
- C. element
- D. electron
- E. ion

4. Which of the following process is a physical change?
- A. Rusting of iron
- B. Burning of paper
- C. Melting of ice
- D. Fermentation of palm wine
- E. Boiling of an egg

5. The organ that removes metabolic wastes from the human body is
- A. anus
- B. heart
- C. kidney

D. rectum
E. spleen

6. The food chain that occurs in nature is
A. green plants → lizards → insects → snakes
B. grasshopper → green plants → lizards → hawks
C. green plants → snakes → hawks → man
D. green plants → grasshopper → lizards → snakes
E. mice→ lizards → hawks → snakes

7. The device that uses an electromagnet in its operation is the
A. electric bell
B. electric iron
C. radio
D. refrigerator
E. sewing machine

8. Which of the following statements about mammals is true?
A. They have hairs
B. They lay eggs
C. They have feathers
D. They have beaks
E. They are cold-blooded

9. The parasite that is found on dogs is
A. capsid
B. guinea worn
C. plasmodium
D. tape worm
E. tick

10. The substance that sublimes when heated is
A. baking powder
B. camphor
C. common salt
D. sugar
E. sulphur

11. In plants, when the male gamete fuses with the egg, the first structure that is formed is the
A. embryo
B. testa
C. fruit
D. seed
E. zygote

12. Clayey soil holds more water than any other type of soil because it has
A. large particle size
B. larger air spaces
C. irregular particle size
D. greater attraction for water
E. reduced air spaces

13. Disposal of urine and faeces into a river used for domestic purposes, may cause the outbreak of
A. bilharzia
B. malaria
C. river blindness
D. tetanus
E. tuberculosis

14. The temperature of 20°C on the Kelvin scale is
A. 253 K
B. 263 K
C. 273 K
D. 283 K
E. 293 K

15. The heavenly body that produces and emits its own light is
A. Mars
B. meteorite
C. the moon
D. the sun
E. Venus

16. Desert plants shed their leaves to reduce the rate of
A. absorption
B. diffusion
C. osmosis
D. respiration
E. transpiration

17. Which of the following materials transforms electrical energy into light energy?
A. Electrical bell
B. Filament bulb
C. Electric guitar
D. Lantern
E. Radio

18. The name of the smallest blood vessel in humans is
A. artery
B. arteriole

C. capillary
D. vein
E. venule

19. Which of the following activities is a reflex action?
A. Eating
B. Fighting
C. Learning
D. Sneezing
E. Writing

20. The method of separation used in the treatment of water at the water works is
A. decantation
B. distillation
C. evaporation
D. magnetization
E. precipitation

21. The gas produced when dilute hydrochloric acid is added to calcium carbonate is
A. ammonia
B. carbon dioxide
C. chlorine
D. hydrogen
E. oxygen

22. Which of the following environmental factors increases sweating in humans?
A. Low temperature
B. Low pressure
C. High pressure
D. High temperature
E. High humidity

23. Which of the following crops is propagated vegetatively?
A. Cassava
B. Coconut
C. Mango
D. Okro
E. Tomato

24. The type of energy stored in food is referred to as
A. chemical energy
B. electrical energy
C. heat energy
D. kinetic energy
E. light energy

25. The end product of protein digestion is

A. amino acids
B. glucose
C. maltose
D. poly-peptide
E. sucrose

26. The attracting power of magnets is greatest at the

A. centre
B. edges
C. poles
D. sides
E. surface

27. The substances which help in the digestion of food materials are called

A. enzymes
B. hormones
C. lymph
D. mucus
E. plasma

28. The transfer of heat through a solid medium is by

I. conduction
II. convection
III. radiation

A. I only
B. II only
C. III only
D. I and II only
E. II and III only

29. Casual and loose sex life can result in the spread of

A. AIDS
B. cholera
C. poliomyelitis
D. malaria
E. tuberculosis

30. Which of the following conditions will cause the same quantity of sugar to dissolve faster in a given volume of water?

A. Stirring the solution
B. Grinding the sugar
C. Heating the solution

D. Dissolving in a bigger container
E. Cooling the solution

31. The raw material for photosynthesis is
A. carbon dioxide
B. chlorophyll
C. mineral salts
D. oxygen
E. sunlight

32. Which of the following heavenly bodies is at the centre of the planetary system?
A. Earth
B. Mercury
C. Pluto
D. Sun
E. Venus

33. Legumes are generally included in crop rotation because they
A. are easy to uproot in preparation for the next planting season
B. are able to fix atmospheric nitrogen in the soil
C. have short life span
D. are good cover crops
E. have high market value

34. The insect which transmits the plasmodium parasite is called
A. butterfly
B. cockroach
C. housefly
D. mosquito
E. tsetse fly

35. The property of water that causes mosquito larvae to float on it is
A. density
B. viscosity
C. surface tension
D. volume
E. capillarity

36. Which of the following practices is a bad habit?
A. Keeping one's surroundings tidy
B. Drinking clean water
C. Covering one's food
D. Using a brother's towel and comb
E. Washing of hands before eating

37. Addition of more solvent to a solution makes the solution more

A. dense
B. dilute
C. concentrated
D. saturated
E. super saturated

38. An atom has 4 protons and 5 neutrons in its nucleus. How many positively charged particles are in the atom?

A. 1
B. 2
C. 3
D. 4
E. 5

39. Which of the following statements about all living things is / are true?

I. They respire
II. They manufacture their own food
III. They grow

A. I only
B. II only
C. I and II only
D. I and III only
E. II and III only

40. Spectacle with converging lenses are used by people suffering from

A. cataract
B. glaucoma
C. headache
D. long-sightedness
E. short-sightedness

August 2000

SCIENCE Section A

SOLUTIONS

OBJECTIVE TEST

1.	B.	fin
2.	A.	galaxy
3.	A.	atom
4.	C.	Melting of ice
5.	C.	kidney
6.	D.	green plants → grasshopper → lizards → snakes
7.	A.	electric bell
8.	A.	They have hairs
9.	E.	tick
10.	B.	camphor
11.	E.	zygote
12.	E.	reduced air spaces
13.	A.	bilharzia
14.	E.	293 K
15.	D.	the sun
16.	E.	transpiration
17.	B.	Filament bulb
18.	C.	capillary
19.	D.	Sneezing

20. A. decantation

21. B. carbon dioxide

22. D. High temperature

23. A. Cassava

24. A. chemical energy

25. A. amino acids

26. C. poles

27. A. enzymes

28. A. I only

29. A. AIDS

30. C. Heating the solution

31. A. carbon dioxide

32. D. Sun

33. B. are able to fix atmospheric nitrogen in the soil

34. D. mosquito

35. C. surface tension

36. D. Using a brother’s towel and comb

37. B. dilute

38. D. 4

39. D. I and III only

40. D. long-sightedness

August 2000

SCIENCE

SECTION B
ESSAY
1 hour

[60 marks]

Answer **three** *questions* **only** *from this section.*
Illustrate your answers wherever possible, with large, clear and fully labelled diagrams.
Credit will be given for clarity of expression and orderly presentation of material.
All questions carry equal marks

1. (a) (i) What is personal hygiene?
(ii) List four ways by which personal hygiene could be maintained

(b) Define each of the following terms and give two examples in each case:
(i) compound
(ii) mixture

(c) (i) What is a machine?
(ii) Give one example each of a
(α) first class lever
(β) second class lever
(γ) third class lever

(d) (i) What is a satellite?
(ii) State one use of satellite

2. (a) What is an electrostatic force?

(b) (i) State the unit of
(α) force
(β) work

(ii) A body of mass 80 kg is lifted vertically through a distance of 5.0 m. Calculate the work done on the body.
[Acceleration due to gravity (g) = 10 ms^{-2}]

(c) Describe briefly how a mixture of common salt and powdered sulphur could be separated.

(d) State three characteristics each of
(i) an insect pollinated flower
(ii) a wind pollinated flower

3. (a) (i) What is fertilization
(ii) Describe briefly the processes that lead to fertilization in humans after mating

(b) (i) Define the term density
(ii) Describe briefly how the density of a piece of stone could be determined

(c) (i) What is an element?
(ii) Classify each of the following substances as an element, compound or mixture:
Water, air, potassium, alcohol, common salt, sugar

4. (a) Define each of the following terms:
(i) alloy
(ii) diffusion
(iii) colloid

(b) Give two examples each of:
(i) alloy
(ii) colloid

(c) (i) What is pollution?
(ii) List two pollutants each of:
(α) air
(β) water
(γ) land

(d) A coconut of weight 50 N hangs 15 m above the ground.
(i) Name the type of energy possessed by the coconut.
(ii) Calculate the value of this energy

August 2000

SCIENCE Section B

SOLUTIONS

ESSAY

1. (a) (i) Personal Hygiene

The practice of cleanliness of a person and his/her physical environment

(ii) Ways by which personal hygiene could be maintained

- Bathing regularly (at least twice a day)
- Brushing one's teeth regularly (advisedly, after each meal)
- Washing one's clothes regularly
- Washing hands with soap after visiting the toilet
- Keeping one's room clean and tidy
- Cleaning one's bathroom and toilet regularly

[any four]

(b) Definition and examples

(i) Compound

a substance formed by the chemical combination of two or more elements in definite proportions

Examples: HCl, or Nacl, or $Ca(OH)_2$, or $CaCl_2$, or NH_3, or $MgCO_3$, etc

(ii) Mixture: A physical combination of two or more substances

Or: a substance consisting of two or more substances that have been combined physically

(c) (i) Machine

A mechanical device that makes work easier and/ or faster

(ii)

			Examples:
(α)	first class lever	-	Scissors, Crowbar, beam balance, pliers
(β)	second class lever	-	Bottle opener, wheel barrow, paper cutter
(γ)	third class lever	-	human forearm, forceps, baseball bat, broom

[any one each]

(d) (i) Satellite

Any heavenly body that orbits (revolves around) a larger one

(ii) Use of artificial satellites

- Weather studies and forecasting
- Telecommunication
- Sending and receiving TV signals
- Taking photographs from space for various purposes
- Internet networking
- Global Positioning System (GPS) - a worldwide navigation system

[any one]

2. (a) Electrostatic force

The force of attraction or repulsion between two stationary electrical charges

(b) (i) Unit of:

(α) force = newton (N)
(β) work = joule (J)

(ii) **Work** = force × distance
= weight × distance
= (m×g) × distance
= 80 × 10 × 5.0 m
= 4000 joules

(c) Separating mixture of common salt and powdered sulphur

- Add water to the mixture and stir to dissolve the salt
- Pour the mixture through a filter paper in a funnel to separate the powdered sulphur (residue) from the salt solution (filtrate)
- Heat the salt solution till all the water evaporates, leaving the salt crystals behind
- Dry the powdered sulphur and the salt obtained

(d) Characteristics of

(i) an insect pollinated flower

- Petals are brightly coloured
- Larger size of flowers
- Nectar present
- Attractive scent
- Relatively fewer but larger pollen grains

[any three]

(ii) a wind pollinated flower

- Dull coloured petals
- Nectar absent
- No or unattractive scent
- Relatively more but smaller pollen grains

[any three]

3. (a) (i) Fertilization:

The fusion of a male sex cell and a female sex cell to form a zygote

(ii) Processes that lead to fertilization in humans after mating

- During mating, semen is deposited in the vagina and /or womb of the female through the penis of the male.
- The ovary of the female would have already released an egg (ovum) which moves into the fallopian tube.
- Some of the sperms in the semen swim into the fallopian tube through the womb.
- One sperm fuses with the released egg to form a zygote in the fallopian tube of the female.

(b) (i) Density

The mass per unit volume of a substance

Or, mathematically, Density = mass / volume

(ii) Stone's density

I. Measure the mass(M) of the stone using a beam balance

II. Fill a measuring cylinder with water to a suitable volume.

III. Record the volume (V_1)

IV. Tie a thread / string around the stone and gently lower it into the water in the cylinder.

V. Record the new volume (V_2)

VI. Find difference in volumes ($V_2 - V_1$)

VII. Volume of metal ball = $(V_2 - V_1)$

VIII. Find the ratio $M / (V_2 - V_1)$ = Density of stone

(c) (i) Element

A substance that is made up of the same kind of atoms

Or: Any substance that cannot be broken down into a simpler one by a chemical reaction

(ii)

ELEMENT	COMPOUND	MIXTURE
Potassium	Water	Air
	Alcohol	
	Common salt	
	Sugar	

4. (a) (i) Alloy

A substance containing two or more metals uniformly physically combined

Or: A homogeneous mixture of two or more metals

(ii) Diffusion

The random movement of atoms, molecules, or ions from one site in a medium to another, resulting in complete mixing

Or:

The movement of particles of a substance from a region of higher concentration to a region of lower concentration

(iii) Colloid

A suspension of tiny particles dispersed in another substance.

(b) Examples of

(i) alloy - bronze, brass, steel,

(ii) colloid - blood, milk, body cream, mayonnaise, emulsion paint

(c) (i) Pollution

Contamination of Earth's environment with materials that interfere with human health, the quality of life, or the natural functioning of *ecosystems*

(ii) **Pollutants**

(α) air: exhaust fumes, smoke, carbon monoxide, sulphur dioxide

(β) water: Sewage, industrial wastes, crude oil spillage and pesticides

(γ) land: chemical fertilizers, fungicides and pesticides

(d) (i) Type of energy = Potential energy

(ii) Potential energy = weight × height

= 50 N × 15 m

= 750 Nm or 750 J

August 1999

SCIENCE

SECTION A
OBJECTIVE TEST
45 minutes

1. A mixture of sand and common salt could be separated by
- A. filtration and distillation
- B. dissolution and evaporation
- C. dissolution, sedimentation and evaporation
- D. dissolution, filtration and evaporation
- E. sedimentation and filtration

2. What structure links the mouth to the stomach
- A. Colon
- B. Duodenum
- C. Oesophagus
- D. Rumen
- E. Trachea

3. Which of the following organisms causes bilharziasis?
- A. Blood fluke?
- B. Bloodworm
- C. Liver fluke
- D. Roundworm
- E. Tapeworm

4. Which of the following processes is a chemical change?
- A. Digestion of starch by saliva
- B. Melting of wax
- C. Dissolution of sugar in water
- D. Boiling of water
- E. Evaporation of sea water

5. An atom has 5 protons and 5 neutrons in its nucleus. How many electrons are present in the atom?
- A. 1
- B. 2
- C. 5
- D. 9

E. 10

6. Which of the following features in plants are used for defence?

A. Buds
B. Leaf spines
C. Pitchers
D. Roots
E. Tendrils

7. The ability of birds to fly depends on

A. Their hollow hones only
B. Their flight feathers only
C. Their feathers covering their bodies
D. Their hollow bones and flight feathers
E. Their light bodies only

8. Gases are more compressible than solids because:

A. Solids have harder surfaces
B. There are wider spaces between gas molecules
C. Gases have fixed volumes
D. The particles of solids are regularly arranged
E. The molecules of gases are randomly arranged

9. Which of the following statements about machines is/are true?

I. The commonest and simplest of all machines is the lever
II. The efficiency of a machine is the ratio of work output to work input
III. The efficiency of a machine is usually less than one
IV. Machines make working much easier and convenient

A. I only
B. I and II only
C. I and III only
D. I, II and III only
E. I, II, III and IV

10. Which of the following substances reacts with ammonium chloride to produce ammonia gas?

A. Calcium hydroxide
B. Calcium chloride
C. Calcium carbonate
D. Calcium sulphate
E. Calcium nitrate

11. Plastic materials may be used to make the handles of cooking utensils because they are

A. Hard
B. Poor conductors of heat

C. Attractive
D. Easy to wash
E. Not poisonous

12. Which of the following substances will turn red litmus paper blue?
A. Hydrochloric acid
B. Sodium chloride solution
C. Sulphuric acid
D. Water
E. Sodium hydroxide

13. The heat of the sun reaches the earth through
A. radiation
B. convection
C. conduction
D. both radiation and conduction
E. both radiation and convection

14. The mineral salts in dead organisms become available to the soil through
A. Decomposition
B. Diffusion
C. Fermentation
D. Leaching
E. Manuring

15. A ray of light is incident at an angle of 30° to the surface of a plane mirror. Calculate the angle of reflection.
A. 30°
B. 60°
C. 90°
D. 120°
E. 180°

16. Salt could be obtained from sea water by
A. boiling
B. condensation
C. decantation
D. distillation
E. evaporation

17. A fertilised ovum is known as
A. a gamete
B. a zygote
C. oocyte
D. an ovary

E. a foetus

18. The smallest planet known in the solar system is

A. Earth
B. Jupiter
C. Mars
D. Mercury
E. Venus

19. The organic constituent of the soil is

A. Air
B. Humus
C. Mineral salts
D. Rock particles
E. Water

20. A car head lamp lights up when connected to a battery. What type of energy is obtained from the battery?

A. Chemical energy
B. Electrical energy
C. Heat energy
D. Light energy
E. Potential energy

21. Which of the following statements about bacteria is true?

I. All infections are caused by bacteria
II. Most bacteria are harmless to man
III. Sterilisation does not kill bacteria

A. I only
B. II only
C. III only
D. II and III only
E. I and II only

22. Which of the following organisms is a parasite on plants?

A. Bee
B. Butterfly
C. Capsid
D. Red ant
E. Tick

23. The main parts of the central nervous system are the

A. brain and nerves
B. spinal cord and the nerves
C. brain and the spinal cord

D. brain, the nerves and the spinal cord
E. brain, the eyes, the ears and the nerves

24. Which of the following pairs of diseases can be spread easily when food is exposed to houseflies?
A. Cholera and Guinea Worm
B. Dysentery and Malaria
C. Dysentery and Guinea Worm
D. Malaria and tuberculosis
E. Dysentery and cholera

25. Which of the following activities is a mode of transmission of AIDS?
A. Shaking hands with an infected person
B. Eating with an infected person
C. Having sex with an infected person
D. Sharing clothes with an infected person
E. Sleeping in the same room with an infected person

26. The eclipse of the moon occurs when the
A. moon and the sun are in a straight line
B. earth and the sun are in a straight line
C. sun is between the earth and the moon
D. earth is between the sun and the moon
E. moon comes between the sun and the earth

27. Spitting in public is a bad social habit because it
A. makes the environment humid
B. reduces the amount of air in the body
C. reduces the amount of water in the body
D. releases disease-causing organisms into the environment
E. reduces the amount of saliva in the mouth

28. The male sex cells in the flower are found in the
A. filament
B. ovary
C. ovule
D. pollen grains
E. stigma

29. Which of the following substances is a pollutant?
A. Domestic waste
B. Fog
C. Nitrogen
D. Oxygen
E. Treated Waste

30. Digestion of protein starts in the

A. duodenum
B. gullet
C. mouth
D. small intestine
E. stomach

31. The sun, the heavenly bodies and the planets form the

A. atmosphere
B. galaxy
C. milky way
D. solar system
E. universe

32. Which of the following practices encourages soil erosion?

A. Planting trees on bare soil
B. Clearing trees and slopes
C. Planting grasses along slopes
D. Ploughing across slopes
E. Preventing bush fires

33. Which of the following metals is an alloy?

A. Aluminium
B. Brass
C. Copper
D. Iron
E. Zinc

34. Which of the following statements about plants is true?

I. Most plants show movement
II. Green plants manufacture their own food
III. All animals depend directly on plants for food

A. I only
B. II only
C. III only
D. I and II only
E. II and III only

35. Which of the following practices will destroy the breeding grounds of mosquitoes?

A. Sleeping in mosquito nets
B. Spraying with insecticides
C. Using insect repellent
D. Introducing fish into ponds
E. Burying all empty cans

36. A plastic pen on rubbing in the hair attracts pieces of paper. What force is responsible for the attraction?

A. Electrostatic force
B. Frictional force
C. Gravitational force
D. Magnetic force
E. Centripetal force

37. When carbon dioxide is passed through lime water

A. the lime water becomes warm
B. the lime water turns milky
C. the lime water remains clear
D. lime is produced
E. the lime water decomposes

38. Which of the following solids melts easily on exposure to sunlight?

A. Animal fat
B. Candle wax
C. Plastics
D. Rubber
E. Shea butter

39. Which of the following structures regulate the amount of light that enters the eye?

A. Choroid
B. Cornea
C. Iris
D. Lens
E. Retina

40. Leguminous crops such as beans are cultivated to

A. add humus to the soil
B. increase the nitrate contents of soils
C. add nitrogen to the soil
D. make the soil loamy
E. add protein to the soil

August 1999

SCIENCE Section A

SOLUTIONS

OBJECTIVE TEST

1.	D.	dissolution, filtration and evaporation
2.	C.	Oesophagus
3.	A.	Blood fluke
4.	A.	Digestion of starch by saliva
5.	C.	5
6.	B.	Leaf spines
7.	D.	Their hollow bones and flight feathers
8.	B.	There are wider spaces between gas molecules
9.	E.	I, II, III and IV
10.	A.	Calcium hydroxide
11.	B.	Poor conductors of heat
12.	E.	Sodium hydroxide
13.	A.	radiation
14.	A.	Decomposition
15.	B.	60°
16.	E.	evaporation
17.	B.	a zygote
18.	D.	Mercury
19.	B.	Humus

20. B. Electrical energy

21. B. II only

22. C. Capsid

23. C. brain and the spinal cord

24. E. Dysentery and cholera

25. C. Having sex with an infected person

26. D. earth is between the sun and the moon

27. D. releases disease-causing organisms into the environment

28. D. pollen grains

29. A. Domestic waste

30. E. stomach

31. E. universe

32. B. Clearing trees and slopes

33. B. Brass

34. B. II only

35. E. Burying all empty cans

36. A. Electrostatic force

37. B. the lime water turns milky

38. E. Shea butter

39. C. Iris

40. C. add nitrogen to the soil

August 1999

SCIENCE

SECTION B
ESSAY
1 hour

[60 marks]

Answer **three** *questions* **only** *from this section.*
Illustrate your answers wherever possible, with large, clear and fully labelled diagrams.
Credit will be given for clarity of expression and orderly presentation of material.
All questions carry equal marks

1. (a) State one function of each of the following organs in humans
- (i) skin
- (ii) lung
- (iii) kidney
- (iv) testis

(b)
- (i) What is vegetative reproduction?
- (ii) Name two plants which reproduce vegetatively

(c) Name four main factors which determine the climate of a place

(d) Define the following:
- (i) work
- (ii) power

(e) Name the separation method used in the production of each of the following substances
- (i) akpeteshie (alcohol)
- (ii) common salt
- (iii) tap water

2. (a) How would you demonstrate that expired air contains carbon dioxide?

(b)
- (i) What is an ion
- (ii) Give two examples of ions

(c) (i) What is capillary action

(ii) State two processes which use capillary action

(d) (i) What is a lever?

(ii) Give two examples of second class levers

(e) (i) Use any three of the following organisms to construct a food chain.
Rat, hawk, okro leaves, toad, cassava, man, grasshopper

(ii) List three ways of keeping the teeth healthy

3. (a) (i) Describe briefly what happens to a morsel of kenkey in the mouth during eating.

(ii) List in order, the parts of the digestive system of humans

(b) A mixture of iron filings and sulphur is heated in a test tube

(i) Give the name of the compound formed

(ii) Write the chemical equation of the reaction which took place

(iii) State two methods of preventing iron from rusting

(c) (i) A boy pulled a load through a distance of 5 m. If he exerts a force of 15 N in the direction of the force, calculate the work done

(ii) Convert the following weights expressed in kilogram force (kgf) to newton (N)

(α) 0.75 kgf

(β) 1.33 kgf

[1 kgf = 10 N]

4. (a) (i) State two reasons which indicate that respiration and burning are similar processes

(ii) What is cross-pollination?

(iii) List four methods by which cholera can be controlled and prevented

(b) (i) Name the three coldest planets

(ii) State one reason which accounts for the coldness of the planets you have stated.

(iii) An object of mass 150 g occupies a volume of 250 cm^3. Calculate its density.

(c) Name a suitable solvent for each of the following substances

(i) grease

(ii) oil paint

(iii) alcohol

(iv) sugar

August 1999

SCIENCE Section B

SOLUTIONS

ESSAY

1. (a)

	Organ	Function
(i)	Skin	• Protects the internal parts of the body • Serves as sense organ for touch / feeling • Excretes metabolic waste substance, sweat
(ii)	Lung	• Excretes carbon dioxide • Absorbs oxygen for respiration
(iii)	Kidney	• Excretes urea • Produces urine
(iv)	testis	• Produces sperms • Produces the hormone testosterone

[any one(1) for each]

(b) (i) Vegetative reproduction

Method by which plants reproduce without the union of cells or nuclei of cells

Or:

The process whereby plants produce offspring from growing parts / buds of the parent plant

(ii) Plants that reproduce vegetatively

Cassava, sugarcane, yam, cocoyam, banana, plantain, ginger, pineapple, etc

[any two]

(c) Main factors that determine the climate of a place

Rainfall pattern, temperature, humidity, altitude (height above sea level), sunshine, vegetation, wind [any four]

(d) (i) Work

Work is done when an applied force moves an object in the direction of the force.

Or:

The transfer of energy, measured as the product of the force applied to a body and the distance moved by that body in the direction of the force

Or:

Work = $f \times d$, where f = force applied

and d = distance moved in the direction of the applied force

(ii) Power

The time rate of doing work

Or (mathematically):

Power = Work done / time

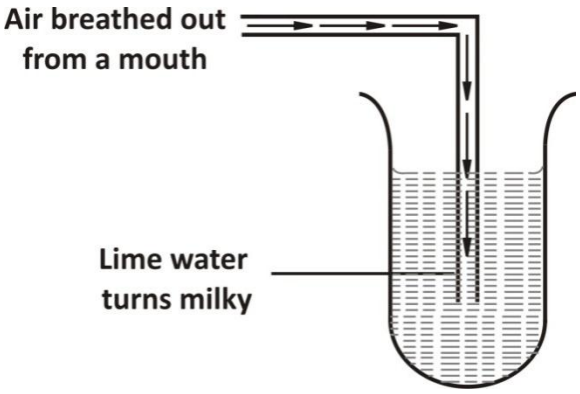
(e) SEPARATION METHOD

(i) akpeteshie (alcohol) - Distillation

(ii) common salt - Evaporation

(iii) tap water - Sedimentation

2. (a) Demonstration that expired air contains carbon dioxide



- Insert straw or other narrow tube into lime water contained in a transparent container,
- Gently breathe out air from the mouth into the lime water
- The lime water turns milky

(b) (i) ion -

An atom which has lost or gained one or more electrons

Or:

A negatively or positively charged atom

(ii) Examples of ions

Al^{3+}, Cl^{-}, O^{2-} , K^{+}, Mg^{2+}

(c) (i) Capillary action

The rising or falling of the surface of a liquid where it is in contact with a solid

(ii) Processes which use capillary action

- Kerosene or oil rising up a wick
- Water rising up in a towel with one end soaked in water
- Ink rising up on a blotting paper
- Water rising up from the lower to the upper layers of soil.

(d) (i) Lever

Simple machine consisting of a rigid bar that rotates about a fixed point, called a fulcrum

(ii) Examples of second class levers

Bottle opener, wheel barrow, paper cutter

(e) (i) Food chain

Okro leaves → grasshopper → man

Or: cassava → rat → man

Or: okro leaves → grasshopper → toad

Or: cassava → rat → hawk

(ii) Ways of keeping the teeth healthy

- Brush regularly, (possibly, after each meal)
- Avoid taking very hot or very cold foods or drinks
- Avoid taking lots of sweets, especially ones that stick to the teeth
- Avoid picking teeth with very sharp/ hard objects
- Regularly eat foods rich in calcium, phosphorus and vitamins
- Visit the dentist regularly for check ups

[any three]

3. (a) (i) Morsel of kenkey in mouth

I. The tongue pushes the kenkey against the teeth, which grinds the kenkey.

II. Three pairs of glands empty saliva into the mouth through ducts to moisten the food.

III. The saliva contains the enzyme ptyalin, which begins break down starch into sugar.

IV. The tongue pushes the softened kenkey—called a bolus—to the back of the mouth for swallowing

(ii) The parts of the digestive system in order

Mouth, throat, gullet (oesophagus), stomach, small intestines (duodenum, jejunum, ileum), large intestines (caecum, colon), rectum, anus

(b) Iron filings and sulphur heated in test tube

(i) FeS or Iron (II) sulphide

(ii) Fe + S → FeS

(iii) Preventing iron from rusting

- Painting
- Greasing
- Alloying
- Galvanizing (coating with zinc)

(c) (i) **Work done** = force × distance
= 15 N × 5 m
= 75 Nm or 75 J

(ii) Conversions

(α) 0.75 kgf = 0.75 × 10 N = <u>7.5 N</u>

(β) 1.33 kgf = 1.33 × 10 N = <u>13.3 N</u>

4. (a) (i) Similarity between respiration and burning

- Both processes use up oxygen and release carbon dioxide
- Both are chemical processes / irreversible
- Both processes release heat energy (exothermic reactions)

(ii) Cross-pollination

The transfer of pollen grains from the anther of a flower to the stigma of another flower of another plant of the same species.

(iii) Methods by which cholera can be controlled and prevented

- Eating hot meals,
- drinking clean potable water,
- washing hands with soap after visiting the toilet
- covering meals to prevent flies from settling on them
- keeping our environment clean and free from filth
- ensuring that rubbish bins are properly covered.

[any four]

(b) (i) Coldest planets - Uranus, Neptune and Pluto

(ii) They are the farthest away from the sun, hence, very little or no heat energy from the sun reaches them.

(iii) Density = mass / volume
= 150 g / 250 cm^3
= 0.6 g/cm^3 or 0.6 gcm^{-3}

(c)

		<u>SUITABLE SOLVENT</u>
(i) grease	-	turpentine / kerosene / petrol

(ii)	oil paint	-	turpentine / kerosene
(iii)	alcohol	-	water
(iv)	sugar	-	water

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SCIENCE

SECTION A
OBJECTIVE TEST
45 minutes

1. Which of the following processes could be used to preserve fish for a long time?
- A. Boiling only
- B. Salting only
- C. Sun drying only
- D. Boiling and sun drying
- E. Salting and sun drying

2. One hectare is equivalent to
- A. 100 m^2
- B. 1,000 m^2
- C. 10,000 m^2
- D. 100,000 m^2
- E. 1,000,000 m^2

3. Which of the following causes ringworm?
- A. Bacteria
- B. Fungi
- C. Protozoa
- D. Ticks
- E. Viruses

4. The central nervous system is made up of the
- A. Brain only
- B. Spinal cord only
- C. Brain and sense organs
- D. Brain and spinal cord
- E. Spinal cord and sense organs

5. An object weighing 60 N is dragged through a distance of 2 m. Calculate the work done
- A. 6 J
- B. 15 J

C. 30 J
D. 120 J
E. 1200 J

6. Which of the following substances is necessary for the formation of food in plants?
A. Carbon dioxide
B. Nitrogen
C. Oxygen
D. Starch
E. Sugar

7. How many times is the sun overhead at the equator within a year?
A. 1
B. 2
C. 3
D. 4
E. 5

8. Which of the following takes the shape of its container?
A. Chalk
B. Charcoal
C. Paper
D. Petrol
E. Stone

9. A group of stars form
A. The universe
B. The atmosphere
C. The solar system
D. The planetary system
E. A galaxy

10. Which of the following types of soil is good for plant growth?
A. Clay
B. Course sand
C. Fine sand
D. Loam
E. Silt

11. The equation Carbon (IV) oxide + water → Glucose + Oxygen, represents
A. Breathing
B. Diffusion
C. Photosynthesis
D. Respiration
E. Transpiration

12. Which of the following pairs of diseases are both contracted through water?

A. Cholera and dysentery
B. Smallpox and dysentery
C. Ringworm and smallpox
D. Tuberculosis and cholera
E. Tuberculosis and dysentery

13. The efficiency of a machine is usually less than one due to

A. friction
B. greasing
C. heat
D. the load used
E. the effort applied

14. Which of the following pairs of organisms are decomposers?

A. Animals and bacteria
B. Animals and fungi
C. Bacteria and plants
D. Bacteria and fungi
E. Plants and fungi

15. A mixture of engine oil and water could be best separated by

A. Evaporation
B. Freezing
C. Decantation
D. Heating
E. Addition of soap

16. Which of the following plants is propagated vegetatively?

A. apple
B. coconut
C. pawpaw
D. pear
E. sweet potato

17. Which of the following diseases is transmitted by close contact with an infected person?

A. AIDS
B. Bilharzia
C. Elephantiasis
D. Measles
E. Polio

18. The breakdown of rocks into smaller particles is known as

A. decomposition

B. erosion
C. ploughing
D. terracing
E. weathering

19. Which of the following energy changes takes place when a church bell is tolled?
A. Chemical energy is changed to sound energy
B. Kinetic energy is changed to sound energy
C. Potential energy is changed to sound energy
D. Potential energy is changed to kinetic energy
E. Thermal energy is changed to sound energy

20. The dodder on milk bush is an example of
A. a plant parasite on animal host
B. a plant parasite on plant host
C. an animal parasite and a plant parasite
D. an animal parasite on an animal host
E. an animal parasite on a plant host

21. Which of the following features is shown by plants only?
A. Feeding
B. Growth
C. Transpiration
D. Reproduction
E. Respiration

22. The structure of the skin that removes waste substances is the
A. fat deposit
B. hair
C. sweat duct
D. sweat gland
E. sweat pore

23. Which of the following statements about light is true?
A. Light is reflected from polished surfaces
B. Light travels through opaque objects
C. Light passes through plane mirrors
D. The angle of incidence is between the normal and the reflected ray
E. The speed of light is less than that of sound

24. The efficiency of a machine is given by the ratio
A. load/effort
B. momentum / velocity
C. distance moved by effort / distance moved by load
D. work input / work output

E. mechanical advantage / velocity ratio

25. About 55% of blood is made up of the plasma while the remaining 45% is made up of
A. Cells
B. Dissolved food substances
C. Dissolved gases
D. Liquid
E. Salt

26. Which of the following is a structure in the ear?
A. Cornea
B. Cochlea
C. Iris
D. Retina
E. Trachea

27. Brass is an alloy made up of
A. Aluminium and zinc
B. Copper and tin
C. Copper and zinc
D. Iron and carbon
E. Steel and tin

28. Which of the following controls the amount of light that enters the eyes?
A. The cornea
B. The iris
C. The lens
D. The pupil
E. The retina

29. A body which moves round the earth is called
A. A meteorite
B. A satellite
C. A star
D. Jupiter
E. The sun

30. Which of the following processes should be carried out first when separating a mixture of sand and common salt?
A. crystallization of the salt
B. dissolution of the salt
C. distillation of the solvent
D. filtration of the mixture
E. sublimation of the solute

31. A temperature controlling device which works using the principles of expansion and contraction is called a .....

A. conductor
B. fuse
C. manometer
D. thermometer
E. thermostat

32. Kerosene and petrol are obtained from crude oil by

A. condensation of the crude oil
B. decantation of the crude oil
C. distillation of the crude oil
D. evaporation of the crude oil
E. filtration of the crude oil

33. The fusion of the male and female gametes is known as:

A. copulation
B. evolution
C. fertilization
D. ovulation
E. reproduction

34. Calculate the voltage across a wire of resistance 4 ohms if 3 amperes of current is passed through it.

A. 0.75 V
B. 1.00 V
C. 1.33 V
D. 7.00 V
E. 12.00 V

35. Which of the following flower parts develops into a seed? The

A. ovule
B. pollen grain
C. stamen
D. stigma
E. style

36. The ability of an organism to resist infection by producing antibodies is known as

A. diagnosis
B. immunity
C. incubation
D. inoculation
E. vaccination

37. The part of cocoyam which is used in propagation is the

A. bulb

B. corm
C. seed
D. sucker
E. rhizome

38. Which of the following diseases affects the lungs?
A. Cholera
B. Measles
C. Tetanus
D. Tuberculosis
E. Typhoid

39. Gaseous exchange between air and blood in the lungs takes place in the
A. alveoli
B. bronchi
C. bronchioles
D. larynx
E. trachea

40. The stretched arm of a JSS student holding a stone in the palm is an example of a
A. First class lever
B. Second class lever
C. Third class lever
D. Fourth class lever
E. Fifth class lever

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SCIENCE Section A

SOLUTIONS

OBJECTIVE TEST

1.	E.	Salting and sun drying
2.	C.	10,000 m^2
3.	B.	Fungi
4.	D.	Brain and spinal cord
5.	D.	120 J
6.	A.	Carbon dioxide
7.	B.	2
8.	D.	Petrol
9.	E.	A galaxy
10.	D.	Loam
11.	C.	Photosynthesis
12.	A.	Cholera and dysentery
13.	A.	friction
14.	D.	Bacteria and fungi
15.	C.	Decantation
16.	E.	sweet potato
17.	D.	Measles
18.	E.	weathering
19.	B.	Kinetic energy is changed to sound energy
20.	B.	a plant parasite on plant host

21. C. Transpiration

22. D. sweat gland

23. A. Light is reflected from polished surfaces

24. E. mechanical advantage / velocity ratio

25. A. Cells

26. B. Cochlea

27. C. Copper and zinc

28. B. The iris

29. B. A satellite

30. B. dissolution of the salt

31. E. thermostat

32. C. distillation of the crude oil

33. C. fertilization

34. E. 12.00 V

35. A. ovule

36. B. immunity

37. B. corm

38. D. Tuberculosis

39. A. alveoli

40. C. Third class lever

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SCIENCE

SECTION B
ESSAY
1 hour

[60 marks]

Answer **three** *questions* **only** *from this section.*
Illustrate your answers wherever possible, with large, clear and fully labelled diagrams.
Credit will be given for clarity of expression and orderly presentation of material.
All questions carry equal marks

1. (a) State two differences between a planet and a star.

(b) (i) What is surface tension.
(ii) State one way in which surface tension can be useful to man.

(c) (i) Give one example each of a rhizome, a corm, a stem tuber, a bulb and a sucker.
(ii) State two reasons why a rhizome, a corm and a stem tuber are considered as stems.

(d) Name the compounds formed when the following elements combine:
(i) zinc and oxygen
(ii) calcium and chlorine
(iii) sodium and bromine
(iv) potassium and iodine

2. (a) (i) What is a fertile soil?
(ii) Name four of the most important nutrients in mineral salts which plants use to form food.

(b) (i) Define density of a substance.
(ii) Describe briefly how you would determine the density of an irregular piece of stone.
(iii) A box with volume 1200.0 m^3 is filled with sand to the brim. If the density of the sand is 1.95 kgm^{-3}, calculate the mass of the sand contained in the box.

(c) (i) Define an atom.
(ii) Why are atoms electrically neutral?

3. (a) Explain the following

(i) Crackling noises are heard from a metal sheet roof when the weather becomes very hot.

(ii) Railway tracks are laid with gaps in between their ends.

(iii) A fitting mechanic washes his greasy hands with petrol and not with water.

(b) What observation will be made if a piece of blue litmus paper is dropped in each of the following:

(i) sodium hydroxide solution

(ii) dilute hydrochloric acid

(iii) lime juice

(iv) distilled water

(v) rain water

(c) State any three ways in which constipation could be avoided.

(d) What are the end products of the digestion of each of the following food substances?

(i) Carbohydrates

(ii) Proteins

(iii) fats

4. (a) State four ways by which plants are able to protect themselves against animals.

(b) (i) Define pollination.

(ii) List four features of insect pollinated flowers.

(c) An atom has three protons and three neutrons.

(i) How many electrons are there in this atom?

(ii) Draw a labeled diagram to show the arrangement of all the particles in the atom.

(d) A body of mass 12.0 kg is placed on an orange tree 20.0 m above the ground

(i) Calculate its potential energy with respect to the ground

(ii) If the body is released from this height, what will be its kinetic energy just before it hits the ground.

[$g = 10\ ms^{-2}$]

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SCIENCE Section B

SOLUTIONS

ESSAY

1. (a) Differences

Planet	Star
Does not produce light	Produces light
Made up of solid and liquid materials	Made up of burning gases
Revolves around a star (sun) in a specific orbit	Does not move / stationary (ie, relative to the planets)

[any two]

(b) (i) Surface tension

A condition existing at the surface of a liquid, resembling a stretched elastic skin

Or: the property of a liquid that makes its surface behave like a stretched elastic skin

(ii) Ways in which surface tension is useful to man

Surface tension property of water ensures that water does not leak through raincoats, umbrellas, tents, canopies or other materials made of canvas.

(c) (i)

		EXAMPLE
Rhizome	-	ginger
Corm	-	cocoyam
Stem tuber	-	yam
Bulb	-	onion
Sucker	-	banana

(d)

			COMPOUNDS FORMED		
(i)	**zinc and oxygen**	-	Zinc oxide	**or**	ZnO
(ii)	**Calcium and chlorine**	-	Calcium chloride	**or**	$CaCl_2$
(iii)	**Sodium and bromine**	-	Sodium bromide	**or**	NaBr
(iv)	**Potassium and iodine**	-	Potassium iodide	**or**	KI

2. (a) (i) Fertile soil

Soil that is rich in all the soil nutrients required for healthy plant growth

(ii) Most important nutrients in mineral salts

Phosphorus, Nitrogen, Calcium, Magnesium, Potassium, Sulphur, Iron

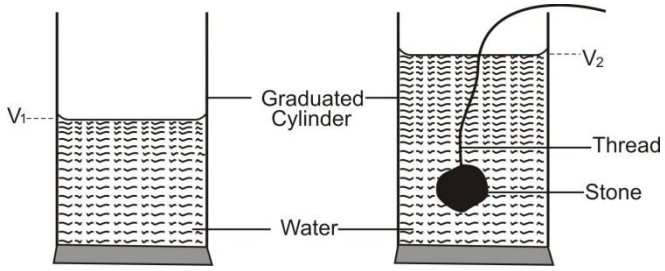
[any four]

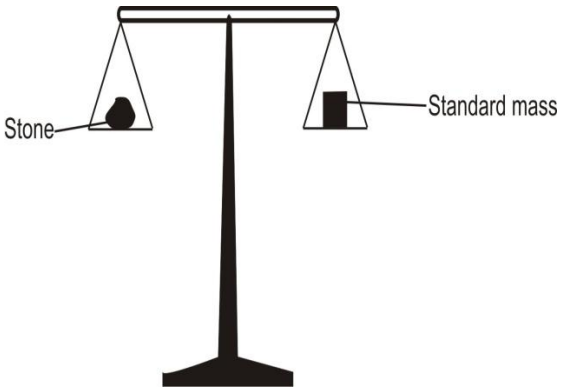
(b) (i) Density of a substance

The mass per unit volume of the substance

Or: Density = mass / volume

(ii) Finding density of an irregular piece of stone





I. Measure the mass(M) of the stone using a beam balance

II. Fill a measuring cylinder with water to a suitable volume.

III. Record the volume (V_1)

IV. Tie a thread / string around the stone and gently lower it into the water in the cylinder.

V. Record the new volume (V_2)

VI. Find difference in volumes ($V_2 - V_1$)

VII. Volume of the stone = $(V_2 - V_1)$

VIII. Find the ratio $M / (V_2 - V_1)$ = Density of stone

(iii) Mass of sand = Density × volume

= $1.95\ kgm^{-3}$ × $1200.0\ m^3$

= 3900 kg

(c) (i) Atom

The smallest part of an element that can take part in a chemical reaction

Or:

The smallest portion into which an element can be divided and still retain its properties

(ii) Why atoms are electrically neutral

The numbers of protons and electrons in an atom are equal. The positively charged protons neutralize the negatively charged electrons, hence, making the atom electrically neutral

3. (a) Explanation

(i) Crackling noises heard from a metal sheet roof when hot

The heat causes the metal sheet roof to expand. The expansion causes the bending of the metal sheet if there is not enough space. The bending causes the crackling noise

(ii) Railway tracks are laid with gaps in between their ends.

The gaps provide allowance for the expansion of the railway tracks, so that they do not bend during hot periods, when expansion occurs.

(iii) A fitting mechanic washes his greasy hands with petrol and not with water

Grease is soluble in petrol, but not soluble in water. Hence, the petrol dissolves the grease on his hands.

(b) Observation made when blue litmus paper is dropped in:

(i)	sodium hydroxide solution	-	colour remains blue
(ii)	dilute hydrochloric acid	-	colour changes to red
(iii)	lime juice	-	colour changes to red
(iv)	distilled water	-	colour changes to purple
(v)	rain water	-	colour changes to red /pink

(c) Ways of reducing constipation

- Drinking a lot of water / liquids
- Eating fruits and vegetables regularly
- Eating foods that contain high amounts of fiber, such as wheat, barley, etc
- Getting enough sleep
- Exercising regularly / daily

[any three]

(d) End products of digestion of:

(i)	**Carbohydrates**	-	glucose
(ii)	**Proteins**	-	amino acids
(iii)	**fats**	-	fatty acids and glycerol

4. (a) Ways by which plants protect themselves against animals

- Unpleasant smell from certain plants makes them repulsive to animals.
- Unpleasant taste of certain plants makes them undesirable to taste / eat.
- Sharp thorns / spines of certain plants make them difficult to touch.
- Poisonous parts of certain plants can cause irritation / pain / infection / death of animals.
- Sticky secretions or parts of certain plants makes them uncomfortable to touch

[any four]

(b) (i) Pollination

The transfer of pollen grains from the anther of a flower to the stigma of another flower of the same species

(ii) Features of insect pollinated flowers

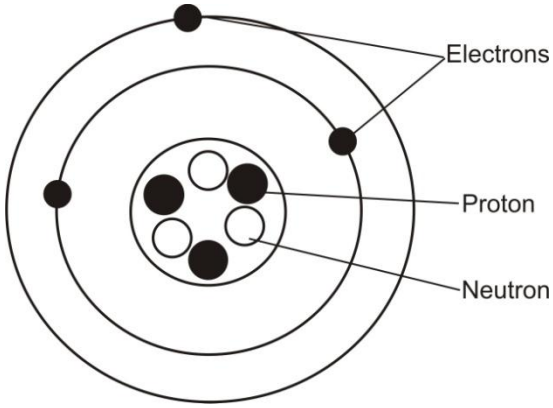
- Petals are brightly coloured
- Larger size of flowers
- Nectar present
- Attractive scent
- Relatively fewer but larger pollen grains

[any four]

(c) Atom with 3 protons and 3 neutrons

(i) 3 electrons

(ii) Labeled diagram



(d) (i) Potential energy = mass × height × acceleration due gravity

= 12 kg × 20 m × 10 ms^{-2}

= <u>2400 kgm^2s^{-2}</u> or <u>2400 J</u>

(ii) Kinetic energy just before it hits the ground = <u>2400 J</u>

August 1997

SCIENCE

SECTION A
OBJECTIVE TEST
45 minutes

1. Digested food is absorbed into the blood stream through the
- A. duodenum
- B. large intestines
- C. rectum
- D. small intestines
- E. stomach

2. Which of the following is parasite?
- A. Fruitfly
- B. Housefly
- C. Mosquito
- D. Tick
- E. Tsetse fly

3. Soil acidity can be reduced by adding
- A. acid solution
- B. ammonium sulphate
- C. fertilizer
- D. lime
- E. rotten leaves

4. The planet earth is located in a galaxy known as
- A. constellation
- B. meteorites
- C. milky way
- D. satellite
- E. super nova

5. The most productive soil in agriculture is
- A. clayey soil
- B. loamy soil
- C. sandy soil
- D. humus

E. silt

6. A chemical that can be used to test for the presence of protein in food substances is
A. Benedict's solution
B. Fehling's solution
C. Eosin solution
D. Iodine solution
E. Million's reagent

7. According to the principle of conservation of energy, 'energy can
A. be created and destroyed
B. be created but not destroyed
C. be destroyed but not created
D. be reduced but not increased
E. neither be created nor destroyed

8. The purpose of vaccination is to
A. replace poisoned tissues
B. kill disease-producing organisms in the body
C. increase the activity of white blood cells
D. induce the production of anti bodies
E. cure the disease

9. Which of the following planets may be seen as a bright star in early mornings or evenings?
A. Jupiter
B. Mars
C. Saturn
D. Uranus
E. Venus

10. Which of the following gives the correct order by which blood circulates continuously through the circulatory system?
A. Heart → veins → organs → arteries → heart
B. Heart → arteries → organs → veins → heart
C. Heart → arteries → veins → organs → heart
D. Heart → organs → arteries → veins → heart
E. Heart → veins → arteries → organs → heart

11. The weight of an object is the
A. force with which the sun pulls it
B. force with which the moon pulls it
C. force of the object when gravity is not acting on it
D. force with which gravity acts on it
E. mass of the object

12. The process of fusion of sperm and ovum is known as

A. fertilization
B. menstruation
C. pollination
D. reproduction
E. lactation

13. How much work is done when a kerosene tin is pushed with a force of 20 N through a distance of 2 m?

A. 0.01 J
B. 0.10 J
C. 10.0 J
D. 20.0 J
E. 40.0 J

14. Which of the following form part of the alimentary canal?

I. Colon
II. Gullet
III. Liver
IV. Pancreas

A. I and II only
B. I and IV only
C. II and III only
D. II and IV only
E. I, II and III only

15. An example of a simple lever whose effort is at its centre is

A. a bottle opener
B. a pair of scissors
C. a pair of forceps
D. the beam balance
E. wheelbarrow

16. Which of the following leaves are sensitive to touch?

A. Cassava leaves
B. Mimosa leaves
C. Orange leaves
D. Palm leaves
E. Tomato leaves

17. Which of the following properties is / are common to both liquids and gases?

A. Fixed shape
B. Fixed shape and fixed volume
C. Fixed size
D. No definite shape

E. No definite shape and no definite volume

18. Which of the following plants has suckers?
A. Onion
B. Orange
C. Plantain
D. Sweet Potato
E. Yam

19. Which of the following activities of man causes both air and water pollution?
A. Afforestation
B. Construction of houses
C. Construction of roads
D. Deforestation
E. Industrialization

20. The number of hydrogen atoms present in **two** molecules of water is
A. 1
B. 2
C. 3
D. 4
E. 5

21. Decay is an important biological process because it brings about the
A. formation of nutrients in living organisms
B. manufacture of cells of living organisms
C. production of oxygen by plants
D. release of nutrients from dead organisms
E. release of waste substances from bacteria

22. The darkest part of a shadow formed during an eclipse is called
A. annular eclipse
B. lunar eclipse
C. penumbra
D. solar eclipse
E. umbra

23. Which of the following particles can be found in the nucleus of an atom?
A. Electrons and protons
B. Electrons and shells
C. Neutrons and electrons
D. Neutrons and protons
E. Shells and neutrons

24. Which of the following substances is used to purify water for town supply?

A. Alcohol
B. Carbon dioxide
C. Chlorine
D. Nitrogen
E. Sulphur

25. The shedding of leaves by plants during the dry season protects them against
A. bush fire
B. cold weather
C. plant-eating animals
D. heat loss
E. water loss

26. The volume of water in a container rises when a piece of stone is put into the container. The change in the volume of water is equal to the
A. mass of the container
B. mass of the stone
C. density of the stone
D. density of the water
E. volume of the stone

27. Which of the following carry deoxygenated blood in the body?
A. Arteries
B. Red blood cells
C. Valves
D. Veins
E. White blood cells

28. Which of the following forces will cause an object to move in a circular path?
A. Centripetal force
B. Gravitational force
C. Inertial force
D. Reaction force
E. Tensional force

29. The paired fins in fishes are used for
A. protection and diving
B. diving and balancing
C. balancing and control
D. steering and diving
E. balancing and steering

30. In which of the following is urine produced?
A. Bladder
B. Kidney

C. Liver
D. Penis and vagina
E. uterus

31. Photosynthesis occurs only in green plants because they
A. absorb water
B. are found in the tropics
C. are very active
D. contain chlorophyll
E. take in carbon dioxide

32. The food we eat undergoes slow oxidation in our bodies to produce heat. In this process
A. chemical energy is converted to heat energy
B. electrical energy is converted to heat energy
C. kinetic energy is converted to heat energy
D. mechanical energy is converted to heat energy
E. potential energy is converted to heat energy

33. Which of the following planets is nearest to the earth?
A. Jupiter
B. Mercury
C. Pluto
D. Saturn
E. Venus

34. Which of the following organs help to improve soil fertility?
A. Earthworms
B. Guinea worms
C. Hookworms
D. Roundworms
E. Threadworms

35. A stone released from a catapult was able to kill a bird because
A. the stone had a lot of energy
B. the stone was sharp
C. there was tension in the stone
D. the rubber of the catapult was strong
E. the stone was big

36. Which of the following is a plant parasite?
A. Capsid
B. Caterpillar
C. Dodder
D. Louse
E. Nematode

37. By the end of digestion, lean meat is turned into

A. amino acids
B. fatty acids
C. glycerol
D. glucose
E. sugar

38. In which of the following electrical appliances is electrical energy mainly converted to heat energy?

A. Cooker
B. Fan
C. Refrigerator
D. Tape recorder
E. Television set

39. The energy for lighting a torchlight bulb comes from the

A. bulb
B. cell
C. filament
D. glass
E. switch

40. The largest component of sweat is

A. glucose
B. oil
C. salt
D. urea
E. water

August 1997

SCIENCE Section A

SOLUTIONS

OBJECTIVE TEST

1.	D.	small intestines
2.	D.	Tick
3.	D.	lime
4.	C.	milky way
5.	B.	loamy soil
6.	E.	Million's reagent
7.	E.	neither be created nor destroyed
8.	D.	induce the production of anti bodies
9.	E.	Venus
10.	B.	Heart → arteries → organs → veins → heart
11.	D.	force with which gravity acts on it
12.	A.	fertilization
13.	E.	40.0 J
14.	A.	I and II only
15.	C.	a pair of forceps
16.	B.	Mimosa leaves
17.	D.	No definite shape
18.	C.	Plantain
19.	E.	Industrialization

20. D. 4

21. D. release of nutrients from dead organisms

22. E. umbra

23. D. Neutrons and protons

24. C. Chlorine

25. E. water loss

26. E. volume of the stone

27. D. Veins

28. A. Centripetal force

29. E. balancing and steering

30. B. Kidney

31. D. contain chlorophyll

32. A. chemical energy is converted to heat energy

33. E. Venus

34. A. Earthworms

35. A. the stone had a lot of energy

36. C. Dodder

37. A. amino acids

38. A. Cooker

39. B. cell

40. E. water

August 1997

SCIENCE

SECTION B
ESSAY
1 hour

[60 marks]

Answer **three** *questions* **only** *from this section.*
Illustrate your answers wherever possible, with large, clear and fully labelled diagrams.
Credit will be given for clarity of expression and orderly presentation of material.
All questions carry equal marks

1. (a) Define the term osmosis

(b) State two examples of everyday observations which can be explained in terms of capillary action.

(c) (i) Name the three processes by which heat can be transferred.
(ii) Name three appliances which use thermostat.

(d) (i) What is a saturated solution?
(ii) State whether each of the following is a solution, suspension or a colloid:
Clay in water
Sugar in water
Ammonia in water
Powdered chalk in water
Potassium permanganate in water
Soap in water

2. (a) Copy the table below and write the usual host of each of the parasites given in the spaces provided.

Parasite	**Host**
Plasmodium	
Capsids	

Tapeworm	
Dodder	

(b) (i) Name three classes of food

(ii) For each of the following food substances, give the major class of food to which it belongs: *starch, milk, margarine, sugar*

(c) Write the chemical formula for each of the following compounds:

(i) calcium hydroxide

(ii) zinc chloride [zinc (II) chloride]

(iii) sodium nitrate [sodium trioxonitrate (V)],

(iv) calcium carbonate [calcium trioxocarbonate (IV)]

(v) potassium sulphate [potassium tetraoxosulphate (VI)]

(d) (i) Define the term viscosity

(ii) Describe an experiment you performed to compare the viscosities of two liquids A and B

3. (a) (i) Define the terms humidity and wind

(ii) Name the instruments used to measure humidity and the speed of wind

(b) (i) What is a galaxy?

(ii) Give one example of galaxy

(c) State three methods by which food can be preserved.

(d) Explain why it is better to wear white clothes than black clothes on a sunny day.

(e) Write down the names of the new substances formed when the following substances are allowed to react:

(i) Sodium hydroxide and dilute sulphuric acid

(ii) Calcium carbonate and dilute hydrochloric acid

(iii) Ammonia and dilute sulphuric acid

4. (a) Define the following terms and give an example each

(i) Herbivore

(ii) Carnivore

(iii) Omnivore

(b) Two dry cells connected in parallel are in turn connected in series with a bulb and switch.

Draw a circuit diagram to illustrate this arrangement

(c) List four garden tools

(d) What property of water causes mosquito larvae to float on it?

(e) Describe an experiment you prepared to show that some part of air is used in burning.

August 1997

SCIENCE Section B

SOLUTIONS

ESSAY

1. (a) Osmosis

The movement of a solvent through a semipermeable membrane from a dilute to a more concentrated solution

Or:

The movement of solvent molecules from a region of low concentration to a region of higher concentration

(b) Examples of everyday observations in terms of capillary action

- Kerosene or oil rising up a wick
- Water rising up in a towel with one end soaked in water
- Ink rising up on a blotting paper
- Water rising up from the lower to the upper layers of soil

[any two]

(c) (i) Processes by which heat can be transferred

Conduction, convection and radiation

(ii) Appliances that use the thermostat

Electric pressing iron, refrigerator, freezer, electric kettle, air-conditioner, room heater

[any three]

(d) (i) A saturated solution

A solution containing the maximum amount of solute that can be absorbed at a given temperature

Or:

A solution that cannot dissolve any more solute at a given temperature

(ii) Classification: solution, suspension or a colloid:

Clay in water	-	suspension
Sugar in water	-	solution
Ammonia in water	-	solution
Powdered chalk in water	-	suspension

Potassium permanganate in water - solution
Soap in water - colloid

2. (a)

Parasite	Host
Plasmodium	man
Capsids	cocoa
Tapeworm	man
Dodder	milk bush

(b) (i) Classes of food

Carbohydrates, proteins, fats and oils, vitamins and mineral
[any three]

(ii)

		CLASS
Starch	-	Carbohydrates
Milk	-	Proteins
Margarine	-	Fats and oils
Sugar	-	Carbohydrates

(c)

		CHEMICAL FORMULA
(i)	**calcium hydroxide**	$Ca(OH)_2$
(ii)	**zinc chloride [zinc (II) chloride]**	$ZnCl_2$
(iii)	**sodium nitrate [sodium trioxonitrate (V)],**	$NaNO_3$
(iv)	**calcium carbonate [calcium trioxocarbonate (IV)]**	$CaCO_3$
(v)	**potassium sulphate [potassium tetraoxosulphate (VI)]**	K_2SO_4

(d) (i) Viscosity

The property of a fluid or semifluid that causes it to resist flowing
Or:
The force existing within a fluid that opposes the movement of its molecules

(ii) An experiment to compare the viscosities of two liquids A and B

Apparatus:

- Two metal balls of same kind, shape and size,
- two measuring cylinders
- two different liquids, A and B
- a stop clock

Method / Process

- The two cylinders are filled to the same level with the two liquids, A and B
- One of the metal balls is dropped gently into liquid A as the stop clock is started at the same time

- The time taken (t_1) for the ball to reach the bottom is recorded and the clock reset.
- The other metal ball is also dropped gently into the other liquid, B as the clock is again started at the same time.
- The time taken (t_2) for the ball to reach the bottom is recorded

Observation:
- One ball gets to the bottom faster than the other.

Conclusion:
- The liquids A and B have different viscosities
- The viscosity of the liquid in which the ball had a lower speed is higher than that of the other liquid.

3. (a) (i) Humidity and wind

Humidity: The amount of moisture (water vapour) in the air

Wind: Air in motion

(b) (i) Galaxy:

A group of billions of stars and their planets that form a unit within the universe

(ii) Example of galaxy

Milky way, Andromeda, Hoag's Object, M84, M86, M100

(c) Methods of preserving food

- Salting
- Frying/ heating/ roasting, etc
- Drying
- Freezing
- Canning [any three]
- Adding preservatives

(d) Why it is better to wear white clothes than black clothes on a sunny day

Black clothes will feel much hotter than white, since black materials absorb and retain much more heat than white materials. Hence, on a sunny day, it is better to wear white than black, since white clothes will feel cooler, because they are poor absorbers of heat

(e) New substances formed

(i) Sodium sulphate and water **or** $Na_2SO_4 + 2H_2O$

(ii) Calcium chloride, carbon dioxide and water **or** $CaCl_2 + CO_2 + H_2O$

(iii) Ammonium sulphate **or** $(NH_4)_2SO_4$

4. **(a)** **Definition and example each**

(i) **Herbivore**

Definition: An animal that feeds only or mainly on plants

Examples: goats, cattle, sheep, camel, giraffe, rabbit

(ii) **Carnivore**

Definition : An animal that feeds only or mainly on other animals

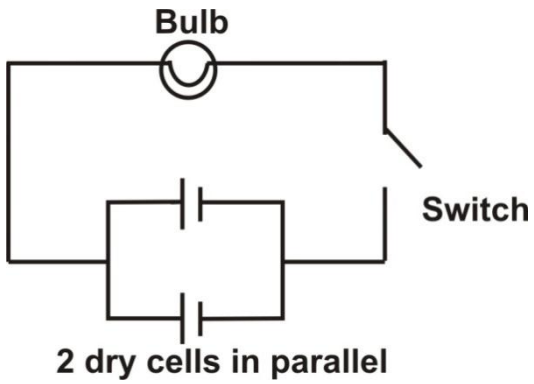
Examples: lion, tiger, crocodile, wolf, hyena, etc,

(iii) **Omnivore**

Definition : An animal that feeds on both plants and animals

Examples: human, chicken, [any one for each]

(b) **Circuit diagram.**



(c) **Garden tools**

Cutlass, fork, watering can, spade, hoe, rake, trowel, and scythe

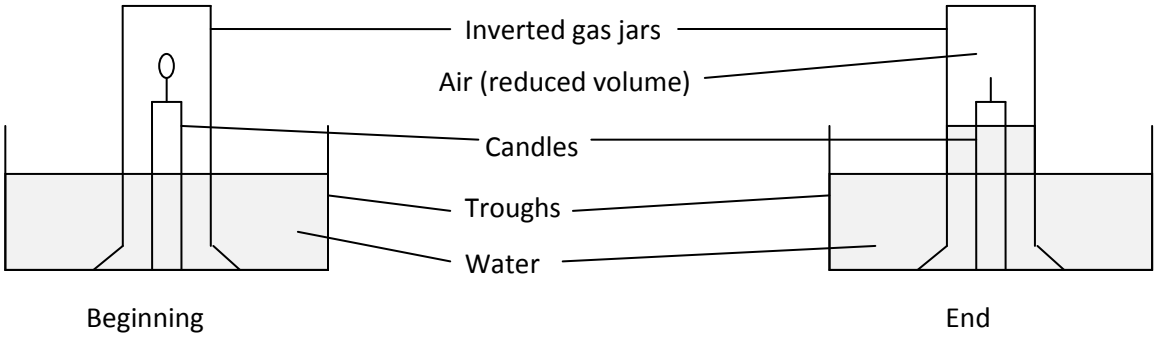
[any four]

(d) **Surface Tension**

A condition existing at the surface of a liquid, resembling a stretched elastic skin

Or: the property of a liquid that makes its surface behave like a stretched elastic skin

(e) An experiment to show that some part of air is used in burning



Apparatus:

- Water trough
- Water
- Gas jar
- 2 Candles
- Matches / fire lighter

Method / Process

- Fix one candle in the trough and fill the trough with water to a suitable level.
- Light the two candles at the same time.
- Gently place an inverted gas jar over one of the lighted candles

Observation:

- The fire of the candle under the inverted gas jar goes off after a short time
- The uncovered candle keeps burning

Conclusion:

- The fire of the candle under the gas jar went off because there was no air supply
- Some part of air is used in burning.

August 1996

SCIENCE

SECTION A
OBJECTIVE TEST
45 minutes

1. Which of the following is a sexually transmitted disease?
- A. Cholera
- B. Rickets
- C. Tuberculosis
- D. Syphilis
- E. Measles

2. The excretory organs in man are the
- A. Kidneys and urethra
- B. Kidneys and the skin
- C. Kidneys, lungs and the skin
- D. Kidneys and the lungs
- E. Kidneys, bladder and urethra

3. Which of the following farming practices is the best way of maintaining soil fertility?
- A. Crop rotation
- B. Seasonal cropping
- C. Constant ploughing
- D. Bush burning
- E. Mono cropping

4. Which of the following is **not** considered as a reflex action?
- A. Blinking of the eye
- B. Moving the hand from a hot object
- C. Jerking of the knee
- D. Talking while eating
- E. Coughing to remove an object from the throat

5. A boy exerts a force of 40.0 N to lift a load into a vehicle. If the work done is 80.0 J, calculate the distance through which the load is lifted.
- A. 0.5 m
- B. 2.0 m
- C. 40.0 m
- D. 120.0 m

E. 3200.0 m

6. Which of the following arrangements is the correct order of increasing complexity of the structure?

A. Cells → systems → tissues → organs
B. Cells → tissues → systems → organs
C. Cells → organs → tissues → systems
D. Cells → systems → organs → tissues
E. Cells → tissues → organs → systems

7. The law of conservation of energy states that energy can

A. be created but not transformed
B. be destroyed but not transformed
C. neither be created nor transformed
D. not be created but can be transformed
E. neither be created nor destroyed, but only be transformed

8. On the moon, there are large holes called

A. valleys
B. gullies
C. spurs
D. craters
E. pits

9. The darkest part of a shadow formed during an eclipse is called

A. Lunar eclipse
B. Annular eclipse
C. Penumbra
D. Umbra
E. Solar eclipse

10. The main waste products formed in plants are

A. water, carbon dioxide and urea
B. water, carbon dioxide and oxygen
C. water, carbon dioxide and mineral salts
D. water, ammonium compounds and oxygen
E. water, ammonium compounds and carbon dioxide

11. A mixture of sand and water can be separated through

A. filtration
B. ionization
C. condensation
D. sedimentation
E. evaporation

12. Which of the following chemical equations is the correct representation of the preparation of carbon dioxide?

A. $CaCO_3 + 2HCl \rightarrow CaO + Cl_2 + H_2 + CO_2$
B. $CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2CO_3$
C. $CaCO_3 + 2HCl \rightarrow CaO + Cl_2 + H_2O + CO_2$
D. $CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2 + CO_2$
E. $CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$

13. Which of the following is not an example of matter?

A. Hydrogen
B. Oxygen
C. Light
D. Water
E. Charcoal

14. A stone with mass 80.0 g was put into a measuring cylinder containing water. If the water level rises by 20.0 cm, calculate the density of the stone.

A. 0.25 gcm^{-3}
B. 1.20 gcm^{-3}
C. 1.80 gcm^{-3}
D. 3.30 gcm^{-3}
E. 4.00 gcm^{-3}

15. The sun is made up of burning

A. gases
B. rocks
C. gold
D. coal
E. sulphur

16. The force overcome by a machine is known as the

A. effort
B. load
C. pivot
D. lever
E. work

17. Calculate the work done when a weight of 20.0 N is lifted vertically through a distance of 10.0 cm

A. 2000.0 J
B. 200.0 J
C. 20.0 J
D. 2.0 J
E. 0.2 J

18. Heat travels through a vacuum by a process known as
A. conduction
B. convection
C. radiation
D. vibration
E. transmission

19. Soil aeration can be improved by
A. Adding mineral to the soil
B. Application of fertilizer to the soil
C. Weathering
D. Activities of earthworms
E. Manuring

20. Which of the following does not play any part in digestion?
A. Mouth
B. duodenum
C. rectum
D. pancreas
E. Stomach

21. Most machines waste energy because of
A. old age
B. force
C. friction
D. load
E. too much effort

22. All the following add nitrates to the soil except
A. Okro
B. Broad beans
C. French beans
D. Groundnuts
E. Pigeon pea

23. The blood in mammals is composed of
A. a red liquid
B. red blood cells and white blood cells
C. liquid plasma and solid corpuscles
D. blood protein and glycerol
E. sugar and mineral salts only

24. In a machine, 200.0 J of energy was lost as heat. If the energy input is 600.0 J, what is the efficiency of the machine?
A. 67%

B. 40%
C. 33%
D. 30%
E. 12%

25. The type of energy stored in fuels is called
A. light energy
B. solar energy
C. electrical energy
D. kinetic energy
E. chemical energy

26. An object at rest may have
A. velocity
B. momentum
C. kinetic energy
D. potential energy
E. acceleration

27. Birds are able to fly because they have
A. hollow bones
B. web legs
C. tail feathers
D. flat chest
E. wide eyes

28. The chief source of energy in the world today is
A. coal
B. crude oil
C. natural gas
D. uranium
E. water

29. Which of the following is used by plants to make its food?
A. carbon monoxide
B. oxygen
C. carbon dioxide
D. hydrogen
E. nitrogen

30. Sleeping sickness is spread by
A. blackfly
B. firefly
C. Mosquito
D. Housefly

E. Tsetse fly

31. An example of a crop that reduces the fertility of the soil rapidly is
A. Beans
B. Legumes
C. Groundnuts
D. Cassava
E. Plantain

32. The blackpod disease which attacks cocoa pods is caused by
A. virus
B. bacterium
C. plasmodium
D. fungus
E. protozoa

33. The organ which enables a fish to breathe in water is the
A. Nostril
B. Pectoral fin
C. Dorsal fin
D. Gall bladder
E. Gill

34. Which of the following part of a flower grows into the seed?
A. Stamen
B. Ovule
C. Ovary
D. Stigma
E. Sepal

35. When the palms of the hand are rubbed together, kinetic energy is changed to
A. thermal energy
B. potential energy
C. solar energy
D. elastic energy
E. chemical energy

36. Trees in desert regions have thick barks which enables them to
A. prevent heat loss
B. absorb more light
C. prevent water loss
D. absorb more water
E. absorb more oxygen

37. Which of the following is true about loamy soil?

I. Is formed from a mixture of sand and clay
II. It is good for plant cultivation
III. Its water retention is poor

A. I only
B. I and II only
C. I and III only
D. II and III only
E. I, II and III

38. In which part of the alimentary canal is food finally changed into a form that can be absorbed into the bloodstream?

A. small intestines
B. oesophagus
C. mouth
D. stomach
E. large intestines

39. In which units are the vast distances between stars in space measured?

A. Knots
B. Kilometres
C. Furlongs
D. Miles
E. Light years

40. Which of the following happens during fertilization in humans?

A. A membrane forms around the egg
B. only one sperm fuses with the egg
C. the egg divides into two
D. the egg moves down into the egg tube
E. the egg moves down into the womb

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SCIENCE Section A

SOLUTIONS

OBJECTIVE TEST

1.	D.	Syphilis
2.	C.	Kidneys, lungs and the skin
3.	A.	Crop rotation
4.	D.	Talking while eating
5.	B.	2.0 m
6.	E.	Cells → tissues → organs → systems
7.	E.	neither be created nor destroyed, but only be transformed
8.	D.	craters
9.	D.	Umbra
10.	D.	water, ammonium compounds and oxygen
11.	A.	filtration
12.	E.	$CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$
13.	C.	Light
14.	E.	$4.00\ gcm^{-3}$
15.	A.	gases
16.	B.	load
17.	D.	2.0 J
18.	C.	radiation
19.	D.	Activities of earthworms

20. C. rectum

21. C. friction

22. A. Okro

23. C. liquid plasma and solid corpuscles

24. A. 67%

25. E. chemical energy

26. D. potential energy

27. A. hollow bones

28. B. crude oil

29. C. carbon dioxide

30. E. Tsetse fly

31. D. Cassava

32. D. fungus

33. E. Gill

34. B. Ovule

35. A. thermal energy

36. C. prevent water loss

37. B. I and II only

38. A. small intestines

39. E. Light years

40. B. only one sperm fuses with the egg

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SCIENCE

SECTION B
ESSAY
1 hour

[60 marks]

Answer **three** *questions* **only** *from this section.*
Illustrate your answers wherever possible, with large, clear and fully labelled diagrams.
Credit will be given for clarity of expression and orderly presentation of material.
All questions carry equal marks

1. (a) Define the following terms:
- (i) erosion
- (ii) reproduction
- (iii) parasite
- (iv) excretion

(b) Give one example each of erosion and parasite.

(c) Draw and label a diagram of the female reproductive organ in a human being.

(d) (i) Define the term work.
(ii) An amount of 300.0 J of work is done when a force moves through a distance of 10.0 m in the direction of the force. Calculate the value of the force.

(e) Write down the names of the following compounds:
- (i) NH_4Cl
- (ii) HCl
- (iii) $CuSO_4$
- (iv) $FeCl_3$

(f) Explain how sulphur can be obtained from a mixture of powdered sulphur and iron filings.

2. (a) Draw and label a diagram to show how the eclipse of the sun is formed.

(b) (i) What is a lever?

(ii) Classify the following as first class lever, second class lever or third class lever by copying and completing the table below:
Beam balance, crowbar, wheel barrow, a pair of scissors, claw hammer, sugar tong, human fore-arm and bottle opener.

First class lever	Second class lever	Third class lever

(c) (i) What is pollination?

(ii) Give two agents of pollination.

(d) Name three excretory organs and the products they excrete.

(e) (i) What is an alloy?

(ii) What are the compositions of each of the following alloys?
Bronze, Brass, Steel and Duralumin.

(f) Sodium chloride is prepared by the reaction between dilute hydrochloric acid and sodium hydroxide solution. Write down the equation for this reaction.

3. (a) An atom has an atomic number of 8 and a mass number of 16. State the number of each of the following in the atom:

(i) Protons

(ii) Electrons

(iii) Neutrons

(b) (i) Draw the structure of the atom in (a) above

(ii) If the atom in (a) above gains two electrons what will be the charge of the ion?

(c) State the end-product of the following food substances:

(i) Carbohydrates

(ii) Proteins

(iii) Fats

(d) (i) What is the preservation of food.

(ii) List four methods of food preservation.
Explain why one of the methods you have listed is used for food preservation.

(e) (i) An object is placed in front of a plane mirror. Draw a ray diagram to show how the image of the object is formed.

(ii) State two characteristics of an image formed by a plane mirror.

4. (a) (i) Give the names of organisms which cause the following diseases:
Sleeping sickness, Cholera, Malaria, Bilharzia

(ii) Which part of the following crops does a farmer usually grow?
Banana, cocoyam, tomato, ginger, yam, cassava

(b) Name the organs used by the following organisms for respiration:
(i) Tilapia
(ii) Toad

(c) (i) What happens to the temperature of water that is boiling in an open container?
(ii) Give two characteristics of a liquid.
(iii) Name the three processes by which heat is transmitted.
(iv) State two applications of the expansion of solids in everyday life.

(d) Explain the following terms:
(i) Physical change
(ii) Chemical change

(e) State whether each of the following is a physical or a chemical change
(i) rusting of iron
(ii) freezing of water
(iii) burning of wood
(iv) fermentation
(v) grinding of chalk
(vi) dissolving sugar in water

(f) Name the compound formed when theh following elements combine:
(i) Iron and sulphur
(ii) Zinc and oxygen
(iii) Sodium and chlorine
(iv) Calcium and chlorine

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SCIENCE Section B

SOLUTIONS

ESSAY

1. (a) (i) Erosion

The washing away of the top soil by agents of erosion

Or:

The removal of soil material by natural processes, principally running water, glaciers, waves, and wind

(ii) Reproduction

The production of offspring or new individuals through a sexual or asexual process

(iii) Parasite

A plant or animal that lives on or in another host organism in a way that harms or is of no advantage to the host

(iv) Excretion

The elimination (removal) of metabolic waste substances from the organism

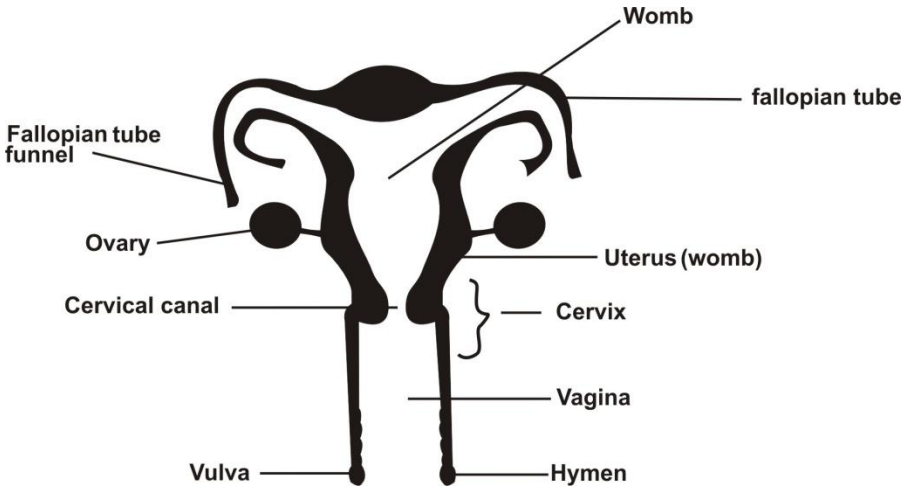
(b) Example

Erosion: sheet, rill, gully

Parasite: plasmodium, dodder, tick, tapeworm, etc

[any one of each]

(c) The female reproductive organ



(d) **(i)** **Work**

Product of the force applied to a body and the distance moved by that body in the direction of the force

Or: Work = ***f*** × ***d***, Where *f* = force applied

and *d* = distance moved in the direction of the applied force

(ii) **Force** = work ÷ distance

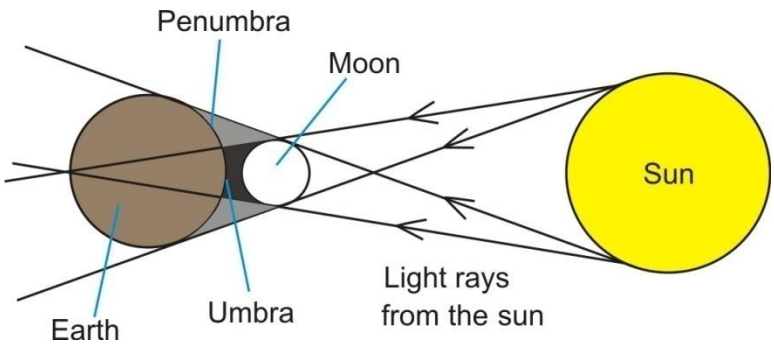
= 300 ÷ 10

= <u>30 N</u>

(e)

		<u>NAMES</u>
(i)	**NH_4Cl**	Ammonium chloride
(ii)	**HCl**	Hydrochloric acid
(iii)	**$CuSO_4$**	Copper (II) tetraoxosulphate (VI) or copper sulphate
(iv)	**$FeCl_3$**	Iron (III) chloride

2. **(a)** **Diagram** to show how the eclipse of the sun is formed



(b) **(i)** **Lever**

Simple machine consisting of a rigid bar that rotates about a fixed point, called a fulcrum

(ii)

First class lever	Second class lever	Third class lever
beam balance crowbar pair of scissors claw hammer	wheel barrow bottle opener	sugar tong human forearm

(c) **(i)** **Pollination**

The transfer of pollen grains from the anther of a flower to the stigma of another flower of the same species.

(ii) **Agents of pollination**

Wind, water, man, birds, insects

(d)

EXCRETORY ORGAN	EXCRETORY PRODUCT
Skin	sweat
Kidney	urea
Lungs	carbon dioxide

(e) **(i)** **Alloy**

Substance containing two or more metals uniformly physically combined

Or: A homogeneous mixture of two or more metals

(ii)

ALLOY	COMPOSITION
Bronze:	Copper and tin
Brass:	Copper and zinc
Steel:	Iron and carbon
Duralumin:	Aluminium, copper and manganese

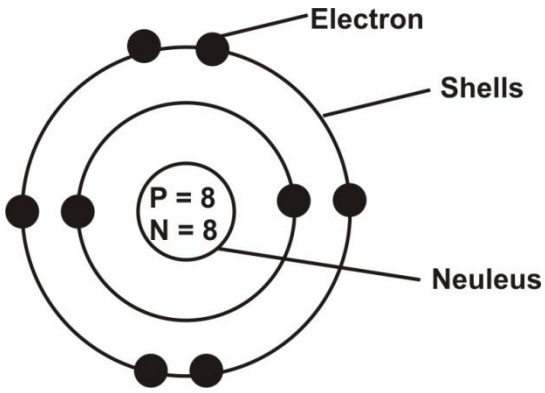
(f) **Equation for reaction**

$HCl + NaOH \rightarrow NaCl + H_2O$

3. **(a)** atomic no. = 8, mass no. = 16;

			Number of:
(i)	Protons	=	8
(ii)	Electrons	=	8
(iii)	Neutrons	=	8

(b) (i) Structure of atom:



(ii) The charge will be = – 2

(c)

			THE END-PRODUCT
(i)	**Carbohydrates**	-	glucose
(ii)	**Proteins**	-	amino acids
(iii)	**Fats**	-	fatty acids and glycerol

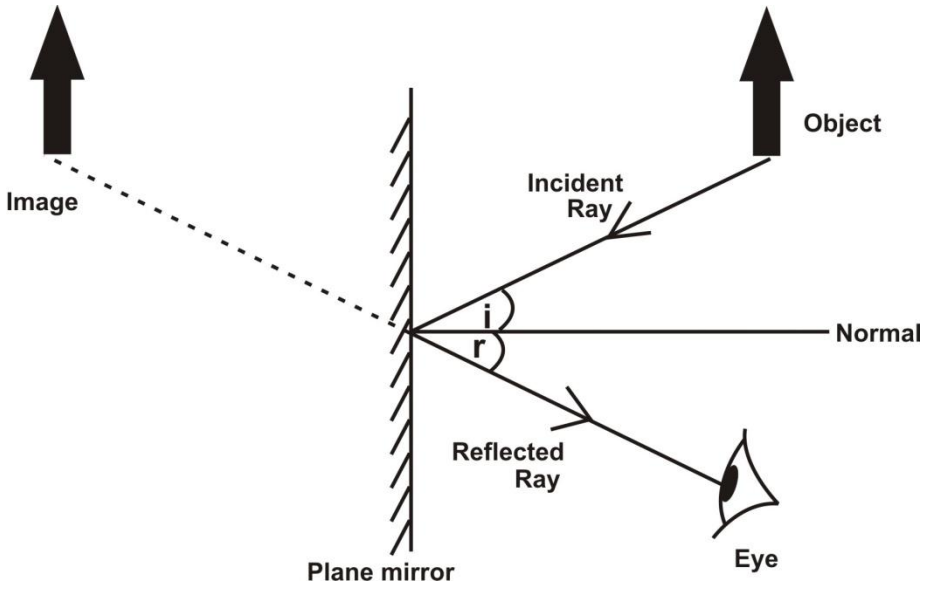
(d) (i) Preservation of food

The processes by which food is made to stay for a long time without going bad

(ii) Methods of food preservation

METHOD	WHY IT IS USED
Salting	Salt kills the bacteria that cause food spoilage and also removes moisture therefore making the bacteria unable to act.
Heating	The heat destroys the bacteria that cause food spoilage
Drying	Removes the water /moisture, hence making the bacteria unable to thrive
Canning	Canning seals the container completely and keeps out moisture and food spoiling bacteria from the food
Freezing	The bacteria are either killed or made unable to act at very low temperatures
Adding artificial preservatives	The artificial preservatives, like salt, kill the food spoiling bacteria or render them unable to act.

(e) (i) Ray diagram



(ii) Characteristics of image formed by plane mirror

- Same size as object
- Same distance from mirror as object
- Virtual
- Erect / upright
- Laterally inverted

[any two]

4. (a) (i)

DISEASE	CAUSATIVE ORGANISM	
Sleeping sickness	Protozoa	(Trypanosoma)
Cholera	bacteria	(vibrio cholerae)
Malaria	protozoa	(plasmodium)
Bilharzia	flatworm	(blood fluke /schistosome)

(ii)

CROP	PART GROWN
Banana	sucker
Cocoyam	corm
Tomato	seeds
Ginger	rhizome
Yam	stem tuber
Cassava	stem cutting

(b) THE ORGANS USED FOR RESPIRATION:

(i) Tilapia - Gills

(ii) Toad - Lungs / mouth / skin

(c) **(i)** **Temperature of water that is boiling in an open container**

The temperature remains constant at 100°C or 373.16K

(ii) **Characteristics of a liquid.**

- Has no definite shape /takes the shape of its container
- Has a definite volume and mass
- Molecules are loosely bound among themselves
- Molecules are move about freely
- Cannot be compressed.

(iii) **Name the three processes by which heat is transmitted.**

Convection, conduction and radiation

(iv) **Two applications of the expansion of solids in everyday life.**

- Thermostats of refrigerators, pressing irons, etc
- Riveting

(d) **Explain the following terms:**

(i) **Physical change**

One in which no new substance is formed and is usually reversible, eg, the melting of ice.

(ii) **Chemical change**

One in which a new substance is formed and is irreversible, eg, the neutralization reaction between HCl and NaOH

(e)

			TYPE OF CHANGE
(i)	**Rusting of iron**	-	Chemical change
(ii)	**Freezing of water**	-	Physical change
(iii)	**Burning of wood**	-	Chemical change
(iv)	**Fermentation**	-	Chemical change
(v)	**Grinding of chalk**	-	Physical change
(vi)	**Dissolving sugar in water**	-	Physical change

(f)

			COMPOUND FORMED		
(i)	**Iron and sulphur**	-	Iron (II) sulphide	or	FeS
(ii)	**Zinc and oxygen**	-	Zinc oxide	or	ZnO
(iii)	**Sodium and chlorine**	-	Sodium chloride	or	NaCl
(iv)	**Calcium and chlorine**	-	Calcium chloride	or	$CaCl_2$

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SCIENCE

SECTION A
OBJECTIVE TEST
45 minutes

1. Which of the following activities does **not** cause soil erosion?
- A. Clearing of vegetation
- B. Bush burning
- C. Heavy rainfall
- D. Over grazing
- E. Crop rotation

2. What is the difference between mass and weight? The
- A. mass of a body is twice the weight
- B. weight of a body is constant but its mass varies
- C. mass of a body varies but its weight does not
- D. weight of a body varies but its mass does not
- E. mass of a body is half the weight

3. By what process is heat transferred from the bottom of a beaker containing water to the top?
- A. convection
- B. conduction
- C. radiation
- D. absorption
- E. transmission

4. In what units are work and energy measured?
- A. work in joules, energy in watts
- B. work in joules, energy in joules
- C. work in newtons, energy in joules
- D. energy in watts, work in watts
- E. energy in newtons, work in joules

5. Which of the following is the organic part of the soil?
- A. Mineral salts
- B. Water
- C. Rock particles
- D. Air

E. Humus

6. A mixture of sugar and water can be separated by
A. Filtration
B. Evaporation
C. Decantation
D. Distillation
E. Condensation

7. The air pressure at great heights can be measured with
A. A thermometer
B. An anemometer
C. A barometer
D. A hydrometer
E. A hygrometer

8. Which of the following animals reproduces without mating?
A. Toad
B. Lizard
C. Dog
D. Man
E. Amoeba

9. Which of the following environmental factors can cause the fastest rate of water loss from leaf surfaces?
A. High humidity
B. Wet wing
C. Still air
D. Low temperature
E. Dry air

10. A fruit adapted for dispersal by wind is likely to be
A. succulent
B. brightly coloured
C. hooked
D. sticky
E. hairy

11. All the following substances are mixtures except
A. ink
B. blood
C. air
D. palm wine
E. carbon dioxide

12. In a first class lever, the load is placed

A. on the effort
B. between the effort arm and the pivot
C. at one end of the load arm
D. between the pivot and the load arm
E. on the pivot

13. Excretion in animals involves the removal of
A. water from the body
B. urine and faeces from the body
C. blood from the body
D. waste materials from the body cells
E. undigested food from the body

14. Which of the following is an indirect source of light?
A. Torch light
B. The sun
C. Fire
D. The moon
E. Candle flame

15. Which of the following is the sense organ for sight?
A. Skin
B. nose
C. eye
D. ear
E. tongue

16. Liquids suitable for use in thermometers must not:
A. wet glass
B. have a low boiling point
C. have a high melting point
D. have a high density
E. expand

17. When living organisms die, the mineral salts in their bodies become available to the soil through the process of
A. evaporation
B. sedimentation
C. defecation
D. decay
E. osmosis

18. Ice floats on water because
A. water is denser than ice
B. ice is denser than water

C. ice has the same temperature as water
D. ice has a lower temperature than water
E. ice is different from water

19. The reason for chewing food before swallowing is to
A. soften the food
B. make digestion faster
C. cause saliva to flow out
D. exercise the jaws
E. make the teeth strong

20. Power is defined as the
A. ability to do work
B. amount of energy
C. rate of distance covered
D. rate of doing work
E. ease of moving a heavy object

21. The force which is able to hold light objects on the surface of liquids is called
A. surface tension
B. friction
C. capillarity
D. viscosity
E. cohesion

22. In the nucleus of an atom, there are
A. neutrons and shells
B. protons and electrons
C. neutrons and protons
D. electrons and shells
E. electrons and neutrons

23. The male sperms fertilize the female eggs to form the
A. embryo
B. ovary
C. ovum
D. placenta
E. uterus

24. Most plants will not do well in the desert because
A. there is no top soil
B. it is too sunny there
C. rainfall is very poor
D. the speed of wind is high
E. the gullies are too deep for plant cultivation

25. What is the work done when a girl lifts a pot of water weighing 40 N upward through a distance of 1 m?

A. 400.0 J
B. 40.0 J
C. 4.0 J
D. 1.0 J
E. 0.4 J

26. Which of the following is **not** characteristic of all living organisms? The ability to

A. break down food to obtain energy
B. produce food using the sun's energy
C. make use of food nutrients to grow
D. respond to changes within the environment
E. give rise to young new individuals

27. The process used in getting salt from sea water is

A. boiling
B. condensation
C. decantation
D. sedimentation
E. evaporation

28. Which of the following bodies is a satellite of the earth?

A. Mars
B. Sun
C. Venus
D. Moon
E. Mercury

29. Which of the following are soil micro-organisms?

A. Termites
B. Viruses
C. Ticks
D. Bacteria
E. Crabs

30. The darkest part of a shadow formed when a large source of light is used is the

A. Partial eclipse
B. Lunar eclipse
C. night
D. penumbra
E. umbra

31. The part of air which supports burning is

A. nitrogen

B. inert gas
C. water vapour
D. oxygen
E. carbon dioxide

32. What structure links the mouth with the stomach? The
A. duodenum
B. oesophagus
C. Ileum
D. trachea
E. colon

33. Which of the following instruments is used in measuring current in an electric circuit?
A. Voltmeter
B. Ammeter
C. Barometer
D. Thermometer
E. Anemometer

34. There are thick forests in part of the world where the climate is
A. Hot and wet
B. cool and wet
C. hot and dry
D. cool and dry
E. warm and dry

35. The outer ear is also referred to as
A. cochlea
B. tympanum
C. pinna
D. incus
E. malleus

36. Gin can be obtained from palm wine by
A. condensation
B. freezing
C. sedimentation
D. distillation
E. filtration

37. Which of the following is likely to happen when the relative humidity is high?
A. It is likely to rain
B. It is likely to be dry
C. It is likely to be sunny
D. The day will be hot

E. It will be windy

38. Which of the following features is common to both the fish and the lizard? The possession of

A. a sticky tongue
B. a lateral line system
C. dorsal fins
D. clawed digits
E. scales on the body

39. Which of the following statements about the solar system is true? The

A. sun moves round the earth
B. planets move round the sun
C. planets move round the moon
D. sun moves in its orbit
E. sun moves round the moon

40. What type of fruit is a mango?

A. Drupe
B. Pepo
C. Berry
D. Caryopsis
E. Pome

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SCIENCE Section A

SOLUTIONS

OBJECTIVE TEST

1.	E.	Crop rotation
2.	D.	weight of a body varies but its mass does not
3.	A.	convection
4.	B.	work in joules, energy in joules
5.	E.	Humus
6.	B.	Evaporation
7.	C.	A barometer
8.	E.	Amoeba
9.	E.	Dry air
10.	E.	hairy
11.	E.	carbon dioxide
12.	C.	at one end of the load arm
13.	D.	waste materials from the body cells
14.	D.	The moon
15.	C.	eye
16.	A.	wet glass
17.	D.	decay
18.	A.	water is denser than ice
19.	A.	soften the food

20. D. rate of doing work

21. A. surface tension

22. C. neutrons and protons

23. A. embryo

24. C. rainfall is very poor

25. B. 40.0 J

26. B. produce food using the sun’s energy

27. E. evaporation

28. D. Moon

29. D. Bacteria

30. E. umbra

31. D. oxygen

32. B. oesophagus

33. B. Ammeter

34. A. Hot and wet

35. C. pinna

36. D. distillation

37. A. It is likely to rain

38. E. scales on the body

39. B. planets move round the sun

40. A. Drupe

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SCIENCE

SECTION B
ESSAY
1 hour

[60 marks]

Answer **three** *questions* **only** *from this section.*
Illustrate your answers wherever possible, with large, clear and fully labelled diagrams.
Credit will be given for clarity of expression and orderly presentation of material.
All questions carry equal marks

1. (a) Explain each of the following terms:
- (i) Diffusion
- (ii) Surface tension
- (iii) Capillarity

(b) Why does water wet glass?

(c)
- (i) State the composition of blood
- (ii) List three functions of blood

(d)
- (i) Describe how ions are formed from atoms
- (ii) State two differences between a proton and an electron.
- (iii) Write the chemical symbol for each of the following:
  - (α) an atom of Lithium
  - (β) a molecule of chlorine
  - (γ) an atom of oxygen

2. (a) Make a labelled diagram of the human eye.

(b)
- (i) Name two defects of the eye.
- (ii) State how each of the defects you have named above can be corrected.

(c)
- (i) What is a satellite?
- (ii) State one difference between a natural and an artificial satellite.
- (iii) State two uses of artificial satellites.

(d) Write down the names of the new substances formed when each of the following substances react with dilute hydrochloric acid.
(i) Ammonia
(ii) Sodium hydroxide
(iii) Calcium carbonate

3. (a) In each of the activities listed below state whether the changes that occur are physical or chemical. Explain your answers.
(i) An egg is boiled for five minutes
(ii) A piece of bread is chewed in the mouth for two minutes
(iii) A piece of meat is chewed in the mouth for two minutes
(iv) A piece of wood is burnt to ashes

(b) Define each of the following
(i) A machine
(ii) Mechanical advantage of a machine
(iii) Velocity ratio of a machine

(c) A system of levers is used to overcome a resistance of 3,000 N when an effort of 150 N is applied to it. Calculate the mechanical advantage of the system.

(d) (i) List three general causes of diseases
(ii) State four ways by which diseases can be prevented.

4. (a) Distinguish between sexual reproduction and asexual reproduction.
(b) State the vegetative part that can be used to propagate each of the following plants:
(i) cassava
(ii) onion
(iii) banana
(iv) yam
(v) ginger
(vi) pineapple

(c) (i) Define density of a substance.
(ii) Describe how you would determine the density of a sample of soil in the laboratory.

(d) State one solvent for each of the following substances
(i) Common salt
(ii) Paint
(iii) Coal tar
(iv) Sucrose
(v) Chlorophyll

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SCIENCE Section B

SOLUTIONS

ESSAY

1. (a) (i) Diffusion

The random movement of atoms, molecules, or ions from one site in a medium to another, resulting in complete mixing

Or:

The movement of particles of a substance from a region of higher concentration to a region of lower concentration

(ii) Surface Tension

A condition existing at the surface of a liquid, resembling a stretched elastic skin

Or:

The property of a liquid that makes its surface behave like a stretched elastic skin

(iii) Capillarity

The rising and falling of a liquid's surface where it is in contact with a solid

(b) Why water wets glass

The adhesive force of attraction between the water molecules and the molecules of glass is greater than the cohesive forces of attraction among the water molecules, hence water tends to cling to the glass or wet the glass.

(c) (i) The composition of blood

- The red blood cells or corpuscles
- The white blood cells or corpuscles
- Blood platelets
- Plasma (containing water; dissolved food substances like glucose, amino acids and fatty acids; hormones, enzymes, metabolic waste substances such as carbon dioxide, salts and urea)

(ii) Functions of blood

- Transports oxygen to the cells of the body
- Transports metabolic waste materials away from the cells of the body
- Transports digested food nutrients to parts of the body
- Helps to maintain a uniform body temperature
- Prevents excessive blood loss by clotting.
- Helps to protect the body against infections and diseases

(d) (i) how ions are formed from atoms

Atoms become charged when they lose or gain electrons in order to be more stable or attain a completely filled outer shell. When atoms gain electrons, they become negatively charged ions, known as anions; when they lose electrons, they become positively charged ions, known as cations

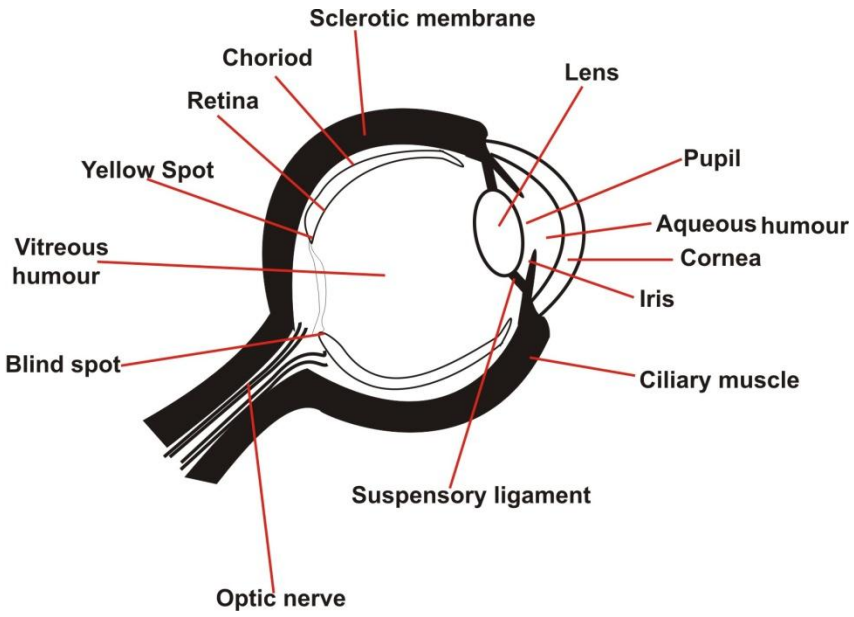
(ii) Difference

PROTONS	ELECTRONS
positively charged	negatively charged
located in the nucleus	located on the shells
has relative mass of 1	has a negligible relative mass

(iii)

		CHEMICAL SYMBOL
(α)	an atom of Lithium	Li
(β)	a molecule of chlorine	Cl_2
(γ)	an atom of oxygen	O

2. (a) Make a labeled diagram of the human eye



(b) **(i)** **Two defects of the eye**

Long sightedness, short sightedness,

(ii) Long sightedness: Is corrected using a convex (converging) lens

Shortsightedness: Is corrected using a concave (diverging) lens

(c) **(i)** **A satellite**

A heavenly body that moves round (orbits) a larger heavenly body

(ii)

Natural Satellite	Artificial Satellite
Created by God (or nature)	Manufactured by man
Has no mechanical or electrical devices	Has man-made mechanical or electrical devices

[any one]

(iii) **Uses of artificial satellites**

- Weather studies and forecasting
- Telecommunication
- Sending and receiving TV signals
- Taking photographs from space for various purposes
- Internet networking
- Global Positioning System (GPS) - a worldwide navigation system

(d)

		SUBSTANCE FORMED on reaction with dilute HCl
(i)	**Ammonia (NH_3)**	Ammonium chloride (NH_4Cl)
(ii)	**Sodium hydroxide (NaOH)**	Sodium chloride (NaCl) and water (H_2O)
(iii)	**Calcium carbonate ($CaCO_3$)**	Calcium chloride ($CaCl_2$), Carbon dioxide (CO_2) and water (H_2O)

3. **(a)** **Type of change and explanation**

(i) **An egg is boiled for five minutes**

Chemical change, since there is a new substance formed; and the change is **not** reversible

(ii) **A piece of bread is chewed in the mouth for two minutes**

Chemical change, since there is a new substance formed; and the change is **not** reversible

(iii) **A piece of meat is chewed in the mouth for two minutes**

Physical change, since no new substance is formed.

(iv) **A piece of wood is burnt to ashes**

Chemical change, since a new substance is formed and the change is **not** reversible

(b) Definitions

(i) A machine

A mechanical device that makes work easier and / or faster

(ii) Mechanical advantage of a machine

The ratio of the load to the effort

Or: = Load / effort

(iii) Velocity ratio of a machine

The ratio of the effort distance to the load distance

= (effort distance) / (load distance)

(c) Mechanical Advantage = Load / Effort

= 3000 N / 150 N

= 20

(NB: the mechanical advantage has no units, since it is a ratio)

(d) (i) General causes of diseases

- Dirty environments
- Lack of personal hygiene
- Lack of proper nutrition (or deficiency of certain food nutrients)
- Contact with or sharing items with person infected with communicable diseases.
- Lack of regular medical checkups

(ii) Ways of preventing diseases

- Cleaning our environments regularly
- Keeping our bodies and personal items clean always
- Eating balanced diets
- Avoiding contact with or sharing items with persons infected with communicable diseases
- Going for medical checkups regularly

4. (a)

SEXUAL REPRODUCTION	ASEXUAL REPRODUCTION
Occurs by the fusion of two separate sex cells – male and female	Occurs by the meiosis of only one cell
Results in offspring with different traits from parents	Results in offspring with same traits as parents
Two parents of opposite sex are involved	Only one parent is involved

(b) PLANT VEGETATIVE PART

(i)	cassava	stem cutting
(ii)	onion	bulb
(iii)	banana	sucker
(iv)	yam	stem tuber
(v)	ginger	rhizome
(vi)	pineapple	sucker / crown

(c) (i) Density of a substance

The mass per unit volume of the substance

Or: Density = mass / volume

(ii) Determination of density of soil

- The mass, M, of a given quantity of the soil is determined using a beam balance or chemical balance or electronic scale
- A measuring cylinder is filled with water to a suitable volume, V_1 and the reading is recorded.
- The soil is gently poured into the measuring cylinder and the new (higher) volume, V_2 is read and recorded
- The volume of the soil is calculated as $V_2 - V_1$
- The density of the soil is calculated as $\frac{Mass}{Volume} = \frac{M}{V_2 - V_1}$

(d)

	SUBSTANCE	**SOLVENT**
(i)	**Common salt**	water
(ii)	**Paint**	water (if emulsion), turpentine (if oil paint)
(iii)	**Coal tar**	petrol / kerosene
(iv)	**Sucrose**	water
(v)	**Chlorophyll**	alcohol

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SCIENCE

SECTION A
OBJECTIVE TEST
45 minutes

1. Which of the following substances can exist in all the three states of matter at 0°C?

A. Petrol
B. Palm oil
C. Milk
D. Alcohol
E. Water

2. Pluto is the darkest planet because it

A. is larger than the sun
B. is the largest planet
C. is beyond the reach of the sun's rays
D. is smaller than the sun
E. falls in the penumbra region of an eclipse

3. In reproduction, the embryo can be referred to as

A. an ovary
B. a fertile egg
C. a fertilized egg
D. a placenta
E. a mature ovum

4. Which of the following statements about second class levers is true?

A. The pivot is between the effort and the load
B. The load is between the effort and the pivot
C. There is no pivot
D. The load is on the pivot
E. The effort is between the pivot and the load

5. Which of the following is **not** a communicable disease?

A. smallpox
B. chicken pox
C. influenza
D. pneumonia

E. Tuberculosis

6. When bread is chewed for a long time, it tastes sweeter because it
A. is made from sugar and flour
B. is digested into simple sugar
C. mixes with saliva which is alkaline
D. is broken into smaller pieces
E. contains yeast which is sweet

7. Which of the following gases is **not** an air pollutant?
A. cigarette fumes
B. carbon dioxide
C. sulphur dioxide
D. exhaust fumes
E. carbon monoxide

8. Dissolved mineral salts from the soil enter the plant by the process of
A. phototropism
B. transpiration
C. diffusion
D. osmosis
E. ingestion

9. Which of the following organs is the center of the Central Nervous System?
A. Kidney
B. Liver
C. Heart
D. Brain
E. Lung

10. Soil fertility can be conserved by all the following methods **except**
A. mulching
B. afforestation
C. deforestation
D. crop rotation
E. green manuring

11. Which of the following insects is an agent of pollination?
A. Mosquito
B. Housefly
C. Bee
D. Ant
E. Cockroach

12. During the dry season most trees shed their leaves to

A. protect the tree
B. avoid being pulled down by strong winds
C. conserve water and nutrients
D. discourage animals from eating the leaves
E. reduce the loss of carbon dioxide

13. The chemical formula for sodium chloride
A. Na_2Cl
B. NaCl
C. Sn_2Cl
D. $NaCl_2$
E. $SnCl_2$

14. Which of the following is the cleanest source of drinking water?
A. Lake
B. Stagnant water
C. River
D. Deep well
E. Surface well

15. The edible part of the mango fruit is called
A. endocarp
B. drupe
C. fibre
D. mesocarp
E. epicarp

16. Which of the following is **not** a characteristic of all living organisms
A. Respiration
B. Reproduction
C. Photosynthesis
D. Feeding
E. Excretion

17. Which of the following conditions in a simple machine will make working difficult? When
A. The load distance is equal to the effort distance
B. The effort distance is twice the load distance
C. The effort distance is three times the load distance
D. Small effort moves through a longer distance
E. The load distance is greater than effort distance.

18. A dam for generating power is constructed across a river. What type of energy is possessed by the water behind the dam?
A. Kinetic energy
B. Potential energy

C. Electrical energy
D. Heat energy
E. Light energy

19. Which of the following is a function of prop roots? They
A. Store starch in the swollen branch roots
B. Take in air for respiration
C. Hold the plant more firmly in the soil
D. Develop into storage roots
E. Develop into fibrous roots

20. Which of the following living organisms is **not** a reptile?
A. Lizard
B. Tortoise
C. Frog
D. Snake
E. Crocodile

21. The following are steps involved in how nutrients in decaying organic matter finally get to man.
I. The mineral in them go into the soil
II. Man goes to eat the plant food
III. They dissolve in soil water
IV. They are used up by plants during photosynthesis

Which of the following arrangements represents the correct order in which the processes occur?

A. I, II, III and IV
B. I, III, IV and II
C. IV, II, III and I
D. II, III, I and IV
E. III, IV, I and II

22. Which of the following is the energy possessed by an object placed on a table?
A. Kinetic energy
B. Electrical energy
C. Potential energy
D. Sound energy
E. Heat energy

23. The change in the shape of the moon is caused by the
A. clouds covering the moon surface
B. earth revolving around the sun
C. moon revolving around the earth
D. sun revolving around the moon
E. earth rotating on its axis

24. The most sensitive region of the eye to light rays is the

A. retina
B. pupil
C. cornea
D. iris
E. lens

25. Digestion is a process by which food is

A. transported in soluble form within the body of animals
B. broken down into tiny pieces in the mouth
C. moved through the alimentary canal
D. broken down to produce energy, water and carbon dioxide
E. broken down into forms which can be easily absorbed

26. Which of the following processes is **not** a physical change

A. Dissolution
B. Melting
C. Boiling
D. Evaporation
E. Burning

27. Which of the following is used to measure the potential difference across a resistor in an electrical circuit?

A. A galvanometer
B. A rheostat
C. A meter bridge
D. A voltmeter
E. An ammeter

28. There are thick forests in some parts of the world because the climate there is

A. Warm and dry
B. Cool and dry
C. Hot and dry
D. Cool and wet
E. Hot and wet

29. Which of the following practices encourages soil erosion?

A. Planting cover crops
B. Mulching
C. Contour ridging
D. Crop rotation
E. Ploughing along the slope

30. A car battery produces electrical energy from

A. Chemical energy
B. Sound energy
C. Kinetic energy
D. Heat energy
E. Potential energy

31. Which of the following substances is **not** found in the excretory products of man?
A. Ammonium salt
B. Common salt
C. Glucose
D. Water
E. Carbon dioxide

32. Which of the following processes can be used to obtain local gin from palm wine?
A. Evaporation
B. Condensation
C. Distillation
D. Filtration
E. Sublimation

33. The linear sequence of living things feeding and being fed on is called
A. linear feeding
B. food chain
C. photosynthesis
D. nutrition
E. transpiration

34. Which of the following organisms causes malaria?
A. Tsetsefly
B. Fungi
C. Virus
D. Plasmodium
E. Bacteria

35. The male and female sex cells of a plant are respectively in the
A. Sperm and egg
B. Pollen grain and ovary
C. Sperm and ovary
D. Ovary and ovule
E. Embryo and seed

36. Which of the following preventive measures is **not** used in the control of tuberculosis?
A. Isolating the patient
B. Spitting carelessly in the environment
C. Keeping separate cutlery, plates, cups for the patient

D. Vaccinating young babies with BCG vaccine
E. Spitting into a container partly filled with disinfectant

37. During an eclipse of the sun, the region that has total darkness is known as
A. umbra
B. orbit
C. penumbra
D. shadow
E. satellite

38. Animals that live in grasslands include
A. lions and giraffes
B. monkeys and tigers
C. antelopes and hares
D. woodpeckers and baboons
E. elephants and leopards

39. Which of the following are products of respiration?
A. Glucose and oxygen
B. Waste food, energy and carbon dioxide
C. Energy, carbon dioxide and water
D. Energy, oxygen and water
E. Carbon dioxide and oxygen

40. Which of the following living things hide during the sunny days in order to conserve water in its body?
A. Rat
B. Lizard
C. Butterfly
D. Toad
E. Bat

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SCIENCE Section A

SOLUTIONS

OBJECTIVE TEST

1.	E.	Water
2.	C.	is beyond the reach of the sun's rays
3.	C.	a fertilized egg
4.	B.	The load is between the effort and the pivot
5.	D.	pneumonia
6.	B.	is digested into simple sugar
7.	B.	carbon dioxide
8.	C.	diffusion
9.	D.	Brain
10.	C.	deforestation
11.	C.	Bee
12.	C.	conserve water and nutrients
13.	B.	NaCl
14.	D.	Deep well
15.	D.	mesocarp
16.	C.	Photosynthesis
17.	E.	The load distance is greater than effort distance.
18.	B.	Potential energy
19.	C.	Hold the plant more firmly in the soil

20. C. Frog

21. B. I, III, IV and II

22. C. Potential energy

23. C. moon revolving around the earth

24. A. retina

25. E. broken down into forms which can be easily absorbed

26. E. Burning

27. D. A voltmeter

28. E. Hot and wet

29. E. Ploughing along the slope

30. A. Chemical energy

31. C. Glucose

32. C. Distillation

33. B. food chain

34. D. Plasmodium

35. B. Pollen grain and ovary

36. B. Spitting carelessly in the environment

37. A. umbra

38. C. antelopes and hares

39. C. Energy, carbon dioxide and water

40. D. Toad

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SCIENCE

SECTION B
ESSAY
1 hour

[60 marks]

Answer **three** *questions* **only** *from this section.*
Illustrate your answers wherever possible, with large, clear and fully labelled diagrams.
Credit will be given for clarity of expression and orderly presentation of material.
All questions carry equal marks

1. (a) Distinguish between the following terms:
- (i) Physical change and chemical change
- (ii) Solution and suspension

(b) Classify the following as either a physical or chemical change
- (i) Common salt in water
- (ii) Burning of charcoal
- (iii) Melting of ice
- (iv) Hydrochloric acid in sodium hydroxide solution

(c)
- (i) What is a parasite?
- (ii) Give two examples of parasites and their hosts.

(d) Give the chemical symbols of the following elements
- (i) Oxygen
- (ii) Chlorine
- (iii) Aluminium
- (iv) Calcium

(e)
- (i) Explain the term self-pollination
- (ii) Name four agents of pollination
- (iii) State the function of the brightly coloured petals in insect pollination

2. (a) (i) Explain the term digestion

(ii) State the end products of digestion of each of the following food substances

(α) carbohydrates;

(β) proteins;

(γ) fats and oils

(iii) Name the enzyme present in saliva and state its function

(b) (i) By what process does the heat of the sun reach the earth?

(ii) An aluminium bowl placed on a table is filled with hot water. Name the processes by which heat is lost by the bowl.

(iii) Give one reason why wood is used as a handle for some cooking utensils.

(c) List separately the vegetative parts and reproductive parts of a flowering plant

(d) (i) What are alloys?

(ii) Give two examples of alloys and state their constituents

3. (a) (i) Explain the term friction

(ii) Give two methods of reducing friction

(iii) A rigid bar is used to lift a stone

(α) Draw a diagram showing the effort, pivot and load positions

(β) If the length of the bar is 100 cm and the pivot is placed 80 cm from the effort, calculate the load distance

(b) (i) Explain the term excretion

(ii) State the organs in human beings which excrete the following waste products:

(α) Sweat

(β) Carbon dioxide

(γ) Urine

(c) (i) What is a food chain?

(ii) Draw a food chain using the following organisms:
Hawk, cassava leaf, lizard and grasshopper

(d) State three conditions necessary for the germination of seed.

(e) (i) Name two diseases which can be transmitted sexually

(ii) Give two features which can be passed on from parents to children

4. (a) (i) List the various planets of the solar system in the order they occur starting with the planet nearest the sun

(ii) State one difference between the eclipse of the sun and the eclipse of the moon.

(b) (i) Define the term force

(ii) State three types of force

(iii) A box is pulled by a force of 20 N through a distance of 8m along a smooth floor. Calculate the work done by the force.

(c) Draw simple electrical circuits showing the following arrangements:

(i) A switch and a light bulb in series with two dry cells in series

(ii) A switch and a light bulb in series with two dry cells in parallel

(d) Name two diseases that can be controlled by immunization

August 1994

SCIENCE Section B

SOLUTIONS

ESSAY

1. (a) Difference

(i)

Physical change	Chemical change
No new substance is formed	A new substance is formed
It is usually reversible	It is irreversible (not reversible)

(ii)

Solution	Suspension
A liquid/fluid which has one or more other substances uniformly dissolved in it	a dispersion of fine solid particles in a liquid or gas

(b)

(i) **Common salt in water** - Physical change
(ii) **Burning of charcoal** - chemical change
(iii) **Melting of ice** - Physical change
(iv) **HCl in NaOH** - chemical change

(c) **(i)** **Parasite**

(ii) **Examples:**

PARASITE	HOST
Tapeworm	human / mammal
Tick	dog / cattle / sheep
Dodder	milk bush / alfalfa /clover
Broomrape	tomato crop
Mistletoe	apple / oak trees
Witch weed	sugarcane / corn
Plasmodium	human

Roundworm	human/ mammal
Hookworm	human/ mammal
Guinea worm	human/ mammal
Schistosome	human/ mammal

[any two]

(d)

	ELEMENT		CHEMICAL SYMBOLS
(i)	**Oxygen**	-	O
(ii)	**Chlorine**	-	Cl
(iii)	**Aluminium**	-	Al
(iv)	**Calcium**	-	Ca

(e) **(i)** **Self-pollination**

The transfer of pollen grains from an anther of a flower to the stigma of the same flower.

(ii) **Agents of pollination**

Wind, water, insects, birds, man,

(iii) **Function of brightly coloured petals**

To attract insects and birds

2. **(a)** **(i)** **Digestion**

The breakdown of food in the body into simpler forms that can be absorbed into the bloodstream and used

(ii)

			END PRODUCTS OF DIGESTION
(α)	**carbohydrates**	-	glucose
(β)	**proteins**	-	amino acids
(γ)	**fats and oils**	-	fatty acids and glycerol

(iii) **Enzyme in saliva and its function**

Ptyalin - break down starch into sugar

(b) **(i)** **process by which heat of the sun reach the earth**

Radiation

(ii) Conduction and convection

(iii) Wood is a bad conductor of heat, hence heat from the cooking utensil does not easily get to the hand.

(c)

Vegetative parts	Reproductive parts
stem roots leaves	flowers (stigma, style, ovary, anther & filament)

(d) (i) Alloys

Substance containing two or more metals physically and uniformly combined

Or: A homogeneous mixture of two or more metals

(ii)

ALLOY	**CONSTITUENTS**
Steel	iron and carbon
Bronze	copper and tin
Brass	copper and zinc
Duralumin	Aluminium, copper and manganese

3. (a) (i) Friction

The force that opposes the relative sliding motion of two bodies in contact

Or:

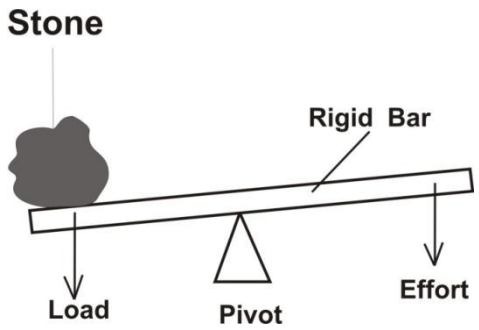
The resistance encountered by an object moving relative to another object with which it is in contact

(ii) Methods of reducing friction

- Applying lubricants such as grease / oil
- Smoothening the contact surfaces
- Using wheels / rollers
- Using ball bearings

(iii)

(α) **Diagram** showing effort, pivot and load positions



(β) **Load distance** = total length – effort distance

= 100 cm – 80 cm

= 20 cm

(b) (i) Excretion

The removal of metabolic waste materials from the cells of a body

Or:

The discharge of waste matter from the body's tissues or organs

(ii)

		EXCRETORY ORGAN
(α)	**Sweat**	Skin
(β)	**Carbon dioxide**	Lung
(γ)	**Urine**	Kidney

(c) (i) Food chain

The linear sequence that shows energy transfer from one living organism to the other by feeding or being fed upon

Or

A hierarchy of different living things, each of which feeds on the one below to obtain energy

(ii) Food chain

Cassava leaf → grasshopper → lizard → hawk

(d) Conditions necessary for germination of seed

Viable seed, suitable temperature, air, moisture / water [any three]

(e) (i) Diseases that can be transmitted sexually

- Acquired Immune Deficiency Syndrom (AIDS)
- Gonorrhea
- Genital warts
- Syphilis
- Hepatitis B
- Genital herpes

[any two]

(ii) Features that can be passed from parents to children

- Intelligence
- Height,
- Colour of eyes,
- Shape/size of nose/ear/mouth,
- Colour of skin,
- Ability to roll tongue [any two]

4. (a) (i) Planets of the solar system in order

Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune, (Pluto)

(ii)

Eclipse of the sun	Eclipse of the moon
The moon comes between the earth and the sun	The earth comes between the moon and the sun
The shadow of the moon is cast on the earth	The shadow of the earth is cast on the moon

[any one]

(b) (i) Force

A force is a push or pull exerted on a body.

Or:

A physical influence that tends to change the position or shape of an object with mass

(ii) Types of force

Gravitational, tensional, centripetal, centrifugal, magnetic, electrostatic, upthrust, cohesive, adhesive

[any three]

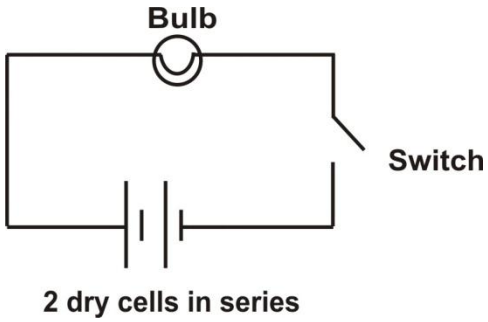
(iii) Work done = force × distance

= 20 N × 8 m

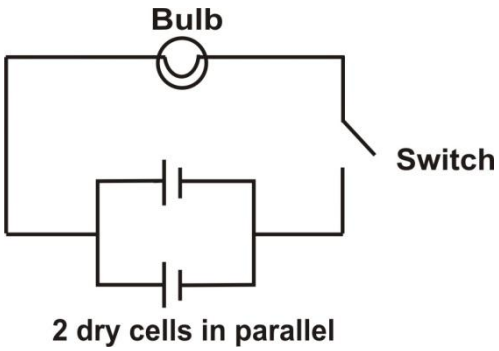
= 160 Nm or 160 J

(c) Electrical circuit showing:

(i) A switch and a light bulb in series with two dry cells in series



(ii) A switch and a light bulb in series with two dry cells in parallel



(d) Diseases that can be controlled by immunization

Smallpox, Measles, Chicken pox, Whooping cough, Tetanus, Poliomyelitis, Tuberculosis, Diphtheria

[any two]

August 1993

SCIENCE

SECTION A
OBJECTIVE TEST
45 minutes

1. The three states of matter are
- A. Water, liquid and gas
- B. Earth, liquid and gas
- C. Solid, liquid and air
- D. Gas, earth and water
- E. Gas, liquid and solid

2. Soil erosion ....
- A. washes away the water content of soil
- B. enables crops to grow better
- C. washes away the top soil
- D. leaves the soil free of pests
- E. increases the fertility of soils

3. What is the average body temperature of a healthy person?
- A. 25°C
- B. 36°C
- C. 37°C
- D. 38°C
- E. 39°C

4. Steel is an alloy of
- A. Iron and Carbon
- B. Zinc and Copper
- C. Tin and Copper
- D. Iron and Zinc
- E. Carbon and Zinc

5. Leguminous crops such as beans may often be cultivated to
- A. Add humus to the soil
- B. Add nitrogen to the soil
- C. Add protein to the soil
- D. Make the soil loamy
- E. Increase the calcium content of the soil

6. Which of the following is an excretory organ?

A. Anus
B. Eye
C. Heart
D. Skin
E. Pancreas

7. Which of the following is the function of the hair in the nose? It

A. keeps the nose warm
B. makes breathing easier
C. keeps the nose moist
D. prevents carbon dioxide from entering the body
E. filters dust from the air breathed in

8. Which of the following is equal to one metre?

A. 1 cm
B. 10 cm
C. 100 cm
D. 500 cm
E. 1000 cm

9. Which of the following statements about feeding relationships is **not** true?

A. Energy is transferred from one organism to another
B. A green plant is the basis of any food chain
C. Bacteria feed on dead organisms for energy
D. A carnivore feeds on green plants for energy
E. Omnivores feed on both plants and animals

10. In a simple machine, the force that is applied to enable work to be done is known as

A. Lever
B. Load
C. Pivot
D. Effort
E. Hinge

11. Which of the following changes does **not** occur in a girl at the age of puberty?

A. The breasts develop
B. The hips widen
C. Menstruation
D. The teeth becomes whiter
E. Hair grows in the pubic area

12. The fixed point of a lever at which turning takes place is the

A. pivot

B. effort
C. load
D. handle
E. machine

13. Which of the following mixtures can be separated by filtration?
A. Salt in water
B. Sugar in water
C. Sand in water
D. Oil in water
E. Omo in water

14. The skin is made up of
A. One layer
B. Two layers
C. Three layers
D. Four layers
E. Six layers

15. The force which acts between planets and keeps them in their orbits is called
A. electrical force
B. magnetic force
C. gravitational force
D. frictional force
E. elastic force

16. Which of the following types of soil allows the fastest rate of drainage?
A. Clayey soil
B. Loamy soil
C. Black soil
D. Sandy soil
E. Humus

17. Which of the following is the percentage composition of oxygen in the air?
A. 0.03
B. 1
C. 21
D. 78
E. 98

18. Disposal of sewage into a river which supplies water for domestic use may lead to an outbreak of
A. malaria
B. cholera
C. guinea worm
D. river blindness

E. tuberculosis

19. The process by which a naphthalene ball (camphor) placed in a box gets smaller in size is termed
A. Melting
B. Diffusion
C. Evaporation
D. Sublimation
E. Condensation

20. Which of the following food substances is not oxidized during respiration?
A. Fatty acids
B. Amino acids
C. Glucose
D. Cellulose
E. Vitamins

21. Kerosene and petrol are obtained from crude oil by
A. distillation
B. condensation
C. filtration
D. evaporation
E. decantation

22. Which of the following encourages the spread of venereal diseases?
A. Poor family planning
B. Drug Abuse
C. Mental ill health
D. Parental wealth
E. Indiscriminate sexual intercourse

23. A torch light bulb lights up when connected to a dry cell. What type of energy is obtained from the cell?
A. Chemical energy
B. Light energy
C. Potential energy
D. Electrical energy
E. Heat energy

24. Which of the following is not a soil nutrient?
A. Hydrogen
B. Nitrogen
C. Potassium
D. Calcium
E. Phosphorus

25. Which of the following organisms causes bilharziasis?

A. Tapeworm
B. Blood fluke
C. Blood worm
D. Round worm
E. Liver fluke

26. A piece of stone sinks in water because
A. its density is the same as that of water
B. it is able to displace its own weight of water
C. its density is less than that of water
D. its density is greater than that of water
E. the water contains air which allows the stone to sink

27. Which of the following is characteristic of reflex action
A. It happens slowly
B. It happens consciously
C. It takes a long path
D. It involves thinking
E. It is automatic

28. All the following are sense organs except the
A. eye
B. brain
C. ear
D. skin
E. tongue

29. A mixture of raw starch and water is best separated by
A. boiling
B. distillation
C. decantation
D. winnowing
E. crystallization

30. Which of the following is a horizontal underground stem?
A. Cocoyam
B. Onion
C. Carrot
D. Yam
E. Ginger

31. Which of the following is a physical change?
A. Heating of sugar
B. Heating of sulphur and iron filing
C. Boiling of water

D. Burning of magnesium ribbon
E. Heating of potassium chlorate

32. In an experiment in which a variegated leaf is used, which of the following is proved to be necessary for photosynthesis?
A. Carbon dioxide
B. Light
C. Chlorophyll
D. Water
E. Oxygen

33. Friction is defined as a force which
A. accompanies motion
B. acts against motion
C. causes motion
D. makes motion easy
E. is created by motion

34. Which of the following is **not** a function of the lens in the human eye? It
A. allows light rays to pass through it
B. refracts the light rays passing through it
C. helps to focus the image onto the retina
D. becomes thinner when focusing a distant object
E. reflects part of the light rays falling on it.

35. Which of the following practices in a community will increase its environmental pollution?
A. Building of schools with workshops
B. Planting of trees on large areas of land
C. Depositing of household rubbish near a source of drinking water
D. Removal of sand and refuse from choked gutters
E. building of an efficient underground sewage system

36. Smoke is an example of a mixture of
A. Gases
B. Liquids in gases
C. Solids in liquids
D. solids in gases
E. solids in solids

37. Which of the following is the best method of preventing skin disease?
A. The use of perfume on the skin
B. The use of bleaching cream on the skin
C. Avoiding contact with diseased person
D. keeping the skin clean by washing it regularly
E. Changing of clothes regularly

38. Which of the following has a different physical state from the others under ordinary conditions?

A. Zinc
B. Lead
C. Iron
D. Mercury
E. Aluminium

39. Heat from the sun reaches the earth by

A. conduction
B. convection
C. radiation
D. reflection
E. refraction

40. During an eclipse of the moon, the

A. Sun and the moon are on one side of the earth
B. Earth and the sun are on one side of the moon
C. Sun comes between earth and moon in a straight line
D. Earth comes between sun and moon in a straight line
E. Moon comes between sun and earth in a straight line

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SCIENCE Section A

SOLUTIONS

OBJECTIVE TEST

1.	E.	Gas, liquid and solid
2.	C.	washes away the top soil
3.	C.	37°C
4.	A.	Iron and Carbon
5.	B.	Add nitrogen to the soil
6.	D.	Skin
7.	E.	filters dust from the air breathed in
8.	C.	100 cm
9.	D.	A carnivore feeds on green plants for energy
10.	D.	Effort
11.	D.	The teeth becomes whiter
12.	A.	pivot
13.	C.	Sand in water
14.	B.	Two layers
15.	C.	gravitational force
16.	D.	Sandy soil
17.	C.	21
18.	B.	cholera
19.	D.	Sublimation

20. E. Vitamins

21. A. distillation

22. E. Indiscriminate sexual intercourse

23. D. Electrical energy

24. A. Hydrogen

25. B. Blood fluke

26. D. its density is greater than that of water

27. E. It is automatic

28. B. brain

29. C. decantation

30. E. Ginger

31. C. Boiling of water

32. C. Chlorophyll

33. B. acts against motion

34. E. reflects part of the light rays falling on it.

35. C. Depositing of household rubbish near a source of drinking water

36. D. solids in gases

37. D. keeping the skin clean by washing it regularly

38. D. Mercury

39. C. radiation

40. D. Earth comes between sun and moon in a straight line

August 1993

SCIENCE

SECTION B
ESSAY
1 hour

[60 marks]

Answer **three** *questions* **only** *from this section.*
Illustrate your answers wherever possible, with large, clear and fully labelled diagrams.
Credit will be given for clarity of expression and orderly presentation of material.
All questions carry equal marks

1. (a) Define the following terms and give one example of each:
(i) Element
(ii) Mixture
(iii) Compound

(b) Explain the following terms:
(i) Work
(ii) Energy

(c) State the units in which each of the following are measured:
(i) Power
(ii) Energy
(iii) Electric current
(iv) Temperature

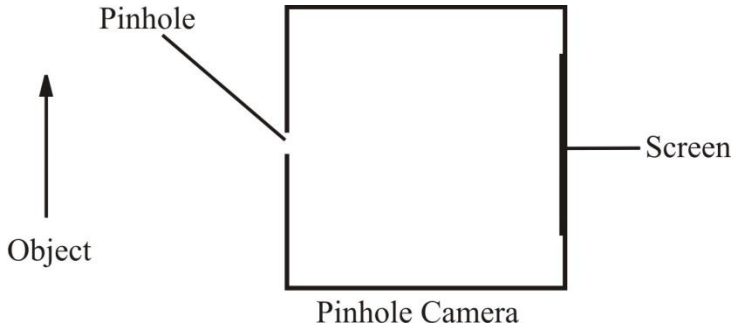
(d) A body of mass 100 kg is placed at a point 2 m above the ground. What is the potential energy of the body with reference to the ground? [Take g = 10 ms^{-2}]

(e) State three characteristics of each of the following:
(i) An insect pollinated flower
(ii) A wind pollinated flower

2. (a) (i) Explain the term balanced diet.

(ii) Name any two classes of food which should be present in a balanced diet.
(iii) Give one function of each class of food named in (a) (ii) above.

(b) An object is placed in front of a pinhole camera as shown in the diagram below:



Draw in your answer book a ray diagram (not to scale) to show how the image is formed on the screen of the camera.

(c) State the two laws of reflection of light.

(d) (i) Describe how ions are formed from atoms.
(ii) State two differences between a proton and an electron.

3. (a) (i) Explain the term electromagnetism.
(ii) Given a nail, insulated wire, key (or switch) and a battery, show by means of a circuit diagram how the nail can be magnetized.

(b) List three symptoms of malaria.

(c) State three ways of keeping the environment clean.

(d) Write down the names of the new substances formed when the following compounds react:
(i) Ammonia and dilute hydrochloric acid;
(ii) Sodium hydroxide and dilute hydrochloric acid;
(iii) Calcium carbonate and dilute hydrochloric acid.

4. (a) (i) What is meant by a satellite?
(ii) State one difference between a natural and an artificial satellite.
(iii) State two uses of artificial satellite.

(b) Give two differences between the mass and weight of a body.

(c) Write the chemical formulae for the following substances.
- (i) Calcium oxide;
- (ii) Water;
- (iii) Copper sulphate

(d) What will be the effect on the colour of blue litmus paper when it is dropped in each of the following solutions?
- (i) Sodium hydroxide solution;
- (ii) Dilute hydrochloric acid;
- (iii) Lime juice;
- (iv) Water

(e)
- (i) Define breathing.
- (ii) Arrange the following parts of the human being respiratory system in order in which air from the atmosphere pass, through them during breathing:
  Trachea, nostrils, lungs and bronchus.
- (iii) Describe briefly what happens to a piece of bread when it reaches the duodenum of a human being.

August 1993

SCIENCE Section B

SOLUTIONS

ESSAY

1. (a) (i) Element

A substance that is made up of the same kind of atoms

Or:

Any substance that cannot be broken down into a simpler one by a chemical reaction

(ii) Mixture

A physical combination of two or more substances

Or:

A substance consisting of two or more substances that have been combined physically

(iii) Compound

A substance formed by the chemical combination of elements in fixed proportions

Or:

A substance containing two or more elements that are chemically combined

(b) Explanation

(i) Work

Work is done when an applied force moves an object in the direction of the force.

Or:

The transfer of energy, measured as the product of the force applied to a body and the distance moved by that body in the direction of the force

Or:

Work $= f \times d$, where f = force applied

and d = distance moved in the direction of the applied force

(ii) Energy

The ability / capacity to do work

(c)

			UNIT
(i)	**Power**	-	watt (W)
(ii)	**Energy**	-	joule (J)
(iii)	**Electric current**	-	ampere (A)
(iv)	**Temperature**	-	Kelvin (K) **or** degree Celsius (°C) **or** degree Fahrenheit (°F)

(d) **Potential Energy** = mass × height × acceleration due to gravity

= 100 kg × 2 m × 10 ms^{-2}

= $2000kgm^2s^{-2}$ or 2000 N

(e) **Characteristics**

(i) **An insect pollinated flower**

- Petals are brightly coloured
- Larger size of flowers
- Nectar present
- Attractive scent
- Relatively fewer but larger pollen grains

[any three]

(ii) **a wind pollinated flower**

- Dull coloured petals
- Nectar absent
- No or unattractive scent
- Relatively more but smaller pollen grains

[any three]

2. **(a)** **(i)** **Balanced diet**

A diet that contains all the essential food nutrients in their right proportions

(ii) **Classes of food**

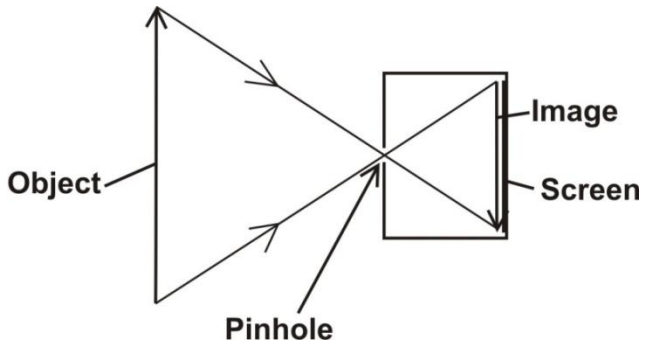
Carbohydrates, proteins, fats and oils, vitamins and minerals

(iii) **Function(s) of each class**

Food Class	Function(s)
Carbohydrates	▪ Provide energy to the body ▪ Glucose helps the brain cells to function properly ▪ Glucose help maintain tissue protein ▪ Glucose help metabolize fats ▪ Glucose fuels the central nervous system
Proteins	▪ Build the body cells / tissues ▪ Repair worn out /damaged body cells/tissues ▪ Help speed up certain chemical reactions in the body ▪ Help provide energy to the body if carbohydrates and fats are insufficient
Fats and oils	▪ Protect the body from extreme cold or heat ▪ Serve as a store of energy for the body ▪ Help in building the membranes that surround the body cells

	▪ Help the body to absorb certain vitamins
Vitamins	▪ Strengthens the body's immune system ▪ Helps the body to fight against diseases ▪ Facilitates the metabolic activities in the body
Minerals	▪ Ensure healthy growth of teeth and bones ▪ Help with enzyme action ▪ Help with muscle contraction ▪ Help with nerve reaction ▪ Help with blood clotting

(b) Diagram to show pinhole camera image



(c) Laws of reflection of light

- The angle of incidence is equal to the angle of reflection
- The incident ray, the reflected ray and the normal, at the point of incidence, all lie in the same plane

(d) (i) How ions are formed from atoms

Atoms become charged when they lose or gain electrons in order to be more stable or attain a completely filled outer shell. When atoms gain electrons, they become negatively charged ions, known as anions; when they lose electrons, they become positively charged ions, known as cations

(ii)

PROTON	ELECTRON
positively charged	negatively charged
located in the nucleus of an atom	located outside the nucleus, on the shells of an atom
relatively heavy	relatively very light

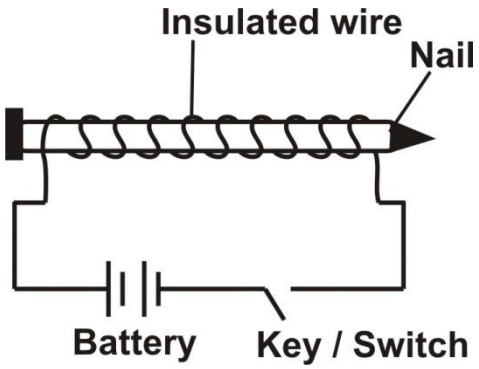
3. (a) (i) Electromagnetism

Magnetism produced by an electric current

or:

The process of making a magnet using an electric current

(ii) Diagram showing how nail can be magnetized



(b) Symptoms of malaria

- fever
- body pains
- feeling cold or hot
- headache
- nausea
- lack of appetite [any three]

(c) Ways of keeping the environment clean

- Clearing choked gutters
- Clearing weedy areas
- Burying empty cans
- Covering rubbish bins
- Sweeping dirty areas
- Scrubbing toilets and bathrooms

(D) New Substance Formed

(i) Ammonia (NH_3) and dilute hydrochloric acid (HCl);

NEW SUBSTANCE: Ammonium chloride (NH_4Cl)

(ii) Sodium hydroxide (NaOH) and dilute hydrochloric acid (HCl);

NEW SUBSTANCE: Sodium chloride (NaCl) and water (H_2O)

(iii) Calcium carbonate ($CaCO_3$) and dilute hydrochloric acid (HCl).

NEW SUBSTANCE: Calcium chloride ($CaCl_2$), Carbon dioxide (CO_2) and water (H_2O)

4. (a) (i) Satellite

A satellite is any heavenly body that orbits (revolves around) a larger one.

(ii)

Natural Satellite	Artificial Satellite
Created by God (or nature)	Manufactured by man
Has no mechanical or electrical devices	Has man-made mechanical or electrical devices

[any one]

(iii) Uses of artificial satellites

- Weather studies and forecasting
- Telecommunication
- Sending and receiving TV signals
- Taking photographs from space for various purposes
- Internet networking
- Global Positioning System (GPS) - a worldwide navigation system

[any two]

(b)

MASS	WEIGHT
the amount of matter a substance contains	the force exerted on a body as a result of gravity
remains constant (unchanging) regardless of location	varies with location (depending on gravitational force acting at that point)
Units = kilogram (kg)	Units = newton (N)

(c) **CHEMICAL FORMULA**

(i) Calcium oxide; CaO

(ii) Water; H_2O

(iii) Copper sulphate $CuSO_4$

(d) **EFFECT ON BLUE LITMUS PAPER**

(i) Sodium hydroxide solution; - Remains blue

(ii) Dilute hydrochloric acid; - Turns red

(iii) Lime juice; - Turns red

(iv) Water - Turns purple

(e) (i) Breathing

The process of taking air into the lungs and pushing it out again

(ii) Arrange in order:

Nostrils, Trachea, bronchus and lungs

(iii) Bread at the duodenum

Enzymes in the pancreatic juice, from the pancreas, break down the bread (sugars and starches) into simple sugars.

August 1992

SCIENCE

SECTION A
OBJECTIVE TEST
45 minutes

1. Which of the following is true about matter? It has
- A. Weight and direction
- B. Weight only
- C. Weight and volume
- D. Volume and direction
- E. Volume only

2. The combining power of an element is called
- A. ion
- B. electron
- C. atom
- D. valency
- E. proton

3. Why is it bad to throw faeces into rivers? Because
- A. Other people collect drinking water from them
- B. It changes the colour of the river
- C. It causes hardness of the water
- D. The faeces cannot dissolve in the water
- E. The river banks will be dirty

4. Which of the following methods will be most suitable for separating a mixture of iron filings and sulphur powder?
- A. Winnowing
- B. Decanting
- C. Magnetization
- D. Evaporation
- E. Distillation

5. During athletics, a boy slipped on a banana peel on the ground and fell. This is because
- A. There was no friction
- B. The ground was hard
- C. The banana peel was soft

D. The ground was sandy
E. The frictional force was too great

6. Which of the following is a rhizome?
A. Potato
B. Ginger
C. Onion
D. Cocoyam
E. Cassava

7. How many different elements are present in the compound $CuCl_2$?
A. Five
B. Four
C. Three
D. Two
E. One

8. Which of the following energy changes takes place when a church bell is tolled?
A. Potential energy to kinetic energy
B. Kinetic energy to sound energy
C. Potential energy to sound energy
D. Chemical energy to heat energy
E. Sound energy to Kinetic energy

9. Living things are different from non-living things because living things
A. are made up of protoplasm
B. do not have definite forms
C. take in liquid foods
D. have chlorophyll
E. vary widely in form and size

10. Which of the following gases enable burning to take place?
A. Oxygen
B. Carbon dioxide
C. Hydrogen
D. Ammonia
E. Nitrogen

11. The work done when a weight of 40 N is lifted up vertically through a distance of 10 m is
A. 400 J
B. 50 J
C. 30 J
D. 4 J
E. 0.25 J

12. Which of the following products is **not** excreted by the skin?

A. Water
B. Sweat
C. Urea
D. Common salt
E. saliva

13. An element is formed by the combination of

A. an atom and a compound
B. a compound and a mixture
C. an atom and a mixture
D. a molecule and a compound
E. the same kind of atoms

14. Energy is measured in

A. Kelvin
B. Joules
C. Watts
D. Metres
E. Amperes

15. Which of the following is **not** a characteristic of all living things?

A. Transpiration
B. Growth
C. Reproduction
D. Feeding
E. Respiration

16. Which of the following substances is a mixture?

A. Sodium chloride
B. Sulphur dioxide
C. Sugar solution
D. Carbon dioxide
E. Ammonium chloride

17. The best reason why a cooking pan should have a wooden handle is that, wood is a

A. good conductor of heat
B. bad conductor of heat
C. good radiator of heat
D. bad radiator of heat
E. good reflector of heat

18. For healthy growth, the food we eat must contain

A. fat, oils and proteins
B. important nutrients in the right proportions

C. a good proportion of carbohydrates and proteins
D. mineral salts, vitamins and water
E. mineral salts in right proportion

19. When the testa of a soaked bean is removed, the seed is seen to be made up of mainly
A. micropyles
B. hilum
C. radicle
D. embryo
E. cotyledons

20. Which of the following human activities causes pollution of the environment?
A. Cutting down timber in the forest for firewood
B. Overgrazing of the field by cattle
C. Burning of coal and petrol in industries
D. Building dams across rivers
E. Destruction of diseased crops

21. Which of the following explains why no work is done when a push is made against a firm wall?
A. The force acts in the opposite direction
B. The wall does not move
C. The force is in the same direction
D. The material of the wall is strong
E. The force is in horizontal direction

22. The process by which a hot saturated solution is cooled to obtain some of the solute is called
A. condensation
B. evaporation
C. crystallization
D. distillation
E. sedimentation

23. Which of the following diseases is caused by a fungus?
A. Typhoid
B. Leprosy
C. Ringworm
D. Measles
E. Dysentery

24. Stem cutting is used in propagating cassava because it
A. gives rise to roots and leaves
B. stores food in the stem
C. gives rise to big tubers
D. gives rise to small tubers

E. stores food in the leaves

25. Which of the following methods is **not** a way of preventing malaria parasites from person to person?
A. Sleeping in mosquito nets
B. Sleeping outside in the open
C. Draining all stagnant pools of water around our homes
D. Taking anti-malaria drugs weekly
E. Burying all empty cans that hold water in our surroundings

26. Faeces is the
A. excretory products resulting from body activities
B. food which is digested
C. food in the mouth which is removed from the body
D. unwanted product of digestion which is removed from the body
E. excess food which is stored in the body

27. Which of the following explains why a wet cloth dries faster on a dry day than on a humid day?
A. The rate of evaporation is very high during a dry day
B. There is much water vapour in the atmosphere during a dry day
C. It is very misty during a dry day
D. there is no evaporation during a humid day
E. The sun does not set during a humid day

28. Urinating in rivers is a common manner of spreading diseases such as
A. cholera
B. bilharzia
C. malaria
D. measles
E. Typhoid

29. Which of the following is **not** a way of preventing soil erosion?
A. Planting trees on bare soil.
B. Clearing trees along slopes
C. Planting grasses along slopes
D. Ploughing across slopes
E. Preventing bush fire

30. Atmospheric pressure is measured with
A. an anemometer
B. a thermometer
C. a galvanometer
D. a barometer
E. a voltmeter

31. Which of the following is not matter?

A. Iced block
B. Air
C. Light
D. Smoke
E. Stone

32. Regular brushing of teeth is primarily done in order to
A. keep the mouth fresh
B. clear away food particles from the teeth
C. remove bad smell in the mouth
D. kill bacteria in the mouth
E. control the flow of saliva

33. On heating a certain solid in a dry container, all the substance was lost without passing through the liquid state. This means the
A. solid can easily melt
B. solid is an alloy
C. solid cannot stand heat
D. solid particles are held together by weak forces
E. solid undergoes sublimation

34. Which of the following forces tends to pull forces towards the centre of the earth?
A. Electrostatic force
B. Magnetic force
C. Force of gravity
D. Force of attraction
E. Centrifugal force

35. Which of the following diseases will result from eating improperly cooked meat?
A. Tapeworm infestation
B. Guinea worm infestation
C. Small pox
D. Leprosy
E. Measles

36. The angle between a ray of light and the surface it strikes is 30°. Calculate the angle of incidence.
A. 210°
B. 120°
C. 90°
D. 60°
E. 30°

37. The Central Nervous System consists of
A. the brain and the spinal cord
B. the brain and the ears

C. the spinal cord and eyes
D. the brain, eyes and ears
E. eyes, ears, skin and spinal cord

38. Which of the following classes of food can be broken down into amino acids?
A. Carbohydrates
B. Vitamins
C. Mineral salts
D. Fats and oils
E. Proteins

39. Which of the following is the name of the darkest part of a shadow formed when a large source of light falls on an object?
A. Partial eclipse
B. Lunar eclipse
C. Total eclipse
D. Penumbra
E. Umbra

40. Which of the following is evidence that particles in gases move faster than those in liquids and solids?
I. The scent of a perfume fills a room when its container is opened
II. Smoke spreads over a large area as burning takes place
III. The wind blows fast when it is about to rain

A. I only
B. III only
C. II only
D. I and II only
E. I, II and III

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SCIENCE Section A

SOLUTIONS

OBJECTIVE TEST

1.	C.	Weight and volume
2.	D.	valency
3.	A.	Other people collect drinking water from them
4.	C.	Magnetization
5.	A.	There was no friction
6.	B.	Ginger
7.	D.	Two
8.	B.	Kinetic energy to sound energy
9.	A.	are made up of protoplasm
10.	A.	Oxygen
11.	A.	400 J
12.	E.	saliva
13.	E.	the same kind of atoms
14.	B.	Joules
15.	A.	Transpiration
16.	C.	Sugar solution
17.	B.	bad conductor of heat
18.	B.	important nutrients in the right proportions

19. E. cotyledons

20. C. Burning of coal and petrol in industries

21. B. The wall does not move

22. C. crystallization

23. C. Ringworm

24. A. gives rise to roots and leaves

25. B. Sleeping outside in the open

26. D. unwanted product of digestion which is removed from the body

27. A. The rate of evaporation is very high during a dry day

28. B. bilharzia

29. B. Clearing trees along slopes

30. D. a barometer

31. C. Light

32. B. clear away food particles from the teeth

33. E. solid undergoes sublimation

34. C. Force of gravity

35. A. Tapeworm infestation

36. D. 60°

37. A. the brain and the spinal cord

38. E. Proteins

39. E. Umbra

40. D. I and II only

August 1992

SCIENCE

SECTION B
ESSAY
1 hour

[60 marks]

Answer **three** *questions* **only** *from this section.*
Illustrate your answers wherever possible, with large, clear and fully labelled diagrams.
Credit will be given for clarity of expression and orderly presentation of material.
All questions carry equal marks

1. (a) (i) Define density of a substance
(ii) Explain how the density of a stone could be determined

(b) A box with a volume of 1000 m^3 is filled with sand to the brim. If the density of sand is 2 kgm^{-3}, calculate the mass of sand in the box.

(c) Copy and complete the table below:

Parasite	**Disease**	**Prevention**
Vibrio cholera	Cholera	Eating hot food
Plasmodium	(i)	(ii)
Louse	(iii)	Personal hygiene
(iv)	Ringworm	(v)

(d) Name the method which could be used to separate each of the following mixtures into their components:
(i) Alcohol and water
(ii) Salt and water
(iii) Powdered charcoal and iron filings

(iv) Powdered chalk and water

2. (a) (i) Name the process by which green plants manufacture their food.
(ii) State three factors necessary for this process to take place.

(b) (i) Mention the device which could be used to carry a heavy body onto a truck.
(ii) Name four simple levers used in everyday life.
(iii) Draw a lever and label the *pivot, effort* and *load* on it

(c) (i) Using any ammonium compound, describe the general preparation of ammonia gas. (Diagram of apparatus is not required)
(ii) Write the chemical equation for the preparation

3. (a) A sample of garden soil is put in a cylinder containing water. It is shaken and allowed to settle. Draw and label the layers formed.

(b) (i) Name four types of forces.
(ii) A boy used a catapult to pluck a mango from a tree. State two forces involved in this activity.
(iii) Where are the forces in (b) (ii) applied?

(c) Describe an experiment to show that there is change in heat energy when a new substance is formed.

4. (a) (i) Mention the three states of matter.
(ii) State the two methods by which one state can be converted to another.

(b) A bag of cement is pulled along a smooth horizontal floor with a horizontal force of 1000 N. If the work done is 100,000 J, calculate the distance through which the bag is pulled.

(c) (i) Draw the side view of a bony fish and label the parts.
(ii) Explain how the fish takes in oxygen.
(iii) For each of the following animals name two parts which help it to live in its environment
(α) Bird
(β) Fish

August 1992

SCIENCE Section B

SOLUTIONS

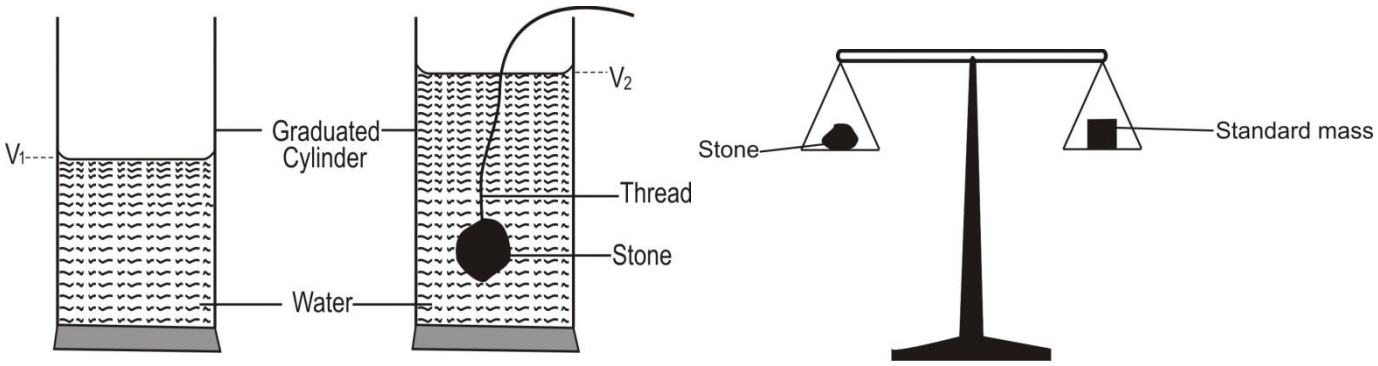
ESSAY

1. (a) (i) Density of a substance

The mass per unit volume of the substance

Or: Density = mass / volume

(ii) Finding density of stone



I. Measure the mass(M) of the stone using a beam balance

II. Fill a measuring cylinder with water to a suitable volume.

III. Record the volume (V_1)

IV. Tie a thread / string around the stone and gently lower it into the water in the cylinder.

V. Record the new volume (V_2)

VI. Find difference in volumes ($V_2 - V_1$)

VII. Volume of metal ball = $(V_2 - V_1)$

VIII. Find the ratio M / $(V_2 - V_1)$ = Density of stone

(b) Mass of sand in the box = Density × volume

= $2\ kgm^{-3} \times 1000\ m^3$

= 2000 kg

(c)

Parasite	Disease	Prevention
Vibrio cholera	Cholera	Eating hot food
Plasmodium	**(i)** Malaria	**(ii)** ▪ Sleeping under insecticide treated mosquito nets, ▪ applying mosquito repellents on the body, ▪ burying empty cans, ▪ clearing bushy areas, ▪ spraying insecticides in room, ▪ clearing choked drains, [any one]
Louse	**(iii)** ▪ relapsing fever ▪ typhus ▪ anaemia [any one]	Personal hygiene
(iv) fungus	Ringworm	**(v)** ▪ Keep skin clean and dry ▪ Avoid contact with infected persons or animals ▪ Avoid sharing personal items like clothes, combs or towels ▪ Use antifungal medication [any one]

(d) **SEPARATION METHOD**

(i)	**Alcohol and water**	-	Distillation
(ii)	**Salt and water**	-	Evaporation
(iii)	**Powdered charcoal and iron filings**	-	magnetization
(iv)	**Powdered chalk and water**	-	filtration

2. **(a)** **(i)** **The process by which green plants manufacture their food**
Photosynthesis

(ii) **Factors necessary for this process to take place**
Carbon dioxide, water, chlorophyll, light (sunlight)

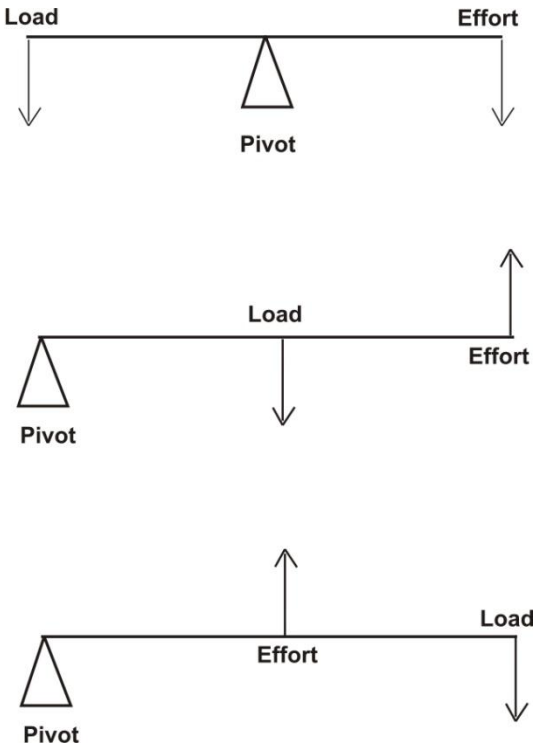
(b) (i) Device that could be used to carry a heavy body onto a truck

Inclined plane, forklift, pulley, crane , [any one]

(ii) Simple levers used in everyday life

Wheel barrow, crowbar, sugar tongs, scissors, knife, forceps, claw hammer, bottle opener, see-saw, human forearm, nutcracker, etc [any four]

(iii) Diagram of lever



[any one]

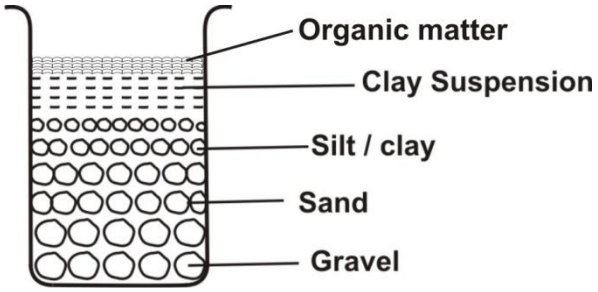
(c) (i) Preparation of ammonia gas

It can be prepared by heating a mixture of ammonium chloride and slaked lime (calcium hydroxide). The ammonium gas produced is passed through calcium oxide to remove moisture (water)

(ii) Chemical equation for ammonia preparation

$2\ NH_4Cl + Ca(OH)_2 \rightarrow CaCl_2 + 2\ NH_3 + 2H_20$

3. (a) Soil profile diagram



(b) (i) Types of forces

Gravitational, frictional, tensional, centripetal, centrifugal, magnetic, electrostatic, upthrust, cohesive, adhesive

(ii) Forces involved in catapult to pluck mango

Tensional force and gravity

(iii) Where the forces are applied

- Tensional force in the stretched elastic of the catapult moved the stone
- Force of gravity pulls the mango to the ground

(c) An experiment to show change in heat energy when new substance is formed

Apparatus:

Beaker, water, calcium chloride, stirrer, thermometer

Method:

- The temperatures of the water and calcium chloride are measured separately and recorded
- The water and calcium chloride are combined in the beaker, stirred continuously and the temperature of the resultant solution measured

Observation:

It is observed that the temperature of the resultant solution is higher than the temperatures of the separate substances.

Conclusion:

There is a change in heat energy when a new substance is formed.

4. (a) (i) States of matter

Solid, liquid and gas

(ii) Methods by which one state can be converted to another

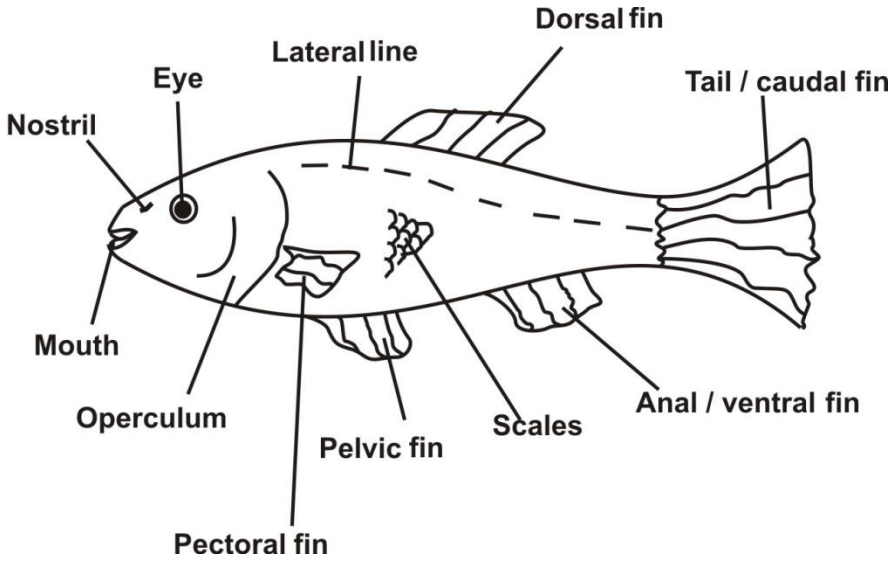
Heating and cooling

(b) Distance = Work / force

= 100,000 J / 1000 N

= <u>100 m</u>

(c) (i) Labelled diagram showing side view of a bony fish



(ii) How the fish takes in oxygen

The fish takes in water through the mouth. The water passes over the gills. The dissolved oxygen in water diffuses into the gills and enters the bloodstream

(iii) Parts that help it to live in environment

(α) Bird

Feathers, hollow bones, beak, wings, streamlined body, air sac, claws

[any two]

(β) Fish

Scales, fins, gills, lateral line, nostril, tail, mouth

[any two]

August 1991

SCIENCE

SECTION A
OBJECTIVE TEST
45 minutes

1. The digestion of carbohydrates starts in the
- A. mouth
- B. throat
- C. stomach
- D. duodenum
- E. ileum

2. Photosynthesis in plants requires chlorophyll, sunlight, water and
- A. oxygen
- B. carbon dioxide
- C. steam
- D. hydrogen
- E. nitrogen

3. Which of the following objects is a good conductor of heat?
- A. Wood
- B. Rubber
- C. Copper
- D. Plastic
- E. Wool

4. What is the work done when a force of 2.5 N moves through a distance of 4 m?
- A. 0.6 J
- B. 1.5 J
- C. 1.6 J
- D. 6.5 J
- E. 10.0 J

5. The person who studies the weather is
- A. a pilot
- B. a physician
- C. an astronaut
- D. an astrologer
- E. meteorologist

6. A child suffering from Kwashiorkor lacks
- A. protein
- B. fats and oils
- C. vitamins
- D. mineral salts
- E. carbohydrates

7. Which of the following substances can dissolve almost all solutes?
- A. Kerosene
- B. Palm oil
- C. Water
- D. Turpentine
- E. Petrol

8. All of the following methods are used to preserve food in our homes except
- A. smoking
- B. salting
- C. boiling
- D. canning
- E. drying

9. The chemical symbol for Potassium is
- A. Fe
- B. K
- C. P
- D. Pb
- E. Zn

10. The part of the banana plant used in reproduction is called
- A. sucker
- B. rhizome
- C. stem
- D. tuber
- E. bulb

11. The process by which kerosene rises up the wick in a lantern is
- A. evaporation
- B. conduction
- C. diffusion
- D. capillarity
- E. convection

12. Which of the following animals can live both in water and on land?
- A. Mouse

B. Rat
C. Frog
D. Grasscutter
E. Mud fish

13. A bar magnet attracts all the following objects except
A. pins
B. pair of scissors
C. nails
D. razor blade
E. glass

14. When an object moves from rough surface onto a smooth one with the same energy, its speed
A. increases
B. decreases
C. remains the same
D. becomes zero
E. cannot be determined

15. How many states of matter do we have?
A. 2
B. 3
C. 4
D. 5
E. 6

16. All of the following animals lay eggs except
A. fish
B. lizard
C. rabbit
D. toad
E. turkey

17. Which of these is not a sense organ?
A. Eye
B. Ear
C. Nose
D. Skin
E. Heart

18. Which of the following processes is used to separate insoluble solids from liquids?
A. Distillation
B. Sublimation
C. Filtration
D. Evaporation

E. Sedimentation

19. An organism living on another organism and causing harm to it is referred to as a
A. host
B. parasite
C. producer
D. consumer
E. vector

20. What sickness are you likely to be suffering from when you have blood in your urine?
A. Typhoid
B. Syphilis
C. Cholera
D. Bilharzia
E. Ulcer

21. Which of these is a vegetable crop?
A. Pineapple
B. Potato
C. Cabbage
D. Cassava
E. Mango

22. Insects can walk on the surface of water because of
A. diffusion
B. capillarity
C. viscosity
D. friction
E. surface tension

23. When a substance is made up of the same kind of atoms it is called
A. a mixture
B. a compound
C. an element
D. a solution
E. a solvent

24. What do human beings obtain from food to enable them work?
A. Blood
B. Energy
C. Vitamins
D. Proteins
E. Minerals

25. Which of these instruments is used in measuring humidity?

A. Hygrometer
B. Thermometer
C. Photometer
D. Potometer
E. Anemometer

26. The part of the plant which is found in the soil is the
A. flower
B. stem
C. leaf
D. root
E. fruit

27. When salt is put into soups, it
A. melts
B. sublimes
C. dissolves
D. evaporates
E. expands

28. The complete development of a human baby in the womb normally takes
A. five months
B. six months
C. seven months
D. nine months
E. eleven months

29. The washing away of the top soil by running water is called
A. erosion
B. rotation
C. plantation
D. weathering
E. irrigation

30. All of the following are solids except
A. aluminium
B. lead
C. silver
D. potassium
E. mercury

31. The process by which living things increase in number is referred to as
A. growth
B. development
C. reproduction

D. multiplication
E. addition

32. Which of the following soil types can hold the least amount of water?
A. Loam
B. Clay
C. Silt
D. Sand
E. Sandy loam

33. When a piece of candle is heated, it changes from
A. liquid to gas
B. liquid to flame
C. solid to flame
D. solid to liquid
E. solid to gas

34. Which of the following metals is used for making roofing sheets?
A. Zinc
B. Silver
C. Mercury
D. Tin
E. Aluminium

35. What is the name of the final product of protein digestion?
A. Glucose
B. Fatty acid
C. Amino acid
D. Glycerol
E. Sucrose

36. The lion is a typical example of
A. a pet
B. a carnivore
C. an omnivore
D. a herbivore
E. a parasite

37. A uniform mixture of two or more metals is called
A. a solute
B. a colloid
C. a compound
D. a suspension
E. an alloy

38. Which of the following activities illustrates distillation?

A. Preparation of soup
B. Preparation of akpeteshie
C. Tapping of palm wine
D. Preparation of starch
E. Freezing of water

39. The hatching of eggs into chicken depends on constant supply of

A. water
B. feed
C. pressure
D. heat
E. light

40. A fertilized ovule results in the formation of

A. an ovary
B. a style
C. a pistil
D. a stigma
E. an embryo

August 1991

SCIENCE Section A

SOLUTIONS

OBJECTIVE TEST

1.	A.	mouth
2.	B.	carbon dioxide
3.	C.	Copper
4.	E.	10.0 J
5.	E.	meteorologist
6.	A.	protein
7.	C.	Water
8.	D.	canning
9.	B.	K
10.	A.	sucker
11.	D.	capillarity
12.	C.	Frog
13.	E.	glass
14.	A.	increases
15.	B.	3
16.	C.	rabbit
17.	E.	Heart
18.	C.	Filtration

19. B. parasite

20. D. Bilharzia

21. C. Cabbage

22. E. surface tension

23. C. an element

24. B. Energy

25. A. Hygrometer

26. D. root

27. C. dissolves

28. D. nine months

29. A. erosion

30. E. mercury

31. C. reproduction

32. D. Sand

33. D. solid to liquid

34. E. Aluminium

35. C. Amino acid

36. B. a carnivore

37. E. an alloy

38. B. Preparation of akpeteshie

39. D. heat

40. E. an embryo

August 1991

SCIENCE

SECTION B
ESSAY
1 hour

[60 marks]

Answer **three** *questions* **only** *from this section.*
Illustrate your answers wherever possible, with large, clear and fully labelled diagrams.
Credit will be given for clarity of expression and orderly presentation of material.
All questions carry equal marks

1. (a) (i) What is the difference between self-pollination and cross-pollination?
(ii) Name two agents of cross-pollination.

(b) A ray of light strikes the surface of a plane mirror at an oblique angle.
Draw a ray diagram to show each of the following:
(i) the normal
(ii) the incident ray
(iii) the reflected ray
(iv) the angle of incidence
(v) the angle of reflection

(c) Give one property of the image formed by a plane mirror.

(d) (i) Carbon dioxide is prepared in the laboratory by pouring some dilute hydrochloric acid on calcium carbonate. Give a balanced equation for this reaction.
(ii) How would you test for carbon dioxide?
(iii) Give one use of carbon dioxide.

2. (a) Define the following processes:
(i) Excretion
(ii) Respiration
(iii) Name one organ in man which is responsible for each of the processes in 2(a) (i) and (ii) above

(b) Mention three ways of protecting people from infection during an outbreak of cholera.

(c) Write down the chemical names of the following compounds.
(i) $MgCl_2$
(ii) FeS
(iii) $CuSO_4$
(iv) NH_4OH

(d) List the names of the planets in order of their distances away from the sun; starting with the one nearest to it.

3. (a) (i) What is a solution?
(ii) Write down balanced chemical equations for the preparation of the following substances.
(α) Sodium Chloride: Prepared from a reaction between Sodium hydroxide and dilute hydrochloric acid
(β) Ammonia: Prepared from a reaction between nitrogen and hydrogen.

(b) Classify the following under voluntary and involuntary actions:
(i) Laughing
(ii) Blinking of the eyes
(iii) Sneezing
(iv) Eating

(c) (i) State four characteristics which are common to all living things.
(ii) Write down one function of each of the following:
(α) Leaves
(β) Roots

(d) (i) Mention two main defects of the human eye.
(ii) How are these defects corrected in each case?

4. (a) (i) Name the gas which is used in both rusting and burning.
(ii) State two ways in which rusting can be prevented in everyday life.

(b) Classify each of the following items as element, compound, mixture or colloid:
(i) Sodium chloride
(ii) Sugar solution
(iii) Fog
(iv) Hydrogen
(v) Sulphur

(c) (i) Use any three of the following to form a food chain:
Man, bird, green grass, goat, grasshopper

(ii) Name two diseases that can be controlled by immunization.

(d) (i) A boy throws a stone vertically up into the air. What are the energy changes that take place as the stone falls to the ground?

(ii) Give two appliances which use electromagnets in their operation

(iii) List two ways in which a magnet can lose its magnetism.

August 1991

SCIENCE Section B

SOLUTIONS

ESSAY

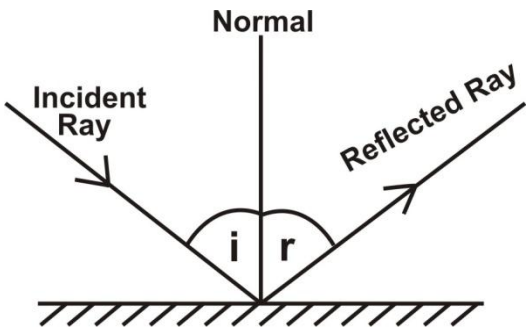
1. **(a)** **(i)**

Self-pollination	Cross-pollination
The transfer of pollen grains from an anther of a flower to the stigma of the same flower.	The transfer of pollen grains from the anther of a flower to the stigma of another flower on another plant of the same kind.

(ii) **Agents of cross-pollination**

Wind, water, insects, birds, man [any two]

(b) **Ray diagram showing**



(c) **Properties of image formed by plane mirror**

- Same size as object
- Same distance from mirror as object
- Virtual
- Erect / upright
- Laterally inverted [any one]

(d) **(i)** **Balanced equation for reaction**

$CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$

(ii) Test for carbon dioxide

Pass the carbon dioxide through lime water [$Ca(OH)_2$]; The lime water turns milky.

(iii) Uses of carbon dioxide

- Fire extinguisher
- Photosynthesis in green plants
- Refrigerant
- Production of soft drinks
- One of the gases used in anaesthesia

[any one]

2. (a) (i) Excretion

The removal of metabolic waste materials from the cells of a body

Or:

The discharge of waste matter from the body's tissues or organs

(ii) Respiration

The breakdown of food substances in an organism to release energy, carbon dioxide, water

(iii) Organ responsible for

Excretion: lung, skin, kidney [any one]

Respiration: lung

(b) Ways of protecting people from infection during an outbreak of cholera

Ensuring that (people)

- Receive immunization
- Eat hot meals,
- drink clean potable water,
- wash hands with soap after visiting the toilet
- cover meals to prevent flies from settling on them
- keep the environment clean and free from filth

[any three]

(c)

		CHEMICAL NAMES
(i)	**$MgCl_2$**	Magnesium chloride
(ii)	**FeS**	Iron (II) sulphide
(iii)	**$CuSO_4$**	Copper (II) sulphate
(iv)	**NH_4OH**	Ammonium hydroxide

(d) The planets in order:

Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune, (Pluto)

3. (a) (i) Solution

A liquid which has one or more other substances uniformly dissolved in it.

Or:
A fluid consisting of two or more substances mixed together and uniformly dispersed

(ii) Balanced chemical equation for:

(α) Sodium Chloride: Prepared from a reaction between Sodium hydroxide and dilute hydrochloric acid

$NaOH + HCl \rightarrow NaCl + H_2O$

(β) Ammonia: Prepared from a reaction between nitrogen and hydrogen

$N_2 + 3H_2 \rightarrow 2NH_3$

(b)

VOLUNTARY	INVOLUNTARY
Laughing	blinking of the eyes
eating	sneezing

(c) (i) Characteristics common to all living things

- They respire
- They excrete
- They feed
- They grow
- They reproduce
- They respond / are sensitive to stimuli

[any four]

(ii) Function of:

(α) Leaves

- Photosynthesis (preparation of food using carbon dioxide and water)
- Transpiration (loss of water vapour through the stomata)
- Respiration (intake of oxygen to produce energy)
- Guttation (oozing out of excess water from the surface of leaf)
- Food storage in certain leaves

[any one]

(β) Roots

- Absorbs water and mineral salts from the soil
- Some roots store food prepared by the plant
- Holds the plant firmly in place

[any one]

(d) (i) Main defects of the human eye

- Long sightedness (seeing from long distances only/ not seeing from a close distance)
- Short sightedness (seeing from short distances only/ not seeing from a long distance)

(ii) Correction of long sightedness

Convex / converging lens are used to correct long sightedness

Correction of short sightedness

Concave / diverging lens are used to correct short-sightedness

4. (a) (i) Gas used in both rusting and burning

Oxygen

(ii) Ways of preventing rusting

- Painting
- Greasing/ oiling
- Galvanizing

(b) Classify as element, compound, mixture or colloid

(i)	Sodium chloride	-	compound
(ii)	Sugar solution	-	mixture
(iii)	Fog	-	colloid
(iv)	Hydrogen	-	element
(v)	Sulphur	-	element

(c) (i) food chain

Green grass → grasshopper → bird

Or: Green grass → goat → Man

Or: grasshopper → bird → Man

(ii) Disease that can be controlled by immunization

Smallpox, Measles, Chicken pox, Whooping cough, Tetanus, Poliomyelitis, Tuberculosis, Diphtheria

(d) (i) Energy changes – stone falling to the ground

Potential → Kinetic → Sound + Heat

(ii) Appliances that use electromagnets

Speakers, motors, computer hard drives, microphones, telephone receivers, electric bells

(iii) Ways by which a magnet can lose its magnetism

- By heating it above a certain temperature (known as the Curie temperature)
- By placing it in a magnetic field of opposite polarity; either from another permanent magnet or a solenoid
- By a physical shock; through dropping it or hitting it with a hammer

[any two]

August 1990

SCIENCE

SECTION A
OBJECTIVE TEST
45 minutes

1. Which of the following is a liquid?
- A. Iron
- B. Mercury
- C. Brass
- D. Gold
- E. Aluminium

2. The space occupied by matter is its
- A. area
- B. length
- C. radius
- D. width
- E. volume

3. Which of the following processes could be used to obtain water from a salt solution?
- A. Filtration
- B. Decantation
- C. Evaporation
- D. Distillation
- E. Sedimentation

4. The portion of air which is used in burning is
- A. nitrogen
- B. water vapour
- C. oxygen
- D. carbon dioxide
- E. inert gas

5. Brine is a saturated solution of
- A. sodium chloride and carbon dioxide
- B. calcium carbonate and water
- C. sodium chloride and water
- D. calcium carbonate and kerosene

E. magnesium and water

6. The chemical symbol for iron is
A. I
B. K
C. Fe
D. Au
E. Cl

7. A torchlight derives its energy from
A. petrol
B. kerosene
C. a magnet
D. gas oil
E. a dry cell

8. A substance which cannot be broken down into simpler substances is called
A. a molecule
B. a compound
C. an element
D. a mixture
E. an atom

9. What type of energy is lost when sweat evaporates from the body?
A. light energy
B. mechanical energy
C. chemical energy
D. Heat energy
E. Electrical energy

10. The process by which steam is changed into liquid is called
A. evaporation
B. condensation
C. filtration
D. melting
E. sublimation

11. Which of the following is **not** true about frictional force? It
A. increases motion
B. opposes motion
C. damages machines
D. produces heat
E. decreases efficiency of machines

12. The instrument used in measuring the speed of wind is called:

A. Thermometer
B. Wind Vane
C. Photometer
D. Anemometer
E. Hygrometer

13. The force which comes into operation when a ball rolls on the ground is:
A. gravitational force
B. frictional force
C. magnetic force
D. electrostatic force
E. Force due to tension

14. Which of the following is true about a first class lever?
A. The load is between the effort and the pivot
B. The pivot is between the effort and the load
C. The effort is between the pivot and the load
D. The load is on the pivot
E. There is no pivot

15. In which of the following situations is work not done? When a
A. truck moves
B. wheelbarrow is pushed
C. fruit falls from a tree
D. football is kicked
E. book lies on a table

16. Which of these structures regulate the amount of light that enters the eye?
A. Retina
B. Cornea
C. Pupil
D. Choroid
E. Lens

17. All of the following are herbivores except
A. sheep
B. dog
C. goat
D. grasscutter
E. rabbit

18. The brain and the spinal cord are part of the
A. digestive system
B. circulatory system
C. respiratory system

D. nervous system
E. excretory system

19. Which of the following is used by green plants for respiration?
A. Carbon dioxide
B. Oxygen
C. Light
D. Water
E. Chlorophyll

20. A fish dies shortly after being removed from water. This is because it
A. cannot swim
B. is streamlined
C. is hot on land
D. has scales
E. cannot respire

21. Which of the following habitats are you likely to find a tadpole?
A. Bush
B. Sea shore
C. Pond
D. Deep well
E. Vegetable garden

22. Which of these is not a characteristic of all living things?
A. Transpiration
B. Growth
C. Reproduction
D. Feeding
E. Respiration

23. Plants with thorns are not usually eaten by animals. The thorns therefore
A. assist the plants to grow
B. make the plant strong
C. render the plant poisonous
D. protect the plant
E. make the plant unattractive

24. Which of the following soils is likely to be found in a waterlogged area?
A. Sand
B. Clay
C. Gravel
D. Black soil
E. loam

25. Which of the following soil organisms improve soil fertility?

A. Ant-lion
B. Red ant
C. Millipede
D. Snail
E. Earthworm

26. Which of the following is cultivated on plantation basis?

A. Okro
B. Tomato
C. Rubber
D. Cassava
E. Maize

27. The shape of a child's nose may resemble that of the mother due to

A. heredity
B. similarity
C. reflection
D. familiarity
E. attraction

28. All the following plants reproduce sexually except

A. coconut
B. sweet potato
C. pawpaw
D. apple
E. pear

29. Which of the following is not a mammal?

A. Whale
B. Bat
C. Cat
D. Hen
E. Mouse

30. Plants lose water through a process called

A. photosynthesis
B. fertilization
C. transpiration
D. respiration
E. evaporation

31. Fertilized ovules develop into

A. fruits
B. cotyledon

C. plumule
D. testa
E. seeds

32. In mammals, sperms are produced in the
A. testis
B. ureter
C. penis
D. urethra
E. ovary

33. Starch is digested and absorbed into the blood as
A. soluble starch
B. water
C. glucose
D. fat
E. sucrose

34. Which of the following organs are connected by the ureter?
A. Liver and kidney
B. Kidney and bladder
C. Bladder and heart
D. Intestines and anus
E. Lungs and bladder

35. A patient who coughs and spits out blood may be suffering from
A. asthma
B. pneumonia
C. measles
D. tuberculosis
E. influenza

36. All of the following promote Primary Health **except**
A. keeping the surroundings clean
B. effective disposal of waste
C. personal hygiene
D. balanced diet
E. bleaching the skin

37. The personnel who see to it that individuals who do not maintain hygienic surroundings are taken to court is the
A. police
B. lawyer
C. sanitary inspector
D. court clerk

E. prison officer

38. Which of the following statements is **not** true about the wet season

A. Bush fires are common
B. Tree and weeds grow fast
C. Small animals increase in number
D. The weather is generally cool
E. Food is in abundance

39. A mirror is used to direct the sun's rays onto the classroom wall by

A. radiation
B. reflection
C. conduction
D. revolution
E. convection

40. The sun and the planets form the

A. universe
B. atmosphere
C. milky way
D. constellation
E. solar system

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SCIENCE Section A

SOLUTIONS

OBJECTIVE TEST

1.	B.	Mercury
2.	E.	volume
3.	D.	Distillation
4.	C.	oxygen
5.	C.	sodium chloride and water
6.	C.	Fe
7.	E.	a dry cell
8.	E.	an atom
9.	D.	Heat energy
10.	B.	condensation
11.	A.	increases motion
12.	D.	Anemometer
13.	B.	frictional force
14.	B.	The pivot is between the effort and the load
15.	E.	book lies on a table
16.	C.	Pupil
17.	B.	dog
18.	D.	nervous system

19. B. Oxygen

20. E. cannot respire

21. C. Pond

22. A. Transpiration

23. D. protect the plant

24. B. Clay

25. E. Earthworm

26. C. Rubber

27. A. heredity

28. B. sweet potato

29. D. Hen

30. C. transpiration

31. E. seeds

32. A. testis

33. C. glucose

34. B. Kidney and bladder

35. D. tuberculosis

36. E. bleaching the skin

37. C. sanitary inspector

38. A. Bush fires are common

39. B. reflection

40. E. solar system

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SCIENCE

SECTION B
ESSAY
1 hour

[60 marks]

Answer **three** *questions* **only** *from this section.*
Illustrate your answers wherever possible, with large, clear and fully labelled diagrams.
Credit will be given for clarity of expression and orderly presentation of material.
All questions carry equal marks

1. (a) Define the following terms and give one example of each:
- (i) Compound
- (ii) Mixture
- (iii) Element
- (iv) Solute

(b) What energy changes take place in the following processes?
- (i) A mango drops from a tree.
- (ii) A carpenter hits a nail with a hammer.
- (iii) A torch is switched on.

(c)
- (i) Draw a well labeled diagram of a flower
- (ii) Of what importance is the flower to the plant?

(d) Name the sense organs and state one function of each.

2. (a)
- (i) What is soil erosion? Give two types of soil erosion.
- (ii) Give three activities of man that can cause soil erosion.
- (iii) List three ways by which soil can be conserved.

(b)
- (i) What is a lever?
- (ii) Classify the following under first class, second class and third class levers.
  - Sugar tong
  - Bottle opener

Paper cutter
A pair of scissors
Fishing rod
Claw hammer

(c) In each of the following activities, state whether the changes that occurred are:
(i) physical or chemical
(ii) reversible or irreversible
(α) Some boiled cassava and plantain were pounded together into fufu.
(β) A piece of ice block is put in a cup. After a while, it is found out that the block had changed into a liquid
(γ) A clean piece of nail was put into a beaker of water. After five days it was observed that the colour of the nail had changed.

3. (a) For each of the following diseases, name the causative organism and one method of prevention.
(i) Cholera
(ii) Bilharzia
(iii) Malaria
(iv) Tuberculosis

(b) Show how you would separate each of the following mixtures:
(i) Common salt and sand
(ii) Muddy water

(c) Explain why the planet and their moons move around the sun but they do not clash.

(d) Give two uses of the sun's energy in everyday life.

4. (a) (i) What is pollution?
(ii) Name two types of pollution. For each type, give one pollutant and its source.

(b) (i) Explain the term vegetative reproduction.
(ii) Give any two examples of plants that reproduce vegetatively.

(c) (i) List the parts of an atom and show the electrical charges of each of the parts.
(ii) Explain the term work.
(iii) Distinguish between heat and temperature and give their units.

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SCIENCE Section B

SOLUTIONS

ESSAY

1. (a)

(i) Compound:

A substance formed by the chemical combination of elements in fixed proportions

Or: a substance containing two or more elements that are chemically combined

(ii) Mixture

A physical combination of two or more substances

Or:

A substance consisting of two or more substances that have been combined physically

(iii) Element

A substance that is made up of the same kind of atoms

Or:

Any substance that cannot be broken down into a simpler one by a chemical reaction

(iv) Solute

A substance that is dissolved in another substance

(b) Energy changes

(i) A mango drops from a tree.

Potential → Kinetic → Sound

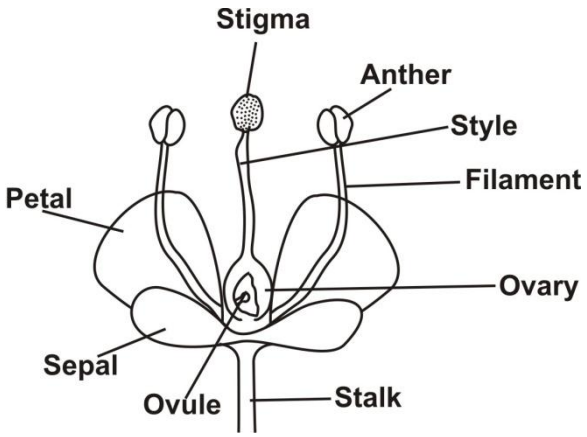
(ii) A carpenter hits a nail with a hammer.

Chemical → Potential → Kinetic → Sound + Heat

(iii) A torch is switched on.

Chemical → Electrical → Light

(c) (i) A well-labeled diagram of a flower



(ii) Importance of flower to plant

- Sexual reproduction by sex cells
- Seed development (ovule develops into seed)
- Fruit production (ovary develops into fruit)
- Attraction of birds and insects by petals / nectar (for cross pollination)

(d)

SENSE ORGAN	FUNCTION
Skin	Touch
Eye	Sight
Nose	Smell
Tongue	Taste
Ear	Hearing

2. (a) (i) Soil erosion

The washing away of the top soil by agents of erosion

Or: The removal of soil material by natural processes, principally running water, glaciers, waves, and wind

Types of soil erosion: Gully, Rill, Sheet and Splash [any two]

(ii) Activities of man that can cause soil erosion

- Removal of cover crops
- Bush burning
- Ploughing along slopes
- Felling of trees
- Sand-winning [any three]

(iii) Ways by which soil can be conserved

- Mulching
- Manuring

- Practicing Crop rotation
- Planting of cover crops
- Planting more trees
- Practicing strip cropping

(b) (i) A lever

Simple machine consisting of a rigid bar that rotates about a fixed point, called a pivot/ fulcrum

FIRST CLASS	SECOND CLASS	THIRD CLASS
A pair of scissors Claw hammer	Bottle opener Paper cutter	Sugar tong fishing rod

(c) (i) Physical or chemical (ii) reversible or irreversible

(α) Some boiled cassava and plantain were pounded together into fufu.

(i) Physical (ii) irreversible

(β) A piece of ice block is put in a cup. After a while, it is found out that the block had changed into a liquid

(i) Physical (ii) reversible

(γ) A clean piece of nail was put into a beaker of water. After five days it was observed that the colour of the nail had changed

(i) chemical (ii) irreversible

3. (a)

DISEASE	CAUSATIVE ORGANISM	METHOD OF PREVENTION
Cholera	bacteria (*Vibrio cholerae*)	▪ Immunization / vaccination ▪ Eating hot meals ▪ drinking clean potable water ▪ washing hands with soap after visiting the toilet and before meals ▪ covering meals to prevent flies from settling on them ▪ keeping our environment clean and free from filth ▪ ensuring that rubbish bins are properly covered
Bilharzia	flatworm (blood fluke /schistosome)	▪ avoid touching water bodies infected with the vector (water snails) ▪ apply chemicals to water bodies infected with water snails to kill the vector

		▪ Avoid polluting water bodies
Malaria	Protozoon (plasmodium)	▪ Using mosquito net ▪ Using mosquito repellent ▪ Using anti-malaria drugs ▪ Clearing mosquito breeding grounds such as choked gutters, stagnant pools of water, etc
Tuberculosis	bacteria (*Mycobacterium tuberculosis)*	▪ Immunization with BCG vaccine ▪ Ensuring adequate ventilation ▪ Avoid sharing cutlery, cups and glasses with infected persons

[any one for each]

(b) How to separate the mixtures:

(i) Common salt and sand

I. Add water to the mixture and stir to dissolve the salt

II. Pour the mixture through a filter paper in a funnel to separate the sand (residue) from the salt solution (filtrate)

III. Heat the salt solution till all the water evaporates, leaving the salt crystals behind

IV. Dry the sand and the salt obtained

(ii) Muddy water

- Fold a filter paper into a funnel placed in a beaker.
- Pour the muddy water through the filter paper in the funnel to separate the suspended mud particles (residue) from the water (filtrate)
- The water is collected in the beaker, leaving the mud on the filter paper.

(c) Why planets and their moons move around the sun without a clash

Each planet and each moon has a separate path (orbit) in which it moves. They are kept in those separate orbits by the gravitational forces of attraction between them and the larger bodies around which they move and hence do not clash.

(d) Uses of sun's energy in everyday life

- Provision of light for seeing
- Drying of clothes
- Preservation of food by drying
- Photosynthesis by green plants
- Electricity generation by solar cells
- Synthesis of vitamin D by the skin

[any two]

4. (a) (i) Pollution

Contamination of Earth's environment with materials that interfere with human health, the quality of life, or the natural functioning of *ecosystems*

(ii)

TYPE OF POLLUTION	POLLUTANT	SOURCE
Air pollution	▪ Smoke ▪ carbon monoxide ▪ sulphur dioxide ▪ Lead	▪ Bush /tyre burning, ▪ Motor-vehicle exhaust, Industrial processes ▪ Sulphuric acid plants, Power-generation facilities that use oil or coal containing sulphur ▪ Motor-vehicle exhaust, battery plants
Water pollution	▪ Sewage ▪ industrial wastes ▪ crude oil spillage ▪ Toxic chemicals	▪ Homes, offices ▪ Industries ▪ Oil tankers, oil drilling rigs, oil refineries ▪ Factories, Over-fertilized farmlands (through washing away by rain)
Land pollution	▪ Rubbish (non-decomposable waste) ▪ chemical fertilizers, ▪ fungicides & pesticides	▪ Homes, offices, factories ▪ Excessive fertilizer use by farmers ▪ Indiscriminate application of fungicides and pesticides

[any two types of pollution] [any one pollutant and its source for the two types]

(b) (i) Vegetative reproduction

Method by which plants reproduce without the union of cells or nuclei of cells

Or:

The process whereby plants produce offspring from growing parts / buds of the parent plant

(ii) Examples of plants that reproduce vegetatively

Cassava, sugarcane, yam, cocoyam, banana, plantain, ginger, pineapple, etc

(c) **(i)**

PART OF ATOM	ELECTRICAL CHARGE
Proton	+ 1
Neutron	0
Electron	– 1

(ii) **Work**

Work is done when an applied force moves an object in the direction of the force.

Or:

The transfer of energy, measured as the product of the force applied to a body and the distance moved by that body in the direction of the force

Or:

Work = $f \times d$, where f = force applied

and d = distance moved in the direction of the applied force

(iii)

HEAT	TEMPERATURE
The energy possessed by a body that results in its hotness or coldness	A measure of the hotness or coldness of the body
Unit = joule (J)	Unit = kelvin (K) or degee Celsius (°C) or degree Fahrenheit (°F)

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